Contents

Chapter 1

Introduction

1.1 From sound law to derivation

Historical linguists ordinarily test an etymology by carrying a reconstructed form through the sound changes that separate it from its alleged reflex. Most such derivations remain implicit. A scholar knows that Proto-Germanic **p* yields Old English *f*, that West Germanic **z* became *r* under the appropriate conditions, and that one change must have preceded another because the reverse order produces the wrong form. For a single word this mental arithmetic presents little difficulty. Across hundreds of words and scores of interacting changes it becomes treacherous. Each step may look familiar while the derivation as a whole is false.

In computer-assisted phonological reconstruction (CAPR), I represent each sound law as a finite-state transducer and compose the transducers in the order argued for below. The resulting cascade applies every change without discretion: the same rule, with the same environment, reaches every eligible form. A proposed reconstruction either yields the Old English comparator or it does not. Moving a rule may repair one derivation and destroy another. Both results deserve attention, for a failure locates the point at which phonology ceases to suffice and philology must decide among analogy, borrowing, dialect mixture, uncertain attestation, or a faulty reconstruction.

No new principle is involved. The Neogrammarians already demanded exceptionless sound laws and an account of apparent exceptions (Osthoff and Brugmann 1878--1910). Their formula was the *Ausnahmslosigkeit der Lautgesetze*. Formalization merely enforces that demand mechanically: every eligible form bears the stated consequences. The computer cannot decide which reconstruction is historically defensible, but it prevents the investigator from forgetting what that reconstruction entails.

I apply this method to the development of Proto-Germanic and early West Germanic forms into Old English. Germanic makes a severe test. Its historical grammar rests on two centuries of philological labor, while Old English offers abundant but orthographically and dialectally varied testimony (Campbell 1959; Hogg 1992; Ringe and Taylor 2014; Fulk 2018). A computational account cannot plead scarcity of evidence. It must reproduce familiar developments, identify the evidence for their order, and explain why its input sometimes differs from the headword printed in an etymological dictionary.

1.2 The formal claim

A sound change defines a relation between strings. For each etymology I supply a reconstructed input to an ordered series of such relations and compare the result with an Old English form. Kaplan and Kay demonstrated that the familiar rewrite rules of phonology admit a finite-state interpretation (Kaplan and Kay 1994); Foma gives that interpretation executable form (Hulden 2009). A toy rule deleting final **z* may be written:

```
define ToyFinalZLoss [{*z} -> 0 || _ .#.];
```

The notation states a target, an output, and an environment. Actual rules also refer to segment classes, stress, syllable weight, word boundaries, and the output of earlier changes. A loose prose formulation can conceal choices that code must resolve. Does the rule affect every vowel or only short vowels? Does a following consonant block it? Does it apply before or after the loss of a final syllable? A transducer forces each question into the open.

Order gives the cascade its historical content. An early change may create the environment for a later one and thus feed it; alternatively, it may remove that environment and bleed it. A word affected by both changes can therefore establish their relative chronology. Other changes commute across the available lexicon: reversing them alters no output in the active dataset. Such a result does not prove simultaneity or historical indifference. It shows only that the present evidence fails to order them. Throughout this book I distinguish chronology compelled by the tested forms from placement adopted on the authority of the historical grammars.

Backward reconstruction requires a further restriction. An unrestricted inverse transducer will propose formally possible strings that no Germanic language could have inherited. I therefore restrict backward reconstruction with an inventory and a statement of permissible ancestral forms. Reconstruction always combines correspondences with a theory of what could count as a word in the *Grundsprache*; this restriction states that theory rather than leaving it tacit.

1.3 Inputs, targets, and success

I compare a selected earlier Germanic form with a selected Old English target. Neither selection is innocent. Dictionaries cite lexemes, but sound change operates on word-forms. The ancestor of an Old English plural, preterite, or oblique case may differ from the reconstructed lemma in precisely the material on which a later sound law acts. Kroonen and Orel provide indispensable lexical reconstructions, while the grammars often supply the paradigm history needed to choose the actual input (Orel 2003; Kroonen 2013; Ringe and Taylor 2014).

I therefore distinguish the citation reconstruction from the transducer input. The former identifies the etymon; the latter represents the paradigm cell or remodeled stem whose history is at issue. Where the two differ, the lexical entry states the difference and argues for it. This prevents a convenient input from masquerading as a received reconstruction. An unmotivated alteration made solely to secure the desired output would empty the exercise of historical meaning.

The distinction is concrete in Old English *sċuldrum* ‘shoulders, dative plural’. Its inflectional history requires an input of the type **skúldramiz* ‘shoulders, reconstructed dative-plural input’; the singular headword represents a different paradigm cell.

The target also requires judgment. Old English spelling varies by date, dialect, manuscript, and editorial practice. The cascade may produce an internal phonological symbol that an orthographic transducer then maps to a normalized written form. A string match at this final stage cannot by itself establish an etymology; conversely, a superficial spelling mismatch need not disprove one. I treat phonological development and orthographic normalization as separate operations so that exact computation does not confuse notation with history.

Within these limits a successful derivation has three senses. It succeeds formally when the output string matches the target. It succeeds philologically when the chosen input and comparator are the proper forms to compare. It succeeds historically when the proposed path agrees with the wider Germanic evidence. The first kind of success is cheap. The argument of the book concerns the conjunction of all three.

1.4 The evidence of failure

Regular sound change makes irregularity legible. If an inherited form refuses to pass through an otherwise successful cascade, the mismatch demands a name. Analogy may have replaced the expected reflex with a form drawn from another paradigm cell. Borrowing may have introduced the word after the relevant changes. A dialectal form may lie outside the modeled West Saxon path. The target may be late, corrupt, or normalized beyond what the manuscript evidence warrants. Finally, either the reconstruction or the rule may be wrong.

These possibilities should not be suppressed by narrow, lexeme-specific "sound laws." A system that derives every target by multiplying exceptions has only encoded its answers. I instead distinguish several classes of non-regular result: attested variants, early and late analogy, reconstructed Old English comparators, known but unmodelled phonological and morphological developments, and unexplained exceptions. The categories are claims, not housekeeping labels. They identify where regular phonology ends and what additional history the evidence requires.

This treatment follows the original Burmish CAPR work, in which resistant forms often disclosed loans or mistaken cognate assignments. Old English shifts the balance toward morphology and analogy, but the methodological advantage remains the same. Failure concentrates inquiry. It tells us which assumption—input, target, environment, order, or lexical history—must bear the explanation.

1.5 Evidence for relative chronology

The chronology chapters combine three kinds of evidence. First come the statements of the standard historical grammars. These establish the received description and often the broad order of developments (Campbell 1959; Hogg 1992; Ringe and Taylor 2014; Fulk 2018). Second come individual witness words. A derivation that succeeds under one order and fails under the reverse order supplies direct lexical evidence for that relation. Third come exhaustive order tests across the active dataset. These reveal whether an apparently decisive relation is local, whether other words contradict it, and how far a rule can move without disturbing any output.

Suppose that a consonant change creates the environment for a later vowel change. Under the received order both apply and the Old English target emerges. Reverse them and the vowel change misses its environment. The word then supports the priority of the consonant change. The converse case is equally informative: if an early rule creates a segment that a later rule would wrongly alter, the creating rule must follow the other. When reversal changes no output, the lexical evidence leaves the order open even if philological considerations still favor one placement.

A note on notation: a superscript dagger placed immediately before an italicized form marks a concrete predicted output generated by a counterfactual manipulation of the rule order (for example, moving a change earlier or later than argued). This dagger indicates that the marked form is a model prediction, not an independently attested historical form; reconstructions remain starred in the usual way.

Sims-Williams argues for mechanizing precisely this kind of reasoning (Sims-Williams 2018). Computation does not replace the historical argument; it makes the extent of that argument measurable. A traditional chronology may prove correct but less tightly constrained than its customary presentation suggests. A relation described as local may in fact rest only on a broad terminus. Such negative results are salutary. They separate what the data demonstrate from what a convenient exposition merely presupposes.

1.6 Rules and words

The book accordingly moves twice through the same history. Part I begins with the ordered rules. For each development it states the historical problem, gives the rule in formal notation, and examines the evidence for placement. The code appears because it is part of the claim, but the code-name is never an explanation. `OEIUmlaut`, for example, is only a rule name; its linguistic content lies in the stated environment, the historical discussion, and the words whose derivations depend upon it.

Part II begins with the words. Each entry identifies the reconstruction, the selected input, the Old English comparator, and the derivational class. A trace shows which rules altered the form; the accompanying prose explains the philological decisions that the trace cannot make. Short entries record ordinary derivations. Longer ones treat doubtful reconstructions, variant attestations, paradigm-cell selection, or analogy. The asymmetry is intentional: equal database rows do not warrant equal historical discussion.

The reader can thus move in either direction. A chronology chapter names the lexical witnesses that constrain a rule; their entries display the complete derivations. A lexical entry invokes a change; the corresponding chapter explains its formulation and place in the cascade. The index verborum provides a third route through the material, gathering reconstructed and attested forms by language.

1.7 Reproducibility and disagreement

An executable derivation identifies the exact point of disagreement. One reader may accept a sound change but reject its environment; another may accept the rule and dispute its order; a third may object to the selected paradigm cell or to the normalization of the Old English target. Each objection addresses a recorded decision. Given the same inputs, rules, and order, the stated outputs follow. Reproducibility here concerns those consequences, not the surrender of philological judgment to an algorithm.

Prose can sound settled while concealing several incompatible derivations. Code is less accommodating. A rule that is too broad damages words it was never designed to explain; one that is too narrow leaves legitimate reflexes untouched. A reconstruction plausible in isolation may fail after three later changes. The resulting embarrassment is a virtue of the method. Applying every rule to the whole lexicon subjects local analyses to a global test.

The converse danger is false precision. A deterministic cascade may tempt the reader to mistake exact strings for exact history. Every result still depends on choices about segmentation, symbol inventory, reconstruction, morphology, dialect, chronology, and orthography. I therefore cite the philological sources, display the selected forms, preserve unresolved exceptions, and distinguish tested order from source-based order. Those choices remain accountable to philology.

1.8 The argument

A book must advance an argument. Mine is that traditional rule-based reconstruction becomes clearer when every proposed derivation can be executed, every rule must face the whole lexicon, and every failure is reported. The Germanic-to-Old-English case shows both the power and the boundary of that claim. Much of the history admits a coherent ordered account. The residue does not disappear: it resolves into morphology, analogy, variation, borrowing, imperfect attestation, and a small number of genuine problems.

The sound-change chapters ask how tightly the lexical evidence constrains the cascade. The lexical chapters ask whether individual etymologies survive that cascade without philological sleight of hand. Neither inquiry suffices alone. A formal system that reaches the targets by hiding doubtful inputs is bad historical linguistics; elegant prose that never reproduces its derivations is an incomplete formal account. Executable derivations make those two standards answer to one another.

Part I

Sound changes, formalization, and relative chronology

Sound-change overview

Scope and orientation

The sequence begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

Numbering note

SC numbers remain the established legacy identifiers. The Version 1 book presents the changes in historical chapter order, which differs from the computational cascade order for several rules.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC077 is unused.

Chapter 2

From Proto-Northwest Germanic to Proto-West Germanic

2.1 Historical interval

This chapter covers the sound changes that took place in the proto-language shared by the West Germanic languages — Old English, Old Frisian, Old Saxon, Old High German, and Old Dutch — before the individual languages diverged. The starting reconstruction is Proto-Northwest Germanic (PNWGmc), the hypothetical common ancestor of North Germanic and West Germanic together; the ending reconstruction is Proto-West Germanic (PWGmc), the immediate common ancestor of the West Germanic languages specifically.

2.2 Scope and internal diversity

Changes in this chapter are not all equally pan-West-Germanic in scope. They may be grouped broadly as follows:

Northwest Germanic innovations (shared by both North and West Germanic): innovations in the unstressed vowel system, certain final-syllable vowel changes, and selected consonant cluster simplifications. Changes labelled `NWGmc` in the CAPR rule names fall here, though rule prefixes are not always reliable guides to historical scope.

Proto-West-Germanic innovations (shared within West Germanic but not in North Germanic): the cluster of morphological and phonological changes that distinguish Old English, Old High German, Old Saxon, and Old Frisian from Old Norse. Changes labelled `PWGmc` in the CAPR rule names generally fall here. They include early apocope rules, certain consonant assimilations, and the West Germanic gemination of consonants before `*j`.

2.3 Major changes

The chapter opens with the root-noun nominative `*-z` loss (SC096), the generalization of endingless nominatives through the athematic consonant stems, complete before Proto-West Germanic: none of the West Germanic daughters shows any ending in this class (Ringe and Taylor 2014, 118). It is the earliest of the three historically distinct final-`*z` developments; the other two (SC020 and SC097) open Chapter 2.

The unstressed `*ai > *ē` development (SC014) represents one of the most pervasive shared NW–West Germanic vowel shifts, turning unstressed endings such as the dative singular and strong-adjective plural to longer vowels. Ringe and Taylor treat this as one of the clearest post-PNWGmc shared developments (Ringe and Taylor 2014, 40–41); Fulk groups it among the North/West-Germanic shared innovations that distinguish the period from Gothic (Fulk 2018, § 5.2). The corresponding stressed monophthongization (SC004) belongs later in the cascade and is treated in Chapter 2.

The West Germanic consonant changes of this chapter — j-gemination (SC010), early i-apocope (SC006), coronal-w assimilation (SC008), and related rules — represent the most productive phonological territory for the CAPR derivations. They feed a large proportion of the distinctive consonant clusters of Old English. Handbooks vary in exactly how they group and name these changes (Campbell 1959, §§ 404, 406; Hogg 1992, § 7.1).

The nasal spirant corridor (SC026–SC027), treated in Chapter 2 at its cascade position, illustrates a type of change common in historical grammars of the "Ingvaeonic" or "North Sea Germanic" area: nasals disappear before voiceless fricatives, with compensatory vowel lengthening (Campbell 1959, §§ 462–463; Hogg 1992, § 7.77). The CAPR model splits this into two ordered steps to make the vowel effect computationally tractable; the book prose explains that split against the handbook tradition, which typically presents the change as a single process.

Chapters in this part of the book follow the executable cascade order, which models the reconstructed chronology itself. Several rules that carry `PWGmc` labels — final bare-`*a` loss (SC041), surviving bimoric `*ō` unrounding (SC042), and Sievers-law syncope (SC050) — execute later in the cascade and are therefore presented in Chapter 3, where their individual sections discuss their historical stage labels. Conversely, one rule with a West Saxon label, the palatal-glide rule (SC016), executes early and is presented in this chapter; its section notes the mismatch, which will be resolved in a later renaming pass.

One historically Proto-Germanic change, Gm-simplification (`SC002 PGmcGmSimplification`), precedes everything in this chapter as a support stage of the cascade. It is documented in the book-entry plan and its literature dossier confirms the source base is narrow (two lexical families: **draugma-* 'dream' and **taugma-* 'team'; (Kroonen 2013, 101, 511)). A reader-facing section for SC002 awaits a stronger explanatory source base and is not yet assembled in the reader-facing sequence.

2.4 A note on source terminology and subgrouping

The literature uses several partly overlapping stage labels for this period:

- Northwest Germanic: the node uniting North and West Germanic.
- Proto-West Germanic: the node uniting only the West Germanic languages.
- North Sea Germanic or Ingvaeonic: a proposed subgroup within West Germanic covering Old English, Old Frisian, and Old Saxon (and sometimes Old Low Franconian), sharing certain innovations over a broader area.
- Anglo-Frisian: a narrower proposed subgroup linking only Old English and Old Frisian.

These labels are not always used consistently across sources. Ringe and Taylor are cautious about reconstructing a discrete Proto-West-Germanic node (Ringe and Taylor 2014, 50–55). Campbell notes that many of the "West Germanic" shared features could alternatively be treated as parallel developments rather than common inheritance (Campbell 1959, §§ 1–5).

CAPR uses `PWGmc` and `NWGmc` as organizing labels for this chapter without claiming to have settled all questions about West Germanic subgrouping. Changes that appear in the literature under "Ingvaeonic" labels but affect the Old English–to-Proto-Germanic derivation chain are treated here as late expressions of the same West Germanic developmental period unless existing CAPR dossier research specifically argues for Anglo-Frisian or English-specific placement.

2.5 Rule names

The CAPR rules in this chapter carry names beginning with `NWGmc` or `PWGmc`. These names are stable internal identifiers. A name beginning with `NWGmc` does not guarantee that the change is exclusive to Northwest Germanic, and a name beginning with `PWGmc` does not guarantee that it is absent from North Germanic. The historical analysis in each sound-change section takes priority over the name prefix.

2.6 Root-noun nominative **-z* loss

Historical discussion

The athematic consonant stems — the "root nouns" of the handbooks — attached the nominative-singular marker directly to a consonant-final root, and the Proto-Germanic outcome of that collision is genuinely uncertain. Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states plainly that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306, §4.3.4). His own paradigm tables carry the uncertainty into print: the nominative of 'foot' appears as "fōts? (fōs?)", while 'mouse' is plain **mūs*, its expected extra sibilant already absorbed by degemination (Ringe 2017, 149, 313). Ringe and Taylor repeat the same three-way agnosticism — the root-noun nominative "either ended in **-s* or **-z*, or was endingless" (Ringe and Taylor 2014, 28, §2.3.1).

The dictionary traditions encode this situation in different notations, and the differences are conventions of citation rather than competing claims of fact. Orel prints morphologically explicit nominatives with final *-z* across the whole class: **bōkz* 'book', **gansz* 'goose', **lūsz* 'louse' (Orel 2003, 52, 126, 252). Kroonen cites the same words as bare stems or endingless forms, **bōk-* and **gans-* (Kroonen 2013, 71–72, 168–69), and Kluge/Seebold print a third variant with voiceless *-s*, as in **bōks* 'Buch' (Kluge 2011, 158). Bammesberger shows why the marker is nonetheless real: for voiced-final stems the overt nominative **-z* is positively reconstructible — **burgz* (Gothic *baúrgs*), **frijōndz* (Gothic *frijonds*) — even though **fōt-z* is "phonotaktisch kaum denkbar", and in West Germanic the ending simply "fiel es ab" (Bammesberger 1990, 190–92, §8.2.3.1). CAPR retains Orel's morphologically explicit forms as its inputs precisely because they record the inflectional marker whose fate this rule describes.

The three focal words are only superficially parallel. The root-final *s* of 'louse' is itself an extension of **luw-* on the model of 'mouse' (Bammesberger 1990, 195, §8.3). For 'goose', Szemerényi's lengthening would give a Proto-Indo-European nominative **ǵʰanss* > **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196, §8.3; Kroonen 2013, 168–69). 'Book' preserves a plain obstruent-final root. What unites them is not a single phonetic history but membership in a paradigm class that generalized the endingless nominative.

The branch evidence dates and localizes that generalization. Gothic keeps its sibilant throughout the class (*baúrgs*, *nahts*, *reiks*) (Fulk 2018, 165–66, §§7.26–7.27). Old Norse redistributes the ending morphologically: masculine root nouns keep *-r* (*fótr*), feminines are endingless (*nótt*, *geit*), while the vocalic-stem feminines *kýr*, *sýr*, *ær* retain *-r* from **-z* with R-umlaut (Fulk 2018, 167, §7.28; Bammesberger 1990, 192–93). West Germanic alone is uniform: there was "no ending in PWGmc, as none of the daughters exhibits any" (Ringe and Taylor 2014, 118, §3.4). Fulk supplies the mechanism: Szemerényi's law removed the nominative sibilant after sonorant-final stems, and endinglessness then spread analogically through the class (Fulk 2018, 143, §7.2). The development is therefore best understood as a morphological generalization enacted differently in each branch — absolute in West Germanic, consonant- and gender-conditioned in North Germanic, absent in Gothic — rather than as one exceptionless sound law.

This change is distinct from the two later final-**z* developments. It was complete before Proto-West Germanic, whereas the loss of **-z* in unstressed syllables (SC020 EAFFinalZDeletion) is itself a Proto-West Germanic change: polysyllabic consonant-stem nominatives such as **fadurz* 'father' — and, in this corpus, **frijōndz* 'friend', **melukz* 'milk', and **mēnōþz* 'month' — kept their ending into Proto-West Germanic and lost it there in an unstressed syllable (Ringe and Taylor 2014, 44–45, §3.1.1). The still later northern loss of **-z* in stressed monosyllables (SC097 MonosyllabicFinalZLoss) affects vowel-final monosyllables like **hwaz* and does not touch the consonant-final root nouns at all, whose ending was gone long before.

SC096. Root-noun nominative **-z* loss (`RootNounNomZLoss`)

```
define RootNounNomZLoss [{*z} -> 0 ||
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
        EnglishStarVocalic+
```

```
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.];
```

The rule deletes word-final **z* after a consonant in a monosyllable. Three claims of different kinds meet here and must be kept apart. The historical claim is morphological: the nominative-singular ending was lost in the athematic root-noun class, so that this one paradigm cell came to lack its marker — a development complete before Proto-West Germanic (Ringe and Taylor 2014, 118, §3.4). The lexical claim belongs to the dictionaries: Orel's citation forms, which supply the corpus inputs, print that marker explicitly as *-z* in **bōkz*, **flauxz*, **gansz*, and **lūsz* (Orel 2003, 52, 105, 126, 252). The executable statement is neither of these but a computational proxy for them: 'delete word-final **z* after a consonant in a monosyllable'. It is not proposed as a Proto-Germanic sound law; it earns its place only because every form the corpus submits to the morphological development is a consonant-final monosyllable, so the narrow phonological statement covers the class exactly. Four corpus derivations witness the rule, each yielding its expected Old English outcome: PGmc **bōkz* 'book' yields OE *bōc* 'book', **gánsz* 'goose' yields *gōs* 'goose', **lūsz* 'louse' yields *lūs* 'louse', and **fláuxz* 'flea' yields *flēah* 'flea'. Should the corpus ever acquire a consonant-final monosyllable in **-z* that is not a root-noun nominative, the proxy and the morphology would come apart, and the rule would need to be re-scoped; the project's regression tests pin the firing population to exactly these four words so that any fifth firing forces that adjudication rather than passing silently.

The rule applies at the head of the English line, before SC009 PWGmcIjContraction. That ordering is fixed by the identity of the process rather than by a wrong form: contraction turns the polysyllabic **fríjōndz* 'friend' into a monosyllable, and if the root-noun rule applied after contraction it would capture **friundz* — yet the ending of 'friend' survived into Proto-West Germanic and fell in an unstressed syllable, the change described under SC020 EAFFinalZDeletion (Ringe and Taylor 2014, 44–45, §3.1.1). Because both rules delete the same segment, moving this rule later changes no Old English output; the early placement keeps the derivation of 'friend' aligned with the historical account rather than with an accident of the cascade.

Negative controls behave as the morphology predicts. Stressed monosyllables whose **z* follows a vowel — the domain of the later northern change — do not meet the post-consonantal environment. Medial **z* is untouched and remains available for rhotacism (SC003 EAFRhotacism): PGmc **déuzą* 'deer' still yields OE *dēor* 'deer' with its rhotacized medial consonant.

2.7 Early unstressed vowel changes

Historical discussion

The first change monophthongizes unstressed **ai*; the second carries early unstressed front-vowel leveling farther in forms such as *weorold* 'world'. Both have a diagnostic later boundary in the dataset.

Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the broad Northwest Germanic reduction of unstressed **ai* to a long mid vowel that merges with unstressed **e*, in final and nonfinal syllables alike (Ringe and Taylor 2014, 37–41). Two dative-singular endings in the dataset, span **spánnai* 'span' and meed **mízdai* 'meed', carry the change. The stressed development of **ái* to **ā* is treated separately as SC004 EAFAiMonophthongization.

SC014. Monophthongization of unstressed **ai* (PNWGmcUnstressedAiMonophthongization)

```
define PNWGmcUnstressedAiMonophthongization [
    {*ai} -> {*ē}
];
```

The dative-singular endings span **spánnai* 'span' and meed **mízdai* 'meed' carry this change; both give a final **ē*. If SC014 PNWGmcUnstressedAiMonophthongization is delayed until after SC072 OEUnstressedLongVowelShortening, the **ē* is no longer present for shortening, so PGmc **spánnai* 'span' yields †*spannē* rather than expected OE *spanne* 'span'. This shows that SC014 PNWGmcUnstressedAiMonophthongization must come before SC072 OEUnstressedLongVowelShortening in the modeled sequence.

Ringe and Taylor's merger of unstressed **ai* with long mid **ē* establishes the historical development, in final and nonfinal syllables alike. The stressed development of **ái* to **ā* is a separate and later change; see SC004 EAFAiMonophthongization.

Historical discussion of early unstressed front-vowel leveling

Campbell treats the merger of unstressed front vowels directly and also records the variation of *weorold* 'world' and *weoruld* 'world' (Campbell 1959, 141–42, 154–55). These forms supply SC015 PNWGmcILowering with a firmer lexical basis than the preceding change.

SC015. Leveling of early unstressed front vowels (PNWGmcILowering)

```
define PNWGmcILowering [
    {*i} -> {*e}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel,
    {*í} -> {*é}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel
];
```

The *weorold* 'world' and *weoruld* 'world' variants turn the general source claim into an ordering test. If SC015 PNWGmcILowering is delayed until after SC036 OEInterStressRaising, PGmc **wíràldu* 'world' yields †*wuruld* rather than expected OE *weorold* 'world'; earlier movement changes no output.

The derivation thus fixes front-vowel leveling before interstress raising while leaving its earlier boundary open.

SC016 OEWsPalatalGlide and SC017 PNWGmcULowering follow with a more tightly constrained local chronology.

2.8 Unstressed **a*-raising before final **m*

Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

SC005. Unstressed **a*-raising before final **m* (PNWGmcAToUBeforeM)

```
define PNWGmcAToUBeforeM [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
     ↪  ({*z})? .#.
];
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 PNWGmcULowering, PGmc **skúldramiz* 'shoulders' yields †*sċoldrum* rather than expected OE *sċuldrum* 'shoulders'; earlier placements converge on the expected output. The scope of the change is established by inflectional evidence across multiple paradigm types: a-stem dative plural ON *dǫgum* 'days', OE *dagum* 'days', OS *dagun* 'days', OHG *tagum* 'days', beside Gothic *dagam* 'days'; strong-adjective dative singular ON *góðum* 'good', OE *gōdum* 'good', OS *gōdum* 'good', beside Gothic *godamma* 'good' (OS also shows variant forms gōdumu and -un); and first-plural present ON *berum* 'we carry', OHG *berumēs* 'we carry', beside Gothic *baíram* 'we carry'. Across these sets, North/West Germanic shows unstressed *-um* where Gothic preserves *-am*. The derivation of *sċuldrum* 'shoulders' supplies a CAPR ordering witness for the relative chronology, but the cognate set for 'shoulder' does not contribute comparative evidence for the rule's historical scope.

2.9 Early i-apocope

Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geoguþ* 'youth' (Brunner 1965, §§ 145–146; Ringe and Taylor 2014, 141). Campbell's *dugup* 'troop' and *geogup* 'youth' examples belong to the same pattern (Campbell 1959, § 332).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

SC006. Early i-apocope (`PWGmcEarlyIApocope`)

```
define PWGmcEarlyIApocope [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
];
```

The absence of umlaut in *geoguþ* 'youth' provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLong-Diphthong, PGmc **skáwōθi* 'shows' yields †*sċēaweþ* rather than expected OE *sċēawaþ* 'shows'.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

2.10 Final **ō*-lowering before **r*

Historical discussion

Ringe and Taylor separate two points here: a broader shortening of vowels before word-final **r* in unstressed syllables (for which kinship **r*-stems such as PGmc **fadér* > PWGmc **fader* are a key diagnostic), and the specific **ō*-before-**r* development needed here (Ringe and Taylor 2014, 58–59). The direct lexical witnesses for **ō* in that environment are two independent etyma: PGmc **fedwōr* 'four' with WGmc reflexes OE *fēower* 'four', OFris *fiuwer* 'four', OS *fiuwar* 'four'; and PGmc **watōr* 'water' with OE *wæter* 'water'.

The rule is historically secure but narrow: final or pre-final **ō* before word-final **r*. The clearest evidence remains concentrated in the `four` and `water` material. No broader environment for **ō* is attested.

SC007. Lowering of final bimoric **ō* before **r* (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering [
    {*ō} -> {*a} || _ {*r} .#.
];
```

OE *wæter* 'water' reveals why lowering must precede SC043 EAFBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* 'water' yields †*water* rather than expected OE *wæter* 'water': brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. Comparative support for the **ō*-before-**r* rule comes from the two lexical witnesses **fedwōr* 'four' and **watōr* 'water' and their WGmc reflexes; kinship **r*-stems belong to the broader pre-**r* shortening context, not to a direct **ō* > **a* control. Within CAPR, *wæter* 'water' is the form that establishes ordering before brightening. No broader lowering of **ō* is attested.

2.11 Coronal-w assimilation

Historical discussion

Ringe and Taylor treat the assimilation of **dw* and **zw* to **ww* as a shared Proto-West-Germanic innovation supported by one example of each input cluster (Ringe and Taylor 2014, 56–57; Stiles 1985, 89–94). The **dw* example is the numeral 'four': PGmc **feðwor* (Gothic *fidwor*) → WGmc **fewwar* → OE *fēower*, Old Frisian *fiuwer*, Old Saxon *fiuwar*. The **zw* example is the second-person plural pronoun, where two oblique case forms show the change: acc./dat. PGmc **izwiz* (Gothic *izwis*) → OE *eow*, Old Frisian *iu*, Old Saxon *iu*, OHG *iu*; and gen. Ringe and Taylor's PGmc **izweraz* (Gothic *izwara*) → OE *eower*, OHG *iuwer* (Ringe and Taylor 2014, 56). Stiles discusses the same pronominal material using his own reconstruction conventions and explicitly treats Gothic *izwara* among the relevant comparanda (Stiles 1985, 89–94). These two case forms belong to a single pronominal paradigm, not to two independent etyma.

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but the evidence for each cluster is confined: the numeral alone supplies the **dw* instance, and the oblique case forms of the second-person plural pronoun supply the **zw* instance. Both clusters are now witnessed in the corpus: 'four' for **dw*, and 'you' — selected in its dat.(-acc.) plural cell **izwiz* — for **zw*, deriving through **iwwi*, apocopated **iww*, to OE *ēow* 'you' (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, § 8.3, pp. 204–205).

SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
];
```

OE *fēower* 'four' exposes a feeding relation: coronal assimilation must create **ww* while simplification can still reduce it. If SC008 PWGmcCoronalWAssimilation is delayed until after SC031 OEWWSimplification, PGmc **fédwōr* 'four' yields †*fēowwer* rather than expected OE *fēower* 'four'. Earlier placements alter no output.

The numeral fixes that relative order. The pronoun now fixes a second one: assimilation must precede rhotacism (SC003 EAFRhotacism). The **z* of **izwiz* stands between vowel and **w*; had rhotacism applied first, it would have produced †*irwiz*, from which OE *ēow* 'you' can never be derived. The executable cascade composes the assimilation well before rhotacism, and the corpus derivation of *ēow* 'you' fails if the two are reversed. 'Four' remains the sole **dw* witness and the sole source of the coronal-assimilation → *ww*-simplification ordering constraint. The earlier boundary of the assimilation remains undetermined.

2.12 *ij*-contraction in *friend*

Historical discussion

Ringe and Taylor describe a change of **ijo* to **iu* in the ancestor of *friend*, with the pathway PGmc **frijōnd-* (Gothic *frijonds*) → PWGmc **friund* → OE *frēond*, Old Frisian *frīund*, Old Saxon *friund*, Old High German *friunt* (Ringe and Taylor 2014, 62). The same source immediately warns that the **ijo* sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62). Luick (printed p. 118) notes that *iu* generalised within several *j*-stem paradigms through a related but differently conditioned loss of *j*, but does not supply a second example of the exact stressed **ijo* sequence (Luick 1914--1940, 118).

The change concerns a rare sequence attested only in the **frijōnd-* etymon and cannot safely be generalized into a broadly productive rule.

SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

```
define PWGmcIjContraction [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

Only the **frijōnd-* etymon tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* ‘friend’ yields †*friund* rather than expected OE *frēond* ‘friend’; moving contraction earlier within the tested range changes no output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond the **frijōnd-* etymon, precisely the reservation made by Ringe and Taylor.

2.13 West Germanic j-gemination

Historical discussion

Fulk treats West Germanic consonant gemination before `*j` after a short vowel as a regular development and illustrates it with forms such as OE *settan* 'set' and *lecgan* 'lay' (Fulk 2018, 127, §6.15).

The change applies specifically after a short vowel before **j*, not to geminate consonants generally.

SC010. West Germanic j-gemination (`PWGmcJGemination`)

```
define PWGmcJGemination [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

OE *nett* 'net' fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátją* 'net' yields †*nete* rather than expected OE *nett* 'net'. Earlier movement of gemination changes no output.

The chronology is phonologically transparent: the consonant must geminate before **j* ceases to be consonantal. The witness establishes no earlier boundary.

2.14 Syllabic j after final-vowel loss

Historical discussion

Ringe and Taylor state directly that after final unstressed `*a` and `*ą` were lost, postconsonantal `*j` became syllabic `*i`, with outcomes behind OE *here* 'army' and *rice* 'kingdom' (Ringe and Taylor 2014, 46).

The sources establish the development, although the lexical evidence supplies little independent support for its position. Its scope is postconsonantal j, not high-vowel vocalization generally.

SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

```
define PWGmcSyllabicJ [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

The same PGmc **nátją* 'net' witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGemination yields †*nete* rather than expected OE *nett* 'net'; moving it later changes no output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* 'army' and *rice* 'kingdom' as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

2.15 *lþ*-voicing

Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan* 'fold', *beald* 'bold', *wuldor* 'glory', and *gylden* 'golden' (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold* 'gold', and *feld* 'field' (Campbell 1959, 169, §414).

The comparative evidence supports *lþ* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

SC012. Northern West Germanic *lþ*-voicing (`EAFLThVoicing`)

```
define EAFLThVoicing [
    {*θ} -> {*d} || {*l} _
];
```

The `field`, `fold`, `gold`, and `wold` families preserve **lþ* to **ld*, but none dates the change against a neighboring rule. Every output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lþ* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

2.16 Dental hardening

Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative *ð became stop *d in all positions (Ringe and Taylor 2014, 43).

The change is systemic across early West Germanic and extends beyond any one lexical family.

SC013. Dental hardening (`PWGmcDentalHardening`)

```
define PWGmcDentalHardening [
    {*ð} -> {*d}
];
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

2.17 Northwest Germanic u-lowering

Historical discussion

Northwest Germanic lowered **u* to **o* when the following syllable contained a non-high vowel. Campbell describes the change and lists *ġeoc* 'yoke' among its regular outcomes (Campbell 1959, 42–43, §115); Fulk gives the same word as a standard example — "OIcel. ok, OE geoc, OHG joh beside juh and OS juk" — and notes the paradigmatic alternation between lowered and unlowered stems that the conditioning produced (Fulk 2018, 56, §4.3). A word-initial **j* does not block the change: the blocking effect of *j* concerns only a consonantal *j* standing between the target vowel and the conditioning vowel, as in the class I weak verbs of the *cnyssan* 'strike' type (Fulk 2018, 56, §4.3). Ringe and Taylor accordingly reconstruct the Proto-West Germanic paradigm of 'yoke' with the lowering applied (Ringe and Taylor 2014, 129).

The clearest corpus witnesses are *ġeoc* 'yoke', *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow'. Where the following syllable kept a high vowel the lowering did not apply, as in *ġeoguþ* 'youth', whose root *u* survived (Brunner 1965, 64–65, §92.1).

SC017. Lowering of **u* before following non-high vowels (PNWGmcULowering)

```
define PNWGmcULowering [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
];
```

Lowering of *u* to *o* is fixed on both sides by *ġeoc* 'yoke', *nosu*, *sċofl* 'shovel', and *sorg*.

The lowering feeds the much later West Saxon palatal-glide spelling (SC016 OEWsPalatalGlide): the *o* that the scribes wrote in *ġeoc* 'yoke' is the output of this change, so PGmc **júką* 'yoke' passes through **joką* on its way to the attested spelling (Fulk 2018, 56, §4.3; Ringe and Taylor 2014, 129). After SC019 PNWGmcFinalLongORaising, PGmc **núsō* 'nose' yields †*nusu* rather than expected *nosu*, PGmc **skúflō* 'shovel' yields †*sċufl* rather than expected *sċofl* 'shovel', and PGmc **súrgō* 'sorrow' yields †*surg* rather than expected *sorg*. These witnesses place SC017 PNWGmcULowering before final long-*o* raising, and the *ġeoc* spelling shows its output surviving into the written record.

2.18 Stressed monosyllable ***ō*-raising

Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' (Campbell 1959, 47, §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

SC018. Raising of final stressed monosyllabic **ō* (PNWGmcStressedMonosyllableORaising)

```
define PNWGmcStressedMonosyllableORaising [
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
];
```

Campbell's *cū* 'cow', *hū* 'how', and *tū* 'two' establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 PNWGmcStressedMonosyllableORaising with neighboring changes leaves every output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

2.19 Raising of final unstressed long **ō*

Historical discussion

Ringe and Taylor describe the change of unstressed final non-nasalized long **ō* to short **u* as a Northwest Germanic development (Ringe and Taylor 2014, 30). It applies in the same final-syllable environment as the subsequent loss of word-final **z*. The derivation of *rœste* 'rest' fixes their local order: SC019 PNWGmcFinalLongORaising must still see final **ō*, and word-final **z*-deletion removes the following **z* only afterward.

The change supplies the final vowel of forms such as *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow'.

SC019. Raising of final unstressed long **ō* (PNWGmcFinalLongORaising)

```
define PNWGmcFinalLongORaising [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
];
```

Two groups of witnesses confine final unstressed long **ō* > **u*. The forms *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow' fix its lower boundary.

Before SC017 PNWGmcULowering, PGmc **núsō* 'nose' yields †*nusu* rather than expected OE *nosu* 'nose', PGmc **skúflō* 'shovel' yields †*sċufl* rather than expected *sċofl* 'shovel', and PGmc **súrgō* 'sorrow' yields †*surg* rather than expected *sorg* 'sorrow'. After word-final **z*-deletion (SC020 EAFFinalZDeletion), PGmc **rástōz* 'rest' yields †*rast* rather than expected *rœste* 'rest'. These failures place SC019 PNWGmcFinalLongORaising after u-lowering and before final *z*-loss.

Chapter 3

From Proto-West Germanic to Anglo-Frisian

3.1 Historical interval

This chapter covers the sound changes that occurred after the Proto-West Germanic period and before, or during the emergence of, the specifically English line. The starting reconstruction is Proto-West Germanic; the end point is the Proto-Anglo-Frisian stage — or more precisely, the cluster of innovations that define the English and Frisian branch within West Germanic.

3.2 A necessary terminological caution

The title of this chapter uses “Anglo-Frisian” as an organizing historical concept. That choice requires an explicit qualification.

The scholarly literature uses several overlapping terms for this developmental period:

- North Sea Germanic and Ingvaeonic: labels used by some scholars for a proposed subgroup comprising Old English, Old Frisian, and Old Saxon (or more narrowly, Old English and Old Frisian only). The innovations associated with this label — especially the nasal spirant changes and certain vowel developments — are sometimes described as diffusion rather than shared inheritance (Campbell 1959, §§ 1–3).
- Anglo-Frisian: a label used specifically for the Old English / Old Frisian branch, or for innovations shared between the two languages. Its use presupposes a tighter relationship between English and Frisian than between either and Old Saxon.
- Proto-Anglo-Frisian (PAF): the strongest interpretation, positing a discrete reconstructed common ancestor for Old English and Old Frisian specifically. This is the position of Ringe and Taylor, who reconstruct a PAF stage between Proto-West Germanic and Proto-Old-English (Ringe and Taylor 2014, 54–68).

CAPR does not commit to a universally accepted discrete PAF node. The chapter title uses “Anglo-Frisian” because the key changes of this period — especially West Germanic rhotacism (SC003), word-final *z deletion (SC020), and Anglo-Frisian brightening (SC043) — are most prominently associated with that label in the handbook literature. But the analysis does not require that every change passed through a single genealogical PAF stage. Some changes may be West Germanic broadly; others may reflect areal diffusion. The existing CAPR dossiers record the source-by-source picture where these distinctions matter.

3.3 Major changes and their historical basis

West Germanic rhotacism (SC003)

The medial change of `*z` to `*r` in environments such as **déuzaz* 'deer', **xúrdaz* 'hoard', and **líznōjaną* 'learn' is historically a post-Proto-West-Germanic development. Ringe and Taylor argue that rhotacism was not inherited from Proto-Northwest Germanic and was not uniform within West Germanic (Ringe and Taylor 2014, 52, 98, 102). Crist separates this change explicitly from the deletion of word-final `*z` and argues that rhotacism must follow the deletion rules (Crist 2001, 104–6; 2002, 1, 4). Hogg gives the standard Old English–facing summary: `*z` yielded `*r` in intervocalic position but was generally lost in final position (Hogg 1992, 37).

The CAPR rule is named `EAFRhotacism`, placing it in the Early Anglo-Frisian corridor, CAPR's operational post-Proto-West-Germanic stage on the English line; the reader-facing chapter label describes the change as a West Germanic rhotacism.

Word-final `*z` deletion (SC020) and the three final-`*z` developments

The loss of word-final `*z` is not one process but three historically distinct developments, and this chapter contains two of them. The central one, SC020, is the Proto-West Germanic loss of `*z` in unstressed syllables, seen in forms such as **rástōz* 'rest (nom.sg.)' and stated for the whole branch by Ringe and Taylor (Ringe and Taylor 2014, 44–45); Crist's analysis distinguishes it both from the earlier NWGmc changes and from the later narrower Ingvaeonic deletion rules (Crist 2002, 1, 4). Earlier still, the consonant-stem root nouns had generalized endingless nominatives before Proto-West Germanic (SC096, Chapter 1). Later, and only in the north, `*z` was lost in stressed monosyllables with compensatory lengthening (SC097, this chapter); the southern dialects instead retained and rhotacized it. The standard handbooks confirm the West Germanic deletion in general terms: Campbell notes that `*z` is "later lost or changed to `r`" (Campbell 1959); Hogg gives a clean statement that Germanic `*z` is generally lost in final position (Hogg 1992, 37); the three-way division refines those summaries rather than contradicting them.

The CAPR rule for the unstressed loss is still named `EAFFinalZDeletion`, an identifier that predates the restaging of the change to Proto-West Germanic; the name is retained as a stable identifier pending a global renaming pass, and the historical stage recorded in the staging metadata takes priority over the name prefix. SC020 remains presented in this chapter, beside rhotacism, because the two changes jointly determine the fate of every remaining `*z`.

Anglo-Frisian ai-monophthongization (SC004)

The monophthongization of stressed/root `*ái` to `*ā`, seen in the soul derivation, is a North Sea Germanic areal development. Versloot argues that it spread in successive waves through a dialect continuum, with Old English among the widest to carry it, so the change is better read as an areal diffusion through the continuum than as a single dated node (Versloot 2017, 281–324). The resulting `*ā` is later fronted to ǣ in the relevant Old English environments.

The CAPR rule is named `EAFAiMonophthongization` and executes at cascade position 28. That position is CAPR's operational home for a North Sea Germanic areal change on the English line; it is a modelling choice, not a claim that the change passed through a discrete Proto-Anglo-Frisian node. The one usable chronological anchor is the `soul` derivation, which requires the monophthongization to precede OE interstress raising (SC036).

The unstressed development `*ai > *ē` (in final and nonfinal syllables) is a separate and earlier Proto-Northwest Germanic change (SC014), discussed in Chapter 1; its corpus witnesses are the dative-singular endings of `span` (**spánnai* 'span' > *spanne* 'span') and `meed` (**mízdai* 'meed' > *meorde* 'meed').

Anglo-Frisian brightening (SC043, treated in Chapter 3)

The fronting of low `*a` to `*æ` outside nasal environments is the defining Anglo-Frisian change. It executes later in the cascade than the changes of this chapter, so its full section appears in Chapter 3; it is introduced here because it anchors the "Anglo-Frisian" label that names this period. Campbell gives the classical statement: "By a very early change Prim. Gmc. `a > æ` in OE and OFris. when not followed by a nasal consonant" (Campbell 1959, §§ 163–165). Hogg gives the most familiar modern label pair: "This vowel normally fronted to /ae/ by the sound change of Anglo-Frisian Brightening (or First Fronting)" (Hogg 1992, § 5.8).

The change is notable for what follows it: OE Breaking presupposes the fronted input; OE a-Restoration partially undoes it in back-vowel environments. The three-change sequence (brightening, breaking, restoration) is one of the clearest relative-chronology chains in the Old English historical grammar.

Campbell notes that English and Frisian may not simply reflect one undifferentiated shared prehistoric event, and Ringe and Taylor leave open whether the wider spread of fronted outcomes happened mainly on the continent or in Britain (Ringe and Taylor 2014, 60–62). CAPR's implementation treats the change as a single rule; the book prose acknowledges the uncertainty about its exact geographical scope.

The current CAPR inventory has this change labeled "Old English" in the pipeline taxonomy (no separate Anglo-Frisian bucket previously existed). The historical staging map places its section in Chapter 3, at its executable cascade position.

3.4 Cascade vs. historical order in this chapter

The changes in this chapter are presented in the order in which they apply in the executable cascade, which models the reconstructed chronology itself: first word-final `*z` deletion in unstressed syllables (SC020), then the later northern monosyllabic `*z`-loss (SC097) that completes the final-`*z` story, then West Germanic rhotacism (SC003), which turns every surviving `*z` into `*ɾ` only after the deletions have run their course. There follow the unstressed `*ō` raising (SC021), the `*mn` dissimilation and n-stem `*n` loss (SC022–SC023), the long-`*ē` developments (SC024–SC025), the nasal-spirant corridor (SC026–SC027) with preconsonantal `*x` loss (SC028), and finally Anglo-Frisian ai-monophthongization (SC004, the North Sea areal vowel change) closing the chapter.

Book order, cascade order, and reconstructed historical order coincide here: in particular, the deletions of final `*z` precede rhotacism both in the sources and in the executable derivation, so no form ever meets rhotacism with a word-final sibilant intact.

3.5 West Germanic final **z*-deletion

Historical discussion

Word-final **z* in unstressed syllables was lost in Proto-West Germanic. Ringe and Taylor state the change for the whole branch and illustrate it with the nominative plural **dagōz* > **dagō* and the consonant-stem nominative **fadurz* > **fadur*, noting that the ending is lost after consonants as well as after vowels (Ringe and Taylor 2014, 44–45, §3.1.1). Crist's handout formulates the same development and its Ingvaeonic sequels (Crist 2002, 2, §§5–6). The change is pan-West-Germanic, not specifically Ingvaeonic: every West Germanic daughter shows the loss, and the Frienstedt comb inscription *kaba* < **kambaz* 'comb' (c. 250–300 CE) supplies early epigraphic confirmation (Fulk 2018, 25, n. 1).

The conditioning segment is specifically the voiced sibilant **z*, never **s*: Ringe and Taylor's near-minimal pair of nominative singular **dagaz* > **dag* beside genitive singular **dagas*, which keeps its sibilant into Old English *dæġes*, shows that the change reads the Verner voicing distinction (Ringe and Taylor 2014, 212, §6.1). Where the handbooks disagree about whether a given ending had **-s* or **-z* — as for the nominative plural **-ōz* — the disagreement matters directly to whether this rule applies (Ringe and Taylor 2014, 115–16, §4.2.1).

This is the middle of three historically distinct final-**z* developments, and Ringe and Taylor explicitly separate it from the later loss in stressed monosyllables, citing Crist's demonstration that they are two changes (Ringe and Taylor 2014, 44–45, §3.1.1). Earlier, the consonant-stem (root-noun) nominatives of monosyllables had already generalized endinglessness before Proto-West Germanic (SC096 RootNounNomZLoss), so forms like **bōkz* 'book' never reach this rule with their marker intact. Later, and only in the north, **z* was lost in stressed monosyllables with compensatory lengthening (SC097 MonosyllabicFinalZLoss); the present rule leaves stressed monosyllables untouched. Older accounts that grouped all of these under one loss of final **z*, such as Campbell's, are superseded by this three-way division (Campbell 1959, 166).

At the boundary with Chapter 1's Northwest Germanic sequence, the derivation of *ræste* 'rest' shows that final **ō*-raising (SC019 PNWGmcFinalLongORaising) must precede this rule: raising applies to **-ō* but not to **-ōz*, whose final vowel is still sheltered by the sibilant when raising runs (Ringe and Taylor 2014, 15–16, 24). On the later side, Ringe and Taylor order the loss of **z* before the loss of word-final bare **-a*, since **dagaz* first becomes **daga* and only then **dag* (Ringe and Taylor 2014, 45–46, §3.1.2).

SC020. West Germanic final **z*-deletion (`EAFFinalZDeletion`)

```
define EAFFinalZDeletion [{*z} -> 0 ||
    .#. ?* EnglishStarVocalic
        [EnglishStarConsonant | EnglishPalatalConsonant]+
        EnglishStarVocalic ?* _ .#.,
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
        EnglishStarVocalic+
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.];
```

The rule deletes word-final **z* in unstressed syllables, stated through two environments. The first clause covers polysyllables, where the final syllable of these corpus forms is unstressed: this is the ordinary case, with 110 corpus derivations, such as PGmc **bárdaz* 'beard' on its way to OE *beard* 'beard' and **rástōz* 'rest' on its way to *ræste* 'rest'. The second clause covers post-consonantal **z* in monosyllables. By the time this rule runs, SC096 RootNounNomZLoss has already removed the genuine root-noun nominative endings, so the only form reaching the second clause is **fríjōndz* 'friend', contracted to monosyllabic **fríundz* by SC009 PWGmcIjContraction; its ending, like that of **fadurz*, stood in an unstressed syllable when the Proto-West Germanic change applied and so belongs here rather than to the root-noun development (Ringe and Taylor 2014, 44–45, §3.1.1). Stressed monosyllables ending in vowel plus **z* meet neither clause and are left for SC097 MonosyllabicFinalZLoss.

The chronology of word-final **z*-loss is unusually well delimited: *ræste* 'rest' supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 PNWGmcFinalLongORaising, PGmc **rástōz* 'rest' yields †*rast* rather than expected OE *ræste* 'rest'. After SC040 OEMedUnstressedULowering, PGmc **bébruz* 'beaver' yields †*befro* rather than expected *befer* 'beaver', PGmc **kwéðuz* 'cud' yields †*cwedo* rather than expected *cwedu* 'cud', and PGmc **félθuz* 'field' yields †*feldo* rather than expected *feld* 'field', alongside eight other newly failing rows. Final *z*-loss therefore follows SC019 PNWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The **rástōz* 'rest' derivation fixes the local relation to SC019 PNWGmcFinalLongORaising. The distant boundary at SC040 OEMedUnstressedULowering shows only that word-final **z*-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final **ō*-raising.

3.6 Early apocope in unstressed words

Historical discussion

Alongside the regular loss of word-final short high vowels in third syllables (SC006 PWGmcEarlyIApocope), Ringe and Taylor identify a second, earlier apocope: “Short high vowels were also lost after heavy syllables in unstressed words” (Ringe and Taylor 2014, 57–58, §3.1.4). The two laws must not be conflated. Fully stressed disyllables kept their final **-i* long enough to cause i-umlaut — OE *ġiest* ‘guest’ < **gastiz* and *fȳr* ‘fire’ < **fūri* require exactly that survival (Ringe and Taylor 2014, 55, §3.1.4) — whereas words that carried no sentence stress lost the vowel already in Proto-West Germanic.

The conditioning is prosodic. Like Verner’s law, the change is governed by accent: it applied in words unstressed in the sentence, and its apparent exceptions are systematic, not sporadic. Forms such as OE *ymbe* ‘around’ and OHG *umbi* kept their final vowel because, as Ringe and Taylor observe, proclitics “were not phonologically word-final” and so stood outside the environment altogether (Ringe and Taylor 2014, 57–58, §3.1.4). Sentence-level accent placement therefore decides which sandhi variant each daughter language continues, and doublets across the family reflect the stressed and unstressed sentence forms of the same word — regular sandhi, not lexical diffusion.

The diagnostic witness is the second-person plural pronoun. Ringe and Taylor print the Proto-West Germanic form as a doublet: PGmc **izwiz* (Gothic *izwis*) → **iwwi* (by SC008 PWGmcCoronalWAssimilation and the loss of final **z*, SC020 EAFFinalZDeletion) → PWGmc **iuwi* ~ **iuw* (Ringe and Taylor 2014, 41–42, §3.1.1). Old English continues the apocopated, unstressed variant, and Ringe and Taylor’s proof is the vocalism itself: “OE iow ‘you (dat. pl.)’ definitely does [show early apocope] (since it does not exhibit i-umlaut)” (Ringe and Taylor 2014, 57–58, §3.1.4). Had the **-i* survived, i-umlaut (SC055 OEIUmlaut) would have fronted the diphthong; West Saxon *ēow* ‘you’ beside early West Saxon and Northumbrian *īow* ‘you’ shows the normal unumlauted development (Campbell 1959, § 702, p. 283).

SC098. Early apocope in unstressed words (PWGmcUnstressedWordFinalIApocope)

```
define PWGmcUnstressedWordFinalIApocope [
    {*i} -> 0 || {*w} {*w} _ .#.
];
```

The corpus transcription does not mark the absence of word stress, so the rule states the law through a proxy environment: word-final **-i* after the geminate **ww* created by coronal-w assimilation, which in the present corpus is exactly coextensive with the law’s unstressed-word domain. The same convention serves SC096 RootNounNomZLoss, where a development whose true conditioning the notation cannot yet express is likewise implemented over an exactly coextensive segmental environment.

The corpus witness is ‘you’: **izwiz* → **iwwiz* (assimilation) → **iwwi* (final **z*-loss) → **iww* (this rule) → OE *ēow* ‘you’. The chronology is fixed on both sides. The rule is fed by the loss of final **z* (SC020 EAFFinalZDeletion), since only that loss makes the **-i* word-final; and it must precede i-umlaut (SC055 OEIUmlaut) — that ordering is Ringe and Taylor’s own dating argument, for an unapocopated **iwwi* ‘you’ surviving to the umlaut period would yield an umlauted diphthong and a form other than the attested *ēow* ‘you’. The rule applies within Proto-West Germanic, before the later northern loss of final **z* in stressed monosyllables (SC097 MonosyllabicFinalZLoss): Ringe and Taylor treat the apocope among the Proto-West Germanic final-syllable developments and print the apocopated variant as a Proto-West Germanic form (Ringe and Taylor 2014, 41–42, 57–58). Fully stressed disyllables are untouched, as the history requires: **gastiz* and **fūri* pass through unchanged and duly umlaut to *ġiest* ‘guest’ and *fȳr* ‘fire’.

The surviving word-final geminate **ww* is then vocalized to a long diphthong (SC033 OEEwLongDiphthong) before geminate simplification (SC031 OEWWSimplification) can destroy it — Ringe and Taylor date that vocalization to Proto-West Germanic itself (**fewwar* → PWGmc **feuwar*) (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, § 8.3, pp. 204–205) — giving **ēoww*, simplified to **ēow*, the attested Old English form.

3.7 Northern monosyllabic final *z-loss

Historical discussion

Long after the Proto-West Germanic loss of final **z* in unstressed syllables (SC020 EAFFinalZDeletion), the northern West Germanic dialects lost word-final **z* in stressed monosyllables as well, with compensatory lengthening of a short nucleus. Ringe and Taylor's witness set is OE *mā* 'more' < **maiz*, the pronouns *wē* 'we', *ġē* 'you', *mē* 'me', *þē* 'thee', *hē* 'he', *hwā* 'who' < **hwaz*, and — hedged in their own print with question marks — *cū* 'cow' (Ringe and Taylor 2014, 86, §3.3.1). The southern dialects retained the sibilant and rhotacized it: Old High German *mir*, *wir*, *mēr*, *er* answer the Old English endingless forms, which is why Fulk counts this loss among the diagnostic Ingvaeonic features (Fulk 2018, 18, n. 6). Ringe and Taylor explicitly treat this as a change separate from the Proto-West Germanic unstressed loss, citing Crist's demonstration that the two must be distinguished (Ringe and Taylor 2014, 44–45, §3.1.1).

The scholarship disagrees about the exact conditioning, and the disagreement is worth recording. An older account, represented by Campbell and going back to Luick, derived the endingless pronouns from unaccented sentence variants rather than from a regular sound change (Campbell 1959, 166; Luick 1914--1940, 819). Ringe and Taylor reject that analysis because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86, §3.3.1). Crist formulates an Ingvaeonic rule in which **z* is lost after front vowels, with compensatory lengthening, covering preconsonantal cases as well; his data contain no word-final back-vowel monosyllables, so forms like **hwaz* and the ancestor of *cū* fall outside what his statement can decide — a documented gap rather than a refutation (Crist 2002, 1, 4, §§1, 10). Kilday narrows the preconsonantal subcase to Old Saxon and Old Frisian while accepting the word-final monosyllabic loss for Old English, contrasting regular *meord* 'reward' with the loanword-influenced *mēd* 'reward' (Kilday 2024, 1–3). CAPR adopts Ringe and Taylor's quality-neutral formulation because it alone generates the back-vowel witnesses, while noting that the front-vowel forms are compatible with both analyses.

Apparent counterexamples are analogical, not phonological: OE *dēor* 'deer', *ār* 'oar', and *gār* 'spear' show final *-r* from levelling out of inflected forms where the sibilant was word-internal and regularly rhotacized, not from retention of word-final **z* (Ringe and Taylor 2014, 86, §3.3.1, n. 24). The change precedes rhotacism (SC003 EAFRhotacism), which Ringe and Taylor place last in this sequence of northern developments (Ringe and Taylor 2014, 87, §3.3.1).

The corpus now witnesses this change directly. The interrogative pronoun 'who' is selected in its nominative singular masculine cell, PGmc **hwaz* (Gothic *hwas*), precisely the form Ringe and Taylor cite for this loss (Ringe and Taylor 2014, 86, §3.3.1): the rule lengthens the short nucleus and deletes the sibilant, giving **hwā*, whence OE *hwā* 'who'. The resulting back vowel never undergoes Anglo-Frisian brightening — "**hwǣ* does not exist", as Campbell puts it (Campbell 1959, § 125, p. 49; Brunner 1965, § 137 Anm. 1, p. 129) — so the derivation ends with the attested form. Other members of Ringe and Taylor's witness set remain outside the corpus because it selects oblique or plural cells for them — 'cow' and 'meed', for instance, enter the cascade in inflected forms whose **z*, where present, is word-internal.

SC097. Northern monosyllabic final *z-loss (MonosyllabicFinalZLoss)

```
define MonosyllabicFinalZLoss [
    {*a} -> {*ā}, {*á} -> {*ā},
    {*e} -> {*ē}, {*é} -> {*ḗ},
    {*i} -> {*ī}, {*í} -> {*ī́},
    {*o} -> {*ō}, {*ó} -> {*ō},
    {*u} -> {*ū}, {*ú} -> {*ū}
        || .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
            _ {*z} .#.
] .o. [
    {*z} -> 0 ||
        .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
```

```
          EnglishStarVocalic+ _ .#.
];
```

The rule first lengthens a short nucleus standing immediately before word-final *z* in a monosyllable, then deletes the *z* after any vowel in a monosyllable. The principal synthetic controls run the change on Ringe and Taylor's own witnesses, fed to the rule in their chronologically correct intermediate shapes. Short-nucleus inputs show both halves of the change at once: **hwaz* yields **hwā* and **hiz* yields **hī*, each with loss of the sibilant and compensatory lengthening (Ringe and Taylor 2014, 86, §3.3.1). A form whose nucleus is already bimoric skips the lengthening step and simply loses the sibilant: **maiz* yields **mai* at this stage, with its diphthong intact; the attested OE *mā* 'more' arises only later, when the stressed monophthongization (SC004 EAFAiMonophthongization) takes **ai* to **ā*. The word 'cow', whose history Ringe and Taylor themselves print with question marks and whose analysis remains disputed, is deliberately not used as a principal control; a long-vowel input of that shape (**kūz* yielding **kū*) merely repeats what **maiz* already demonstrates, and the word's evidentiary weight is discussed in the historical dossier rather than leaned on here.

The corpus derivation of *hwā* 'who' now fixes this rule's position empirically as well as philologically. It stands after SC020 EAFFinalZDeletion, since the two losses are historically distinct changes with the unstressed loss earlier (Ringe and Taylor 2014, 44–45, §3.1.1). Rhotacism (SC003 EAFRhotacism) follows in the executable cascade as in the historical account: Ringe and Taylor place rhotacism after this loss, at the end of the sequence of *z-losses (Ringe and Taylor 2014, 87, §3.3.1), and the cascade composes it immediately after this rule, so a sibilant removed here can never surface as *-r* — were the order reversed, **hwaz* would rhotacize to **hwar* and 'who' could never be derived; the negative controls below confirm this. Consonant-final monosyllables are untouched: their nominative **-z*, where it ever existed, was eliminated before Proto-West Germanic under SC096 RootNounNomZLoss. Word-internal *z, as in PGmc **déuzą* 'deer' on its way to OE *dēor* 'deer', does not meet the environment of this rule and duly rhotacizes.

3.8 West Germanic rhotacism

Historical discussion

Hogg states that Germanic **z* yielded **r* in intervocalic position in Old English, while final **z* was generally lost (Hogg 1992, 37). Ringe and Taylor argue that this merger of **z* with **r* was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic **z*-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

The internal identifier SC003 EAFRhotacism places the change in CAPR's Early Anglo-Frisian corridor, the operational post-Proto-West-Germanic stage on the English line; historically the change is a West Germanic rhotacism, later than Proto-Germanic. It is also distinct from SC020 EAFFinalZDeletion, which removes final **z* before the surviving medial consonant becomes **r*.

SC003. West Germanic rhotacism (`EAFRhotacism`)

```
define EAFRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc **líznōjaną* 'learn' yields †*lirnian* rather than expected OE *liornian* 'learn', PGmc **líznōθi* 'learns' yields †*lirnaþ* rather than expected *liornaþ* 'learns', PGmc **líznô* 'learn' yields †*lirna* rather than expected *liorna* 'learn', and PGmc **mízdai* 'meed' yields †*merde* rather than expected OE *meorde* 'meed'. Moving rhotacism earlier within the tested range changes no output.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. The lower boundary rests on the historical analyses cited above: Ringe and Taylor put rhotacism at the end of the sequence of **z*-losses — "first **z* was lost in a variety of environments …, then all surviving **z* became **r*" (Ringe and Taylor 2014, 87, §3.3.1) — and Crist observes that the deletions distinguish **z* from **r* and so must precede the merger (Crist 2002, 2–3). The rule is accordingly ordered after the three final-**z* losses (SC096 RootNounNomZLoss, SC020 EAFFinalZDeletion, and SC097 MonosyllabicFinalZLoss) and before breaking, so that the derivation follows the reconstructed chronology.

3.9 *mn*-dissimilation

Historical discussion

In the inherited *n*-stem paradigm the zero-grade oblique cells brought *m* and *n* into direct contact, and in that adjacent cluster the labial nasal dissimilated to a labial spirant: *mn* > *βn* (surfacing as *fn*). Old Norse preserves the older paradigmatic distribution, with the labial confined to the oblique cluster (*himinn* 'heaven' beside dative *hifni*); Old English and Old Saxon generalized it. Fulk treats the cluster change among developments common to Germanic, while warning that its surface results are irregular and that reverse *bn* > *mn* is later well attested in Northwest Germanic (Fulk 2018, 121, §6.11). The relevant *heofon* 'heaven' and *mōnaþ* 'month' material is discussed by Campbell (Campbell 1959, 189, 195, §§470, 484).

The underlying cluster change is therefore late Proto-Germanic / Common Germanic, not securely a pan-Northwest-Germanic innovation; `PNWGmc` remains a stable executable identifier only. The lexical evidence does not constrain a positive local cascade position.

SC022. Dissimilation of adjacent *mn* (`PNWGmcMnDissimilation`)

```
define PNWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _ {*n}
];
```

The rule fires only where *m* stands directly before *n*. It supplies the labial of *stefn* 'stem, trunk' from the *mn*-cluster of the *stamn*-family, and it is the historical change behind the labial of *heofon* 'heaven', which was generalized from the oblique cluster into the vowel-bearing stem before the Old English vocalic changes. (An earlier cross-syllable formulation that labialized an intervocalic *m* before a later nasal has been retired: it simulated paradigm levelling rather than a sound law.)

Moving SC022 PNWGmcMnDissimilation earlier or later leaves every output unchanged. Its executable place in this holding zone is therefore editorial/computational, while its historical classification rests on the handbook account of *mn*-dissimilation.

3.10 Word-final *n*-loss

Historical discussion

The change isolated here is far older than its position in the cascade suggests: it is the general (pre-)Proto-Germanic loss of word-final **n*, with nasalization of the preceding vowel, in polysyllables. Ringe's proof set for the law spans the whole grammar — nouns such as **yugón* > **juką* 'yoke', pronouns such as **tón* > **þanǭ*, and even the verb form **dedǭ* 'I did' — so it is general phonology, not a fact about any one declension (Ringe 2017, 101–3). Gothic *tuggo* shares the weak nominative-singular outcome, and the nasalized reflex **-ǭ* remained contrastive into Proto-West Germanic before yielding OE *-e* (Ringe and Taylor 2014, 54–55, 58–59).

Within the present corpus the change surfaces in exactly one shape: the weak nouns are cited in the stem form **-ōn-*, and this rule carries them to the Proto-Germanic nominative singular in **-ǭ*, as in **túngōn* > **túngǭ* > *tunge* 'tongue', alongside *eorþe* 'earth', *heorte* 'heart', *nǣdre* 'adder', and thirteen further weak nouns. The masculine weak nominative singular in trimoric **-ô* never had a final **-n* to lose, and Proto-Germanic **sebun* 'seven', **nigun* 'nine', and **tehun* 'ten' kept their *-n* by lexical analogy among the numerals (Ringe 2017, 103); the rule's narrow **-ōn* environment leaves all of these correctly untouched.

SC023. Loss of word-final **n* after **ō* (`PNWGmcNStemNLoss`)

```
define PNWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

The verb *dōn* 'do' supplies the negative, counterfeeding witness for the chronology. PGmc **dōną* 'do' passes this rule untouched; only SC047 OEHeavySyllableNasalApocope later strips the final **ą* and creates a new word-final **-ōn*. That secondary **-n* survives into *dōn* precisely because the old loss was no longer active: if SC023 PNWGmcNStemNLoss is displaced after the apocope, it consumes the new nasal and the derivation collapses entirely (+*?*).

The retained *-n* of *dōn* 'do' therefore supplies a terminus ante quem for the loss — it must be dead before the apocope — while the seventeen weak nouns above are its positive witnesses; the lower boundary remains unattested within the cascade, as befits a change already complete in Proto-Germanic.

3.11 Long *ē*-lowering

Historical discussion

The later West Saxon forms *sċēap* ‘sheep’ and *ġēar* ‘year’ imply an earlier lowering of long *ē* before the palatal diphthongal outcomes described more fully later in the sequence. Campbell and Ringe and Taylor discuss those later West Saxon outputs directly (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1).

The change is historically recognizable, but the lexical evidence establishes only a later boundary.

SC024. Lowering of long *ē* before non-nasal consonants (PNWGmcLongELowering)

```
define PNWGmcLongELowering [
    {*ē} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal],
    {*ḗ} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

After SC056 OEWsPalatalDiphthongization, long *ē* > *ǣ* can no longer produce the expected West Saxon forms: PGmc **skḗpą* ‘sheep’ yields †*sċīep* rather than OE *sċēap* ‘sheep’, and PGmc **jḗrą* ‘year’ yields †*ġīer* rather than *ġēar* ‘year’. Earlier placement changes no output, so SC024 PNWGmcLongELowering has a secure upper boundary.

Its lower boundary remains a matter of handbook chronology.

3.12 Long *ē* nasal-rounding

Historical discussion

Before nasals, older long *ē* can round toward the *ō*-vocalism seen later in *mōnaþ* ‘month’ and *mōna* ‘moon’ / *mōn* ‘moon’-type material. Campbell treats this split directly in his discussion of Germanic long *ē* before nasal consonants (Campbell 1959, 53, §129).

The change is historically recognizable, but the tested forms supply no close relative chronology.

SC025. Rounding of long *ē* before nasals (PNWGmcLongENasalRounding)

```
define PNWGmcLongENasalRounding [
    {*ē} -> {*ō} || _ EnglishStarNasal,
    {*ḗ} -> {*ō} || _ EnglishStarNasal
];
```

Reversing SC025 PNWGmcLongENasalRounding with neighboring changes leaves every output unchanged. Its position beside the other *ē*-developments therefore follows the handbooks.

3.13 Nasal spirant changes

Historical discussion

The two rules state successive phases of a single development. Campbell describes nasal loss before voiceless spirants with compensatory lengthening and nasalization of the preceding vowel. Ringe and Taylor assign the same outcomes to inherited northern West Germanic, before late Old English (Campbell 1959, 47, §121; Ringe and Taylor 2014, 140–41).

SC026 EAFNasalSpirantLengthening adjusts the vowel while the nasal-plus-spirant sequence remains present; SC027 EAFNasalSpirantLoss then removes the nasal. The first rule must therefore precede the second.

SC026. North Sea Germanic nasal-spirant lengthening (EAFNasalSpirantLengthening)

```
define EAFNasalSpirantLengthening [
    {*a} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*e} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*o} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*æ} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*á} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*é} -> {*ḗ} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ó} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative
];
```

All three witnesses require the vowel adjustment while the nasal is still present. If SC026 EAFNasalSpirantLengthening follows SC027 EAFNasalSpirantLoss, PGmc **fúnxstiz* 'fist' yields †*fyst* rather than expected OE *fȳst* 'fist', PGmc **gánsz* 'goose' yields †*ġeas* rather than expected *gōs* 'goose', and PGmc **júgunθ* 'youth' yields †*ġeogoþ* rather than expected *ġeoguþ* 'youth'. Earlier placement changes no output. The evidence requires lengthening to precede nasal loss without supplying a lower boundary, in agreement with the handbook treatment of the two as successive phases.

SC027. North Sea Germanic nasal-spirant loss (EAFNasalSpirantLoss)

```
define EAFNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarVoicelessFricative
];
```

The converse test fixes the same boundary: placing SC027 EAFNasalSpirantLoss before SC026 EAFNasalSpirantLengthening produces the same errors in *fȳst* 'fist', *gōs* 'goose', and *ġeoguþ* 'youth'. Later placement changes no output. These forms prove that the vowel was adjusted before the nasal disappeared; they provide no upper boundary for the loss.

3.14 Preconsonantal **x*-loss

Historical discussion

Campbell explicitly treats loss of *x* and gives forms such as *fléam* 'flight' and *hēla* 'heel' as examples of the same broad development (Campbell 1959, 186, §461).

The historical evidence is firmer than the chronology: the lexical evidence does not constrain the rule's position.

SC028. Loss of preconsonantal **x* (`PNWGmcPreconsonantalXLoss`)

```
define PNWGmcPreconsonantalXLoss [
    {*x} -> 0 || _ {*s} EnglishStarConsonant
];
```

No witness word dates preconsonantal **x*-loss before **s* plus another consonant: moving SC028 PNWGmcPreconsonantalXLoss in either direction leaves every output unchanged. Its position within this stretch therefore rests on the handbook chronology for *x*-loss.

3.15 Anglo-Frisian ai-monophthongization

Historical discussion

Inherited stressed **ái* monophthongized to **ā* across the North Sea Germanic area. Ringe and Taylor place the monophthongization of **ai* among the widespread early vowel developments of the English line (Ringe and Taylor 2014, 40–41). Versloot shows that the stressed development spread in successive waves through a dialect continuum, with Old English among the widest to carry it (Versloot 2017, 281–324). The Old English **ā* is later fronted to *ǣ* in the relevant environments.

The change is areal in character. It is shared with Frisian, and its spread through the continuum gives it a range of dates and no single sharp moment.

All twenty-four corpus witnesses carry stressed **ái*; loam **láimą* 'loam' is one of them, stressed in its Old English protoform. The unstressed development **ai* > **ē* is the separate earlier change SC014 PNWGmcUnstressedAiMonophthongization.

SC004. Anglo-Frisian ai-monophthongization (EAFAiMonophthongization)

```
define EAFAiMonophthongization [
    {*ái} -> {*ā}
];
```

The soul form fixes the relation to interstress raising. If the monophthongization is delayed until after that change, PGmc **sáiwalō* 'soul' yields †*sāwel* rather than expected OE *sāwol* 'soul'. An earlier placement changes no output. This shows that SC004 EAFAiMonophthongization must come before SC036 OEInterStressRaising in the modeled sequence.

The unstressed development **ai* > **ē* in final and nonfinal syllables is a separate and earlier change; see SC014 PNWGmcUnstressedAiMonophthongization.

Chapter 4

From Anglo-Frisian to Old English

4.1 Historical interval

This chapter covers the sound changes that occurred within the Old English period: the changes that produced attested Old English from the prehistoric English forms that emerged from the Anglo-Frisian stage. The starting point is the end of the Anglo-Frisian changes of Chapter 2; the ending point is attested West Saxon Old English, the primary dialect of the CAPR corpus.

4.2 Scope and dialect variation

Not every change in this chapter has pan-Old-English scope. Some changes — most notably West Saxon palatal umlaut (SC060), the back-mutation rules (SC059), and the West Saxon diphthong chain (SC031–SC034) — are specifically West Saxon or more broadly southern Old English phenomena. (The West Saxon palatal-glide spellings, SC016, belong to the written surface of Old English and are treated in Chapter 4.)

The CAPR derivations target West Saxon Old English citation forms as the default comparator. Changes that belong to other dialects, or that are absent from West Saxon, may appear in lexical entries as comparanda rather than as derivational steps.

The existing reader-facing sound-change sections record which changes have pan-Old-English scope and which are specifically West Saxon or Anglian (Campbell 1959, §§ 1–10; Hogg 1992, §§ 1.1–1.15).

4.3 Chapter structure

The changes in this chapter fall into several natural historical subgroups, though the boundaries between them are not always sharp:

Early Old English changes linked to the Anglo-Frisian inheritance: Changes that feed directly on, or are closely related to, Anglo-Frisian brightening (SC043), whose section opens the vowel corridor of this chapter. The `*awj` glide formation (SC029), `*au` fronting (SC030), and the West Saxon diphthong chain (SC031–SC034) all operate on the vowel inventory shaped by Anglo-Frisian brightening. OE Breaking (SC044) and a-Restoration (SC046) similarly presuppose the fronted *æ input.

Old English consonantal changes: Velar palatalization (SC052), palatalization of `*sk` (SC051), j-cluster coalescence (SC057), and related changes produce the characteristically Old English consonant phonemes. Hogg discusses these as OE consonant changes that are not broadly West Germanic (Hogg 1992, §§ 7.18–7.23).

Old English i-umlaut and its context: The i-umlaut (SC055) is one of the most productive changes in the Old English nominal and verbal morphology. Its relative chronology in relation to breaking, palatalization, and back-mutation is carefully documented in the existing CAPR chronology evidence audit and individual dossiers (Campbell 1959, §§ 193–204; Hogg 1992, §§ 5.62–5.68).

Late Old English syllabic reduction and apocope: High-vowel apocope (SC063), medial syncope (SC065), and the cluster of late unstressed-vowel changes (SC069–SC078) represent the later stage of Old English phonological history, when the syllabic structure of the language began to shift toward the more reduced profile of Middle English.

4.4 Cascade positions and historical order

Chapters in this part of the book follow the executable cascade order, which models the reconstructed chronology, so every section in this chapter appears at its cascade position. A few rules presented here carry stage labels from earlier periods; their individual sections discuss the label and its history, and a later renaming pass will resolve the residue:

- SC041 (PWGmc Final Bare-`*a` Loss) and SC042 (Surviving Bimoric `*ō` Unrounding) carry Proto-West Germanic labels but execute in this stretch of the cascade.
- SC064 (NWGmc `*-n` Stem `*n` Loss) carries a Northwest Germanic label but executes after OE High-Vowel Apocope (SC063).
- SC049 (PGmc B Allophony) carries a Proto-Germanic label but executes here.

4.5 Sources

Campbell's *Old English Grammar* is the primary source for the dating and scope of individual changes in this chapter (Campbell 1959). Hogg's *Grammar of Old English* provides modern reassessments and additional relative-chronology evidence (Hogg 1992). Ringe and Taylor supply the most detailed relative-chronology analysis for the earlier portion of the chapter, through back-mutation (Ringe and Taylor 2014, 70–160). Fulk's *Comparative Grammar* provides additional coverage for morphological conditioning (Fulk 2018). For individual changes, source-specific citations appear in the relevant sound-change sections.

4.6 Awj glide formation and au-fronting

Historical discussion

The *hīeġ* 'hay' and *strīeġan* 'strew' material undergoes both changes. Glide formation reshapes the older *awj* sequence, and fronting then affects the resulting *au*. Campbell's discussion of these outcomes and Ringe and Taylor's derivations of *hīeġ* and *strīeġan* describe the same sequence (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

Glide formation creates the input to fronting; diphthong leveling follows both.

Historical discussion of awj glide formation

Older *awj* sequences are the source of forms such as *hīeġ* 'hay' and *strīeġan* 'strew'. Campbell treats the relevant developments directly, and Ringe and Taylor likewise trace the same material through intermediate *auj*-type stages (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

The sources establish glide formation, while the witness forms supply only a later boundary.

SC029. Glide formation in **awj* (`OEAwjGlideFormation`)

```
define OEAwjGlideFormation [
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j},
    {*á} {*w}      {*j} -> {*áu} {*j},
```

```
    {*a} {*w}      {*j} -> {*au} {*j}
];
```

The *hīeġ* 'hay' and *strīeġan* 'strew' derivations show that *awj* reshaping prepared the input to fronting. If fronting is applied first, PGmc **xáwwją* 'hay' yields †*hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* 'strew' yields †*strauian* rather than expected *strīeġan* 'strew'. Earlier placement of glide formation changes no output, so these forms supply an upper boundary without a corresponding lower one.

Historical discussion of au-fronting

Once the glide sequence is in place, *au*-fronting produces the fronted diphthongal outcomes of the broader West Saxon vowel history. Campbell describes *au* > *ēa* (Campbell 1959, 53–54, §135).

Fronting must follow glide formation and precede diphthong leveling, which applies to a wider set of derivations.

SC030. Fronting of **au* (`OEAuFronting`)

```
define OEAuFronting [
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
];
```

Two distinct failure sets confine fronting. Placed before glide formation, it produces the wrong forms: PGmc **xáwwją* 'hay' yields †*hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* 'strew' yields †*strauian* rather than expected *strīeġan* 'strew'. Placed after diphthong leveling, PGmc **galáubijaną* 'believe', **bráudą* 'bread', and **dráugmaz* 'dream', together with sixteen other derivations, fail to produce output at all (+*?*) instead of yielding expected OE *ġelīefan* 'believe', *brēad* 'bread', and *drēam* 'dream'. The lexical errors require fronting to follow glide formation, while the failed derivations require it to precede diphthong leveling.

The later failure set consists of failed derivations, not competing Old English surface forms.

4.7 West Saxon diphthong sequence

Historical discussion

Four distinct developments shape the West Saxon diphthongal field. Campbell discusses inherited *aw*/*ew* outcomes, palatal-triggered diphthongization, and later Anglian smoothing in connected but separate parts of the vowel history; Hogg likewise distinguishes the palatal-diphthongal developments (Campbell 1959, 46, 53–54, 65–70, 95–96, §§120, 135–136, 170–176, 185, 223–227; Hogg 1992, 106–7, 111–12).

The closest interaction joins *ww*-simplification and long-*aw* diphthongization, which together shape *dēaw* 'dew' and *hēawan* 'hew'. Diphthong leveling regularizes a wider field, while long-*ew* diphthongization carries *ēow* into the later environment of breaking.

Historical discussion of WW simplification

West Germanic *ww* sequences lie behind forms such as *dēaw* 'dew' and *hēawan* 'hew', and Campbell treats them as part of the early West Germanic diphthong history (Campbell 1959, 46, §120).

SC031 OEWWSimplification precedes the later long-diphthong outcomes.

SC031. Simplification of **ww* sequences (`OEWWSimplification`)

```
define OEWWSimplification [
    {*w} {*w} -> {*w}
];
```

The *dēaw* 'dew' and *hēawan* 'hew' derivations establish that doubled *w* was simplified before the long *ēaw* development. If SC031 OEWWSimplification follows SC034 OEAwLongDiphthong, PGmc **dáwwō* 'dew' yields †*dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* 'hew' yields †*hawan* rather than expected *hēawan* 'hew'. Earlier placement changes no output. The witnesses require simplification before the long-diphthong change and leave the lower boundary to the broader West Saxon chronology.

Historical discussion of diphthong leveling

Forms such as *hēafod* 'head' reflect the redistribution of diphthongal outcomes across a wider set of words. Campbell describes smoothing and related later monophthongization, although the rule below is more narrowly conditioned than any single textbook label (Campbell 1959, 95–96, §§223–227).

The evidence for SC032 OEDiphthongLeveling is less self-contained than that for the *dēaw* 'dew' / *hēawan* 'hew' developments.

SC032. Leveling of diphthongal outputs (`OEDiphthongLeveling`)

```
define OEDiphthongLeveling [
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
];
```

The two edges of this interval fail differently. Before SC030 OEAuFronting, PGmc **galáubijaną* 'believe', **báug* 'bow', and **bráudą* 'bread' produce no output (+?) instead of expected OE *ġelīefan* 'believe', *bēag* 'bow', and *brēad* 'bread', alongside fifteen other failed derivations. After SC040 OEMedUnstressedULowering, PGmc **xáubudą* 'head' yields †*hēafud* rather than expected *hēafod* 'head'. Absence at the lower edge places diphthong leveling after fronting; the wrong surface form at the upper edge places it before medial unstressed-*u* lowering.

Historical discussion of long *ēow*

The long *ēow* forms of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' form part of the West Saxon vowel history, although their clearest ordering relation points forward. Campbell describes early *eu* in Old English, and Ringe and Taylor give the corresponding examples from chew, four, and knee (Campbell 1959, 53–54, §136; Ringe and Taylor 2014, 188, 202).

The only boundary established by the lexical evidence for SC033 OEEwLongDiphthong lies ahead at SC044 OEBreaking.

SC033. Long *ēow* before following vowels and weak endings (OEEwLongDiphthong)

```
define OEEwLongDiphthong [
    {*e} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*i} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*é} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*í} {*w} -> {*ēo} {*w} || _ OEEwLongContext
];
```

The long *ēow* of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' supplies only a terminus ante quem. If SC033 OEEwLongDiphthong follows SC044 OEBreaking, PGmc **kéwwaną* 'chew' yields †*ċeowan* rather than expected OE *ċēowan* 'chew', PGmc **fédwōr* 'four' yields †*feower* rather than expected *fēower* 'four', and PGmc **knéwą* 'knee' yields †*cneow* rather than expected *cnēow* 'knee'. Earlier placement changes no output. The sources associate *ew* and *iw* with the same diphthongal history but furnish no lower boundary.

Historical discussion of long *ēaw*

After SC031 OEWWSimplification has reduced *ww* to single *w*, the remaining *aw* sequence can develop into the long *ēaw* seen in *dēaw* 'dew' and *hēawan* 'hew'. Campbell treats these outputs in the early diphthong history of West Germanic and Old English (Campbell 1959, 46, 53–54, §§120, 135–136). The resulting long diphthong is *ēaw*.

SC034 OEAwLongDiphthong follows SC031 OEWWSimplification locally and must also precede SC043 EAFBrightening.

SC034. Long *ēaw* before following vowels (`OEAwLongDiphthong`)

```
define OEAwLongDiphthong [
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    {*á} {*w} -> {*ḗa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

A local feeding relation and a later vowel change confine *aw* > *ēaw*. Before SC031 OEWWSimplification, PGmc **dáwwō* 'dew' yields †*dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* 'hew' yields †*hawan* rather than expected *hēawan* 'hew'. After SC043 EAFBrightening, PGmc **skáwōjaną* 'show' yields †*sċawian* rather than expected OE *sċēawian* 'show', PGmc **skáwōθi* 'shows' yields †*sċawaþ* rather than expected *sċēawaþ* 'shows', and PGmc **stráwą* 'straw' yields †*stræw* rather than expected *strēaw* ‘straw’. The *dēaw* and *hēawan* forms require long-diphthong formation after simplification, while *sċēawian* requires it before brightening; the handbooks assign the same interval to the West Saxon development.

4.8 Prefix and compound adjustments

Historical discussion of prefixal **a*-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* 'believe' supplies the comparative witness for the same development (Campbell 1959, 147, §354; Ringe and Taylor 2014, 245; 2014, 267).

The rule has a narrow historical range and gives prefixed forms the weak vowel inherited by later vocalic changes.

SC035. Reduction of prefixal **a* (`OEPrefixAReduction`)

```
define OEPrefixAReduction [
    {*a} -> {*ĕ}
        || .#. {*g} _
           [EnglishStarConsonant | EnglishPalatalConsonant]
           EnglishStarVocalic
];
```

The prefix of *ġelīefan* 'believe' supplies the upper boundary for **ga-* > **ge-*. If SC035 OEPrefixAReduction follows SC043 EAFBrightening, PGmc **galáubijaną* 'believe' yields †*ġealīefan* rather than expected OE *ġelīefan* 'believe'. Earlier placement changes no output, so the witness dates prefix reduction before brightening without locating its beginning.

Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell's discussion of *weorold* 'world' / *weoruld* 'world' and Ringe and Taylor's derivation of **weraldu* 'world' > **weruldu* 'world' > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

SC036. Raising of medial **a* between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising [
    {*a} -> {*u}
        || PGmcStarVowel EnglishStarConsonant* _
           [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
    {*à} -> {*u}
];
```

The two boundaries have unequal force. Before SC019 PNWGmcFinalLongORaising, PGmc **sáiwalō* 'soul' yields †*sāwel* rather than expected OE *sāwol* 'soul'; after SC040 OEMedUnstressedULowering, it yields †*sāwul* rather than *sāwol*, while PGmc **wír-àldu* 'world' yields †*weoruld* rather than *weorold* 'world'. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In handbook terms, medial **a* > **u* belongs to the *world*- and *soul*-type low-stress vocalism that followed the earlier final-vowel changes.

Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as *reġnboga* 'rainbow'. The only boundary the lexical evidence supplies is the immediately following technical stress-stripping stage, which is not a sound change.

SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope [
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _
           OEAnyConsonant+ PGmcStarGraveVowel
];
```

The *reġnboga* 'rainbow' test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc **régna-bùgô* 'rainbow' yields †*reġnefoga* rather than expected OE *reġnboga* 'rainbow', because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

4.9 Medial unstressed vowel changes

Historical discussion

The history of *wuduwe* ‘widow’ orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* ‘world’ / *weoruld* ‘world’ alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu* ‘world’, and **jugunþi* ‘youth’ (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

```
define OEWICombinativeUUmlaut [
    {*í} -> {*ú}
        || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

The *wuduwe* ‘widow’ derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut follows SC040 OEMedUnstressedULowering, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe*; earlier placement changes no output. The witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering [
    {*u} -> {*o}
        || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _
           [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

The two witnesses date medial unstressed **u* > **o* at very different scales. Before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe* ‘widow’; after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* ‘youth’ yields †*ġeogoþ* rather than expected *ġeoguþ* ‘youth’. The local *weorold* ‘world’ and widow evidence places lowering after combinative u-umlaut, while the youth form supplies only the distant requirement that lowering precede unstressed long-vowel shortening.

4.10 Final bare-*a* loss

Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

```
define PWGmcFinalBareALoss [
    {*a} -> 0 || _ .#.
];
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final *z*-deletion, the change gives the wrong outputs: PGmc **bárdaz* ‘beard’ yields †*bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* ‘comb’ yields †*camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* ‘craft’ yields †*craft* rather than expected OE *cræft* ‘craft’, and PGmc **dágaz* ‘day’ yields †*dag* rather than expected *dæġ* ‘day’. The distant lower limit follows final *z*-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

4.11 Surviving bimoric *ō unrounding

Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* 'rest' nevertheless undergoes unrounding before SC043 EAFBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90).

The sole witness establishes a local relation to brightening but supports no broader generalization.

SC042. Unrounding of the surviving bimoric *ō (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding [
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
];
```

The single *ræste* 'rest' derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 EAFFinalZDeletion or after SC043 EAFBrightening, PGmc **rástōz* 'rest' yields †*rasta* rather than expected OE *ræste*. Unrounding must therefore follow final *z*-loss and precede brightening, although only the relation to brightening is local.

4.12 Anglo-Frisian brightening

Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (Campbell 1959, 52, §131; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

SC043. Fronting of low **a* outside nasal environments (EAFBrightening)

```
define EAFBrightening [
    AngloFrisianBrighteningUnstressed .o.
    AngloFrisianBrighteningStressed .o.
    AngloFrisianBrighteningLongFinal
];
```

Two derivations place low **a* > **æ* between unrounding and breaking. Before SC042 PWGmc-SurvivingBimoricOUnrounding, PGmc **rástōz* ‘rest’ yields †*rasta* rather than expected OE *ræste* ‘rest’. After SC044 OEBreaking, PGmc **sláxaną* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. The first witness requires brightening to receive the outcome of the surviving-bimoric **ō* development; the second requires breaking to receive the fronted vowel.

4.13 Breaking and velar-fricative palatalization

Historical discussion

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

Breaking has the fuller handbook treatment, while velar-fricative palatalization follows it locally in the *feoh* 'cattle' and *feohtan* 'fight' type derivations.

SC044. Breaking before *h*, *rC*, and *lC* (`OEBreaking`)

```
define OEBreaking OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI;
```

Breaking must encounter the vowel created by brightening and must precede the fricative change seen in *feoh* 'fee' and *feohtan* 'fight'. Before SC043 EAFBrightening, PGmc **sláxaną* 'slay' yields *sleaan* | *slēaan* rather than expected OE *slēan* 'slay'. After SC045 OEVelarFricativePalatalization, PGmc **féxu* 'cattle' yields †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* 'fight' yields †*fehtan* rather than expected *feohtan*. The two feeding relations place breaking between brightening and velar-fricative palatalization.

SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization [
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*ɣ} -> {*j} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*ɣ} -> {*j} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j},
    {*ɣ} -> {*j} || _ {*j}
]
    .o. EnglishStarAlphabet*;
```

The local chronology comes from *feoh* 'cattle' and *feohtan* 'fight'. Before SC044 OEBreaking, palatalization of **x* and **ɣ* beside front vowels or **j* makes PGmc **féxu* 'cattle' yield †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* 'fight' yield †*fehtan* rather than expected *feohtan*. The distant upper limit comes from *six* 'six': after SC060 OEWsPalatalUmlaut, PGmc **séxs* 'six' yields †*sihs* rather than expected OE *six*. Breaking therefore feeds velar-fricative palatalization directly, while palatal umlaut supplies only the broader upper limit.

4.14 A-restoration and nasal changes

Historical discussion of A-restoration

Campbell's restoration of *a* before following back vowels and Ringe and Taylor's later retraction describe the same post-brightening development (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration (
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 EAFBrightening, PGmc **bákaną* 'bake' yields †*bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* 'fare' yields †*færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákaną* again yields †*bæcan* instead of *bacan*, while PGmc **wádaną* 'wade' yields †*wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

SC047. Heavy-syllable nasal apocope of final **ą* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope [
    {*ą} -> 0 || OEAnyConsonant _ .#.
];
```

The evidence for final nasalized **ą* loss is sharply asymmetric. Before SC034 OEAwLongDiphthong, the single PGmc witness **stráwą* 'straw' yields †*strœw* rather than expected OE *strēaw* 'straw'. After SC048 OESecondaryNasalization, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan* 'bind', alongside a broad *-en* failure set. One lower witness places apocope after long-diphthong formation; many reciprocal upper failures place it before secondary nasalization.

SC048. Secondary nasalization before final **_*n_** (OESecondaryNasalization)

```
define OESecondaryNasalization [
    {*a} -> {*ą} || _ {*n} .#.
];
```

The broad *-an*/*-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan* 'bind'. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* 'steal' yields †*steolan* rather than expected OE *stelan* 'steal', and PGmc **wébaną* 'weave' yields †*weofan* rather than expected *wefan* 'weave'. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

4.15 B allophony

Historical discussion

The positional alternation of Germanic **b* is a Proto-Germanic distributional feature. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

The distribution is narrow, but later changes presuppose the stop-fricative alternation. CAPR implements the rule at a late cascade position for computational reasons: the alternation must interact with consonant environments shaped by intermediate rule applications. Its historical stage is Proto-Germanic.

SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reġnboga* 'rainbow' develops as †*reġnfoga* rather than expected OE *reġnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

4.16 Sievers-law syncope

Historical discussion

Sievers' Law concerns a prosodic and morphological adjustment in heavy stems. It is a distributional rule distinct from b-allophony (SC049 PGmcBAllophony). Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

SC050. Sievers-law syncope (`SieversLawSyncope`)

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

The Sievers-law reduction **-CijV-** > **-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* 'stretch' yields †*strecċan* rather than expected OE *streċċan* 'stretch'; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

4.17 Palatalization of **sk* to **sc*

Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992, 106–7, 111–12). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

The non-fronted vowels of *flasce* 'flask' and *wascan* 'wash' fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* 'flask' and *wæscan* 'wash' rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sċeaft* 'shaft', *sċēar* 'shear', *sċēaþ* 'sheath', *sċēap* 'sheep', and *sċield* 'shield' before SC056 OEWsPalatalDiphthongization. The **sċea-* 'sea'/*sċie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

4.18 Velar palatalization before front vowels

Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading "Frühe Verschiebungen in palataler Richtung," he treats English k and g before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* 'town', *geaf* 'gave', *giefan* 'give', and *giest* 'guest', already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914--1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone sk cluster. His remark that "[sk] is more prone to palatalization and assibilation than [k]" is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* 'child', *dæg* 'day', *giefan* 'give', and *giest* 'guest', which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant j (Hogg 1992, 103–4). This formulation replaces a broad list of palatal outcomes with a phonological environment defined by adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that "after initial velars and **sk* had been palatalized" West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* 'wake', *licgan* 'lie', *lecgan* 'lay', *secg* 'retainer', *ecg* 'edge', *wicg* 'horse', and *brycg* 'bridge', illustrate the same point in lexical detail: front vowels and j create the palatal environment in which plain k and g cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Luick describes a broad early movement; Campbell distinguishes plain velars from the sk complex; Hogg specifies the adjacency and syllable conditions; and Ringe and Taylor order the plain-velar change before West-Saxon diphthongization. Plain-velar palatalization thus forms part of a wider palatalizing environment without being identical to its neighboring changes.

SC052. Palatalization of **k* before front vowels and **j* (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*í},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*í} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*í} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;
```

The *weccan* 'wake', *licgan* 'lie', and *lecgan* 'lay' set identifies front vowels and j as the environment for palatalization of k (Ringe and Taylor 2014, 213–14, §6.4.1). These forms establish the conditioning; different witnesses establish the chronology.

Applied before Sievers-law syncope, PGmc **strákkijaną* ‘stretch’ yields †*strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after i-umlaut fronting, PGmc **kūi* ‘cow’ and **lúnganjō* ‘lungs’ yield *ċȳ* ‘cows’ and *lunġen* ‘lungs’ rather than expected OE *cȳ* ‘cows’ and *lungen* ‘lungs’. The front-vowel `k` change therefore follows Sievers-law syncope and precedes i-umlaut fronting.

SC052. Velar palatalization before front vowels (OEVelarPalatalization)

```
define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

Plain `k` and `g` palatalization in front-vocalic and `j`-adjacent environments follows `sk`-palatalization and occupies a sharply defined pre-umlaut interval. Applied before Sievers-law syncope, PGmc **strákkijaną* ‘stretch’ yields †*strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after general i-umlaut, PGmc **kūi* ‘cow’ yields †*ċȳ* rather than expected *cȳ* ‘cows’, and PGmc **lúnganjō* ‘lungs’ yields †*lunġen* rather than expected *lungen* ‘lungs’. These witnesses place velar palatalization after Sievers-law syncope and before umlaut.

Luick, Campbell, and Ringe and Taylor place *cild* ‘child’ and *dæg* ‘day’ in a consonantal palatalization that precedes later vowel fronting (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1). The umlautal developments therefore receive plain `k` and `g` already reshaped beside front vowels and `j`.

The `sk` change belongs to the same palatalizing region with a separate scope. The *streċċan* ‘stretch’ evidence establishes a specific dependency on earlier syncope; it does not merge the two changes into one process.

4.19 Post-velar **w*-loss and loss of **w* before final **i*

Historical discussion

The first rule is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor derive PGmc **singwan* 'sing' to Old English *singan* 'sing' (Ringe and Taylor 2014, 214, §6.4.2). This comparative evidence establishes the change, although no lexical evidence fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* 'sea' from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The first rule is restricted to the **ngw* sequence; the second has a specific lexical witness and defined earlier and later limits.

SC053. Loss of **w* after velars (`OEPostVelarWLoss`)

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

The comparative development `*singwan > singan` establishes narrow post-velar **w*-loss in the **ngw* sequence, yielding *singan* 'sing'. Moving SC053 OEPostVelarWLoss earlier or later leaves every output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

SC054. Loss of **w* before final **i* (`OEWLossBeforeI`)

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

The history of *sǣ* 'sea' explains why non-initial **w* disappeared before final unstressed **i*. Campbell describes the loss, Ringe and Taylor derive the form from **saiwi-*/**sawi-*, and Luick gives the parallel trajectory (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 EAFFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields †*sǣw* rather than expected OE *sǣ* 'sea'. The loss must therefore follow final *z*-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

4.20 The Old English i-umlaut and West Saxon palatal diphthongization

Historical discussion

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that “the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE” (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* ‘guest’, *giefan* ‘give’, *hierde* ‘shepherd’, and *ieldra* ‘older’ (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: “we come now to a change which is almost as uncontroversial as it is important” (Hogg 1992, 112). His examples, such as *bryd* ‘bride’, *trymman* ‘strengthen’, *bedd* ‘bed’, *ciest* ‘chest’, and *wiersa* ‘worse’, likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it “diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)” (Fulk 2018, 74, §4.13). Ringe and Taylor’s examples such as *gieldan* ‘pay’, *scield* ‘shield’, and *scieppan* ‘create’ show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* ‘follow’, *gylden* ‘golden’, *wyrm* ‘worm’, and *giest* ‘guest’ exemplify the same `i`- or `j`-conditioned fronting across different vowels (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The cow and lung forms establish the lower boundary. If fronting precedes velar palatalization, PGmc **kūi* ‘cow’ yields †*ċȳ* rather than expected OE *cȳ* ‘cows’, and **lúnganjō* ‘lungs’ yields †*lunġen* rather than expected OE *lungen* ‘lungs’. The consonantal change must therefore precede fronting.

The gift and sheath forms establish the upper boundary. If West Saxon palatal diphthongization precedes fronting, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected OE *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. Fronting consequently follows velar palatalization and precedes the West Saxon change; the other components of i-umlaut share those bounds.

SC055. Raising under i-umlaut (`OEIUmlautRaising`)

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted `æ` to `e` continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The same four forms fix both boundaries. If raising precedes velar palatalization, **kūi* 'cow' yields †*ċȳ* instead of expected *cȳ* 'cows' and **lúnganjō* 'lungs' yields †*lunġen* instead of expected *lungen* 'lungs'. If West Saxon palatal diphthongization precedes raising, **géftiz* 'gift' yields †*ġieft* rather than expected *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. These forms place raising after velar palatalization and before West Saxon palatal diphthongization.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before `m` and `n` in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon `ie` forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

The chronology comes from the cow/lung and gift/sheath contrasts. Placed before velar palatalization, diphthongal mutation over-palatalizes **kūi* 'cow' and **lúnganjō* 'lungs'; placed after West Saxon palatal diphthongization, it yields †*ġieft* and †*sċǣþ* instead of expected *ġift* 'gift' and *sċēaþ* 'sheath'. These failures place diphthongal mutation after velar palatalization and before West Saxon palatal diphthongization.

SC055. The composite i-umlaut rule (`OEIUmlaut`)

```
define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
```

```
    .o. OEIUmlautDiphthong;
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The lower boundary is consonantal. If general umlaut precedes velar palatalization, PGmc **kūi* 'cow' yields †*ċȳ* rather than expected *cȳ* 'cows', and PGmc **lúnganjō* 'lungs' yields †*lunġen* rather than expected *lungen* 'lungs'. These over-palatalized forms place general umlaut after velar palatalization.

The upper boundary separates general umlaut from the narrower West Saxon process. If West Saxon palatal diphthongization precedes umlaut, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected OE *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. Together the two witness pairs place general umlaut after velar palatalization and before the West Saxon process.

SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.]
];
```

West Saxon *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show diphthongization after an already palatal consonant (Ringe and Taylor 2014, 215–16, §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg's *giefan* 'give' and *sceap* 'sheep' belong to the same palatal-consonant environment (Hogg 1992, 108–9). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (Fulk 2018, 74, §4.13).

The forms *ġift* 'gift' and *sċēaþ* 'sheath' fix the lower boundary. If West Saxon palatal diphthongization precedes general i-umlaut, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected *ġift*, and PGmc **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ*. These witnesses place West Saxon palatal diphthongization after general umlaut; no tested lexical item supplies a later terminus ante quem.

The one-sided chronology reflects the difference in scale. General umlaut reorganizes the vowel system, whereas West Saxon palatal diphthongization affects a narrower dialectal class after palatal consonants. Its exact later placement remains undemonstrated by the present lexicon.

4.21 J-cluster coalescence

Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīeġan* 'bend' and *sēċan* 'seek' without assigning this later cluster adjustment the status of a major sound law (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ʤ}]
    .o. [{*k} {*j} -> {*ʧ}]
);
```

The forms *bīeġan* 'bend' and *sēċan* 'seek' determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc **báugijaną* 'bow' yields †*bēaġan* rather than expected OE *bīeġan*, and PGmc **sōkijaną* 'seek' yields †*sōċan* rather than expected *sēċan*. This demonstrates that velar palatalization preceded coalescence. Nothing in the present lexicon supplies a terminus ante quem.

4.22 Back mutation

Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* ‘gave’ and *weofan* ‘weave’. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (Ringe and Taylor 2014, 319, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (Campbell 1959, 86, §207), while Hogg draws the instructive comparison with breaking (Hogg 1992, 112). Fulk accordingly separates back mutation from the earlier umlautal changes (Fulk 2018, 69, §4.8).

SC059. Back mutation before labials and liquids (`OEBackMutation`)

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébaną* ‘give’ produce *ġeofan* ‘give’; the expected form is *ġiefan* ‘give’. **stélaną* ‘steal’ likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébaną* ‘weave’ yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

4.23 West Saxon palatal umlaut

Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage †*mieht* and †*nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

4.24 Weak-tail nasal loss

Historical discussion

The pathway from **dōnǭ* ‘do’ to *dōn* ‘do’ supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120–21; Fulk 2018, 91, §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

Final weak-tail **-ną* and **-mą* accordingly yield plain **-n* and **-m*.

Only *dōn* ‘do’ constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* ‘do’. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

4.25 High-vowel apocope

Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
```

```
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    {*u} -> 0 || {*x} _ .#.,
    {*ų} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
];
```

Final **i*, **u*, and **ų* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* ‘cow’ yields †*cū* rather than expected OE *cȳ* ‘cow’, and PGmc **brūdiz* ‘bride’ yields †*brūd* rather than expected OE *brȳd* ‘bride’. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrht* rather than expected OE *fyrhte* ‘fright’. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

4.26 Post-apocope **n*-loss and medial syncope

Historical discussion

Evidence for post-apocope reduction is strikingly uneven. The inherited feminine *in*-stem represented by Gothic *faurhtei*, OE *fyrhtu*, and oblique OE *fyrhte* supplies the relevant evidence (Orel 2003, 120; Ringe and Taylor 2014, 380–81; Campbell 1959, 236, §589.7). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

Only final **n* after long **ī* is at issue, as in the inherited *in*-stem behind OE *fyrhte* ‘fright’.

CAPR models the oblique OE form through the Proto-Germanic genitive singular **fúrxtīnaz* ‘fright’, following the project convention of using an appropriate non-nominative paradigm cell when the nominative does not supply the required derivation. Within this selected genitive derivation, the same input fixes both ordering boundaries. Before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. After SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* again yields †*fyrhten* rather than expected *fyrhte* ‘fright’. I therefore order final bare-a loss, stem-final n-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the selected genitive derivation and depend on one inherited lexeme/paradigm.

SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

```
define OEMedialSyncope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
];
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

4.27 Late syncope and degemination

Historical discussion

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *netele* 'nettle' supplies a concrete lexical type (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Brunner 1965, 144–45, §§158–159). Fulk places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope [
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátilōn* 'nettle' yields †*nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* 'spindle' yields †*spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields †*spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation [
    {*θ} -> 0 || {*t} _
];
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (Hogg 1992, 120–21; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN;
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* 'spindle' (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* 'spindle' yields †*spinnl* rather than expected OE *spinl* 'spindle'. Syncope must therefore create the cluster before degemination simplifies it.

Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

4.28 Early o-shortening

Historical discussion

After the principal palatal and umlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 PNWGmcNStemNLoss, PGmc **nḗdrōn* ‘adder’ yields †*nǣdran* rather than expected OE *nǣdre* ‘adder’, PGmc **érθōn* ‘earth’ yields †*eorþan* rather than expected *eorþe* ‘earth’, and PGmc **fláskōn* ‘flask’ yields †*flascan* rather than expected *flasce* ‘flask’. The same earlier shift also disrupts forms such as *heorte* ‘heart’ and *līne* ‘line’. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 PNWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

4.29 Early unstressed fronting and later o-shortening

Historical discussion

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no lexical evidence fixes its later boundary.

SC070. Early fronting of unstressed ***a** (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

The rule fronts unstressed **a* to **æ* after the earlier shortening has created a frontable vowel but before the later shortening of unstressed **ō*. It produces endings such as OE *-en* in *lungen* ‘lungs’.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* ‘lungs’ yields †*lunġen* rather than expected OE *lungen* ‘lungs’. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* ‘bears’ yields †*boreþ* rather than expected OE *boraþ* ‘bears’, and PGmc **mḗnōθz* ‘month’ yields †*mōneþ* rather than expected *mōnaþ* ‘month’. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

SC071. Later shortening of unstressed ***ō** (`OELateOShortening`)

The following rule handles the later shortening stage.

```
define OELateOShortening [
    {*ō} -> {*o} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
];
```

The rule shortens the remaining unstressed long **ō* after fronting. The shortened vowel is then resolved by the following medial/final distribution, not directly as *a* (Stausland Johnsen 2015, 28–31).

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* ‘bears’ yield †*boreþ* rather than expected OE *boraþ* ‘bears’, and PGmc **líznōθi* ‘learns’ yield †*liorneþ* rather than expected *liornaþ* ‘learns’. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly.

SC099. Medial raising of shortened unstressed ***o** (OEMedUnstressedORaising)

```
define OEMedUnstressedORaising [
    {*o} -> {*u} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]* EnglishStarVocalic
];
```

After SC071 OELateOShortening, the shortened vowel gives *u* in an unstressed medial syllable. The rule encodes Stausland Johnsen's statistically supported account of West Saxon ō-verb pasts, not a general rule for inherited short **o* or for nominal morphology (Stausland Johnsen 2015, 28–31, 36). His diagnostic derivation is PGmc **wúndōdē* 'wounded' > †*wundode* > OE *wundude* 'wounded' (Stausland Johnsen 2015, 28–29).

SC100. Final lowering of shortened unstressed **o* (OEFinalUnstressedOLowering)

```
define OEFinalUnstressedOLowering [
    {*o} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]* .#.
];
```

In a final syllable the same shortened vowel gives *a*. Thus the existing month control continues PGmc **mḗnōθz* 'month' through shortened **o* to OE *mōnaþ* 'month', while **wúndōdē* 'wounded' takes SC099 OEMedUnstressedORaising instead. The medial/final contrast and its chronology after long-vowel shortening are Stausland Johnsen's analysis (Stausland Johnsen 2015, 28–31).

4.30 Unstressed long-vowel shortening and ae-merger

Historical discussion

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **æ* with **e* (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8;
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nḗdrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

SC073. Merger of unstressed **æ* with **e* (`OEUnstressedAEMerger`)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

The rule merges unstressed **æ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **nḗdrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The lexical evidence fixes the local order after SC072 OEUnstressedLongVowelShortening and places the merger before the later h-loss and contraction.

4.31 Medial unstressed-i lowering

Historical discussion

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (Hogg 1992, 120–21; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *sċilling* 'shilling' type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 [
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _
];
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* 'fright' yields †*fyrhti* rather than expected OE *fyrhte* 'fright'. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* 'shilling' yields †*sċilleng* rather than expected *sċilling* 'shilling'. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering [
    {*e} -> {*i} || _ {*n} {*g}
];
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* 'shilling'.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skíllingaz* 'shilling' yield †*sċilleng* rather than expected OE *sċilling* 'shilling'. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

4.32 Prefix i-reduction

Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk's discussion of prefix vowels accounts for OE **be-* and **ne-* (Fulk 2018, 97, §5.7). Hogg and Ringe and Taylor place such weakening within the broader late history of unstressed vowels (Hogg 1992, 120–21; Ringe and Taylor 2014, 298–332, §§6.8.3–6.9.6).

The tested forms do not determine the rule's position relative to a neighboring change.

SC076. Reduction of prefixal **i* in unstressed position (OEPrefixIReduction)

```
define OEPrefixIReduction [
    {*i} -> {*ĕ} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] EnglishStarVocalic
];
```

The rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. The development accounts for later prefix spellings such as OE **be-* and **ne-*.

If the rule is moved earlier or later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not place SC076 OEPrefixIReduction before or after any specific neighboring change.

The handbooks attest late prefix-vowel weakening, but the precise placement remains approximate. No lexical failure fixes it.

4.33 Weak-tail reduction

Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc **bákaną* ‘bake’ yields †*bacen* rather than expected OE *bacan* ‘bake’, and PGmc **bíndaną* ‘bind’ yields †*binden* rather than expected *bindan* ‘bind’, alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc **fléuxaną* ‘flee’ yields †*flēoan* rather than expected OE *flēon* ‘flee’, and PGmc **sláxaną* ‘slay’ yields †*sleaan* rather than expected *slēan* ‘slay’.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

4.34 Final-j loss and final geminate simplification

Historical discussion

After SC079 OEJLossAfterHeavy removes **j* in heavy environments, forms such as *lungen* 'lungs' acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final *nn* outcome in the following derivation.

SC079. Loss of **j* after heavy syllables (`OEJLossAfterHeavy`)

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

The rule removes **j* after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is **j*.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* 'believe' yields †*ġelēafan* rather than expected OE *ġelīefan* 'believe', PGmc **báugijaną* 'bow' yields †*bēaġan* rather than expected *bīeġan* 'bow', and PGmc **fúlgijaną* 'follow' yields †*fulġan* rather than expected *fylġan* 'follow'. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* 'lungs' yields †*lungenn* rather than expected OE *lungen* 'lungs'. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* 'lungs' yield †*lungenn* rather than expected OE *lungen* 'lungs'. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

4.35 J-strengthening, vocalization, and ei-contraction

Historical discussion

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then vocalizes the remaining intervocalic **j*, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ḗa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjaną* 'strew' yields †*strēaġan* rather than expected OE *strīeġan* 'strew'. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields †*strīeian* rather than *strīeġan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīeġan* 'strew' derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

SC082. Intervocalic vocalization of **j* (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule vocalizes intervocalic **j* to **i*, creating the *ei*-like sequence later removed by SC083 OEUnstressedEIContraction in many weak verb forms.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjaną* 'strew' yield †*strīeian* rather than expected OE *strīeġan* 'strew'. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjaną* 'bore' yield †*boreian* rather than expected OE *borian* 'bore', PGmc **xándlōjaną* 'handle' yield †*handleian* rather than expected *handlian* 'handle', and PGmc **mákōjaną* 'make' yield †*maceian* rather than expected *macian* 'make'. The witness forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ {*i}
```

```
];
```

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* 'bore' and *liccian* 'lick'.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **búrōjaną* 'bore' yield †*boreian* rather than expected OE *borian* 'bore', PGmc **líznōjaną* 'learn' yield †*liorneian* rather than expected *liornian* 'learn', and PGmc **líkkōjaną* 'lick' yield †*licceian* rather than expected *liccian* 'lick'. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

4.36 H-loss and contraction

Historical discussion

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (Fulk 2018, 270, §12.21), and Luick describes the corresponding West Germanic contractions (Luick 1914--1940, 165).

SC085. Loss of intervocalic **h* (`OEHLoss`)

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* 'toe' yields †*tāæ* rather than expected OE *tā* 'toe'. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* 'flee' yields †*flēoan* rather than expected OE *flēon* 'flee', PGmc **sláxaną* 'slay' yields †*sleaan* rather than expected *slēan* 'slay', PGmc **téxun* 'draw' yields †*teoon* rather than expected *tēon* 'draw', and PGmc **táixōn* yields †*tāe* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*ḗo},
    {*ḗo} {*o} -> {*ḗo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*ḗ} {*a} -> {*ḗ},
    {*ḗ} {*e} -> {*ḗ},
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*ī́} {*a} -> {*ī́},
    {*ī́} {*e} -> {*ī́},
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
```

```
    {*ū} {*e} -> {*ū}
];
```

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* ‘flee’, *slēan* ‘slay’, and *tēon* ‘draw’.

Moving contraction before SC085 OEHLoss makes PGmc **fléuxaną* ‘flee’ yield †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxaną* ‘slay’ yield †*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yield †*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* ‘toe’ yield †*tāe* rather than expected *tā* ‘toe’. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

4.37 R-metathesis

Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (Brunner 1965, 159, §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (Luick 1914--1940, 201).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstaną* ‘burst’ yield †*beorstan* rather than expected OE *berstan* ‘burst’. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters no output.

The lexical evidence fixes the earlier relation but does not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

Chapter 5

Old English orthography and the written surface

5.1 Historical interval

This short chapter stands apart from the derivational chapters that precede it. The changes of Chapters 1–3 are sound changes: they altered the spoken form of the language. The material treated here belongs instead to the written surface of Old English — scribal conventions that determined how the results of the completed phonological history were committed to parchment.

In the executable model these conventions apply after every phonological rule, at the very end of the cascade, because that is where they belong historically: a spelling practice can only render forms that the spoken language had already produced.

5.2 Scope

The one rule treated here is the West Saxon palatal-glide spelling (SC016), by which back vowels following word-initial [j] — spelled *g* — came to be written with a preceding front glide letter, as in *geoc* 'yoke' for spoken [jok] and *geoguþ* 'youth' for a form whose root vowel remained [u]. Ringe and Taylor state the modern assessment directly: the *eo* of *geoc* is a spelling convention, and the word was pronounced [jok] (Ringe and Taylor 2014, 5). Hogg reaches the same verdict for the back-vowel cases generally (Hogg 1992, 112). The older handbooks — Campbell, Brunner, Bülbring, Luick — analysed the same spellings as rising diphthongs; the section below presents both views (Campbell 1959, 17, § 44; Brunner 1965, 64–65, § 92).

5.3 Sources

Ringe and Taylor provide the modern phonological interpretation (Ringe and Taylor 2014, 5). Campbell (Campbell 1959, 17, 64–67, §§ 44, 170–176), Brunner (Brunner 1965, 64–65, § 92), and Bülbring (Bülbring 1902, 120, §§ 298–299) document the distribution of the spellings; Hogg supplies the critical reassessment (Hogg 1992, 112).

5.4 West Saxon palatal-glide spelling before back vowels

Historical discussion

West Saxon spellings such as *ġeoc* 'yoke', *ġeong* 'young', and *ġeoguþ* 'youth' write a front glide letter between a word-initial palatal and a following back vowel. Campbell describes the phenomenon as the development of rising diphthongs when "palatal glides developed before back vowels” and cites *ġeoc* directly (Campbell 1959, 17, §44); Brunner separates the *u*-cases (*ġeong*, *ġeoguþ*) from the *o*-cases (*ġioc* ‘yoke’, *ġeoc*) (Brunner 1965, 64–65, §92.1); Bülbring likewise treats *iuguð* ‘youth’ and *iuc* under *ju* but derives *ġioc*, *ġeoc* from West Germanic **jok* (Bülbring 1902, 120, §§298–299); and Luick groups all of these under his “schwebende Diphthonge” after palatal onsets (Luick 1914--1940, 158–59, §169).

The phonological interpretation of these spellings is disputed. The older handbook tradition — Campbell, Brunner, Bülbring, Luick — reads them as genuine rising diphthongs. The modern assessment is orthographic: Ringe and Taylor state flatly that *ġeoc* “is /jok/”, the digraph being a spelling convention that became universal after word-initial /j/ (Ringe and Taylor 2014, 5), and Hogg concludes that the back-vowel cases were “never anything more than an orthographic variation”, judging Campbell’s arguments to the contrary “insubstantial” (Hogg 1992, 112; Campbell 1959, 66–67, §176). This model follows Ringe and Taylor and Hogg: the rule is a spelling convention applied to the finished phonology, and it therefore stands at the end of the derivation, in the written-surface stage of the cascade.

Its position also settles a relative chronology. The *o* of *ġeoc* ‘yoke’ is itself the product of Northwest Germanic u-lowering (SC017 PNWGmcULowering): Fulk lists *ġeoc* as a regular lowering example beside OIcel *ok* and OHG *joh* (Fulk 2018, 56, §4.3), and Campbell gives *ġeoc* among the regular *u* > *o* words (Campbell 1959, 43, §115). The lowering therefore feeds the spelling: first **juk-* became **jok-* in Northwest Germanic, and only much later did West Saxon scribes write the result as *ġeoc*. Where lowering did not apply, as in *ġeoguþ* ‘youth’, whose root *u* was protected by the high vowel of the following syllable, the same convention wrote the retained *u* with the same digraph (Brunner 1965, 64–65, §92.1).

SC016. West Saxon palatal-glide spelling before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
    {*ó} -> {*éo} || .#. ġ _ ,
    {*ú} -> {*éo} || .#. ġ _ ,
    {*o} -> {*eo} || .#. ġ _ ,
    {*u} -> {*eo} || .#. ġ _
];
```

The rule rewrites a back vowel after word-initial *ġ* as the digraph spelling, covering both the lowered *o*-cases (*ġeoc* ‘yoke’) and the retained *u*-cases (*ġeoguþ* ‘youth’). Because it is a convention of the written language, it applies after every phonological change; in particular it follows SC017 PNWGmcULowering, which supplies the *o* of *ġeoc*. If the spelling rule were placed before u-lowering, the derivation would have to treat an Old English scribal practice as a Northwest Germanic sound change, an ordering that no source supports. The witnesses *ġeoc* and *ġeoguþ* between them fix both faces of the rule: one shows the convention applied to lowered *o*, the other to unlowered *u*. The handbook domain is broader (it also includes *a*/*ā*/*ō* contexts after word-initial palatals), but this executable rule is intentionally complete for the currently selected corpus witnesses rather than a maximal dialectal enumeration.

Part II

Lexical derivations

Word-by-word derivations

Introduction

The catalogue groups words by the kind of historical explanation they require. Regular derivations establish the reach of the sound laws. The remaining classes mark, without concealment, the places where attestation, morphology, analogy, or an unresolved mismatch intervenes.

The catalogue proceeds from words; the preceding part proceeds from rules. Together they allow each proposed sound law to be tested against its lexical witnesses and each etymology against the complete ordered history.

Data and sources

This volume assembles the lexical corpus from the aligned Germanic dataset and the compact derivation traces that accompany each entry. Comparative dictionaries, Old English dictionaries, and historical grammars are cited in the prose where they bear on particular lexical arguments.

The result is a lexical catalogue rather than a separate report on citation method or trace machinery.

Transducer and derivation method

Four objects must be distinguished in every derivation: the citation reconstruction, the selected input, the transducer outcome, and the Old English target. The summary identifies them where they differ; the boxed trace then divides the changes into Earlier Germanic and Old English stages.

Derivation classes

The lexical catalogue is ordered by seven derivation classes in the current manifest. Counts in this alpha are:

- Regular derivations: 76
- Attested variants: 4
- Early analogy: 36
- Late analogy: 27
- Reconstructed Old English comparators: 3
- Known but unmodelled developments: 2
- Unexplained or deliberately unmodelled exceptions: 5

Chapter 6

Regular derivations

In these entries the selected Germanic input yields the Old English reflex by regular sound change. They establish the standard against which the following analogical and exceptional histories must be judged.

6.1 adder — OE *nǣdre*

Derivation: **nḗdrōn* > *nǣdre* (regular).

Derivation trace

Proto input: **nḗdrōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**nǣdræ*
PNWGmc N Stem N Loss	**nḗdrǭ*	OE Unstressed AE Merger	**nǣdre*
PNWGmc Long E Lowering	**nǣdrǭ*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdre*

Reconstruction and comparative evidence

Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nǣdre* 'adder', *næddre* 'adder' under the latter (Kroonen 2013, 426). Orel likewise points from the masculine entry to a feminine **nēdrōn* 'adder' ~ **nadrōn* 'adder' type (Orel 2003, 325).

The derivational input therefore is not a reshaped convenience form. It is the comparative reconstruction that specifically underlies the Old English noun.

Old English evidence

The Old English word is securely represented by *nǣdre* 'adder', with *næddre* 'adder' as a secondary variant. Clark Hall cross-references *næddre* to *nædre* 'adder', and Fulk treats *næddre* as the later geminated form beside the older base (Clark Hall 1960, 225; Fulk 2018, 149).

Development to Old English

From **nḗdrōn* ‘adder’, the stressed long mid vowel develops to Old English *nǣdre* ‘adder’, and the weak feminine ending remains as final *-e*, giving *nǣdre*. The doubled consonant of *nǣddre* ‘adder’ is secondary and does not alter the inherited base form.

6.2 bake — OE *bacan*

Derivation: **bákaną* > *bacan* (regular).

Derivation trace

Proto input: **bákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**bækaną*
[no change]	OE A Restoration	**bakaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**bakan*
[no change]	OE Secondary Nasalization	**bakąn*
	OE Weak Tail Reduction	**bakan*

Old English form: *bacan*

Reconstruction and comparative evidence

Orel reconstructs the verb as **bakanan* ‘bake’ and cites Old English *bacan* ‘bake’ beside Old High German *backan, bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* ‘bake’ to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall both record *bacan* ‘bake’ as the ordinary Old English verb (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English

From **bákaną* ‘bake’, Anglo-Frisian brightening first gives **bækaną* ‘bake’. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* ‘bake’ (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: **bákaną* > *bacan*.

6.3 beech — OE *bōc*

Derivation: **bōkō* > *bōc* (regular).

Derivation trace

Proto input: **bōkō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bōk*
PNWGmc Final Long O Raising	**bōku*		
Early Anglo-Frisian			
[no change]			

Old English form: *bōc*

Reconstruction and comparative evidence

Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc* 'beech', *bēce* 'beech' among its reflexes (Kroonen 2013). The derivational input **bōkō* 'beech' is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence

Kroonen's Old English evidence already separates the paradigm material: *boc* 'beech' as the nominative form and *bēce* 'beech' as an oblique form (Kroonen 2013). The relevant comparator is therefore *bōc* 'beech'; *bēċe* 'beech' remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English

With nominative input **bōkō* 'beech', the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc* 'beech'. The regular comparison is therefore **bōkō* > *bōc*.

6.4 begin — OE *beġinnan*

Derivation: **bigínnaną* > *beġinnan* (regular).

Derivation trace

Proto input: **bigínnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**bigínnan*
[no change]	OE Secondary Nasalization	**bigínnąn*
Early Anglo-Frisian	OE Velar Palatalization	**bidʒínnąn*
[no change]	OE Prefix I Reduction	**bĕdʒínnąn*
	OE Weak Tail Reduction	**bĕdʒínnan*

Old English form: *beġinnan*

Reconstruction and comparative evidence

The verb is modeled here as inherited **bigínnaną* 'begin'. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (Ringe and Taylor 2014), and Campbell lists *ginnan* 'begin' among familiar examples of palatal *g* in this verb family (Campbell 1959, 174).

Old English evidence

Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* 'begin' / *beginnan* 'begin' (Bosworth and Toller 1898, 84; Clark Hall 1960). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan* 'begin'.

Development note

The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (Ringe and Taylor 2014).

Development to Old English

From **bigínnanq̨* ‘begin’, heavy-syllable nasal apocope yields **bigínnan* ‘begin’. Intervocalic **g* between front vowels then palatalizes to *ġ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan* ‘begin’.

6.5 bier — OE *bǣr*

Derivation: **bḗrō* > *bǣr* (regular).

Derivation trace

Proto input: **bḗrō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bǣr*
PNWGmc Final Long O Raising	**bḗru*		
PNWGmc Long E Lowering	**bǣru*		
Early Anglo-Frisian			
[no change]			

Old English form: *bǣr*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **bērō-* f. ‘bier’ and cites Old English *bar* ‘bier’, *bær* ‘bier’ among the reflexes (Kroonen 2013, 717). The derivational input **bḗrō* ‘bier’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller lemmatize the noun as *bær* ‘bier’, and Kroonen also records *bar* ‘bier’ beside it (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717). The target *bǣr* ‘bier’ is therefore a normalized long-vowel spelling of the same noun.

Source note

Lexicographic spellings vary between *bær* ‘bier’ and *bar* ‘bier’. The normalized target *bǣr* ‘bier’ represents the same long vowel (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717).

Development to Old English

From **bḗrō* ‘bier’, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *ǣ*, and high-vowel apocope yields *bǣr* ‘bier’. The resulting noun matches the normalized Old English target.

6.6 birth — OE *byrd*

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace

Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**byrdi*
EAF Final Z Deletion	**búrdi*	OE High Vowel Apocope	**byrd*
Early Anglo-Frisian			
[no change]			

Old English form: *byrd*

Reconstruction and comparative evidence

Kroonen cites the noun under stem-level **burdi-* and gives Old English *(ge-)byrd* among the reflexes (Kroonen 2013). The derivational input **búrdiz* 'birth' is the nominative-style form that stands behind that stem label.

Old English evidence

Clark Hall and Bosworth-Toller both attest simplex *byrd* 'birth' as an Old English noun meaning 'birth' (Clark Hall 1960; Bosworth and Toller 1898, 125). The prefixed form *gebyrd* 'birth' is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* 'birth' entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (Kroonen 2013; Bosworth and Toller 1898, 125; Campbell 1959).

Form note

The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Hogg's discussion of deverbal feminines provides the broader derivational setting (Hogg 1992).

Development to Old English

From **búrdiz* 'birth', loss of final *z* gives **búrdi* 'birth'. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

6.7 bone — OE *bān*

Derivation: **báiną* > *bān* (regular).

Derivation trace

Proto input: **báiną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**bān*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**bāną*		

Old English form: *bān*

Reconstruction and comparative evidence

Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* 'bone' (Kroonen 2013; Orel 2003). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān* 'bone'.

Old English evidence

Clark Hall and Bosworth-Toller record *bān* 'bone' as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898). Bright's glossary also distinguishes citation-form *bān* 'bone' from oblique *bāne* 'bone' (Bright 1971).

Source note

The comparative headwords **baina-* and **bainan* 'bone' provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną* 'bone'.

Development to Old English

West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną* 'bone'; heavy-syllable nasal apocope then yields *bān* 'bone'. The resulting form matches the attested Old English citation noun.

6.8 book — OE *bōc*

Derivation: **bōkz* > *bōc* (regular).

Derivation trace

Proto input: **bōkz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *bōc*

Reconstruction and comparative evidence

The noun is an athematic consonant stem (root noun), and the sources print its nominative singular in three different notations that encode the same morphology. Orel's citation form is **bōkz* 'book', with the explicit nominative marker **-z* that he writes across the whole root-noun class (Orel 2003, 52). Kroonen cites the bare stem **bōk-* 'book, beech' and treats Gothic *bōka* 'letter' as a separate ō-stem formation (Kroonen 2013, 71–72). Kluge/Seebold print a third variant with voiceless sibilant, **bōks* 'book' (Kluge 2011, 158). These are citation conventions, not competing facts: Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306). No scholar is simply "wrong" here; the notations differ in whether they make the inflectional marker visible.

The derivational input **bōkz* 'book' follows Orel's morphologically explicit convention: it records the nominative marker whose historical elimination the derivation then models. The word is distinct from the ō-stem 'beech' word, whose derivational input here is the separate nominative **bōkō* 'beech'.

Old English evidence

Old English *bōc* ‘book’ is the endingless nominative-accusative singular of a root noun, with the mutated plural *bēċ* ‘books’ preserving the class’s characteristic i-umlaut alternation (Fulk 2018, 165–66). Comparative root-noun evidence shows the branch-by-branch fate of the ending: Gothic keeps its sibilant in this class (*baúrgs* ‘city’, *nahts* ‘night’), while Old Norse *bók* ‘book’ keeps a reflex of **-z* as *-r* only in the masculine root nouns and drops it in feminines (Fulk 2018, 165–67). West Germanic alone is uniform: none of its daughters shows any ending in the root-noun nominative (Ringe and Taylor 2014, 118).

Development to Old English

The nominative marker of the consonant stems was eliminated before Proto-West Germanic: endingless nominatives arose by Szemerényi’s law after sonorant-final stems and spread analogically through the class (Fulk 2018, 143), so that no West Germanic daughter inherits an ending here (Ringe and Taylor 2014, 118). From the derivational input **bōkz* ‘book’ the loss of the marker gives **bōk* ‘book’, which is already the Old English form: *bōc* ‘book’ differs only in spelling. This early morphological loss is a different development from the later West Germanic loss of final **z* in unstressed syllables (as in **bárdaz* ‘beard’) and from the still later northern loss in stressed monosyllables (as in the pronouns); the root-noun ending was gone before either of those changes applied.

6.9 both — OE *bū*

Derivation: **bō* > *bū* (regular).

Derivation trace

Proto input: **bō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Stressed Monosyllable O Raising	**bū*	
Early Anglo-Frisian		
[no change]		

Old English form: *bū*

Reconstruction and comparative evidence

Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai* ‘both’, **bans* ‘both (pl.)’, **bōz* ‘both’/ **bōns* ‘both’, **bō* ‘both’, with Old English *bēġen* ‘both’, *bā* ‘both’, and neuter *bū* ‘both’ (Kroonen 2013, 47). For the present entry, the relevant inherited form is the unextended neuter dual **bō* ‘both’.

The older explanation of *bēġen* ‘both’ derives it from **bō-jen-*, and Orel still gives OE *bezen* ‘both’ (< **bō-jenō* ‘both’) beside ON *báðir* ‘both’, OFris *bēthe* ‘both’, OS *be-thia* ‘both’, and OHG *bēde* ‘both’ (Orel 2003, 65). Fulk reports that explanation cautiously and notes Seebold’s preference for a **bō-þ-* analysis instead (Fulk 2018, § 10.1). That debate concerns *bēġen* and the extended forms behind Modern English *both* ‘both’, German *beide* ‘both’, and Dutch *beide* ‘both’; it does not displace the inherited neuter **bō* ‘both’ > *bū* ‘both’ treated here.

Old English evidence

The Old English dual paradigm is well established. Brunner gives masculine *bēġen* ‘both’, feminine *bā* ‘both’, and neuter *bū* ‘both’ beside *bā*, with compounds such as *bā twā* ‘both (two)’, *bū* *tū* 'both (two, neut.)', and *bām* 'both' *twām* 'two' (Brunner 1965, § 324 Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēġen* 'both', feminine *bā*, neuter *bā*, *bū*, genitive *bēġra* 'both', *bēġ(e)a*, and dative *bǣm* 'both' (Campbell 1959, § 683; Fulk 2018, § 10.1).

bū 'both' is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēġen* 'both' belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* 'both' remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English

**bō* 'both' is a stressed monosyllabic form. Campbell cites *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (Campbell 1959, § 122). Brunner states the same development more directly: Auslautendes *ō* erscheint als û in *bū* … cu … *hū*, *tū* (Brunner 1965, § 69).

The development is therefore straightforward: **bō* 'both' > *bū* 'both'.

Form comparison

The comparison below sets the relevant forms side by side. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
**bō* > *bū*	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry
bēġen	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the Old English form here
bā	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator
báðir, German *beide*, Dutch *beide*, Modern English *both*	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE *bū*

6.10 bow — OE *bīeġan*

Derivation: **báugijaną* > *bīeġan* (regular).

Derivation trace

Proto input: **báugijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Fronting	**báeugījanq̨*
[no change]	OE Diphthong Leveling	**bēagījanq̨*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**bēagījan*
[no change]	OE Secondary Nasalization	**bēagījąn*
	Sievers Law Syncope	**bēagjąn*
	OE Velar Palatalization	**bēadʒjąn*
	OE I Umlaut	**bīedʒjąn*
	OE Weak Tail Reduction	**bīedʒjan*
	OE J Loss After Heavy	**bīedʒan*

Old English form: *bīeġan*

Reconstruction and comparative evidence

Kroonen reconstructs the weak verb as **baugjan-* 'to (make) bend' and cites Old English *biegan* 'bow, bend' among its reflexes (Kroonen 2013). Ringe and Taylor give the northwest Germanic preform **baugijana* 'bow', with later **béagjan* 'bow' behind West Saxon Old English *biegan* 'bow, bend' (Ringe and Taylor 2014). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence

Clark Hall lemmatizes *biegan* 'bow, bend' as an Old English weak verb, and Bosworth-Toller records *bigan* 'bow, bend' as an attested form (Clark Hall 1960; Bosworth and Toller 1898, 102). The form *bīeġan* 'bow, bend' used here is a normalized spelling of that attested Old English weak verb.

Development to Old English

From **báugijaną* 'bow', the stem reaches pre-Old-English **bēagjan* 'bow', after which palatalization of **gj* and i-umlaut yield West Saxon *biegan* 'bow'; Campbell lists *biegan* 'bow' among the regular *ie* outcomes of **éa* under i-umlaut (Ringe and Taylor 2014; Campbell 1959, 80). The development is therefore straightforward: **báugijaną* > *bīeġan* 'bow'.

6.11 breeches — OE *brēċ*

Derivation: **brōkiz* > *brēċ* (regular).

Derivation trace

Proto input: **brōkiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Palatalization	**brōtʃi*
EAF Final Z Deletion	**brōki*	OE I Umlaut	**brētʃi*
Early Anglo-Frisian		OE High Vowel Apocope	**brētʃ*
[no change]			

Old English form: *brēċ*

Reconstruction and comparative evidence

Kroonen cites the noun under **brōk-*, with Old English *brōc* 'breeches' and plural 'breeches' among its reflexes (Kroonen 2013). Ringe and Taylor give the plural development directly as PNWGmc **brōkiz* 'breeches' > **breeci* 'breeches' > OE *bréc* 'breeches' (Ringe and Taylor 2014).

The deeper verbal base belongs to the noun's etymological background, while the derivational input here is the plural noun form **brōkiz* 'breeches'.

Old English evidence

Bright notes *brōc* 'breeches' with plural *brēc* 'breeches', and Clark Hall gives *brēc* fp. breeches while also listing *broc* 'breeches' as a feminine noun probably represented chiefly in the plural (Bright 1971; Clark Hall 1960, 64). I write *brēċ* 'breeches' for the long vowel and palatal consonant; the attested plural is *brēc*.

Development to Old English

After loss of final -*z*, the stem ends in -*ki*, so the velar palatalizes and *ō* 'breeches' undergoes i-umlaut to *ē* 'breeches'; final high-vowel apocope then yields *brēċ* 'breeches' (Ringe and Taylor 2014). The development is therefore regular: **brōkiz* 'breeches' > *brēċ* 'breeches'.

6.12 calf — OE *ċealf*

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace

Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kálb*
EAF Final Z Deletion	**kálba*	EAF Brightening	**kælb*
Early Anglo-Frisian		OE Breaking	**kealb*
[no change]		PGmc B Allophony	**kealβ*
		OE Velar Palatalization	**ʧealβ*

Old English form: *ċealf*

Reconstruction and comparative evidence

Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz* 'calf' with plural **kalbizō* 'calves', while Orel cites **kalbaz* 'calf' as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz*, **kalbiz-* (Kroonen 2013; Orel 2003, 248; Ringe and Taylor 2014, 220). The derivational input here is the singular **kálbaz* 'calf', since the entry concerns the citation-form noun.

Old English evidence

Clark Hall gives *cealf* 'calf' I. (æ, e) nm. (nap. *cealfru* 'calves'), and Bosworth-Toller likewise records *Caelf* / *Cealf* 'calf' beside plural forms such as *calfur* and *cealfru* 'calves' (Clark Hall 1960; Bosworth and Toller 1898, 131). Campbell and Fulk show the same singular-plus--*r*- plural pattern (Campbell 1959; Fulk 2018, 193). I write *ċealf* 'calf' for the palatalized initial; the attested dictionary headword is *cealf* 'calf'.

Development to Old English

After loss of final -*z* and bare -*a*, Anglo-Frisian brightening gives **kælb* 'calf', and breaking before *l* plus consonant yields **kealb* 'calf'. Ringe and Taylor's account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *ċ*-, so **kálbaz* 'calf' develops regularly to *ċealf* 'calf' (Ringe and Taylor 2014, 220).

6.13 corn — OE *corn*

Derivation: **kúrną* > *corn* (regular).

Derivation trace

Proto input: **kúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**kórn*
PNWGmc U Lowering	**kórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *corn*

Reconstruction and comparative evidence

Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan* ‘corn’, both with Old English *corn* ‘corn, grain’ among the reflexes (Kroonen 2013; Orel 2003, 264). The singular form **kúrną* ‘corn’ is the nominative-accusative singular appropriate to the citation noun.

Old English evidence

Clark Hall gives *corn n. ‘corn,’ grain*, Bright’s glossary lists *corn, n.* with genitive singular *cornes*, and Bosworth-Toller treats *corn* ‘corn, grain’ as an ordinary noun headword (Clark Hall 1960; Bright 1971, 347; Bosworth and Toller 1898, 144). The target is therefore an attested citation form, while forms such as *cornes* simply provide paradigm background.

Development to Old English

With northwest Germanic lowering, **kúrną* ‘corn’ becomes **kórną* ‘corn’, and later loss of final nasal after a heavy syllable yields **kórn* ‘corn’, whence *corn* ‘corn, grain’. The oblique form **kurnăn* ‘corn’ belongs to comparative background rather than to the derivational input of this entry.

6.14 deed — OE *dǣd*

Derivation: **dḗdiz* > *dǣd* (regular).

Derivation trace

Proto input: **dḗdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**dǣd*
EAF Final Z Deletion	**dḗdi*		
PNWGmc Long E Lowering	**dǣdi*		
Early Anglo-Frisian			
[no change]			

Old English form: *dǣd*

Reconstruction and comparative evidence

Orel reconstructs the noun as **dēdiz* 'deed', and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* 'deed' through northwest Germanic **dadiz* 'deed' (Orel 2003; Ringe and Taylor 2014). The acute accent in **dḗdiz* 'deed' marks stress on the same reconstructed long vowel.

Old English evidence

Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* 'deed' beside non-West-Saxon *dēd* 'deed' (Campbell 1959; Brunner 1965). Clark Hall likewise lists *dæd* 'deed' and cross-refers Anglian *dēd* to it (Clark Hall 1960). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English

From inherited **dēdiz* 'deed', loss of final *-z* and the West Saxon lowering of stressed long *ē* yield *dǣd* 'deed'; Anglian *dēd* 'deed' preserves the non-West-Saxon outcome (Campbell 1959; Brunner 1965). The development treated here is therefore the regular West Saxon line.

6.15 door — OE *dor*

Derivation: **dúrą* > *dor* (regular).

Derivation trace

Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**dór*
PNWGmc U Lowering	**dórą*		
Early Anglo-Frisian			
[no change]			

Old English form: *dor*

Reconstruction and comparative evidence

Kroonen reconstructs a neuter **dura-* 'gate, (single) door' and cites Old English *dor* 'door' among its reflexes. In the same entry he separates Old English *duru* 'door', Old Frisian *dore*, and Old High German *tura* 'door' as reflexes of **durō-* instead (Kroonen 2013).

Old English evidence

Clark Hall records *dor* 'door' as a neuter noun and separately records feminine *duru* 'door' with its own inflection (Clark Hall 1960). Ringe and Taylor likewise treat *duru* 'door' as an early Old English u-stem, originally a root noun shifted into that class (Ringe and Taylor 2014). The Old English form here is therefore the attested neuter *dor* 'door', while *duru* 'door' remains a parallel Old English reflex from another stem history.

Development to Old English

From **dúrą* 'door', Northwest Germanic u-lowering gives **dórą* 'door', and heavy-syllable nasal apocope then yields *dor* 'door'. The regular development treated in this entry is therefore **dúrą* >

dor; the feminine *duru* ‘door’ belongs to the separate line identified by Kroonen and Ringe-Taylor (Kroonen 2013; Ringe and Taylor 2014).

6.16 fare — OE *faran*

Derivation: **fáraną* > *faran* (regular).

Derivation trace

Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**færaną*
[no change]	OE A Restoration	**faraną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**faran*
[no change]	OE Secondary Nasalization	**farąn*
	OE Weak Tail Reduction	**faran*

Old English form: *faran*

Reconstruction and comparative evidence

Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan* ‘fare’, both with Old English *faran* ‘fare’ among the reflexes (Kroonen 2013; Orel 2003, 132). Campbell also uses *faran* ‘fare’ as a standard example of Old English A-restoration (Campbell 1959, 61).

Old English evidence

Clark Hall lemmatizes the strong verb as *faran* ‘fare’ and separately records weak *færan* ‘to frighten’; *fære* ‘fare’, *færst* ‘fare’, and *færð* ‘fare’ belong to present-tense forms of *faran* rather than to the infinitive itself (Clark Hall 1960). Bosworth-Toller preserves the same distinction (Bosworth and Toller 1898, 108). The Old English form here is therefore the attested citation infinitive *faran* ‘fare’.

Development to Old English

From **fáraną* ‘fare’, Anglo-Frisian brightening first gives **færaną* ‘fare’, but A-restoration before single *r* returns **faraną* ‘fare’; later apocope and weak-tail reduction yield *faran* ‘fare’ (Campbell 1959, 61). Fulk’s contrast with participial faren- < **farœn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (Fulk 2018).

6.17 fell — OE *fell*

Derivation: **féllą* > *fell* (regular).

Derivation trace

Proto input: **féllą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**féll*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *fell*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **fella-* ‘membrane, skin, hide’ and cites Old English *fell* ‘skin, fell’ beside Dutch *vel* ‘skin, fell’ and German *Fell* (Kroonen 2013). The derivational input **féllą* ‘fell’ is the derivable singular form of that same inherited noun.

Old English evidence

Clark Hall records *fell* ‘skin, fell’ as the noun ‘fell, skin, hide’, and Bright’s glossary likewise gives *fell* ‘skin, fell’ with inflected forms such as accusative singular *fel* and dative plural *fellum* (Clark Hall 1960; Bright 1971). The target is therefore the attested noun *fell* ‘skin, fell’, not the verb *fellan* or the preterite *feoll*.

Development to Old English

With **féllą* ‘fell’, no special earlier reshaping is needed: heavy-syllable nasal apocope yields **féll* ‘fell’, surfacing as *fell* ‘skin, fell’. The regular development treated here is therefore **féllą* > *fell*.

6.18 fern — OE *fearn*

Derivation: **fárnaz* > *fearn* (regular).

Derivation trace

Proto input: **fárnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fárn*
EAF Final Z Deletion	**fárna*	EAF Brightening	**færn*
Early Anglo-Frisian		OE Breaking	**fearn*
[no change]			

Old English form: *fearn*

Reconstruction and comparative evidence

Kroonen cites the noun as masculine **farna-* and gives Old English *fearn*, *fern*, while Orel gives the same lexeme as neuter **farnan* ‘fern’ with Old English *fearn* ‘fern’ (Kroonen 2013; Orel 2003, 133). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style **fárnaz* ‘fern’.

Old English evidence

Clark Hall gives *fearn* ‘fern’ as an Old English noun, and Bosworth-Toller records *fearn* ‘fern’ with inflected forms such as *fearnes*, *fearna*, and *fearne* (Clark Hall 1960; Bosworth and Toller 1898, 219). Kroonen also records *fern* ‘fern’, but the local lexical sources give stronger support to *fearn* ‘fern’ as the citation target (Kroonen 2013).

Development to Old English

From **fárnaz* ‘fern’, loss of final *-z* and final *-a* gives **fárn* ‘fern’; Anglo-Frisian brightening then yields **færn* ‘fern’, and breaking before *r* plus consonant gives *fearn* ‘fern’ (Campbell 1959; Ringe and Taylor 2014). The development treated here is therefore the regular *rC*-breaking line.

6.19 field — OE *feld*

Derivation: **félθuz* > *feld* (regular).

Derivation trace

Proto input: **félθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**féld*
EAFL Th Voicing	**félduz*		
EAF Final Z Deletion	**féldu*		
Early Anglo-Frisian			
[no change]			

Old English form: *feld*

Reconstruction and comparative evidence

Ringe and Taylor treat Old English *feld* 'field' as one of the cases where earlier **felþu-* ~ **feldu-* may reflect either inherited *þ* ~ **d* alternation or the regular West Germanic development **lþ* > *ld* (Ringe and Taylor 2014, 170). The broader proto label **félθuz* 'field' can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence

Clark Hall records *feld* 'field' with oblique forms such as *felda* and *felde*, and Campbell notes early place-name spellings in *-felth* beside the later standard form (Clark Hall 1960, 114; Campbell 1959, 169). The Old English form here is therefore the attested citation noun *feld* 'field', with the older *-felth* spellings as historical support rather than as rival targets.

Development to Old English

In the modeled pathway, medial **lþ* becomes *ld*, final *-z* is lost, and high-vowel apocope then yields *feld* 'field'. Whether the voiced dental ultimately reflects inherited alternation or the regular **lþ* > *ld* development, both accounts converge on the same Old English form (Ringe and Taylor 2014, 170).

6.20 flea — OE *flēah*

Derivation: **fláuxz* > *flēah* (regular).

Derivation trace

Proto input: **fláuxz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Fronting	**fláeux*
[no change]		OE Diphthong Leveling	**flēax*
Early Anglo-Frisian			
[no change]			

Old English form: *flēah*

Reconstruction and comparative evidence

This word carries a genuine stem-class dispute, not merely a notational one. Orel reconstructs a root noun with his class-wide explicit nominative marker, **flauxz* 'flea' (Orel 2003, 105). Kroonen instead reconstructs an ō-stem **flauhō-* 'flea', with no *-z* at any layer of the analysis (Kroonen 2013, 145). Kluge and Seebold offer yet a third analysis, a masculine a-stem **flauha-* 'flea' (Kluge 2011, 304), whose nominative singular **flauhaz* would have carried its *-z* in an unstressed syllable and lost it under the West Germanic unstressed loss rather than with the root nouns. The attested forms do not decide the question: Old English *flēah* 'flea' and Old Norse *fló* 'flea' are compatible with all three reconstructions. This is a case where reputable sources disagree about the morphological class itself, and the disagreement is documented here rather than silently normalized away.

The derivational input **fláuxz* 'flea' adopts Orel's root-noun analysis with its morphologically explicit nominative marker. Under Kroonen's ō-stem analysis the nominative would instead have ended in **-ō* and developed like the other ō-stems; the choice of Orel's form keeps the word in the root-noun class alongside 'book', 'goose', and 'louse', whose endingless West Germanic nominatives it shares (Ringe and Taylor 2014, 118).

Old English evidence

Old English *flēah* 'flea' is the endingless nominative-accusative singular. The root-noun class to which Orel assigns it shows no nominative ending anywhere in West Germanic (Ringe and Taylor 2014, 118); on Kroonen's ō-stem analysis the endingless form would instead reflect the regular fate of the ō-stem nominative in this shape. Either way the attested Old English form is the expected one, which is precisely why the class question remains open.

Development to Old English

On the selected root-noun analysis, the nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **fláux* 'flea' from the derivational input **fláuxz* 'flea'. The diphthong is then fronted and levelled in Old English, yielding **flēax* 'flea', spelled *flēah* 'flea' with final *h* for the fricative. The early loss of the root-noun marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the derivational input is a stressed monosyllable, but its **-z* stands after a consonant and was gone long before the northern change, whose domain is vowel-final monosyllables.

6.21 fly — OE *flēogan*

Derivation: **fléuganą* > *flēogan* (regular).

Derivation trace

Proto input: **fléuganą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**flēoganą*
[no change]	OE Heavy Syllable Nasal Apocope	**flēogan*
Early Anglo-Frisian	OE Secondary Nasalization	**flēogąn*
[no change]	OE Weak Tail Reduction	**flēogan*

Old English form: *flēogan*

Reconstruction and comparative evidence

Ringe and Taylor derive the verb as **fleugana* 'fly' > OE *fléogan* 'fly' and elsewhere contrast West Saxon *fléogan* 'fly' with Anglian *flégan* 'fly', alongside related forms noted in the form note below (Ringe and Taylor 2014). The derivational input **fléuganą* 'fly' represents that inherited strong verb in the notation used here.

Old English evidence

Clark Hall and Bosworth-Toller record *flēogan* 'fly' as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag* 'flew', *flugon* 'flew', *flogen* 'flown' with present *fleogeð* 'flies' (Clark Hall 1960; Bosworth and Toller 1898; Bright 1971). The target in this entry is therefore the attested verbal infinitive itself.

Form note

Ringe and Taylor also list related *fléoge* 'fly' / *flége* 'fly' and Anglian *flégan* 'fly', which belong to the same family but do not replace the infinitive *flēogan* 'fly' treated here (Ringe and Taylor 2014).

Development to Old English

From **fléuganą* 'fly', Old English diphthong leveling gives **flēoganą* 'fly'; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* 'fly' (Ringe and Taylor 2014). The development is therefore regular: **fléuganą* > *flēogan* 'fly'.

6.22 forlorn — OE *lēosan*

Derivation: **léusaną* > *lēosan* (regular).

Derivation trace

Proto input: **léusaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**lēosaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēosan*
Early Anglo-Frisian	OE Secondary Nasalization	**lēosąn*
[no change]	OE Weak Tail Reduction	**lēosan*

Old English form: *lēosan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fraliusan* 'lose' and Old English *for-lēosan* 'lose, forlorn'; Orel likewise gives Old English *for-leósan* 'lose' (Kroonen 2013; Orel 2003). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence

The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* 'lose, forlorn' and especially in the participle *forloren* 'forlorn', recorded by Ringe and Taylor and in the dictionaries (Ringe and Taylor 2014; Clark Hall 1960; Bosworth and Toller 1898). The simplex infinitive *lēosan* 'lose' represents the verbal base itself.

Form note

As a base-form comparison, the simplex infinitive is *lēosan* ‘lose’, while the English adjective continues the prefixed Old English family *forlēosan* ‘lose’ / *forloren* ‘forlorn’ (Ringe and Taylor 2014).

Development to Old English

From **léusaną* ‘lose’, Old English diphthong leveling gives **lēosaną* ‘lose’, and later nasal apocope and weak-tail reduction yield *lēosan* ‘lose’ (Ringe and Taylor 2014). The prefixed forms follow the same verbal base with added *for-*.

6.23 gang — OE *gang*

Derivation: **gángaz* > *gang* (regular).

Derivation trace

Proto input: **gángaz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		PWGmc Final Bare A Loss	**gáng*
EAF Final Z Deletion	**gánga*		
Early Anglo-Frisian			
[no change]			

Old English form: *gang*

Reconstruction and comparative evidence

Orel reconstructs the noun as **gangaz* ‘gang’ and cites Old English *gang* ‘going, way’ beside Old Norse *gangr*, Old Frisian *gang* / *gong*, Old Saxon *gang* ‘going, way’, and Old High German *gang* ‘going, way’ (Orel 2003). The derivational input **gángaz* ‘gang’ is the same lexeme in the accent notation used here.

Old English evidence

Clark Hall and Bosworth-Toller both record *gang* ‘going, way’ as the noun ‘going, journey, way’, and Bright’s glossary gives *gong (gang), m., path, course* (Clark Hall 1960; Bosworth and Toller 1898, 159; Bright 1971, 392). The target is therefore the attested noun headword itself.

Form note

This entry concerns the noun *gang* ‘going, way’, not the separate verb *gangan* (Clark Hall 1960; Bosworth and Toller 1898, 159).

Development to Old English

From **gángaz* ‘gang’, loss of final *-z* gives **gánga* ‘gang’, and later loss of final bare *-a* yields *gang* ‘going, way’. The development is therefore regular: **gángaz* > *gang*.

6.24 give — OE *ġiefan*

Derivation: **gébaną* > *ġiefan* (regular).

Derivation trace

Proto input: **gébaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**géban*
[no change]		OE Secondary Nasalization	**gébąn*
Early Anglo-Frisian		PGmc B Allophony	**géβąn*
[no change]		OE Velar Palatalization	**ʤéβąn*
		OE Ws Palatal Diphthongization	**ʤíeβąn*
		OE Weak Tail Reduction	**ʤíeβan*

Old English form: *ġiefan*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* 'give' among its reflexes (Kroonen 2013). Ringe and Taylor contrast West Saxon *giefan* 'give' with Mercian *for-geofan* 'give' and Northumbrian *geafa* 'give', showing that the inherited verb takes different later dialectal shapes (Ringe and Taylor 2014).

Old English evidence

Campbell gives *gefan* (W-S *giefan* 'give') among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* 'give' with forms such as *geaf* and *giefen* (Campbell 1959; Clark Hall 1960). I write *ġiefan* 'give' for the palatal initial.

Dialect note

West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* 'give' or *for-geofan* 'give' continue the same verb without the West Saxon vocalism (Ringe and Taylor 2014).

Development to Old English

From **gébaną* 'give', initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* 'give' (Campbell 1959; Ringe and Taylor 2014). The result is therefore the regular West Saxon infinitive.

6.25 gold — OE *gold*

Derivation: **gúlθą* > *gold* (regular).

Derivation trace

Proto input: **gúlθą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**góld*
EAFL Th Voicing	**gúldą*		
PNWGmc U Lowering	**góldą*		
Early Anglo-Frisian			
[no change]			

Old English form: *gold*

Reconstruction and comparative evidence

Ringe and Taylor cite the noun as **gulþa-* / **gulda-*, and Kroonen gives the same pair (Ringe and Taylor 2014, 42; Kroonen 2013). The derivational input **gúlθą* 'gold' preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence

Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (Bosworth and Toller 1898, 121; Clark Hall 1960, 152). The target is therefore the attested citation form itself.

Development note

Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa-* / **gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (Ringe and Taylor 2014, 42).

Development to Old English

From **gúlθą* 'gold', the regular consonant development gives **gúldą* 'gold'; Northwest Germanic / Old English lowering then yields **góldą* 'gold', and apocope gives *gold* (Campbell 1959; Ringe and Taylor 2014, 42).

6.26 goose — OE *gōs*

Derivation: **gánsz* > *gōs* (regular).

Derivation trace

Proto input: **gánsz*

Earlier Germanic changes		**Old English changes**
Northwest and West Germanic		[no change]
EAF Nasal Spirant Lengthening	**gōns*	
EAF Nasal Spirant Loss	**gōs*	
Early Anglo-Frisian		
[no change]		

Old English form: *gōs*

Reconstruction and comparative evidence

The 'goose' word is a root noun whose nominative singular has an unusually layered history. Orel's citation form is **gansz* 'goose', with his class-wide explicit nominative marker **-z* stacked on the root-final *s* (Orel 2003, 126). Kroonen reconstructs an endingless nominative **gans* 'goose' with Verner-voiced *z* only in the oblique cells (genitive **gunzaz* 'goose (gen.)') (Kroonen 2013, 168–69), and Kluge/Seebold treat the German cognate under the same root noun (Kluge 2011, 331). Bammesberger supplies the deeper layer: by Szemerényi's law the Proto-Indo-European nominative **ǵ^{h}anss* yielded **ǵ^{h}ān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196). A mainstream specialist thus reconstructs a Proto-Germanic 'goose' nominative with a final voiced sibilant; Orel's notation is not idiosyncratic, and Kroonen's endingless citation reflects a different analytical layer of the same paradigm rather than a contradictory fact.

The derivational input **gánsz* ‘goose’ follows Orel’s morphologically explicit convention, keeping the nominative marker visible so that its historical elimination is modelled rather than presupposed.

Old English evidence

Old English *gōs* ‘goose’ is the endingless nominative-accusative singular, with mutated plural *ġēs* ‘geese’ (Fulk 2018, 165–66). The comparative picture matches the rest of the root-noun class: Gothic replaced the word (its ‘goose’ is the u-stem remodeling behind Spanish *ganso* ‘goose’ (Orel 2003, 126)), and no West Germanic daughter shows any nominative ending (Ringe and Taylor 2014, 118).

Development to Old English

The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **gáns* ‘goose’ from the derivational input **gánsz* ‘goose’. The remaining development is the familiar Ingvaeonic nasal-spirant sequence: the vowel lengthens and rounds before the nasal + voiceless fricative cluster and the nasal is then lost, giving **gōs* ‘goose’, which is the attested Old English form. The early loss of the nominative marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; had the marker survived to the later monosyllabic change, a lengthened vowel before *s* would not be the expected outcome of this shape, since the root-noun ending was already gone.

6.27 grave — OE *grafan*

Derivation: **grábaną* > *grafan* (regular).

Derivation trace

Proto input: **grábaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**græbaną*
[no change]	OE A Restoration	**grabaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**graban*
[no change]	OE Secondary Nasalization	**grabąn*
	PGmc B Allophony	**graβąn*
	OE Weak Tail Reduction	**graβan*

Old English form: *grafan*

Reconstruction and comparative evidence

Campbell gives *grafan* ‘dig, grave’ among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence

Clark Hall records *grafan* ‘dig, grave’ as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (Clark Hall 1960). The target here is the attested infinitive headword of the verb.

Development to Old English

From **grábaną* ‘grave’, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* ‘dig, grave’ (Campbell 1959, 61; Ringe and Taylor 2014).

Form note

Noun *græf* ‘grave’ and verbal forms such as *græfð* ‘graves’ or past participial *græfen* ‘graven’ belong to other lexical or paradigm positions and do not replace the infinitive *grafan* ‘grave’ as the target here (Clark Hall 1960).

6.28 guest — OE *ġiest*

Derivation: **gástiz* > *ġiest* (regular).

Derivation trace

Proto input: **gástiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**gæsti*
EAF Final Z Deletion	**gásti*	OE Velar Palatalization	**ʤæsti*
Early Anglo-Frisian		OE I Umlaut	**ʤesti*
		OE Ws Palatal Diphthongization	**ʤiesti*
[no change]		OE High Vowel Apocope	**ʤiest*

Old English form: *ġiest*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* ‘guest’ type (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record the word under forms such as *gist* ‘guest’, *gest* ‘guest’, *giest* ‘guest’, and *gyst* ‘guest’ (Bosworth and Toller 1898; Clark Hall 1960). The Old English form here, *ġiest*, ‘guest’ is the normalized West Saxon form within that attested family.

Development to Old English

From **gástiz* ‘guest’, Anglo-Frisian brightening gives a *gæst-* stage, and i-mutation affects the front vowel before the lost high-vocalic ending. In West Saxon the initial palatal environment then produces *ie*, so the regular outcome is *ġiest* ‘guest’ (Campbell 1959; Ringe and Taylor 2014).

Dialect note

West Saxon *ġiest* ‘guest’ is the Old English form here. Anglian *gest* ‘guest’ and related spellings remain real Old English comparators rather than corrections to that choice (Ringe and Taylor 2014; Bosworth and Toller 1898).

6.29 hair — OE *hǣr*

Derivation: **xḗrą* > *hǣr* (regular).

Derivation trace

Proto input: **xḗrą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çǣrą*
PNWGmc Long E Lowering	**xǣrą*	OE Heavy Syllable Nasal Apocope	**çǣr*
Early Anglo-Frisian			
[no change]			

Old English form: *hǣr*

Reconstruction and comparative evidence

Kroonen cites the ordinary Proto-Germanic hair word as **hēra-* (Kroonen 2013). The derivational input **xḗrą* 'hair' represents that same long-ē stem in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hær* 'hair' / *hǣr* 'hair' as the ordinary Old English noun (Clark Hall 1960, 158; Bosworth and Toller 1898, 510). The target is therefore the attested headword itself.

Development to Old English

From **xḗrą* 'hair', Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hǣr* 'hair'. The development treated here is straightforward and does not require any special paradigm choice.

Form note

Older references to **xazwăz* 'hair' belong to a different lexeme, and the separate *haddr* / *heordan* / *hād-* material does not displace the ordinary simplex *hǣr* 'hair' treated here (Kroonen 2013; Clark Hall 1960, 158).

6.30 harvest — OE *hierfest*

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace

Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xærbistu*
EAF Final Z Deletion	**xárbistu*	OE Breaking	**xearbistu*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çearbistu*
[no change]		PGmc B Allophony	**çearβistu*
		OE I Umlaut	**çierβistu*
		OE High Vowel Apocope	**çierβist*
		OE Med Unstressed I Lowering1	**çierβest*

Old English form: *hierfest*

Reconstruction and comparative evidence

Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest* / *hyrfest* type (Bammesberger 1997; Ringe and Taylor 2014).

Old English evidence

Bosworth-Toller and Clark Hall record *hærfest* 'harvest', with *herfest* 'harvest' as a variant in the lexical tradition (Bosworth and Toller 1898; Clark Hall 1960). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English

From **xárbistuz* 'harvest', Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest* 'harvest'. That is the regular West Saxon development treated here (Ringe and Taylor 2014; Campbell 1959).

Source note

The Old English form here, *hierfest*, represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest* 'harvest' / *herfest* 'harvest', commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (Bammesberger 1997; Ringe and Taylor 2014).

6.31 hedge — OE *heġġ*

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace

Proto input: **xágjaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xággj*
PWGmc J Gemination	**xággjaz*	EAF Brightening	**xæggj*
EAF Final Z Deletion	**xággja*	OE Velar Fricative Palatalization	**çæggj*
Early Anglo-Frisian		OE Velar Palatalization	**çædʒdʒj*
[no change]		OE I Umlaut	**çedʒdʒj*
		OE J Loss After Heavy	**çedʒdʒ*

Old English form: *heġġ*

Reconstruction and comparative evidence

The derivational input models a palatal **-gj-** 'hedge' noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heġġ* 'hedge' type (Campbell 1959).

Old English evidence

Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* 'hedge' / *heċġ* 'hedge' (Bosworth and Toller 1898; Clark Hall 1960). The lexical item itself is therefore well attested even though the form compared here is normalized.

Development to Old English

From **xágjaz* ‘hedge’, West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heġġ* ‘hedge’. The development is treated as regular rather than exceptional.

Form note

Standard dictionary spelling is *heċġ* ‘hedge’ or *hecg* ‘hedge’. Normalized *heġġ* ‘hedge’ is the Old English form here, while the ordinary lexicographic forms remain the main Old English citation evidence (Bosworth and Toller 1898; Clark Hall 1960).

6.32 helm — OE *helm*

Derivation: **xélmaz* > *helm* (regular).

Derivation trace

Proto input: **xélmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xélm*
EAF Final Z Deletion	**xélma*	OE Velar Fricative Palatalization	**çélm*
Early Anglo-Frisian			
[no change]			

Old English form: *helm*

Reconstruction and comparative evidence

Kroonen cites the helmet noun as **helma-* and separately distinguishes a different lexeme **helman-* ‘rudder’ (Kroonen 2013). The derivational input **xélmaz* ‘helm’ is the nominative-style form used for the helmet noun itself.

Old English evidence

Clark Hall and Bosworth-Toller record *helm* ‘helm, helmet’ as the ordinary Old English noun for ‘helmet’, while *helma* ‘helm’ belongs to a separate rudder lexeme (Clark Hall 1960; Bosworth and Toller 1898, 542).

Development to Old English

From **xélmaz* ‘helm’, loss of final *z* and later loss of the short final vowel yield *helm* ‘helm, helmet’. The development is therefore a straightforward citation-form match.

Form note

Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma* ‘helm’, which is a different noun meaning ‘helm, rudder’ (Kroonen 2013; Clark Hall 1960).

6.33 help — OE *helpan*

Derivation: **xélpaną* > *helpan* (regular).

Derivation trace

Proto input: **xélpaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çélpaną*
[no change]		OE Heavy Syllable Nasal Apocope	**çélpan*
Early Anglo-Frisian		OE Secondary Nasalization	**çélpąn*
[no change]		OE Weak Tail Reduction	**çélpan*

Old English form: *helpan*

Reconstruction and comparative evidence

The entry treats the strong verb itself rather than the separate noun *help* 'help'. Bright's principal parts *helpan, healp, hulpon, holpen* show the ordinary Old English strong-verb family continued by this input (Bright 1971).

Old English evidence

Clark Hall and Bosworth-Toller record *helpan* 'help' as the verbal headword 'to help' (Clark Hall 1960; Bosworth and Toller 1898, 542). The target is therefore the attested infinitive citation form.

Development to Old English

From **xélpaną* 'help', no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* 'help' directly.

Form note

Noun *help* 'help' belongs to a separate lexical line and should not replace verbal *helpan* 'help' as the target here (Clark Hall 1960; Bosworth and Toller 1898, 542).

6.34 hind — OE *hind*

Derivation: **xéndjō* > *hind* (regular).

Derivation trace

Proto input: **xéndjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çéndju*
PNWGmc Final Long O Raising	**xéndju*	OE I Umlaut	**çindju*
Early Anglo-Frisian		OE High Vowel Apocope	**çindj*
[no change]		OE J Loss After Heavy	**çind*

Old English form: *hind*

Reconstruction and comparative evidence

Kroonen cites the animal name as **hindō-* f. 'hind' (Kroonen 2013). The derivational input **xéndjō* 'hind' represents that same noun in the present derivation.

Old English evidence

Clark Hall and Bosworth-Toller record *hind* ‘hind, doe’ as the noun ‘hind, female deer’ (Clark Hall 1960; Bosworth and Toller 1898, 554). The target is therefore the attested lexical item itself.

Development to Old English

From **xéndjō* ‘hind’, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note

hindan ‘from behind, behind’ is a different Old English lexeme and does not belong to the noun history of *hind* (Clark Hall 1960; Bosworth and Toller 1898, 554).

6.35 hold — OE *healdan*

Derivation: **xáldaną* > *healdan* (regular).

Derivation trace

Proto input: **xáldaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xœldaną*
[no change]	OE Breaking	**xealdaną*
Early Anglo-Frisian	OE Velar Fricative Palatalization	**çealdaną*
[no change]	OE Heavy Syllable Nasal Apocope	**çealdan*
	OE Secondary Nasalization	**çealdąn*
	OE Weak Tail Reduction	**çealdan*

Old English form: *healdan*

Reconstruction and comparative evidence

Campbell and Ringe-Taylor treat the verb as a regular *a* + *lC* breaking case, with West Saxon *healdan* ‘hold’ opposed to Anglian and Mercian *haldan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Old English evidence

Bright gives the ordinary strong-verb citation form and principal parts *healdan, heold, heoldon, healden* (Bright 1971). The target is therefore the attested infinitive headword itself.

Development to Old English

From **xáldaną* ‘hold’, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Dialect note

West Saxon *healdan* ‘hold’ is the Old English form here. Anglian and Mercian *haldan* ‘hold’ are genuine non-West-Saxon doublets rather than corrections to that choice (Campbell 1959; Ringe and Taylor 2014).

6.36 horn — OE *horn*

Derivation: **xúrną* > *horn* (regular).

Derivation trace

Proto input: **xúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**xórn*
PNWGmc U Lowering	**xórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *horn*

Reconstruction and comparative evidence

Kroonen and Orel cite lemma-style Proto-Germanic headwords of the **hurna-* / **xurnan* 'horn' type for this noun (Kroonen 2013; Orel 2003, 234). The derivational input **xúrną* 'horn' is the nominative-style form used in the derivation here.

Old English evidence

Clark Hall, Bosworth-Toller, and Bright all record *horn* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898, 108; Bright 1971). The target is therefore the attested citation form.

Development to Old English

From **xúrną* 'horn', Northwest Germanic u-lowering gives **xórną* 'horn', and later loss of the final nasal leaves *horn*. The development treated here is fully regular.

Form note

The note's oblique **xurnăn* 'horn' belongs to comparative stem background only. It does not replace the derivational input **xúrną* 'horn' as the derivational form used here (Kroonen 2013; Orel 2003, 234).

6.37 lead — OE *lǣdan*

Derivation: **láidijaną* > *lǣdan* (regular).

Derivation trace

Proto input: **láidijaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lādijan*
[no change]		OE Secondary Nasalization	**lādiją̄n*
Early Anglo-Frisian		Sievers Law Syncope	**lādją̄n*
EAF Ai Monophthongization	**lādijaną*	OE I Umlaut	**lǣdją̄n*
		OE Weak Tail Reduction	**lǣdjan*
		OE J Loss After Heavy	**lǣdan*

Old English form: *lǣdan*

Reconstruction and comparative evidence

Ringe and Taylor derive Old English *lǣdan* ‘lead’ from Proto-Germanic **laidijaną* ‘lead’, and Kroonen likewise cites a weak verb of the **laidjan-* type for ‘lead’ (Ringe and Taylor 2014; Kroonen 2013, 363).

Old English evidence

Clark Hall and Bosworth-Toller both record *lædan* ‘lead’ / *lǣdan* ‘lead’ as the ordinary Old English verb ‘to lead, guide, conduct’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **láidijaną* ‘lead’, monophthongization of **ai* first gives a **lād-* stage. Later syncope, i-mutation, weak-tail reduction, and loss of *j* after a heavy stem yield *lǣdan* ‘lead’, so the development represented here is fully regular (Ringe and Taylor 2014).

6.38 learn — OE *liornian*

Derivation: **líznōjaną* > *liornian* (regular).

Derivation trace

Proto input: **líznōjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**líornōjaną*
Rhotacism	**lírnōjaną*	OE Heavy Syllable Nasal Apocope	**líornōjan*
Early Anglo-Frisian		OE Secondary Nasalization	**líornōjąn*
[no change]		OE I Umlaut	**líornējąn*
		OE Unstressed Long Vowel Shortening	**líornejąn*
		OE Weak Tail Reduction	**líornejan*
		OE Intervocalic J Vocalization	**líorneian*
		OE Unstressed EI Contraction	**líornian*

Old English form: *liornian*

Reconstruction and comparative evidence

Ringe and Taylor place the verb in a class-II weak family of the **liznō-* type (Ringe and Taylor 2014), and Kroonen keeps the same comparative base for the Germanic lexeme (Kroonen 2013). The derivational input therefore requires no change of stem class or paradigm cell.

The non-obvious issue is dialectal. Campbell records Northumbrian *liornian* ‘learn’ beside *leornian* ‘learn’, and he states that the *eo* of *leornian* ‘learn’ is secondary, while Northumbrian preserves forms with *io*, where original *eo* and *io* remain distinct (Campbell 1959, § 123 n. 2).

Old English evidence

**liornian* ‘learn’ is the Northumbrian member of the Old English family. Dictionary practice more often privileges *leornian* ‘learn’ as the ordinary headword (Clark Hall 1960; Bright 1971), but Campbell’s dialect evidence shows that *liornian* ‘learn’ is a genuine Old English form (Campbell 1959, § 123 n. 2).

This entry therefore remains compact. The point is to state clearly that the Old English form here belongs to the Northumbrian side of the OE evidence rather than to the leveled *leornian* ‘learn’ headword tradition.

Development to Old English

From **líznōjaną* ‘learn’, the expected Old English developments include rhotacism of *z*, followed by breaking before *r* plus consonant, and the ordinary reduction of the weak verbal ending. The result is *liornian* ‘learn’, preserving the *io* spelling that Campbell associates with Northumbrian where original *eo* and *io* remain distinct (Campbell 1959, § 123 n. 2).

Leornian reflects the later *eo* development that Campbell treats as secondary [Campbell (1959), §123 n. 2; §296]. The selected Northumbrian target is therefore the regular comparison form for the *i*-grade member of the family.

Form comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular Northumbrian form modeled here from the better-known West Saxon headword.

Form	Status	Relevance to this entry
**líznōjaną* > *liornian*	computed regular output; attested Northumbrian comparison form	selected comparison
leornian	attested later *eo* form and dictionary headword	useful control, but not the target of this entry

6.39 lid — OE *hlid*

Derivation: **xlídą* > *hlid* (regular).

Derivation trace

Proto input: **xlídą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xlíd*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hlid*

Reconstruction and comparative evidence

Orel cites a neuter lexeme of the **xliđ-* type with Old English *hlid* ‘lid, cover’, and Lloyd includes OE *hlid* ‘lid’ beside ON *hliþó* ‘lid’ and OHG *(h)lit* ‘lid’ among forms that retain *i* (Orel 2003; Lloyd 1966).

Old English evidence

Clark Hall and Bosworth-Toller record *hlid* ‘lid, cover’ as the noun ‘lid, cover, door, gate’ (Clark Hall 1960; Bosworth and Toller 1898, 563).

Development to Old English

The derivational input already represents the later Germanic *hliđ-* stage used for the derivation here. From **xlídą* 'lid', heavy-syllable apocope yields *hlid* 'lid', and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (Lloyd 1966).

Form note

An earlier etymological stage **liþuz* 'lid' belongs to comparative background only. The form represented here is the later **xlídą* 'lid' > *hlid* 'lid' line that matches the attested Old English noun (Orel 2003; Lloyd 1966).

6.40 light — OE *līehtan*

Derivation: **léuxtijaną* > *līehtan* (regular).

Derivation trace

Proto input: **léuxtijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**lēoxtijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēoxtijan*
Early Anglo-Frisian	OE Secondary Nasalization	**lēoxtijąn*
[no change]	Sievers Law Syncope	**lēoxtjąn*
	OE I Umlaut	**līextjąn*
	OE Weak Tail Reduction	**līextjan*
	OE J Loss After Heavy	**līextan*

Old English form: *līehtan*

Reconstruction and comparative evidence

Fulk gives Proto-Germanic **liuxtijanan* 'light' with Old English *līehtan* 'illuminate', and Ringe and Taylor likewise derive West Saxon *liehtan* 'light, illuminate' from the same weak-verb formation (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan* 'illuminate', *lihtan* 'light, illuminate', and *līhtan* 'illuminate', distinct from the related noun *lēoht* 'light' and adjective *leoht* / *liht* 'light' (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **léuxtijaną* 'light', the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* 'illuminate' stage, and is represented here by normalized *līehtan* 'illuminate'. The word treated in this entry is therefore the verb 'to light, illuminate', not the related noun from **leuxtą* 'light' (Fulk 2018; Ringe and Taylor 2014).

Dialect note

Ringe and Taylor and Campbell distinguish West Saxon *liehtan* 'light, illuminate' from Anglian *lihtan* 'light, illuminate', while later West Saxon also shows *lyhtan* 'light, illuminate' (Ringe and Taylor 2014; Campbell 1959).

6.41 linden — OE *lind*

Derivation: **líndō* > *lind* (regular).

Derivation trace

Proto input: **líndō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**línd*
PNWGmc Final Long O Raising	**líndu*		
Early Anglo-Frisian			
[no change]			

Old English form: *lind*

Reconstruction and comparative evidence

Kroonen cites **lindō-* 'lime tree' and gives Old English *lind* 'linden, lime tree' as the relevant reflex (Kroonen 2013).

Old English evidence

Clark Hall and Bosworth-Toller both record *lind* 'linden, lime tree' as the noun 'lime-tree, linden' (Clark Hall 1960; Bosworth and Toller 1898, 630).

Development to Old English

From **líndō* 'linden', Northwest Germanic final **ō* raising first gives a **líndu* 'linden' stage, and later high-vowel apocope yields *lind* 'linden, lime tree'. The development is therefore straightforward and regular.

Form note

The Old English noun represented here is *lind* 'linden, lime tree'. Clark Hall also has a separate adjectival *linden* 'made of linden-wood', but that is not the noun counterpart for this entry (Clark Hall 1960).

6.42 louse — OE *lūs*

Derivation: **lūsz* > *lūs* (regular).

Derivation trace

Proto input: **lūsz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *lūs*

Reconstruction and comparative evidence

Orel's citation form is **lusz* 'louse', with the explicit nominative marker **-z* he writes across the root-noun class; his entry points to Bammesberger and Griepentrog for the morphology (Orel 2003, 252). Kroonen has no 'louse' entry, so his dictionary cannot be cited for this word (Kroonen 2013, 345). Kluge/Seebold carry the German cognate as the same root noun (Kluge 2011, 563). Bammesberger shows that the root-final *s* is itself secondary: the root **luw-* 'louse' was extended with *s* on the model of the 'mouse' word (Bammesberger 1990, 195). The 'mouse' parallel also shows why no extra sibilant is audible in the nominative: Ringe reconstructs endingless **mūs* 'mouse', with the expected double sibilant of the nominative removed by degemination (Ringe 2017, 149).

The derivational input **lūsz* 'louse' follows Orel's morphologically explicit convention. The final **-z* is an inflectional marker made visible in the citation form, not a claim that a phonetic cluster *[sz]* was ever pronounced; the notation records the morphology whose elimination the derivation models.

Old English evidence

Old English *lūs* 'louse' is the endingless nominative-accusative singular of a root noun, with mutated plural *lȳs* 'lice' (Fulk 2018, 165–66). Old Norse *lús* 'louse' is likewise endingless, in keeping with the North Germanic redistribution that drops the reflex of **-z* in feminine root nouns (Fulk 2018, 167). No West Germanic daughter shows any ending in this class (Ringe and Taylor 2014, 118).

Development to Old English

The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118). From the derivational input **lūsz* 'louse' the loss of the marker gives **lūs* 'louse', which is already the attested Old English form. Because the root itself ends in *s*, the marker's elimination here is best read together with the degemination that Ringe describes for 'mouse': at no stage does the reconstruction require an audible *-sz* sequence. This early morphological development is distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the root-noun ending was gone before either applied, and the root-final *s* of *lūs* 'louse' was never eligible for either change.

6.43 milk — OE *meoloc*

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace

Proto input: **mélukz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Med Unstressed U Lowering	**mélok*
EAF Final Z Deletion	**méluk*	OE Back Mutation	**méolok*
Early Anglo-Frisian			
[no change]			

Old English form: *meoloc*

Reconstruction and comparative evidence

Kroonen and Orel reconstruct the noun with stem notation **meluk-* and nominative **melukz* ‘milk’, and the nominative-style input used here is **mélukz* ‘milk’ (Kroonen 2013; Orel 2003, 306).

Old English evidence

Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* ‘milk’ < *meoluc* ‘milk’ < **mélukz* ‘milk’, Campbell likewise discusses *meoluc* ‘milk’ and *meoloc* ‘milk’, and Anglian shows *milc* ‘milk’ (Ringe and Taylor 2014; Campbell 1959).

Development to Old English

The unsyncopated line from **mélukz* ‘milk’ loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc* ‘milk’. That fuller unsyncopated outcome is the form represented here (Ringe and Taylor 2014).

Form comparison

Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc* / *meoluc* preserves the fuller nominal shape (Campbell 1959; Ringe and Taylor 2014).

6.44 mother — OE *mōder*

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace

Proto input: **mōdēr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**mōdœr*
PNWGmc Long E Lowering	**mōdǣr*	OE Unstressed AE Merger	**mōder*
Early Anglo-Frisian			
[no change]			

Old English form: *mōder*

Reconstruction and comparative evidence

Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder-* with citation **mōdēr* ‘mother’ (Kroonen 2013; Orel 2003).

Old English evidence

The transmitted Old English headword tradition is *mōdor* ‘mother’ / *modor* ‘mother’, with oblique *mēder* ‘mother’ in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (Clark Hall 1960; Campbell 1959; Ringe and Taylor 2014).

Development to Old English

From **mōdēr* ‘mother’, the regular suffixal development yields *mōder* ‘mother’. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* ‘mother’ reflects later levelling within the r-stem paradigm (Campbell 1959; Ringe and Taylor 2014).

Form comparison

The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* ‘mother’ is the regularized nominative represented here, but dictionaries usually print *mōdor* / *modor* ‘mother’, and the oblique evidence survives in *mēder* ‘mother’ (Clark Hall 1960; Ringe and Taylor 2014).

6.45 net — OE *nett*

Derivation: **nátją* > *nett* (regular).

Derivation trace

Proto input: **nátją*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**nættją*
PWGmc J Gemination	**náttją*	OE Heavy Syllable Nasal Apocope	**nættj*
Early Anglo-Frisian		OE I Umlaut	**nettj*
[no change]		OE J Loss After Heavy	**nett*

Old English form: *nett*

Reconstruction and comparative evidence

Orel gives **natjan* ‘net’ with Old English *nett* ‘net’, and Fulk’s account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (Orel 2003; Fulk 2018).

Old English evidence

Clark Hall and Bosworth-Toller record *nett* ‘net’ as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (Clark Hall 1960; Bosworth and Toller 1898, 29; Campbell 1959).

Development to Old English

From **nátją* ‘net’, West Germanic j-gemination first gives **náttją* ‘net’. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett* ‘net’, so the development represented here is regular (Fulk 2018).

Form note

Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* ‘net’ (Campbell 1959; Orel 2003).

6.46 nightmare — OE *mare*

Derivation: **márōn* > *mare* (regular).

Derivation trace

Proto input: **márōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mærǭ*
PNWGmc N Stem N Loss	**márǭ*	OE A Restoration	**marǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**marœ*
[no change]		OE Unstressed AE Merger	**mare*

Old English form: *mare*

Reconstruction and comparative evidence

Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **marōn-*, with Old English *mare, maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *mare* 'nightmare, monster' and also preserves related variant forms *mera* / *mere* (Clark Hall 1960, 213).

Development to Old English

The selected simplex input **márōn* 'nightmare' regularly gives *mare* 'nightmare' after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (Ringe and Taylor 2014).

Form note

The concept corresponds to an unattested compound **nihtmare* 'nightmare', but the Old English lexical evidence is for simplex *mare* 'nightmare', with oblique *maran* and variant *mere* / *mera* (Ringe and Taylor 2014; Clark Hall 1960, 213).

6.47 coat — OE *rocc*

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace

Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**rókk*
PNWGmc U Lowering	**rókkaz*		
EAF Final Z Deletion	**rókka*		
Early Anglo-Frisian			
[no change]			

Old English form: *rocc*

Reconstruction and comparative evidence

Orel cites a masculine **rukkaz* ‘coat’ for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (Orel 2003; Kroonen 2013, 290).

Old English evidence

Clark Hall and Bosworth-Toller record *rocc* ‘coat’ as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* ‘breast-coat’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

With **rúkkaz* ‘coat’ as the derivational input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note

This entry concerns the garment noun only. The stone word seen in *stānrocc* ‘stone-coat’ belongs to a different lexical history (Clark Hall 1960; Bosworth and Toller 1898).

6.48 sheep — OE *sċēap*

Derivation: **skḗpą* > *sċēap* (regular).

Derivation trace

Proto input: **skḗpą*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**skǣp*
PNWGmc Long E Lowering	**skǣpą*	OE Sk Palatalization	**ʃǣp*
Early Anglo-Frisian		OE Ws Palatal Diphthongization	**ʃēap*
[no change]			

Old English form: *sċēap*

Reconstruction and comparative evidence

Ringe and Taylor cite a later West Germanic **skap* ‘sheep’ > WS *scéap* ‘sheep’, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (Ringe and Taylor 2014; Orel 2003).

Old English evidence

Clark Hall records *scēap* ‘sheep’ with spelling variation, and Campbell likewise lists West Saxon *scéap* ‘sheep’ among the palatal-diphthongized forms (Clark Hall 1960; Campbell 1959).

Development to Old English

From **skḗpą* ‘sheep’, Northwest Germanic lowering gives **skǣpą* ‘sheep’; after apocope and palatalization the West Saxon branch diphthongizes to *sċēap* ‘sheep’. The development represented here is therefore fully regular.

Dialect note

Ringe and Taylor contrast West Saxon *scéap* ‘sheep’ with Mercian and Kentish *scép* ‘sheep’, and Campbell also notes Northumbrian *scip* ‘sheep’. The form represented here is the West Saxon headword (Ringe and Taylor 2014; Campbell 1959).

6.49 shilling — OE *sċilling*

Derivation: **skíllingaz* > *sċilling* (regular).

Derivation trace

Proto input: **skíllingaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**skílling*
EAF Final Z Deletion	**skíllinga*	OE Sk Palatalization	**ʃílling*
Early Anglo-Frisian		OE Med Unstressed I Lowering1	**ʃílleng*
[no change]		OE Med Unstressed I Lowering	**ʃílling*

Old English form: *sċilling*

Reconstruction and comparative evidence

Kroonen treats the cognate set under **skellinga-* ~ **skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* ‘shilling’ among the reflexes (Kroonen 2013; Orel 2003). The derivational input **skíllingaz* ‘shilling’ is the nominative-style form used here to represent that inherited **-ing-* derivative.

Old English evidence

Clark Hall records *scilling* ‘shilling’, and Campbell cites it among nouns with unstressed *i* in derivational *-ing* (Clark Hall 1960; Campbell 1959). The target is the ordinary OE citation form, normalized as *sċilling* ‘shilling’.

Development to Old English

From **skíllingaz* ‘shilling’, loss of final *-az* yields **skílling* ‘shilling’. Old English palatalization of initial *sk* before front vocalism then gives *sċilling* ‘shilling’. The *i* of derivational *-ing-* remains, so the regular outcome is *sċilling* ‘shilling’, not **sċilleng* ‘shilling’ (Campbell 1959; Hogg 1992).

Form note

Kroonen’s **skellinga-* ~ **skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The derivational input **skíllingaz* ‘shilling’ is the specific form used for the derivation represented here (Kroonen 2013).

6.50 show — OE *sċēawian*

Derivation: **skáwōjaną* > *sċēawian* (regular).

Derivation trace

Proto input: **skáwōjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Aw Long Diphthong	**skḗawōjaną*
[no change]		OE Heavy Syllable Nasal Apocope	**skḗawōjan*
Early Anglo-Frisian		OE Secondary Nasalization	**skḗawōjąn*
[no change]		OE Sk Palatalization	**ʃḗawōjąn*
		OE I Umlaut	**ʃḗawējąn*
		OE Unstressed Long Vowel Shortening	**ʃḗawejąn*
		OE Weak Tail Reduction	**ʃḗawejan*
		OE Intervocalic J Vocalization	**ʃḗaweian*
		OE Unstressed EI Contraction	**ʃḗawian*

Old English form: *sċēawian*

Reconstruction and comparative evidence

Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* ‘show’ among the reflexes (Orel 2003; Kroonen 2013, 482). Brunner likewise records the Old English family as *scēawian* ‘show’, *scāwian* ‘show’, confirming that it belongs to the ordinary show-verb set rather than to a special finite-cell formation (Brunner 1965).

Old English evidence

Bright lists *scēawian* ‘show’ (W. II.) and also the related form *scēawa* ‘show’ (Bright 1971). The sources therefore use *scēawian* ‘show’, while *sċēawian* ‘show’ supplies a normalized spelling.

Development to Old English

From **skáwōjaną* ‘show’, the inherited *aw* sequence before a following vowel develops into the *ēaw* diphthong, and **ō* survives between **w* and **j* in the Class II suffix. The development therefore runs regularly to *sċēawian* ‘show’, without the direct *aw+j* problem seen in other verb types (Campbell 1959; Orel 2003).

Form note

The difference between *scēawian* ‘show’ and *sċēawian* ‘show’ is orthographic normalization of initial <*sc*>, not a difference of lexeme or paradigm cell (Campbell 1959; Hogg 1992).

6.51 sleep — OE *slǣpan*

Derivation: **slḗpaną* > *slǣpan* (regular).

Derivation trace

Proto input: **slḗpaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**slǣpan*
PNWGmc Long E Lowering	**slǣpaną*	OE Secondary Nasalization	**slǣpąn*
Early Anglo-Frisian		OE Weak Tail Reduction	**slǣpan*
[no change]			

Old English form: *slǣpan*

Reconstruction and comparative evidence

Kroonen preserves the comparative verb as **slēpan-*, and Fulk cites the same family under root **slēb-* (Kroonen 2013; Fulk 2018, 120). The derivational input **slḗpaną* 'sleep' is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence

Clark Hall gives *slǣpan* 'sleep' with preterite *slēp* 'slept', *slēap* 'slept', and Bright likewise lists *slǣpan* 'sleep' (*slāpan* 'sleep'), *slēp* 'slept', *slēpon* 'slept', *slēpen* 'slept' (Clark Hall 1960; Bright 1971, 435). The target represented here is therefore the normalized infinitive *slǣ́pan* 'sleep', not the preterite forms and not the separate noun *slǣ́p* 'sleep'.

Development to Old English

From **slḗpaną* 'sleep', Northwest Germanic lowering gives **slǣ́paną* 'sleep'. The later OE tail developments then yield *slǣ́pan* 'sleep' regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan* 'sleep' / *slǣpan* 'sleep' and Anglian or Kentish *slēpan* 'sleep', but those do not displace the infinitive chosen here (Brunner 1965; Bülbring 1902).

Form note

The note concerns lemma type rather than a special derivational problem: this row represents the verb *slǣ́pan* 'sleep', whereas *slǣ́p* 'sleep' belongs to noun or lookup background and *slēp* 'slept' / *slēap* 'slept' are preterite forms (Clark Hall 1960; Bright 1971, 435).

6.52 smear — OE *smierwan*

Derivation: **smérwijaną* > *smierwan* (regular).

Derivation trace

Proto input: **smérwijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**sméorwijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**sméorwijan*
Early Anglo-Frisian	OE Secondary Nasalization	**sméorwijąn*
[no change]	Sievers Law Syncope	**sméorwjąn*
	OE I Umlaut	**smíerwjąn*
	OE Weak Tail Reduction	**smíerwjan*
	OE J Loss After Heavy	**smíerwan*

Old English form: *smierwan*

Reconstruction and comparative evidence

Kroonen gives the comparative headword as **smerwjan-* (Kroonen 2013). Ringe and Taylor instead cite a later-stage **smirwijana* 'smear', from which they derive West Saxon *smierwan* 'smear', Mercian *smirwan* 'smear', and Northumbrian *smiriga* (Ringe and Taylor 2014). The derivational input **smérwijaną* 'smear' therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence

The target represented here is the West Saxon citation form *smierwan* ‘smear’. Campbell’s Anglian discussion explains the contrasting *smirwan* ‘smear’, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (Campbell 1959; Clark Hall 1960; Brunner 1965; Bright 1971).

Development to Old English

From **smérwijaną* ‘smear’, breaking before *r* + *consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan* ‘smear’. Anglian *smirwan* ‘smear’ reflects the well-known failure of breaking in this environment, not a different lexeme (Campbell 1959; Ringe and Taylor 2014).

Dialect note

The entry therefore represents the West Saxon member of a broader OE family: *smierwan* ‘smear’ in West Saxon, *smirwan* ‘smear’ in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (Ringe and Taylor 2014; Clark Hall 1960).

6.53 span — OE *spannan*

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace

Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**spánnan*
[no change]	OE Secondary Nasalization	**spánnąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**spánnan*
[no change]		

Old English form: *spannan*

Reconstruction and comparative evidence

Kroonen cites the inherited verb as **spannan-*, with OE *spannan* ‘span, fasten’ among the reflexes (Kroonen 2013). The derivational input **spánnaną* ‘span’ is the infinitive-style form used here for that same verbal lexeme.

Old English evidence

Clark Hall lists noun *spann* and verb *spannan* ‘span, fasten’ as separate headwords, and Brunner likewise records *sponnan, spannan stv.* (Clark Hall 1960; Brunner 1965). The target is the strong-verb infinitive, not the noun.

Development to Old English

From **spánnaną* ‘span’, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan* ‘span, fasten’. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note

English *span* can also reach the noun *spann* in local lookup material. The form represented here is the verb *spannan* 'span, fasten', with the noun treated elsewhere (Clark Hall 1960).

6.54 spar — OE *spearra*

Derivation: **spárrô* > *spearra* (regular).

Derivation trace

Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærrô*
[no change]	OE Breaking	**spearrô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**spearra*
[no change]		

Old English form: *spearra*

Reconstruction and comparative evidence

Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* 'spar, rafter' (Kroonen 2013; Orel 2003). The derivational input **spárrô* 'spar' is the OE-facing nominal form used here for that same lexeme.

Old English evidence

The noun is *spearra* 'spar, rafter'. English gloss overlap also reaches the unrelated verb *sperran* 'to bar', which does not belong to this row.

Development to Old English

From **spárrô* 'spar', Anglo-Frisian brightening gives **spærrô* 'spar', and OE breaking before geminate *rr* yields **spearrô* 'spar', later *spearra* 'spar, rafter'. The development is therefore regular for a breaking-conditioned noun of this type (Luick 1914--1940).

Form note

This entry concerns the noun *spearra* 'spar, rafter' only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (Kroonen 2013; Orel 2003).

6.55 still — OE *stillan*

Derivation: **stéllijaną* > *stillan* (regular).

Derivation trace

Proto input: **stéllijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**stéllijan*
[no change]	OE Secondary Nasalization	**stéllijąn*
Early Anglo-Frisian	Sievers Law Syncope	**stélljąn*
[no change]	OE I Umlaut	**stilljąn*
	OE Weak Tail Reduction	**stilljan*
	OE J Loss After Heavy	**stillan*

Old English form: *stillan*

Reconstruction and comparative evidence

The wider West Germanic family includes adjective *still* 'still, quiet' and verb *stillen* (Kluge 2011). The derivational input **stéllijaną* 'still' represents the verbal j-formation used for the OE row.

Old English evidence

Clark Hall gives *stillan* as the verb and separately *stille* 'still, quiet' as the adjective (Clark Hall 1960). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* 'still, stop' and related forms (Bosworth and Toller 1898, 724). The Old English form here is the verb *stillan*, not the adjective.

Development to Old English

As a heavy-stem Class I weak verb, **stéllijaną* 'still' undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note

The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* 'still, quiet' belongs to the related adjectival branch of the family (Clark Hall 1960; Kluge 2011).

6.56 summer — OE *sumer*

Derivation: **súmaraz* > *sumer* (regular).

Derivation trace

Proto input: **súmaraz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**súmar*
EAF Final Z Deletion	**súmara*	EAF Brightening	**súmær*
Early Anglo-Frisian		OE Unstressed AE Merger	**súmer*
[no change]			

Old English form: *sumer*

Reconstruction and comparative evidence

Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz* 'summer', while Orel preserves an alternate **sumeraz* 'summer' (Kroonen 2013; Ringe and Taylor 2014; Orel 2003, 425). The derivational input **súmaraz* 'summer' follows the **a* vocalism that underlies the regular development represented here.

Old English evidence

Clark Hall gives *sumor* 'summer' m., gs. *sumeres* 'summer', ds. *sumera* 'summer' / *sumere* 'summer', and Bright likewise lists *sumor (sumer)* with genitive *sumeres* 'summer (gen.)' (Clark Hall 1960; Bright 1971, 440). The tradition therefore preserves both *sumor* 'summer' and *sumer* 'summer', with the oblique forms strongly supporting second-syllable *e*.

Development to Old English

From **súmaraz* 'summer', loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer* 'summer'. The form compared here is the regularized *e*-form, while the common citation form *sumor* 'summer' remains part of the attested OE tradition (Ringe and Taylor 2014).

Form note

The entry does not deny *sumor* 'summer'. It represents *sumer* 'summer' as the regular outcome chosen here, while *sumor* 'summer' remains a common headword spelling and *sumeres* 'summer (gen.)' / *sumere* 'summer (dat.)' show that the *e*-vocalism was also real in Old English (Clark Hall 1960; Bright 1971, 440).

6.57 sunder — OE *sundrian*

Derivation: **súndrōjaną* > *sundrian* (regular).

Derivation trace

Proto input: **súndrōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**súndrōjan*
[no change]	OE Secondary Nasalization	**súndrōjąn*
Early Anglo-Frisian	OE I Umlaut	**súndrējąn*
[no change]	OE Unstressed Long Vowel Shortening	**súndrejąn*
	OE Weak Tail Reduction	**súndrejan*
	OE Intervocalic J Vocalization	**súndreian*
	OE Unstressed EI Contraction	**súndrian*

Old English form: *sundrian*

Reconstruction and comparative evidence

Orel distinguishes three related formations: adverbial **sunþraz* 'asunder' > *sundor* 'apart, asunder', Class I verbal **sunþrjanan* 'sunder' > *syndrian* 'separate, sunder', and Class II verbal **sunþrōjanan* 'sunder' > *sundrian* 'sunder' (Orel 2003).

Kluge-Seebold aligns the cognate set with German *sondern* 'separate' and OE *gesundrian* 'separate', so this entry belongs with the Class II verb, not the adverb (Kluge 2011).

Old English evidence

Clark Hall and Bosworth-Toller list *sundrian* 'sunder' and *syndrian* 'sunder' separately from adverbial *sundor* 'asunder'; both also record the prefixed verbal family *ā-sundrian* 'sunder apart' (Clark Hall 1960, 296; Bosworth and Toller 1898). The target is therefore the weak verb *sundrian* 'sunder'.

Development to Old English

From **súndrōjaną* ‘sunder’, the Class II weak-verb suffix yields regular OE *-ian*, producing *sundrian* ‘sunder’. Because this is the **-ōjan-* verb and not the Class I **-jan-* formation, the form represented here does not belong to the umlauted *syndrian* ‘separate’ branch.

Form note

The earlier confusion was lexical, not phonological: *sundor* ‘apart’ is the separate adverb, and *syndrian* ‘separate’ is a related but different verb. The verb treated here is the Class II verb *sundrian* (Orel 2003; Clark Hall 1960, 296).

6.58 swallow — OE *swealwe*

Derivation: **swálwōn* > *swealwe* (regular).

Derivation trace

Proto input: **swálwōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**swælwǭ*
PNWGmc N Stem N Loss	**swálwǭ*	OE Breaking	**swealwǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**swealwæ*
[no change]		OE Unstressed AE Merger	**swealwe*

Old English form: *swealwe*

Reconstruction and comparative evidence

Kroonen gives the bird name as **swalwōn-*, and Ringe and Taylor cite the later West Germanic stage **swalwa* ‘swallow’, from which West Saxon *swealwe* ‘swallow’ and Mercian *swalwe* ‘swallow’ develop (Kroonen 2013, 535; Ringe and Taylor 2014, 200). The selected etymological comparison belongs to the swallow-bird family, not to the verb *swelgan*.

Old English evidence

Clark Hall records *swealwe (a, o)* as the noun headword (Clark Hall 1960). Campbell and Brunner also preserve later or oblique-family forms such as *swaluwe*, *swalewan*, and *swealuwe*, but those belong to wider variation around the noun rather than to the citation form represented here (Campbell 1959; Brunner 1965).

Development to Old English

From **swálwōn* ‘swallow’, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (Ringe and Taylor 2014, 200; Kroonen 2013, 535).

6.59 swine — OE *swīn*

Derivation: citation reconstruction **swī́ną*; form followed here **swī́ną* > *swīn* (regular).

Derivation trace

Proto input: **swī́ną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**swī́n*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *swīn*

Reconstruction and comparative evidence

Kroonen cites the noun as **swina-* ‘pig’ (Kroonen 2013). The selected comparative form here is the bare neuter citation cell **swī́ną* ‘swine’, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence

Clark Hall records *swin (y)* as the ordinary noun headword (Clark Hall 1960). The target here is that singular citation form *swīn* ‘swine’, not a plural glossed in Modern English as *swine*.

Development to Old English

From derivational input **swī́ną* ‘swine’, loss of the final nasal vowel yields *swīn* ‘swine, pig’. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note

The derivational input writes stressed long *ī* as **ī́*, so comparative **swī́ną* ‘swine’ and derivational **swī́ną* ‘swine’ represent the same lexical form.

6.60 think — OE *þenċan*

Derivation: **θánkijaną* > *þenċan* (regular).

Derivation trace

Proto input: **θánkijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**θánkijan*
[no change]	OE Secondary Nasalization	**θánkijąn*
Early Anglo-Frisian	Sievers Law Syncope	**θánkjąn*
[no change]	OE Velar Palatalization	**θántʃjąn*
	OE I Umlaut	**θentʃjąn*
	OE Weak Tail Reduction	**θentʃjan*
	OE J Loss After Heavy	**θentʃan*

Old English form: *þenċan*

Reconstruction and comparative evidence

Kroonen gives the verb as **þankjan-* ‘to think’, and Ringe and Taylor cite fully inflected **þanki-janą* ‘think’ beside OE *þenċan* ‘think’ (Kroonen 2013; Ringe and Taylor 2014). The noun **þankaz* ‘think’ belongs only to the wider derivational background.

Old English evidence

Bosworth-Toller preserves the verb under _þencan_ ‘think’ / _geþencan_ ‘think’, and the citation form here is the ordinary infinitive _þenċan_ ‘think’ (Bosworth and Toller 1898).

Development to Old English

From _*θánkijaną_ ‘think’, palatalization before _*j_ and i-umlaut produce _þenċan_ ‘think’. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note

Campbell’s assibilation discussion uses the same verb _þencan_ ‘think’; the class-III relic _hycgan_ is a different lexeme (Campbell 1959; Hogg 1992).

6.61 thorn — OE _þorn_

Derivation: _*θúrnaz_ > _þorn_ (regular).

Derivation trace

Proto input: _*θúrnaz_

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		PWGmc Final Bare A Loss	_*θórn_
PNWGmc U Lowering	_*θórnaz_		
EAF Final Z Deletion	_*θórna_		
Early Anglo-Frisian			
[no change]			

Old English form: _þorn_

Reconstruction and comparative evidence

Kroonen gives _*þurna-_ ‘thorn, briar’, while Orel preserves the masculine pair _*þurnuz_ ‘thorn’ and _*þurnaz_ ‘thorn’ (Kroonen 2013; Orel 2003). The derivational input _*θúrnaz_ ‘thorn’ belongs to that same comparative family.

Old English evidence

Bright lists _þorn_ ‘thorn’ (m.) and Clark Hall likewise treats _þorn_ ‘thorn’ as the ordinary noun headword (Bright 1971; Clark Hall 1960).

Development to Old English

The inherited stem shows regular lowering of _u_ to _o_ before _r_, and final loss yields _þorn_ ‘thorn’. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note

The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form _þorn_ ‘thorn’.

6.62 tide — OE *tīd*

Derivation: citation reconstruction *tī́diz; form followed here *tī́diz > *tīd* (regular).

Derivation trace

Proto input: *tī́diz

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	*tī́d
EAF Final Z Deletion	*tī́di		
Early Anglo-Frisian			
[no change]			

Old English form: *tīd*

Reconstruction and comparative evidence

Kroonen gives *tīdi- ‘time’, and Orel’s *tīđiz ‘tide’ points to the same feminine noun (Kroonen 2013; Orel 2003). The related verb *tīdan* ‘happen’ is separate.

Old English evidence

Bright records *tīd* ‘tide’ with singular *tīde* ‘tide’ and plural *tīda* ‘tides’, and Clark Hall treats *tīd* ‘tide’ as the ordinary noun ‘time, period, season’ (Bright 1971; Clark Hall 1960, 309).

Development to Old English

From *tī́diz ‘tide’, final *z* is lost and the high final vowel drops, leaving *tīd* ‘tide’. The development is straightforward for a feminine i-stem.

Lexical note

English *tide* can also lead to the separate weak verb *tīdan* ‘happen’; the noun is *tīd* ‘tide’.

6.63 token — OE *tācn*

Derivation: *táiknų > *tācn* (regular).

Derivation trace

Proto input: *táiknų

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	*tākn
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	*tāknų		

Old English form: *tācn*

Reconstruction and comparative evidence

Kroonen cites **taikna-* and Orel **taiknan* 'token' for the noun 'sign, token' (Kroonen 2013; Orel 2003, 438). The derivational input **táikną* 'token' is the simple citation-form noun used for the derivation.

Old English evidence

Campbell and Sievers-Brunner preserve both unbroken *tācn* 'token' and broken *tācen* 'token', with oblique *tācnes* 'token' remaining unbroken (Campbell 1959; Brunner 1965).

Development to Old English

Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn* 'token'. The unbroken form is therefore a regular Old English outcome.

Form note

tācn 'token' is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen* 'token', but that does not displace the older unbroken form (Campbell 1959; Brunner 1965).

6.64 town — OE *tūn*

Derivation: **tūną* > *tūn* (regular).

Derivation trace

Proto input: **tūną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tūn*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *tūn*

Reconstruction and comparative evidence

Kroonen cites **tūna-* 'fenced area', while Orel gives **tūnan* 'town' and **tūnaz* 'town' (Kroonen 2013; Orel 2003, 452). The derivational input **tūną* 'town' is the simple citation-form noun used in the derivation.

Old English evidence

Clark Hall records *tūn* 'town' as the ordinary headword 'enclosure, yard, village, town' (Clark Hall 1960).

Development to Old English

The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* 'town' regularly (Brunner 1965).

Source note

The comparative headwords vary, but the Old English target here is the direct citation form *tūn* ‘town’, not an oblique **tūnăn* ‘town’.

6.65 wade — OE *wadan*

Derivation: **wádaną* > *wadan* (regular).

Derivation trace

Proto input: **wádaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wædaną*
[no change]	OE A Restoration	**wadaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**wadan*
[no change]	OE Secondary Nasalization	**wadąn*
	OE Weak Tail Reduction	**wadan*

Old English form: *wadan*

Reconstruction and comparative evidence

Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* ‘wade, go through’ among the standard open-syllable examples (Campbell 1959; Ringe and Taylor 2014; Luick 1914--1940, 239).

Old English evidence

Clark Hall gives *wadan* ‘wade, go through’ as the verb ‘to go, move, stride, advance’, and Bright lists the same infinitive in the strong-verb paradigm (Clark Hall 1960; Bright 1971).

Development to Old English

From **wádaną* ‘wade’, Anglo-Frisian brightening first gives **wædaną* ‘wade’. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* ‘wade, go through’ (Campbell 1959; Ringe and Taylor 2014).

Development note

The infinitive belongs to the A-restoration class. The citation form is therefore *wadan* ‘wade’, not a fronted *wæden* ‘waded’-type output.

6.66 warp — OE *weorpan*

Derivation: **wérpaną* > *weorpan* (regular).

Derivation trace

Proto input: **wérpaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**wéorpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**wéorpan*
Early Anglo-Frisian	OE Secondary Nasalization	**wéorpąn*
[no change]	OE Weak Tail Reduction	**wéorpan*

Old English form: *weorpan*

Reconstruction and comparative evidence

Ringe and Taylor distinguish preterite **warp* 'warp' from infinitive **werpana* 'warp', and the derivational input here is the verbal form **wérpaną* 'warp' (Ringe and Taylor 2014).

Old English evidence

Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan, wearp, wurpon, worpen* (Clark Hall 1960; Bright 1971).

Development to Old English

Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (Campbell 1959; Hogg 1992).

Lexical note

English *warp* also points to related *wearp* material. The target is the infinitive *weorpan*.

6.67 wash — OE *wascan*

Derivation: **wáskaną* > *wascan* (regular).

Derivation trace

Proto input: **wáskaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæskaną*
[no change]	OE A Restoration	**waskaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**waskan*
[no change]	OE Secondary Nasalization	**waskąn*
	OE Weak Tail Reduction	**waskan*

Old English form: *wascan*

Reconstruction and comparative evidence

Kroonen cites **waskan-*, Orel **waskanan* 'wash', and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (Kroonen 2013; Orel 2003, 489; Ringe and Taylor 2014, 142).

Old English evidence

Clark Hall heads the verb as *wascan* 'wash', while Sievers-Brunner also notes the variant *wæscan* 'wash' (Clark Hall 1960; Brunner 1965).

Development to Old English

From **wáskaną* ‘wash’, brightening gives **wæskaną* ‘wash’. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (Campbell 1959; Ringe and Taylor 2014, 142).

Form note

The conservative citation form *wascan* ‘wash’ is selected here. Spellings such as *wæscan* ‘wash’ or *wasċan* ‘wash’ belong to variant or normalized background rather than to the target of this entry.

6.68 wax — OE *weaxan*

Derivation: **wáxsaną* > *weaxan* (regular).

Derivation trace

Proto input: **wáxsaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæxsaną*
[no change]	OE Breaking	**weaxsaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**weaxsan*
[no change]	OE Secondary Nasalization	**weaxsąn*
	OE Weak Tail Reduction	**weaxsan*
	OE Xs Merge	**weaXSan*

Old English form: *weaxan*

Reconstruction and comparative evidence

Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanan* ‘wax’, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (Kroonen 2013; Orel 2003, 478; Ringe and Taylor 2014).

Old English evidence

Clark Hall gives *weaxan* ‘wax’ as the verb headword and separately records bare *wax* ‘wax’ as a preterite form; Bright likewise treats *weaxan* ‘wax’ as the infinitive (Clark Hall 1960; Bright 1971).

Development to Old English

From **wáxsaną* ‘wax’, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs* > *s* belongs to forms where another consonant follows, such as *wæstm* ‘growth’, not to the infinitive itself (Campbell 1959; Brunner 1965).

Lexical note

The target here is the infinitive *weaxan* ‘wax’. Noun *weax* ‘wax’ and preterite *wax* / *wēox* ‘waxed’ belong to different lexical or paradigm slots.

6.69 way — OE *weġ*

Derivation: **wégaz* > *weġ* (regular).

Derivation trace

Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wég*
EAF Final Z Deletion	**wéga*	OE Velar Palatalization	**wéʤ*
Early Anglo-Frisian			
[no change]			

Old English form: *weġ*

Reconstruction and comparative evidence

Kroonen cites the noun as **wega-* 'way, road', while the selected derivational form here is nominative-singular **wégaz* 'way' (Kroonen 2013). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weġ* 'way' and inflected *wegas* / *wegum* (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Old English evidence

The Old English singular is the ordinary noun *weg* 'way', here normalized as *weġ* 'way' to show the palatal final. The contrasting plural and oblique forms *wegas* 'ways', *wegum* 'ways' keep a velar stop before the following back vowel (Campbell 1959; Ringe and Taylor 2014, 341).

Development to Old English

From **wégaz* 'way', final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *weġ* 'way', whereas *wegas* 'ways' remains velar because the following *a* blocks the same outcome (Campbell 1959; Hogg 1992; Ringe and Taylor 2014, 341).

Form note

Normalized *weġ* 'way' and dictionary *weg* 'way' represent the same noun. No Old English form *wē* 'we' is supported by the evidence for 'way'.

6.70 weapon — OE *wǣpn*

Derivation: **wḗpną* > *wǣpn* (regular).

Derivation trace

Proto input: **wḗpną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**wǣpn*
PNWGmc Long E Lowering	**wǣpną*		
Early Anglo-Frisian			
[no change]			

Old English form: *wǣpn*

Reconstruction and comparative evidence

Kroonen reconstructs a double-stem noun **wēbna-* ~ **wēpna-* and cites OE *wæpn* 'weapon' among its reflexes (Kroonen 2013, 617). The derivational input **wḗpną* 'weapon' represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence

Campbell's cluster-noun discussion preserves unbroken *wépn* 'weapon' beside broken *wépen* 'weapon'-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen* 'weapon' / *wapen* 'weapon' with unbroken oblique *wǣpnes* 'weapon', while Clark Hall lemmatizes the noun under *wapen* 'weapon' and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

Development to Old English

Northwest Germanic lowering gives *wǣpn* 'weapon', and loss of the final nasal vowel leaves the unbroken cluster word-finally. The Old English form here is the attested unbroken form *wǣpn* 'weapon'.

Form note

The ordinary late West Saxon simplex headword is *wǣpen* 'weapon', but Campbell's noun-class discussion also preserves unbroken *wépn* 'weapon' beside broken *wépen* 'weapon'-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* 'weapon' remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

6.71 will — OE *willa*

Derivation: **wéljô* > *willa* (regular).

Derivation trace

Proto input: **wéljô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**willjô*
PWGmc J Gemination	**wélljô*	OE Unstressed Long Vowel Shortening	**willja*
Early Anglo-Frisian		OE J Loss After Heavy	**willa*
[no change]			

Old English form: *willa*

Reconstruction and comparative evidence

Kroonen separates noun **weljan* 'will, wish' from verb **weljan* 'to want', while Orel and Kluge represent the noun as **weljōn* 'will' (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form **wéljô* 'will' is the noun-side input used for this row.

Old English evidence

Clark Hall lemmatizes noun *willa m.* separately from verb *willan* 'will, want' (Clark Hall 1960, 368). The Old English form here is the noun citation form, not the related verb.

Development to Old English

From **wéljô* ‘will’, j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa* ‘will, desire’. The noun is therefore a regular weak masculine outcome.

Lexical note

The target here is the noun *willa* ‘will, wish’. Related verb *willan* ‘will, want’ belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

6.72 wind — OE *windan*

Derivation: **wíndaną* > *windan* (regular).

Derivation trace

Proto input: **wíndaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**wíndan*
[no change]	OE Secondary Nasalization	**wíndąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**wíndan*
[no change]		

Old English form: *windan*

Reconstruction and comparative evidence

Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (Kroonen 2013). Fulk and Ringe and Taylor derive the dental directly from PIE **wendh-*, not from a Verner alternant (Fulk 2018; Ringe and Taylor 2014).

Old English evidence

Clark Hall and Bosworth-Toller record *windan* ‘wind, coil’ as the verb headword (Clark Hall 1960; Bosworth and Toller 1898, 101). The Old English form here is the ordinary infinitive of the strong verb.

Development to Old English

The derivational input **wíndaną* ‘wind’ yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note

English *wind* also names the noun. This row represents the class-III verb, not the noun (Kroonen 2013; Clark Hall 1960).

6.73 wold — OE *weald*

Derivation: **wálθuz* > *weald* (regular).

Derivation trace

Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**wældu*
EAFL Th Voicing	**wálduz*	OE Breaking	**wealdu*
EAF Final Z Deletion	**wáldu*	OE High Vowel Apocope	**weald*
Early Anglo-Frisian			
[no change]			

Old English form: *weald*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **walþu-* and gives OE *weald* 'forest, woodland' beside other West Germanic *wald* 'forest' forms (Kroonen 2013). The derivational input **wálθuz* 'wold' is the nominative singular used for the derivation.

Old English evidence

Clark Hall makes *weald* 'forest, woodland' the main noun headword and cross-refers *wald* 'forest' and *wold* 'plain, open country' to it (Clark Hall 1960). Clark Hall glosses *wald* as *(N,VPs)* — Northumbrian and Vespasian Psalter (Mercian) — making this an explicitly dialect-labelled Anglian form.

Development to Old English

**lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald* 'forest, woodland'. The noun is therefore a regular breaking outcome.

Dialect note

wald 'forest' survives as an Anglian variant in the same family (Clark Hall *wald (N,VPs)*: Northumbrian and Vespasian Psalter). The Old English form here is normalized *weald* 'forest, woodland', not the variant form (Clark Hall 1960; Ringe and Taylor 2014).

6.74 yarn — OE *ġearn*

Derivation: **gárną* > *ġearn* (regular).

Derivation trace

Proto input: **gárną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gærną*
[no change]	OE Breaking	**gearną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**gearn*
[no change]	OE Velar Palatalization	**ʤearn*

Old English form: *ġearn*

Reconstruction and comparative evidence

Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* 'yarn' > **geern* 'yarn' > **gearn* 'yarn' > OE *gearn* 'yarn' (Kroonen 2013; Ringe and Taylor 2014, 220). The derivational input **gárną* 'yarn' is the nominal citation form used here, while oblique **garnăn* 'yarn' belongs only to comparative background.

Old English evidence

Clark Hall records *gearn (e) n.* 'yarn, spun wool', and Bosworth-Toller glosses *gearn* 'yarn' with Latin *filatum* 'spun thread' (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English

From **gárną* 'yarn', brightening and breaking before *rn* yield *gearn* 'yarn'; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn* 'yarn'. The derivation is regular.

Form note

Dictionary *gearn* 'yarn' and normalized *ġearn* 'yarn' refer to the same noun. The comparative stem **garna-* and oblique **garnăn* 'yarn' do not replace the derivational input **gárną* 'yarn'.

6.75 who — OE *hwā*

Derivation: **xwáz* > *hwā* (regular).

Derivation trace

Proto input: **xwáz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
Monosyllabic Final Z Loss	**xwā*	
Early Anglo-Frisian		
[no change]		

Old English form: *hwā*

Reconstruction and comparative evidence

The derivational input is the nominative singular masculine of the interrogative pronoun, PGmc **xwáz* 'who' (traditionally *hwaz; Gothic ƕas 'who'). Ringe and Taylor cite exactly this form among the witnesses for the northern West Germanic loss of word-final *-z in stressed monosyllables: "*hwaz 'who' > OE hwā" (Ringe and Taylor 2014, 86). The nominative is the natural citation cell of the interrogative and the direct ancestor of the Old English headword form, so no paradigm-cell retargeting is involved.

Old English evidence

West Saxon *hwā* 'who' is the standard headword form (Campbell 1959, § 125, p. 49). The vowel is decisive for the derivation: Campbell states that the back vowel of *hwā* never underwent Anglo-Frisian brightening — "*hwǣ does not exist" — and Sievers-Brunner records the same restriction (Campbell 1959, § 125, p. 49; Brunner 1965, § 137 Anm. 1, p. 129).

Development to Old English

The derivation is two historical steps. First, the later northern West Germanic loss of word-final *-z in stressed monosyllables removed the sibilant with compensatory lengthening of the short nucleus: **xwáz* 'who' > **hwā* 'who' (Ringe and Taylor 2014, 86). Old High German instead rhotacized the sibilant (cf. OHG wer 'who' type inflection), which is why this loss counts among the diagnostic Ingvaeonic features (Fulk 2018, 18). Second, the resulting long back *ā stayed back: Anglo-Frisian brightening of long final *ā requires a preceding nucleus in the word, and in this monosyllable there is none, so the form surfaces directly as OE *hwā* (Campbell 1959, § 125, p. 49).

An alternative account derives the endingless pronoun from a generalized unaccented sentence variant *hwa with later lengthening (Campbell 1959, 166; Brunner 1965, § 182, p. 160). Ringe and Taylor reject the unaccented-variant analysis for this witness set because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86). The project follows Ringe and Taylor; the alternative is recorded, not adopted.

Why this row is in the corpus

This row makes SC097, the monosyllabic final *-z loss, corpus-witnessed: before its addition the change was validated only by synthetic controls. It also witnesses the ordering SC097 before SC003 rhotacism — a rhotacized* †hwar *could never yield* hwā* — and exercises the narrowed long-final clause of Anglo-Frisian brightening. See `docs/sound_changes/audits/corpus-maturation-01-candidate-adjudication.md` §1.

6.76 you — OE *ēow*

Derivation: **ízwiz* > *ēow* (regular).

Derivation trace

Proto input: **ízwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Ew Long Diphthong	**ēoww*
PWGmc Coronal W Assimilation	**íwwiz*	OE WW Simplification	**ēow*
EAF Final Z Deletion	**íwwi*		
PWGmc Unstressed Word Final I Apocope	**íww*		
Early Anglo-Frisian			
[no change]			

Old English form: *ēow*

Reconstruction and comparative evidence

The derivational input is the dative(-accusative) plural of the second-person plural pronoun, PGmc **ízwiz* 'you (dat. pl.)' (Gothic izwis 'you'). Ringe and Taylor print the development in full: "PGmc *izwiz 'you (dat. pl.)' (Goth. izwis) > *iwwi > PWGmc *iuwi ~ *iuw (see 3.1.4) > OE īow, OF iū, OS, OHG iu" (Ringe and Taylor 2014, 41–42), following Stiles's demonstration of the coronal-w assimilation (Stiles 1985, 89–94). Fulk corroborates the *-zw- > *-ww- assimilation (Fulk 2018, § 8.3, pp. 204–205). The oblique plural is the cell that survives as the Old English pronoun *ēow*; the modern English word continues this oblique form.

Old English evidence

West Saxon *ēow* 'you' beside early West Saxon and Northumbrian *īow* 'you' shows the normal West Saxon treatment of the diphthong (Campbell 1959, § 702, p. 283). The corpus targets the West Saxon citation form.

Development to Old English

Four historical steps carry the form. (1) Coronal-w assimilation: **ízwiz* 'you' > **íwwiz* 'you' (Ringe and Taylor 2014, 41–42; Stiles 1985, 89–94). (2) Proto-West Germanic loss of word-final *-z in unstressed syllables: > **íwwi* 'you' (Ringe and Taylor 2014, 44–45). (3) Early apocope in unstressed words: > **íww* 'you', the apocopated member of Ringe and Taylor's PWGmc doublet *iuwi ~ *iuw; the absence of i-umlaut in the Old English reflex is Ringe and Taylor's own proof that the trigger vowel fell before umlaut (Ringe and Taylor 2014, 57–58). (4) Vocalization of the geminate *ww to the long diphthong, dated by Ringe and Taylor to Proto-West Germanic itself (*fewwar > *feuwar), followed by geminate simplification: > OE *ēow* 'you' (Ringe and Taylor 2014, 41–42; Fulk 2018, § 8.3, pp. 204–205).

Why this row is in the corpus

This row is a chronology witness of unusual power for basic vocabulary. It forces coronal-w assimilation (SC008) to precede rhotacism (SC003): had rhotacism applied first, *izwiz would have become †*irwiz*, from which *ēow* is underivable. It is the corpus witness for SC098, the early apocope in unstressed words, and for its bleeding of i-umlaut (SC055), and it fixes the order of geminate-w vocalization (SC033) before degemination (SC031). See `docs/sound_changes/audits/corpus-maturation-01-candidate-adjudication.md` §2 and `docs/sound_changes/audits/sc098-dossier-unstressed-word-final-i-apocope.md`.

Chapter 7

Attested variants and comparison forms

Here the Old English comparator belongs to a documented set of variants. The variation is part of the evidence and not an inconvenience to be normalized away.

7.1 cud — OE _cwedu_

Derivation: citation reconstruction _*kwíθuz_; form followed here _*kwéðuz_ > _cwedu_ (attested variant).

Derivation trace

Proto input: _*kwéðuz_

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PWGmc Dental Hardening	_*kwéduz_	
EAF Final Z Deletion	_*kwédu_	
Early Anglo-Frisian		
[no change]		

Old English form: _cwedu_

Reconstruction and comparative evidence

Kroonen reconstructs the resin word as _*kwedu-2_ ‘cud’ and gives Old English variants _cwidu_ ‘cud’, _cweodu_ ‘cud’, and _c(w)udu_ ‘cud’ (Kroonen 2013, 355). Orel likewise lists _cwidu_ under the cognate set (Orel 2003, 266). The derivational input _*kwéðuz_ ‘cud’ therefore represents the older e-grade, voiced-dental form behind the chosen variant _cwedu_ ‘cud’.

Old English evidence

The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss _cwidu_ ‘cud’ > _cwudu_ ‘cud’ > _cudu_ ‘cud’ and also note late West Saxon _cweodu_ ‘cud’; Clark Hall gives _cwudu_ ‘cud’, _cweodu_ ‘cud’, and _cudu_ ‘cud’ (Ringe and Taylor 2014, 338; Clark Hall 1960, 84). Attested _cwedu_ ‘cud’ is treated here as the conservative variant within that set.

Development to Old English

From **kwéðuz* ‘cud’, the West Germanic voiced dental hardens in the expected way and the regular Old English development yields *cwedu* ‘cud’. The other Old English spellings belong to the same lexical family, but reflect later leveling, back-umlaut, or further reduction rather than a need to replace the selected input.

Variant comparison

Variant type	Old English form	Comment
conservative target	*cwedu*	selected attested variant represented here
leveled i-grade form	*cwidu*	common lexical variant in the same family
back-umlauted forms	*cweodu*, *cwudu*	later developments within the same OE tradition
reduced form	*cudu*	further reduced member of the same variant set

7.2 ten — OE *tēon*

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace

Proto input: **téxun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Med Unstressed U Lowering	**téxon*
[no change]	OE Breaking	**téoxon*
Early Anglo-Frisian	OE H Loss	**téoon*
[no change]	OE Contraction	**tḗon*

Old English form: *tēon*

Reconstruction and comparative evidence

**tēon* ‘ten’ is the bare cardinal form. Fulk states that Old English *tien* ‘ten’ shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *tēon* ‘ten’ (Fulk 2018, § 10.2). Brunner gives the same contrast more broadly: **tēon* ‘ten’ develops from **tëhun* ‘ten’, while West Saxon *tien* ‘ten’, *tȳn* ‘ten’ belong to a different, umlauted branch of the numeral history (Brunner 1965, §§ 129.2, 129 Anm. 6, 234).

The comparison form *tēon* ‘ten’ therefore represents the bare cardinal’s un-umlauted line, not the later umlauted simplex tradition.

Old English evidence

The attested simplex forms are varied. Campbell gives *tien* ‘ten’, north-western West Saxon *tēn* ‘ten’, and late Northumbrian *tēo* ‘ten’, *tēa* ‘ten’ (Campbell 1959, § 682). Brunner likewise lists West Saxon *tien* ‘ten’, *tȳn* ‘ten’ beside *tēn* ‘ten’, *tēo* ‘ten’, *tēa* ‘ten’ in other dialects (Brunner 1965, § 325).

Exact simplex *tēon* ‘ten’ is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* ‘tenth’ and *tēontig* ‘ten’ (Brunner 1965, § 129.2;

Fulk 2018, § 10.2). The comparison form *tēon* ‘ten’ is therefore a normalized spelling of that un-umlauted base.

Development to Old English

From **téxun* ‘ten’, lowering of medial unstressed *u* gives **téxon* ‘ten’, breaking gives **téoxon* ‘ten’, loss of intervocalic *h/x* yields **téoon* ‘ten’, and contraction produces **tḗon* ‘ten’, written *tēon* ‘ten’. This is the regular bare-cardinal path.

The umlauted forms *tien* / *tīen* ‘ten’ belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (Fulk 2018, § 10.2; Brunner 1965, § 129 Anm. 6).

Variant comparison

The comparison below sets the relevant forms side by side. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
tēon	normalized un-umlauted comparison form; trace-supported	Old English form here
tien / *tīen*	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here
tēn / *tēo* / *tēa*	attested un-umlauted simplex variants in other dialects	support the same branch as the comparison form

7.3 three — OE *þrīe*

Derivation: **θréjez* > *þrīe* (attested variant).

Derivation trace

Proto input: **θréjez*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**θrije*
EAF Final Z Deletion	**θréje*	OE Intervocalic J Vocalization	**θriie*
Early Anglo-Frisian		OE Contraction	**θrīe*
[no change]			

Old English form: *þrīe*

Reconstruction and comparative evidence

Kroonen cites the numeral under a broader stem-style reconstruction rather than under one Old English-ready paradigm cell (Kroonen 2013, 586). The input **θréjez* ‘three’ is therefore best understood as the inherited masculine nominative-accusative singular.

The Old English numeral has no uniform citation form across the paradigm. The masculine singular line must be kept apart from feminine-neuter *þrēo* ‘three’ and from later reduced spellings of the masculine form.

Old English evidence

Campbell gives masculine nominative-accusative *þrīe* 'three', feminine and neuter nominative-accusative *þrēo* 'three', genitive *þrēora* 'three', and dative *þrim* 'three', adding that late West Saxon has *þry* 'three' / *þri* 'three' for masculine nominative-accusative beside *þrīe* 'three' (Campbell 1959, § 683). Fulk presents the same masculine *þrīe* 'three' beside the wider numeral paradigm (Fulk 2018, § 10.1).

The target is therefore an attested Old English paradigm cell. *þrī* 'three' / *þry* 'three' belongs to later reduction or headword-style citation, whereas *þrīe* 'three' is the conservative masculine nominative-accusative form.

Development to Old English

From **θréjez* 'three', loss of final *-z* leaves **θréje* 'three' as an intermediate form. The following **θrije* 'three' fronts the stem vowel, then vocalizes between vowels, and contraction yields **þrīe* 'three'. The compact trace records the same sequence as θrije > θriie > *þrīe*.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the selected masculine cell from the later reduced form and from the rest of the numeral paradigm.

Form	Status	Relevance to this entry
þrīe	attested masculine nom./acc.; regular output	Old English form here
þrī / *þry*	later reduced masculine variant	genuine OE variant, but not the conservative comparison form
þrēo	attested feminine-neuter nom./acc.	same numeral, different paradigm cell
þrēora, *þrim*	attested genitive and dative forms	confirm the wider paradigm, not the cell compared here

7.4 wasp — OE *wæfs*

Derivation: **wábsaz* > *wæfs* (attested variant).

Derivation trace

Proto input: **wábsaz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wábs*
EAF Final Z Deletion	**wábsa*	EAF Brightening	**wæbs*
Early Anglo-Frisian		PGmc B Allophony	**wæβs*
[no change]			

Old English form: *wæfs*

Reconstruction and comparative evidence

The Proto-Germanic form **wábsaz* 'wasp' reaches Old English without any special change of stem or paradigm cell. The question in this entry is instead which attested Old English member of the variant set should serve as the comparison form.

Fulk presents the Old English forms together as *wæfs* 'wasp' with variants *wæsp* 'wasp' and *wæps* 'wasp' (Fulk 2018, § 6.5). Bülbring and Brunner then make the chronology more explicit by deriving later *wæps* and late West Saxon *wasp* 'wasp' from earlier *waefs* / *wæfs* through restricted metatheses (Bülbring 1902, § 484 Anm. 3; Brunner 1965, §§ 193, 204).

Old English evidence

The earliest directly cited Old English form is *wæfs* 'wasp', written *waefs* 'wasp' in the Épinal-Corpus material discussed by Bülbring and Brunner (Bülbring 1902, § 484 Anm. 3; Brunner 1965, § 193). Later Old English also shows *wæps* 'wasp' and *wæsp* 'wasp' / *wasp* 'wasp', and dictionary practice often favors *wæps* or later spellings as headwords (Clark Hall 1960, 341).

This entry therefore distinguishes chronological priority from headword habit. *wæfs* 'wasp' is not a convenient reconstruction: it is an attested Old English form and also the one that matches the regular development most closely.

Development to Old English

From **wábsaz* 'wasp', the regular Old English path passes through loss of final *z*, Anglo-Frisian fronting, and the allophonic development of *b* to a fricative before *s*, yielding *wæfs* 'wasp'.

The later forms *wæps* 'wasp' and *wæsp* 'wasp' / *wasp* 'wasp' belong to subsequent, lexically restricted metatheses. They are genuine Old English forms, but they are later within the variant history.

Variant comparison

The comparison below sets the relevant forms side by side. It separates the earliest attested and regular form from the later metathesized doublets.

Form	Status	Relevance to this entry
wæfs	earliest attested OE form; regular output	Old English form here
wæps	later attested metathesized variant	genuine OE doublet, but secondary
wæsp / *wasp*	later West Saxon metathesized variant	genuine OE doublet, but not the form compared here

Chapter 8

Early analogy and pre-Old-English input selection

Here an analogical change already separates the transducer input from the lexeme's citation reconstruction before the specifically Old English changes apply. Each entry must therefore justify the remodeled input independently of the successful output.

8.1 bottom — OE *botm*

Derivation: citation reconstruction **búdmaz*; form followed here **búttmaz* > *botm* (early analogy).

Derivation trace

Proto input: **búttmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**bóttm*
PNWGmc U Lowering	**bóttmaz*	OE Preconsonantal Degemination	**bótm*
EAF Final Z Deletion	**bóttma*		
Early Anglo-Frisian			
[no change]			

Old English form: *botm*

Reconstruction and comparative evidence

Kroonen reconstructs the word as a stem complex **budmō* 'bottom' with genitive **buttaz* 'bottom', summarized as **budman-* ~ **buttman-*, and gives Old English *botm* 'bottom' as the reflex (Kroonen 2013, 120). The comparative label **búdmaz* 'bottom' names the lexeme-level stem complex, while the derivational input **búttmaz* 'bottom' represents the pre-Old-English form with oblique **butt-* generalized into the nominative formation.

Orel likewise preserves both sides of the comparison under **budmaz* 'bottom' and **butmaz* 'bottom' (Orel 2003, 100). The derivational input is thus a historical stem choice, not an arbitrary respelling.

Old English evidence

The Old English noun itself is secure. Clark Hall gives *botm* 'bottom' (Clark Hall 1960, 63). Bosworth-Toller cross-references *bodan* 'bottom, floor' to *botm* 'bottom', showing the wider reflex family without weakening the attested lemma (Bosworth and Toller 1898, 112).

Development to Old English

Once the oblique **butt-* stem has been generalized, the derivational input **búttmaz* 'bottom' develops regularly to *botm* 'bottom'. The analogical step is therefore early: it belongs to pre-Old-English stem formation rather than to a later choice among Old English paradigm cells.

8.2 brand — OE *brandes*

Derivation: citation reconstruction **brándaz*; form followed here **brándas* > *brandes* (early analogy).

Derivation trace

Proto input: **brándas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**brándæs*
[no change]	OE Unstressed AE Merger	**brándes*
Early Anglo-Frisian		
[no change]		

Old English form: *brandes*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **brándaz* 'brand', continued by Old English *brand* 'brand' and its continental cognates (Orel 2003, 53). The selected input **brándas* 'brand' is not a different lexeme but the genitive singular of that same a-stem noun.

Both forms belong to the same root and stem class but occupy different inherited inflectional cells. The derivational input is the oblique cell.

Old English evidence

Old English dictionaries lemmatize the noun as *brand* 'brand' (Clark Hall 1960, 49; Bosworth and Toller 1898, 116). Bosworth-Toller also records inflectional forms such as *brandas* 'brand', *branda* 'brand', and *brandum* 'brand' under the same entry (Bosworth and Toller 1898, 116).

The specific comparison form in this entry, *brandes* 'brand', is the expected genitive singular of that a-stem noun. It is therefore an inferred Old English paradigm form rather than the ordinary dictionary headword.

Development to Old English

From **brándas* 'brand', the regular Old English development passes through the usual unstressed-vowel weakening of the inflectional ending, yielding *brandes* 'brand'. Nothing in the stem itself requires a special repair. The root consonants and the stressed vowel are the same as in the citation lemma *brand* 'brand'.

The analytical weight of the entry lies in the ending. By choosing the oblique singular rather than the nominative citation form, the entry presents the same lexeme in a different inherited cell.

Form comparison

The comparison below sets the relevant forms side by side. It separates the citation lemma from the selected oblique singular.

Form / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	*brándaz	expected OE lemma *brand*	*brand*	regular headword-level outcome
genitive singular	*brándas	regular output: *brandes*	*brandes*	exact match for the oblique cell

The noun itself is straightforwardly inherited. The main point of the entry is that *brandes* ‘brand (gen.)’ belongs to the same regular a-stem paradigm as *brand* ‘brand’, even though the citation lemma remains the nominative singular.

8.3 breast — OE *brēost*

Derivation: citation reconstruction *brústz; form followed here *bréustą > *brēost* (early analogy).

Derivation trace

Proto input: *bréustą

Earlier Germanic changes	**Old English changes**	
Northwest and West Germanic	OE Diphthong Leveling	*brēostą
[no change]	OE Heavy Syllable Nasal Apocope	*brēost
Early Anglo-Frisian		
[no change]		

Old English form: *brēost*

Reconstruction and comparative evidence

The word family shows two related but distinct Proto-Germanic formations. The root noun *brust- lies behind forms such as Gothic *brusts* ‘breast’, whereas Old English *brēost* ‘breast’ belongs to a thematic formation *breusta-, alongside Old Norse *brjóst* ‘breast’ and Old Saxon *briost* ‘breast’ (Kroonen 2013, 114; Orel 2003, 95; Ringe and Taylor 2014, 43).

The derivational input *bréustą ‘breast’ therefore differs from the citation label *brústz ‘breast’ because Old English reflects the thematic branch rather than the root noun. The morphological choice comes before the Old English sound changes themselves.

Old English evidence

Clark Hall records the noun as *brēost* ‘breast’ / *breóst* ‘breast’ (Clark Hall 1960, 65). The form is an established Old English lexeme, not a reconstructed target assembled from comparative evidence alone.

What requires explanation is not the Old English attestation but the relation between that attested noun and the broader Germanic word family. The relevant comparison form is therefore the thematic Old English noun *brēost* ‘breast’.

Development to Old English

From **bréustą* ‘breast’, the regular Old English development gives *brēost* ‘breast’, with the expected *eu* > *ēo* vowel history (Campbell 1959, § 115). No special repair is needed once the correct thematic formation is chosen.

The earlier mismatch arose only if the word was forced into the root-noun line. The Old English noun itself continues the thematic branch cleanly and directly.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader root-noun family label from the thematic formation actually continued in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader root-noun family	**brústz*	root-noun type outcomes outside OE	non-OE comparanda	useful family label, but not the direct source of *brēost*
selected thematic formation	**bréustą*	regular output: *brēost*	*brēost*	exact match between formation and attested OE noun

The relevant point is the formation split. *brēost* ‘breast’ is the regular Old English outcome of the thematic **breusta-* branch, not of the root noun **brust-*.

8.4 craft — OE *cræft*

Derivation: citation reconstruction **kráftiz*; form followed here **kráftaz* > *cræft* (early analogy).

Derivation trace

Proto input: **kráftaz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kráft*
EAF Final Z Deletion	**kráfta*	EAF Brightening	**krǽft*
Early Anglo-Frisian			
[no change]			

Old English form: *cræft*

Reconstruction and comparative evidence

Comparative sources disagree about the older stem class. Kroonen gives a u-stem **kraftu-*, while Orel prints **kraftiz* ‘craft’ and **kraftuz* ‘craft’ (Kroonen 2013, 340; Orel 2003, 259). The comparative label **kráftiz* ‘craft’ remains in view as a lexeme-level shorthand, while **kráftaz* ‘craft’ is the pre-Old-English form used for the Old English derivation.

Old English evidence

The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *cræft* ‘craft’ as the headword (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

Development to Old English

The comparison is between possible pre-Old-English inputs. The i-stem comparator **kráftiz* ‘craft’ gives *creft*, while the u-stem comparator **kráftuz* ‘craft’ gives *craft*. The a-stem-shaped input **kráftaz* ‘craft’ yields *cræft* ‘craft’ and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of **-iz* ‘craft’ and without the back-vowel outcome associated with the u-stem comparator.

Form comparison

Candidate input	OE output	Result
*kráftiz	*creft*	non-match; i-stem comparator
*kráftuz	*craft*	non-match; u-stem comparator
*kráftaz	*cræft*	exact match; selected pre-OE input

8.5 dill — OE *dile*

Derivation: citation reconstruction **déljaz*; form followed here **déliz* > *dile* (early analogy).

Derivation trace

Proto input: **déliz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE I Umlaut	**dili*
EAF Final Z Deletion	**déli*	OE Med Unstressed I Lowering1	**dile*
Early Anglo-Frisian			
[no change]			

Old English form: *dile*

Reconstruction and comparative evidence

Comparative evidence preserves both an i-stem and a ja-stem formation, with Old English *dile* ‘dill’ on one side and continental forms such as Old Saxon *dilli* ‘dill’ and Old High German *tilli* ‘dill’ on the other (Fulk 2018, 170). The derivational input **déliz* ‘dill’ therefore represents the i-stem side of the paradigm, whereas the citation label **déljaz* ‘dill’ is a broader comparative headword.

The stem class determines the Old English consonant shape. A ja-stem with **-lj-* would be expected to produce gemination, but the Old English noun shows a single *l*. Fulk’s discussion of ja-stems transferred to the i-stems provides the relevant morphological background for the OE side (Fulk 2018, 170).

Old English evidence

Old English dictionaries record the plant name as *dile* ‘dill’, alongside the variant *dili* ‘dill’ (Bosworth and Toller 1898, 164; Clark Hall 1960, 95). The form discussed here is therefore an attested Old English noun with single *l*.

The Old English evidence is the relevant point. Whatever broader comparative headword is chosen for the family, the inherited form reflected in OE is the i-stem type *dile* 'dill', not a geminated *dill* 'dill' outcome.

Development to Old English

From **déliz* 'dill', regular loss of final *z* and the later lowering of unstressed *i* yield *dile* 'dill'. The stem itself remains ungeminated throughout that path.

The contrast is morphological rather than phonological. If the word were forced through a ja-stem **-lj-* pathway, the expected result would show *ll*. The attested Old English noun instead matches the i-stem development.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the stem class actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative ja-stem label	**déljaz*	ja-stem type outcome with gemination	*dill*-type comparison	useful comparative label, but not the OE form
selected i-stem formation	**déliz*	regular output: *dile*	*dile*	exact match between formation and attested OE noun

The single *l* is the decisive diagnostic. It identifies *dile* 'dill' with the i-stem formation rather than with the continental ja-stem branch.

8.6 fast — OE *festan*

Derivation: citation reconstruction **fastēną*; form followed here **fástijaną* > *festan* (early analogy).

Derivation trace

Proto input: **fástijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**fæstijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**fæstijan*
Early Anglo-Frisian	OE Secondary Nasalization	**fæstiją̄n*
[no change]	Sievers Law Syncope	**fæstją̄n*
	OE I Umlaut	**festją̄n*
	OE Weak Tail Reduction	**festjan*
	OE J Loss After Heavy	**festan*

Old English form: *festan*

Reconstruction and comparative evidence

Kroonen places the verb within the wider **fastu-* adjective family and its derived **fasten-* verbal line, the comparative background behind Old English 'to fast' (Kroonen 2013, 171). Ringe and Taylor, however, distinguish the Old English verb more closely: they treat OE 'to fast' as originally a class-I weak verb that later acquired the stative meaning through lexical confusion (Ringe and Taylor 2014, 110).

The derivational input **fástijanq* 'fast' therefore represents the inherited class-I formation reflected in Old English, whereas the citation label **fastēną* 'fast' belongs to the broader comparative presentation of the lexeme.

Old English evidence

Old English dictionaries record forms such as *festan* 'fast', alongside related *fæstan* 'fast' / *fǣstan* 'fast' spellings and meanings (Bosworth and Toller 1898, 213). The form selected here is *festan*, which fits the regular class-I phonological development.

The *æ*-forms remain relevant, but they do not control the entry. In the present analysis they belong to a later analogical reshaping under the adjective *fæst* 'fast', whereas *festan* 'fast' is the regular inherited class-I comparison form.

Development to Old English

From **fástijanq* 'fast', Anglo-Frisian brightening and subsequent i-umlaut produce the fronted vowel seen in *festan* 'fast'. The later weak-tail reductions and loss of *j* after a heavy syllable complete the regular Old English outcome.

What makes the entry non-regular is not the phonology of *festan* 'fast' itself, but the choice of formation. Old English continues the class-I verb, even though the comparative headword is often given under the parallel **fastēn-* family.

Class comparison

The comparison below sets the relevant forms side by side. It distinguishes the comparative class-III headword from the class-I formation actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative class-III headword	**fastēną*	class-III type outcome, not *festan*	wider family context	useful family label, but not the direct source of the target
selected class-I weak verb	**fástijaną*	regular output: *festan*	*festan*	exact match between formation and attested OE verb
later analogical reshaping	adjective-driven *fæst* influence	*fæstan* / *fǣstan* type spellings	fæstan-type evidence	genuine later OE reshaping, but secondary to the Old English form here

The relevant point is the class split. *festan* 'fast' is the regular Old English outcome of the class-I formation, while the better-known *æ*-forms belong to a later analogical layer.

8.7 flask — OE *flasce*

Derivation: citation reconstruction **flaskō*; form followed here **fláskōn* > *flasce* (early analogy).

Derivation trace

Proto input: **fláskōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**flæskǭ*
PNWGmc N Stem N Loss	**fláskǭ*	OE A Restoration	**flaskǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**flaskœ*
[no change]		OE Unstressed AE Merger	**flaske*

Old English form: *flasce*

Reconstruction and comparative evidence

The wider Germanic family is often cited under a form such as **flaskō* 'flask', but the evidence relevant for Old English points instead to a weak feminine formation **fláskōn* 'flask' / **flaskǭ* 'flask' (Orel 2003, 104). That distinction is crucial for the suffixal history of the noun.

The derivational input therefore differs from the citation label in stem class. Old English *flasce* 'flask' belongs with the weak feminine line, and the plural or oblique forms *flascan* 'flask' support that analysis (Ringe and Taylor 2014, 192).

Old English evidence

Old English dictionaries record the noun as *flasce* 'flask', with inflectional support from forms such as *flascan* 'flask'; a later West Saxon *flaxe* 'flask' is also noted as a secondary variant (Bosworth and Toller 1898, 235; Clark Hall 1960, 121).

The relevant comparison form is therefore the weak feminine noun *flasce* 'flask'. The plural and oblique evidence helps explain why the vowel and ending are preserved as they are in the singular.

Development to Old English

From **fláskōn* 'flask', the weak feminine passes through the expected loss of *n* and the later Old English development of the unstressed ending, reaching *flasce* 'flask'. Campbell cites restored *a* in exactly this environment, including *flasce* 'flask' after inflected *flascan* 'flask' (Campbell 1959, § 158). Once the weak feminine formation is chosen, the noun follows a regular path to its Old English shape.

The decisive issue is morphological rather than phonological. A simple strong feminine citation form does not capture the OE weak noun as cleanly as the selected **fláskōn* 'flask' does.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak feminine formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader comparative headword	**flaskō*	broader family label	wider family context	useful lexeme label, but not the cleanest OE-facing derivation
selected weak feminine formation	**fláskōn*	regular output: *flasce*	*flasce*	exact match between formation and attested OE noun

The weak feminine suffix is the relevant point. It aligns the inherited form with attested *flasce* ‘flask’ and its supporting paradigm forms.

8.8 follow — OE *fylġan*

Derivation: citation reconstruction **fulgēną*; form followed here **fúlgijaną* > *fylġan* (early analogy).

Derivation trace

Proto input: **fúlgijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**fúlgijan*
[no change]	OE Secondary Nasalization	**fúlgiją̃n*
	Sievers Law Syncope	**fúlgją̃n*
Early Anglo-Frisian	OE Velar Palatalization	**fúlʤją̃n*
[no change]	OE I Umlaut	**fylʤją̃n*
	OE Weak Tail Reduction	**fylʤjan*
	OE J Loss After Heavy	**fylʤan*

Old English form: *fylġan*

Reconstruction and comparative evidence

Kroonen reconstructs the verb as **fulgen-* and gives Old English *fylgan* ‘follow’, *folgian* ‘follow’, adding that Old Norse *fylgja* ‘follow’ and Old English *fylg(e)an* continue a formation **fulgjan-* (Kroonen 2013, 159). The comparative headword and the class-I formation are therefore related but not identical.

Ringe and Taylor distinguish PNWGmc **fulgija-* ‘follow’ ~ **fulgai-* ‘follow’ > OE *fylgan* ‘follow’ ~ *folgian* ‘follow’ and describe it as a dual formation that probably reflects an older alternation between j-present and e-stative (Ringe and Taylor 2014, 293–94). This is a stem-class choice, not a spelling choice. The derivational input **fúlgijaną* ‘follow’ belongs to the class-I **fulgija-* / **fulgjan-* branch; the citation form **fulgēną* ‘follow’ belongs to the parallel class-II history behind *folgian* ‘follow’.

Old English evidence

The Old English evidence preserves both formations. Clark Hall lists *fylgan* ‘follow’ with variant spellings *fylgian* ‘follow’ and *fyligan* ‘follow’ (Clark Hall 1960, 125). Bosworth-Toller likewise has a separate *fylgean* ‘follow’ entry (Bosworth and Toller 1898, 275).

Bright notes traces of the older conjugation in *fylg(e)an* ‘follow’ (Bright 1971, 77) and lists *folgian* ‘follow’ (*fylgean* ‘follow’) in the glossary (Bright 1971, 364). The relevant comparison form in this entry is therefore the class-I verb *fylgan* ‘follow’ / *fylgean*, here normalized as *fylġan* ‘follow’. The spelling with ġ represents the palatalized velar before a front-vocalic environment.

Development to Old English

**fúlgijaną* ‘follow’ is a class-I weak-verb formation. In the class-I branch the **j* blocks NWGmc lowering of *u* to *o*, since Ringe and Taylor formulate that lowering for environments in which no **j* intervened (Ringe and Taylor 2014, 96). The same front-vocalic environment then triggers i-umlaut, so *u* becomes *y* (Ringe and Taylor 2014, § 6.6.2).

The subsequent Old English developments are palatalization of the velar, weak-tail reduction, and loss of *j* after a heavy syllable, yielding *fylġan* ‘follow’. This is the regular outcome of the class-I formation. The class-II form *folgian* ‘follow’ belongs to the parallel **-ē-* / **-ai-* branch and is not the form modeled here.

Class comparison

A class comparison identifies which inherited formation corresponds to the established Old English form under discussion. The comparison below is manual; no full automatic class probe is presented here.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation class-II formation	**fulgēną*	probe output: *folgon*	*folgian* ‘follow’	mismatch: the regular output is not the remodeled infinitive *folgian*
parallel class-II branch	PNWGmc **fulgai-*	Ringe-Taylor: OE *folgian* ‘follow’	*folgian* ‘follow’	documents the separate class-II branch, but not the target of this entry
selected class-I formation	**fúlgijaną*	regular output: *fylġan* ‘follow’	*fylġan* ‘follow’ / *fylgan* ‘follow’	exact match between input, output, and class

The relevant point is the class split. *fylġan* ‘follow’ is the regular Old English outcome of the class-I **fulgija-* / **fulgjan-* formation, whereas *folgian* ‘follow’ belongs to the parallel class-II branch.

8.9 gall — OE *ġealla*

Derivation: citation reconstruction **gállą*; form followed here **gállô* > *ġealla* (early analogy).

Derivation trace

Proto input: **gállô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gællô*
[no change]	OE Breaking	**geallô*
Early Anglo-Frisian	OE Velar Palatalization	**ʤeallô*
[no change]	OE Unstressed Long Vowel Shortening	**ʤealla*

Old English form: *ġealla*

Reconstruction and comparative evidence

The wider cognate family can be presented under a form such as **gállą* 'gall', but the Old English noun itself belongs with a weak noun **gallōn-* 'gall', cited here as **gállô* 'gall' (Kroonen 2013, 165). The derivational input therefore differs from the broader comparative headword in stem class.

The stem class determines the Old English shape. The weak masculine pathway preserves the ending needed for *ġealla* 'gall', whereas a simple strong-noun headword does not align as closely with the attested OE noun.

Old English evidence

Old English dictionaries record the noun as *gealla* 'gall', and Bright also gives the dative *geallan* 'gall', confirming a weak-noun paradigm (Bosworth and Toller 1898, 297; Clark Hall 1960, 145; Bright 1971, 372). The normalized spelling *ġealla* 'gall' uses ġ for the palatal consonant.

Campbell also notes dialectal variation, contrasting West Saxon or Kentish *gealla* 'gall' with Anglian *galla* 'gall' (Campbell 1959, § 486). The target of this entry is the West Saxon type *ġealla* 'gall'.

Development to Old English

From **gállô* 'gall', the weak noun develops through the expected Old English history of the suffix and the regular breaking environment before *ll*, yielding *ġealla* 'gall' (Campbell 1959, § 486). Once the weak masculine input is chosen, the noun follows a regular path to its attested Old English form.

The decisive issue is therefore morphological. Old English reflects the weak noun, while the broader family label belongs to a different way of presenting the cognate set.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak noun formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader family label	**gállą*	broader cognate-set headword	wider family context	useful lexeme label, but not the direct source of *ġealla*
selected weak noun	**gállô*	regular output: *ġealla* 'gall'	*ġealla* 'gall'	exact match between formation and attested OE noun

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
dialectal Anglian continuation	weak noun branch	Anglian *galla* 'gall' type	*galla* 'gall'	genuine OE variant, but not the West Saxon form used here

The weak-noun stem class is the relevant point. It gives a direct route to attested *ġealla* 'gall', while the broader comparative label serves only as a family heading.

8.10 heaven — OE *heofon*

Derivation: citation reconstruction **xémenaz*; form followed here **xébun* > *heofon* (early analogy).

Derivation trace

Proto input: **xébun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Med Unstressed U Lowering	**xébon*
[no change]	OE Velar Fricative Palatalization	**çébon*
Early Anglo-Frisian	PGmc B Allophony	**çéβon*
[no change]	OE Back Mutation	**çéoβon*

Old English form: *heofon*

A. Selected CAPR derivation

Item	Value
lexical item	heaven
citation reconstruction / lexeme label	*xémenaz
form followed here	*xébun (northern WGmc / pre-OE* hebun)
Old English target	heofon
classification	early_analogy
derivational result	*xébun > heofon (multiplicity 1, exact match)

Compact live trace (corrected cascade, literal adjacent-*mn* SC022):

step	form
proto input	**xébun*
OE Med Unstressed U Lowering	**xébon*
OE Velar Fricative Palatalization	**çébon*
PGmc B Allophony	**çéβon*
OE Back Mutation	**çéoβon*
Old English Orthography	*heofon*

No rule in this span creates the labial: it enters with the input *b* ($\rightarrow \beta \rightarrow f$). Only vocalic and surface changes are modelled.

B. Why *xébun is not arbitrary

Ringe and Taylor reconstruct northern West Germanic **hebun* 'sky, heaven', gen. **hebunas* 'heaven', whence West Saxon and Northumbrian *heofon* 'heaven', Mercian *heofen* 'heaven' (Ringe and Taylor 2014, 272, 324). R&T's *hebun* is the lexeme CAPR writes **xébun* (canonical *x* for source *h*), dated to that northern West Germanic stage, not Proto-Germanic; §D explains its labial.

The word is an inherited mn-stem (Kroonen 2013, 220): PIE *$h_2$ék-mōn, gen.* h_2k-mn-ós, remodelled in Germanic to nom. **hemō* 'heaven', gen. **hemnaz* 'heaven'. In the zero-grade oblique cluster cells the sequence *-mn- is adjacent, and the change* mn > βn/bn/fn* is regular there (Fulk §6.14 (Fulk 2018, 121)). Old Norse preserves exactly this paradigmatic distribution — *himinn* 'heaven' (nominative, with *m*) beside dative *hifni* 'heaven' (oblique, with *f* from the syncopated cluster) — while Old English and Old Saxon (OS *heban* 'heaven', *hebenes* 'heaven' (Fulk 2018)) generalized the labial into the vowel-bearing stem. The *b* of *xébun is therefore analogical* within this nominative input*, but its historical source is regular and recoverable — back to the inherited genitive (see §E).

C. The unresolved *u*

The medial *u* is the less secure element and must not be treated as certain merely because the cascade needs a back vowel to trigger back mutation. It is the n-stem suffixal vowel, the genuine conditioning environment for West Saxon back umlaut of *e* (parallel to *seven*, §D), and it is supported by Ringe–Taylor's reconstruction and by the Old English back-mutation pattern (Ringe and Taylor 2014, 324). But its exact deeper ablaut grade — *u* here, against the *i* of the cognate vowel-stem **hemina-* (*himinn* 'heaven', Go. *himins* 'heaven') — is not fully recoverable. It is the irreducible residue of the reconstruction; the *b* is not.

D. Regular modelled span

The northern-WGmc / pre-OE → Old English development **xébun* > *heofon* is regular and is the exact structural parallel of the independently regular numeral *seven*, **sébun* > *seofon* (row 2174, *DERIVATION_CLASS* regular). Both have a medial *u* and a labial, and both undergo West Saxon back umlaut of *e* before *u* across that labial (Ringe and Taylor 2014, 324; Hogg 1992). The live traces run in lock-step (medial *u*-lowering, labial spirantization, back mutation, orthography). Ringe and Taylor explicitly treat *heaven* and *seven* as parallel back-umlaut examples, which is why *seven* is retained here as the control.

E. Deep oblique controls

The obliques go deeper than the selected nominative and derive the labial themselves. Live outputs under the corrected literal adjacent-*mn* cascade, with evidential status marked:

cell / form	type	corrected-cascade output	status	note
nom. citation **xémenaz*	reconstructed (citation)	*hemen*	counterfactual (no labial)	citation a-stem; no cluster, so no regular *f*
gen.sg cluster **xémnas*	reconstructed (PIE *h_2k-mn-ós)	*hefnes*	reconstructed, **not attested in OE**	labial **regular** here (mn > βn)
dat.sg cluster **xémni*	reconstructed	*hifn*	attested only as ON *hifni* 'heaven'	labial + i-umlaut regular
dat.pl cluster **xémnum*	reconstructed	*hefnum*	reconstructed, **not attested in OE**	labial regular

cell / form	type	corrected-cascade output	status	note
remodelled stem **xébun*	reconstructed (R&T *hebun)	*heofon*	**attested** (selected row)	labial pre-supplied
R&T genitive **hebunas*	reconstructed	*heofones*	**attested**	labial pre-supplied
OE oblique *heofnes* / *heofnum*	**attested**	—	actual OE forms	carry *eo*; from the re-vowelled stem + late syncope
ON *himinn* : *hifni* 'heaven'	**attested** (North Gmc)	—	attested cluster line	keeps the *m* : *f* alternation
OS *heban* / *hebenes*	**attested** (Old Saxon)	—	labial generalized	vowel-bearing throughout

Distinctions kept explicit: attested OE forms (*heofon*, *heofones*, *heofnes*, *heofnum*) descend from the re-vowelled stem; the cluster outputs *hefnes*/*hifn*/*hefnum* are reconstructed regular outputs, and for Old English they are counterfactual — the actual attested obliques carry *eo* from back mutation on the retained/restored medial vowel, then late syncope (*heofonum* > *heofnum*), not from a bare *-fn-* cluster. The genuine attested witness of the pure cluster line is Old Norse *hifni* 'heaven'.

F. Why the oblique is not the selected row

The oblique lines are historically essential — they show that the labial has a regular origin, all the way back to the inherited genitive *h_2k-mn-ós — but they are inferior as the single CAPR input→target datum. In Old English every oblique cluster form encounters morphological re-vowelling (restoration of a medial vowel into the labialized* -βn- *stem) before it reaches an attested Old English shape. That re-vowelling is a non-regular event; it interrupts any single deterministic path from a cluster oblique to an attested OE form. The nominative input* xébun begins after that restructuring and then proceeds by regular sound change to a secure attested target — the longest clean modellable span.

G. Attestation warning

Bare Old English *hefn* is not securely attested and must not be reintroduced as an attested comparator. The earliest witnesses (Cædmon's Hymn) are *hefun* 'heaven', *hefen* 'heaven', and (Moore Bede) *heben* 'heaven' — all with a medial vowel. The reconstructed cluster output *hefn* (§E) is a reconstruction, never a manuscript form.

H. Classification rationale

The row is *early_analogy*, not *late_analogy* and not *known_unmodelled.*

- The selected row models exactly one form, **xébun → heofon*. The necessary paradigm restructuring (labial levelling + suffix-vowel generalization) precedes the derivational input; once that input is chosen, the modelled development is regular. This is the definition of *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply).
- It is not *late_analogy*: the analogy is in the input, not a later paradigm-cell selection; the later history of *other* paradigm cells (oblique syncope, *heo-* levelling) does not touch the modelled nominative derivation.
- It is not *known_unmodelled*: that label (a legitimate, conservative alternative discussed in `dossier-heaven-paradigm-history-2026.md`) would discard a real, defensible regular span. **xébun* > *heofon is* a clean regular derivation to a secure attested target, so CAPR models it.

The old cross-syllable *mV...n* proxy in SC022 — which formerly fabricated the labial from an intervocalic *m* so that the previous input **xémonų* could reach *heofon* — has been retired in favour of the literal historical adjacent *mn* > *βn*. See `dossier-heaven-paradigm-history-2026.md` §§13–15 and the implementation audit `audits/heaven-sc022-implementation-2026.md`.

8.11 knight — OE *cniht*

Derivation: citation reconstruction **kníxtaz*; form followed here **knéxtaz* > *cniht* (early analogy).

Derivation trace

Proto input: **knéxtaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**knéxt*
EAF Final Z Deletion	**knéxta*	OE Breaking	**knéoxt*
Early Anglo-Frisian		OE Ws Palatal Umlaut	**knixt*
[no change]			

Old English form: *cniht*

Reconstruction and comparative evidence

The comparative sources align on an *e*-grade reconstruction for this noun. Ringe and Taylor cite **kneht* 'knight', and Orel gives **knextaz* 'knight' (Ringe and Taylor 2014, 142; Orel 2003, 256). Kluge-Seebold likewise points to **knehta-* (Kluge 2011, 506). The derivational input **knéxtaz* 'knight' follows that comparative evidence.

A competing citation reconstruction **kníxtaz* 'knight' remains possible as a label for the word family, but it is not the reconstruction followed here. The Old English development discussed below is based on **knéxtaz* 'knight'.

Old English evidence

Old English dictionaries record the noun as *cniht* 'knight' (Clark Hall 1960, 63; Bosworth and Toller 1898, 71). Campbell cites plural *cneohtas* 'knights' among the broken forms, showing the same vowel environment from another point in the paradigm (Campbell 1959, § 146).

The target is therefore an ordinary attested Old English noun. No reconstructed OE comparator is needed here.

Development to Old English

From **knéxtaz* 'knight', the relevant Old English changes include breaking before the velar cluster and then the later reduction that yields *cniht* 'knight'. Campbell later notes the early West-Saxon alternation *cniht* 'knight' beside plural *cneohtas* 'knights' (Campbell 1959, § 305). Sievers-Brunner gives the same contrast as *cniht ... cneohtas* (Brunner 1965, § 122). With that corrected input, the derivation is straightforward.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the handbook-supported *e*-grade input from a competing citation reconstruction.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing citation reconstruction	**kníxtaz*	not the reconstruction followed here	broader citation tradition	useful as a competing label, but not the source-based choice used for the OE derivation
handbook-supported reconstruction	**knéxtaz*	regular output: *cniht* ‘knight’	*cniht* ‘knight’	exact match between comparative reconstruction and attested OE noun
related plural evidence	same stem family	plural *cneohtas* ‘knights’ type background	*cneohtas* ‘knights’	supports the vowel environment, but not the Old English form here cell

8.12 lade — OE *hladan*

Derivation: citation reconstruction **laθōjanę̨*; form followed here **xláðaną* > *hladan* (early analogy).

Derivation trace

Proto input: **xláðaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xlædaną*
PWGmc Dental Hardening	**xládaną*	OE A Restoration	**xladaną*
Early Anglo-Frisian		OE Heavy Syllable Nasal Apocope	**xladan*
		OE Secondary Nasalization	**xladąn*
[no change]		OE Weak Tail Reduction	**xladan*

Old English form: *hladan*

Reconstruction and comparative evidence

Ringe and Taylor cite the strong verb *hladan* ‘lade, load’ directly (Ringe and Taylor 2014, 248). The citation label **laθōjaną* ‘lade’ is used here only as a broader family heading, not as the direct source of the OE derivation.

The derivational input therefore marks an early stem choice. The entry follows the strong Verner-grade form that reaches Old English *hladan* directly.

Old English evidence

Bosworth-Toller records the verb as *hladan* and preserves the expected strong-verb paradigm material around it (Bosworth and Toller 1898, 559). Clark Hall likewise records *hladan* (Clark Hall 1960, 159). The target is an attested infinitive rather than a reconstructed paradigm cell.

For this entry the relevant comparison form is the infinitive *hladan* itself. The question is how that attested strong verb relates to the broader comparative family.

Development to Old English

From **xláðaną* ‘lade’, the verb passes through the expected early voiced stop stage, Anglo-Frisian brightening, and the later A-restoration that returns the root vowel to *a* before the full infinitival ending. Campbell treats verbs such as *hladan* as showing restored *a* in the present system (Campbell 1959, § 744). Ringe and Taylor likewise derive *hladan* from the voiced strong-verb line (Ringe and Taylor 2014, 248). The resulting Old English infinitive is *hladan*.

Class comparison

The comparison below sets the relevant forms side by side. It separates the wider weak-verb family label from the strong verb actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative weak-verb label	*laθōjaną	wider family background	broader family context	useful family label, but not the direct source of *hladan*
selected strong Verner-grade input	*xláðaną	regular output: *hladan*	*hladan*	exact match between formation and attested OE infinitive

8.13 lap — OE *lappa*

Derivation: citation reconstruction **lábbaz*; form followed here **láppô* > *lappa* (early analogy).

Derivation trace

Proto input: **láppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**læppô*
[no change]	OE A Restoration	**lappô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**lappa*
[no change]		

Old English form: *lappa*

Reconstruction and comparative evidence

The comparative sources point to a weak noun with *pp*: Orel gives **lappōn* ‘lap’, and the Old English dictionary tradition preserves variant *læppa* ‘lap’ (Orel 2003, 236; Clark Hall 1960, 180). The derivational input **láppô* ‘lap’ follows that evidence.

A competing comparative label **lábbaz* ‘lap’ has also circulated for the word family, but the cited handbooks do not make it the direct source of the Old English weak noun. The form relevant to the OE development is the weak masculine input **láppô* ‘lap’.

Old English evidence

Campbell cites *lappa* ‘lap’ as a case of restored *a* (Campbell 1959, § 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ (Brunner 1965, § 10). The dictionary tradition also preserves *læppa* ‘lap’ (Clark Hall 1960, 180; Bosworth and Toller 1898, 613).

The target of this entry is the restored singular *lappa* ‘lap’. The variant *læppa* ‘lap’ and the oblique or plural *leappan* remain part of the Old English record and help frame the noun’s vowel history.

Development to Old English

Campbell explicitly lists *lappa* ‘lap’ among the forms with restored *a* (Campbell 1959, § 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ at the same Old English stage (Brunner 1965, § 10). With the weak masculine input chosen, the selected *lappa* ‘lap’ outcome is therefore the regular Old English comparison.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the weak masculine formation from a competing voiced comparative label.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing voiced comparative label	*lábbaz	not the form followed for the OE weak-noun derivation	broader comparative background	useful as a competing label, but not the source-based choice used here
selected weak masculine noun	*láppô	regular output: *lappa*	*lappa*	exact match between formation and attested OE noun
attested OE variant line	same noun family	*læppa*, *leappan*	*læppa* / *leappan*	useful control forms within the same OE tradition

8.14 laugh — OE *hliehhan*

Derivation: citation reconstruction **lákaną*; form followed here **xláxjaną* > *hliehhan* (early analogy).

Derivation trace

Proto input: **xláxjaną*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		EAF Brightening	**xlæxxjaną*
PWGmc J Gemination	**xláxxjaną*	OE Breaking	**xleaxxjaną*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**xleaxçjaną*
[no change]		OE Heavy Syllable Nasal Apocope	**xleaxçjan*
		OE Secondary Nasalization	**xleaxçją n*
		OE I Umlaut	**xliexçjąn*
		OE Weak Tail Reduction	**xliexçjan*
		OE J Loss After Heavy	**xliexçan*

Old English form: *hliehhan*

Reconstruction and comparative evidence

The wider Germanic family includes a non-j branch represented here by the citation label **lákaną* 'laugh', while the derivational input **xláxjaną* 'laugh' reflects the j-present line behind Old English *hliehhan* 'laugh'.

This branch supplies the geminate fricative and the vowel development characteristic of the Old English verb. The comparative family label and the OE-facing input are therefore related but not identical.

Old English evidence

Bosworth-Toller records *hlihhan* 'laugh' as the verb 'to laugh' (Bosworth and Toller 1898, 551). Clark Hall cross-references *hlæhan* 'laugh', *hlehhan* 'laugh', and *hlihhan* 'laugh' to *hliehhan* 'laugh' (Clark Hall 1960, 160–61). Bright's glossary likewise gives *hlihhan (hliehhan, hlyhhan)* (Bright 1971, 315). The target of this entry is the West Saxon *hliehhan* 'laugh'.

The variants belong to the same background, but the attested lemma *hliehhan* 'laugh' supplies the evidence followed here.

Development to Old English

From **xláxjaną* 'laugh', West Germanic j-gemination yields the doubled consonant (Campbell 1959, § 407). Ringe and Taylor derive Old English *hliehhan* 'laugh' from the j-present branch via breaking before the palatalized geminate (Ringe and Taylor 2014, 240). They separately compare the related noun *hleahtor* as the outcome of **hlahtraz* 'laugh' (Ringe and Taylor 2014, 328).

Branch comparison

The comparison below sets the relevant forms side by side. It separates the wider non-j family label from the j-present branch actually reflected in Old English.

Formation / branch	Candidate input	Expected or documented OE outcome	OE comparison form	Result
wider non-j family	*lákaną	comparative background outside the selected OE line	wider family context	useful family label, but not the direct source of *hliehhan*
selected j-present branch	*xláxjaną	regular output: *hliehhan*	*hliehhan*	exact match between branch and attested OE lemma
attested OE variants	same OE verb line	*hlæhhan*, *hlehhan*	*hlæhhan* / *hlehhan*	genuine variant evidence, but secondary to the form compared here

8.15 loam — OE *lām*

Derivation: citation reconstruction **laimōn*; form followed here **láimą* > *lām* (early analogy).

Derivation trace

Proto input: **láimą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lām*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**lāmą*		

Old English form: *lām*

Reconstruction and comparative evidence

The inherited comparative noun is given as **laimōn* 'loam' or **laiman-*, and both Orel and Kroonen identify Old English *lām* 'loam' as a neuter reflex of that family (Orel 2003, 272; Kroonen 2013, 363). The form followed here, **láimą* 'loam', differs from the comparative headword because it represents the stem class that matches the Old English noun most directly.

This is therefore a class shift within the history of the English branch rather than a dispute about the OE target itself.

Old English evidence

Old English dictionaries record the noun as *lām* 'loam', a neuter word for 'loam, clay, mud' (Bosworth and Toller 1898, 604; Clark Hall 1960, 196). The target is an attested citation form rather than a reconstructed comparator.

The relevant question is not whether *lām* 'loam' is Old English, but which inherited formation best accounts for that attested neuter noun.

Development to Old English

From **láimą* 'loam', regular monophthongization of *ai* and the later loss of the final nasal syllable yield *lām* 'loam'. With that OE-facing input, the phonological development is straightforward.

Class comparison

The comparison below sets the inherited comparative n-stem label beside the OE-facing stem class used to derive the attested noun.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited comparative noun	*laimōn	comparative family background	wider family context	useful headword, but not the direct OE-facing input
OE-facing stem class followed here	*láimą	regular output: *lām*	*lām*	exact match between the form followed here and the attested OE noun

8.16 lung — OE *lungen*

Derivation: citation reconstruction **lungō*; form followed here **lúnganjō* > *lungen* (early analogy).

Derivation trace

Proto input: **lúnganjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**lúngennju*
PWGmc J Gemination	**lúngannjō*	OE High Vowel Apocope	**lúngennj*
PNWGmc Final Long O Raising	**lúngannju*	OE J Loss After Heavy	**lúngenn*
Early Anglo-Frisian		OE Final Geminate Simplification	**lúngen*
[no change]			

Old English form: *lungen*

Reconstruction and comparative evidence

Kroonen treats the basic noun as **lungōn-* and also cites an OE-facing derivative **lungunjō-*, continued by Old English *lungen* 'lung' and close West Germanic cognates (Kroonen 2013, 384). The derivational input **lúnganjō* 'lung' models that derived feminine formation rather than the base noun. The notation differs slightly from Kroonen's **lungunjō-*, but both point to the same derived feminine line.

The difference between the citation label and the derivational input is therefore derivational. Old English *lungen* 'lung' is not a direct reflex of the bare base noun **lungō* 'lung'; it belongs to an expanded feminine formation.

Old English evidence

Old English dictionaries record the noun as *lungen* 'lung', with inflected forms such as *lungenne* and *lungena* (Bosworth and Toller 1898, 634). Clark Hall also preserves a small family of compounds such as *lungenādl* 'lung-disease', *lungensealf*, and *lungenwyrt* (Clark Hall 1960, 191).

The target is an attested Old English lexeme with its own paradigm, not a rescued inflectional cell.

Development to Old English

From the selected derived input, the expected derivational consonant and vowel adjustments lead to *lungen* 'lung'. Once the expanded feminine formation is chosen, the Old English outcome is regular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the base noun from the derived feminine formation reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
base noun	*lungō	base-noun outcome without the OE derivative suffix	broader family context	useful headword, but not the direct source of *lungen*

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
derived OE-facing formation	*lúnganjō	regular output: *lungen*	*lungen*	exact match between the derived formation and the attested OE noun
Kroonen's cited derivative	**lungunjō-*	comparative support for the same OE-facing formation	*lungen* and cognate set	supports the derived feminine formation, with notation differing from the normalized input form used here

8.17 navel — OE *nafola*

Derivation: citation reconstruction **nablô*; form followed here **nábulô* > *nafola* (early analogy).

Derivation trace

Proto input: **nábulô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Med Unstressed U Lowering	**nábolô*
[no change]	EAF Brightening	**næbolô*
Early Anglo-Frisian	OE A Restoration	**nabolô*
[no change]	PGmc B Allophony	**naβolô*
	OE Unstressed Long Vowel Shortening	**naβola*

Old English form: *nafola*

Reconstruction and comparative evidence

Kroonen instead gives a nasal-suffix navel formation with Old English *nafela* 'navel' among its reflexes (Kroonen 2013, 420), while Ringe and Taylor give the derivational pathway **nabulō* 'navel' > **næbula* 'navel' > *nafola* 'navel' (Ringe and Taylor 2014, 270). The difference is one of stage and notation rather than of lexeme identity: the derivational input **nábulô* 'navel' is the pre-syncope form needed for the Old English development.

For the Old English comparison, the crucial point is simply that the pre-OE form still contains a medial vowel.

Old English evidence

Ringe and Taylor note the early West Saxon shift *nafola* 'navel' > *nafela* 'navel' (Ringe and Taylor 2014, 336). Campbell likewise records *nafela* 'navel' beside Corpus *nabula* 'navel' (Campbell 1959, § 159). The target of this entry is the nominative singular *nafola* 'navel', the form that matches the selected derivational pathway most directly.

nafela 'navel' is the better-known later West Saxon spelling, while *nabula* 'navel' preserves a less reduced medial vowel. These forms belong to the same lexical history, but this entry is centered on *nafola* 'navel'.

Development to Old English

Ringe and Taylor give the pre-OE line **nabulō* 'navel' > **næbula* 'navel' > OE *nafola* 'navel' (Ringe and Taylor 2014, 270). The trace represents that same development with stress-marked notation and explicit intermediate weakening. Intervocalic *b* then surfaces as *f*, and final weak-tail shortening gives *nafola* 'navel'.

The medial vowel is still present when A-restoration applies. That is why the selected pre-syncope input differs from the syncopated comparative headword.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative citation form from the pre-syncope input and from the later OE spellings.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
syncopated comparative headword	**nablô*	reduced *næfla*-type outcome rather than *nafola* 'navel'	not the Old English form here	useful citation form, but too reduced for the pathway modeled here
selected pre-syncope input	**nábulô*	regular output: *nafola* 'navel'	*nafola* 'navel'	exact match between derivational input and target
later OE reduction stages	same lexical history	attested *nafela* 'navel'; Corpus *nabula* 'navel'	*nafela* 'navel' / *nabula* 'navel'	related OE spellings, but not the chosen comparator

8.18 neck — OE *hnecca*

Derivation: citation reconstruction **xnákkaz*; form followed here **xnékkô* > *hnecca* (early analogy).

Derivation trace

Proto input: **xnékkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**xnékka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hnecca*

Reconstruction and comparative evidence

The noun belongs to an ablauting n-stem family. Kroonen reconstructs a paradigm with nominative **hnekkō* 'neck', genitive **hnukkaz* 'neck', and accusative plural **hnakkuns* 'neck', and he places Old English *hnecca* 'neck, nape' among the e-grade descendants (Kroonen 2011, 167). Kluge-Seebold likewise identifies *ae. hnecca* as an ablaut partner of the a-grade *Nacken* family (Kluge 2011, 347).

A competing comparative label **xnákkaz* 'neck' belongs to the wider family, and Orel also gives an a-grade headword line (Orel 2003, 218). The derivational input **xnékkô* 'neck', however, is the form that matches the Old English branch.

Old English evidence

Clark Hall records the weak masculine noun *hnecca* 'neck, nape' (Clark Hall 1960, 162). Bosworth-Toller likewise records *hnecca* 'neck, nape' (Bosworth and Toller 1898, 567). The target is therefore an attested citation form, not an oblique cell or a reconstructed lemma.

The phonological question is upstream of the Old English evidence. The attested noun already shows that the branch continued an e-grade form rather than the a-grade seen in much of the continental material.

Development to Old English

From **xnékkô* 'neck', the derivation is straightforward. The trace shortens the final long vowel to **xnékka* 'neck', and Old English orthography gives *hnecca* 'neck'.

The derivation depends on the earlier selection of the e-grade weak-noun form continued by Old English.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the wider a-grade family from the selected e-grade Old English branch.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing comparative label	**xnákkaz*	broader a-grade family rather than the selected OE source	continental *Nacken* line	useful family label, but not the input followed for the Old English derivation
weak noun with a-grade	**xnakkô*	expected *hnacca* type outcome	*hnacca*	fixes the class, but not the vowel grade
selected e-grade nominative	**xnékkô*	regular output: *hnecca*	*hnecca*	exact match between derivational input and attested OE noun
oblique paradigm background	**hnukkaz*, **hnakkuns*	ON/OHG/German a-grade continuation	*hnakki* / *Nacken*	shows the wider ablaut family, but not the chosen OE branch

8.19 needle — OE *nǣdl*

Derivation: citation reconstruction **nḗθlō*; form followed here **nḗðlō* > *nǣdl* (early analogy).

Derivation trace

Proto input: **nḗðlō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**nǣdl*
PWGmc Dental Hardening	**nḗdlō*		
PNWGmc Final Long O Raising	**nḗdlu*		
PNWGmc Long E Lowering	**nǣdlu*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdl*

Reconstruction and comparative evidence

Ringe and Taylor treat the word as a voiced/voiceless alternant, citing **nēþlō* 'needle' ~ **nēdlō* 'needle' > OE *nédl* 'needle' (Ringe and Taylor 2014, 329). The form followed here, **nḗðlō* 'needle', is the voiced Verner-grade form used for the Old English comparison, while the citation form **nḗθlō* 'needle' remains the broader lexeme label.

The development discussed here follows the Ringe-Taylor alternant framework.

Old English evidence

Clark Hall records the attested citation form *nǣdl* 'needle' (Clark Hall 1960, 210). Campbell lists *nédl* 'needle' among the expected unbroken forms after *t* and *d* (Campbell 1959, § 367). Hogg also includes *nidi* / *nǣdl* 'needle' in the same broader cluster history (Hogg 1992, 95).

The target is therefore an attested citation form. No oblique-cell substitution is involved in this entry.

Development to Old English

Ringe and Taylor give the historical line **nēþlō* 'needle' ~ **nēdlō* 'needle' > OE *nédl* 'needle' (Ringe and Taylor 2014, 329). Campbell likewise lists *nédl* 'needle' among the expected unbroken forms after *t* and *d* (Campbell 1959, § 367). The trace expresses the same pathway with the voiced alternant followed here for the Old English comparison.

The essential choice lies in which Proto-Germanic alternant is taken as the starting point. Once the voiced form is chosen, the rest of the pathway is regular.

Alternant comparison

The comparison below sets the relevant forms side by side. It separates the broader citation headword from the voiced alternant used for Old English.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative voiceless headword	**nḗθlō*	broader word-family label rather than the OE-facing alternant	**nēþlō* line	useful citation form, but not the derivational input for the Old English comparison

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
voiced Verner alternant followed here	**nḗðlō*	regular output: *nǣdl*	*nǣdl*	exact match between the form followed here and the attested OE noun
later hardening stage	*nḗdlō	intermediate pre-OE stage in the same derivation	*nǣdl*	genuine stage in the pathway, but not the Proto-Germanic form followed here

8.20 nose — OE *nosu*

Derivation: citation reconstruction **nasō*; form followed here **núsō* > *nosu* (early analogy).

Derivation trace

Proto input: **núsō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc U Lowering	**nósō*	
PNWGmc Final Long O Raising	**nósu*	
Early Anglo-Frisian		
[no change]		

Old English form: *nosu*

Reconstruction and comparative evidence

Kroonen reconstructs a Germanic ablaut pair **nasō-* ~ **nusō-* and adds that the root **nus-* is likely to have arisen as a secondary zero grade after a remodeling of the older paradigm (Kroonen 2013, 423). Campbell is more specific for Old English, citing *nosu* ‘nose’ < **nusō* ‘nose’ (Campbell 1959, 44).

The citation reconstruction **nasō* ‘nose’ is therefore best treated as the full-grade comparative headword, while the derivational input **núsō* ‘nose’ represents the remodeled zero-grade line continued by the Old English form discussed here. Orel’s **nasō* ‘nose’ alongside OE *nasu* ‘nose’ preserves the competing full-grade notation and shows that the two lines should not be collapsed without comment (Orel 2003, 320).

Old English evidence

Nosu ‘nose’ is an attested Old English noun. Ringe and Taylor list it among the few surviving early Old English feminine u-stems (Ringe and Taylor 2014, 385). Clark Hall likewise gives *nosu f.* ‘nose’, with genitive-dative singular *nosa*, and cross-refers *nasu* ‘nose’ to *nosu* ‘nose’ (Clark Hall 1960, 810).

The selected OE target is therefore attested *nosu* ‘nose’, not a reconstructed placeholder. The lexicographical record also gives *nasu* ‘nose’ for the full-grade side of the tradition.

Development to Old English

From **núsō* 'nose', the regular path is the one documented by the current trace: **núsō* 'nose' > **nósō* 'nose' > **nósu* 'nose' > *nosu* 'nose'. The early special step lies in the choice of the zero-grade input, not in any late Old English repair.

With that input chosen, the OE development is straightforward. The full-grade line behind **nasō* 'nose' instead points toward *nasu* 'nose', not to the form treated here.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the full-grade comparative line from the remodeled zero-grade input that yields the Old English form.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
full-grade comparative line	*nasō	expected full-grade continuation *nasu*	*nasu*	useful comparative background, but not the Old English-facing input
remodeled zero-grade line	*núsō	regular output: *nosu*	*nosu*	exact match between derivational input and attested OE noun

8.21 sap — OE *sæp*

Derivation: citation reconstruction **sapōn*; form followed here **sápą* > *sæp* (early analogy).

Derivation trace

Proto input: **sápą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**sæpą*
[no change]	OE Heavy Syllable Nasal Apocope	**sæp*
Early Anglo-Frisian		
[no change]		

Old English form: *sæp*

Reconstruction and comparative evidence

The comparative sources do not give one uniform inherited stem. Kroonen preserves the word family as **saf* 'sap' / **ppan-*, with Old English *sæp* 'sap' (Kroonen 2013, 420). Orel preserves the comparative notation **sapōn* 'sap' and **sapan* 'sap' (Orel 2003, 319).

The derivational input **sápą* 'sap' therefore does not replace those comparative labels. It identifies the OE-facing stem shape that yields the attested noun treated here.

Old English evidence

Clark Hall records *sæp* ‘sap’ (e) n. (Clark Hall 1960, 247). The target is therefore an attested neuter Old English noun. Orel’s plain *sap* ‘sap, juice’ notation belongs to comparative normalization, not to the spelling adopted here for the Old English form (Orel 2003, 319).

Development to Old English

From **sápą* ‘sap’, Anglo-Frisian brightening yields *sæ* ‘sap’, and heavy-syllable nasal apocope then produces *sæp* ‘sap’. That is the regular path documented by the current trace.

The competing comparative lines do not give the same result. The inherited n-stem notation **sapōn* ‘sap’ yields *sape* ‘sap’, while an i-stem continuation from the **sapi-* line leads to *sep* ‘sap’ / *sepe* ‘sap’ rather than to *sæp* ‘sap’. The special step in this entry is therefore the early stem choice, not a late OE paradigm-cell selection.

Stem comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative stem lines from the Old English-facing input.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative n-stem line	**sapōn*	local comparator output: *sape* ‘sap’	*sape* ‘sap’	useful comparative background, but not the source of attested *sæp*
inferred i-stem comparator from **sapi-*	**sapiz*	local comparator output: *sepe* ‘sap’	*sepe* ‘sap’	confirms that an i-triggering stem does not reach the target
selected a-stem input	**sápą*	regular output: *sæp* ‘sap’	*sæp* ‘sap’	exact match between derivational input and attested OE noun

8.22 sea — OE *sǣ*

Derivation: citation reconstruction **sái*; form followed here **sáiwiz* > *sǣ* (early analogy).

Derivation trace

Proto input: **sáiwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE W Loss Before I	**sāi*
EAF Final Z Deletion	**sáiwi*	OE I Umlaut	**sǣi*
Early Anglo-Frisian		OE High Vowel Apocope	**sǣ*
EAF Ai Monophthongization	**sāwi*		

Old English form: *sǣ*

Reconstruction and comparative evidence

Kroonen gives the noun in stem notation as **saiwi-*, an i-stem whose English reflex is cited as OE *sœ* 'sea' (Kroonen 2013, 423). Ringe and Taylor write the fuller form **saiwiz* 'sea' and derive it through **sawi* 'sea' > **sei* 'sea' > OE *sǣ* 'sea' (Ringe and Taylor 2014, § 6.7.1). The comparative headword is therefore shorter than the form required for the English history: **sái* 'sea' names the lexeme, but **sáiwiz* 'sea' preserves the medial **w* and the final high vowel that control the later development.

Old English evidence

The Old English noun is the ordinary word for 'sea'. Kroonen cites it as *sœ* 'sea'; the normalized form here is *sǣ* 'sea' (Kroonen 2013, 423). Campbell likewise treats *sea* as continuing the same **saiui* 'sea' > **sǣi* 'sea' history, with loss of *u*/*w* before *i* (Campbell 1959, § 406).

Development to Old English

Once the fuller i-stem input is chosen, the development is regular. After Proto-West-Germanic monophthongization **sáiwiz* 'sea' > **sāwiz* 'sea' and final **-z* loss **sāwiz* 'sea' > **sāwi* 'sea', the non-initial **w* disappears before unstressed **i*, and the following high vowel fronts the root vowel before final apocope. The documented chain is **sáiwiz* 'sea' > **sāwiz* 'sea' > **sāwi* 'sea' > **sāi* 'sea' > **sǣi* 'sea' > *sǣ* 'sea' (Ringe and Taylor 2014, § 6.7.1; Campbell 1959, § 406).

Stem and stage comparison

The comparison below sets the relevant forms side by side. It separates the abbreviated comparative headword from the fuller i-stem input that yields the Old English form.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
abbreviated comparative headword	**sái*	too short to preserve the **w* … **i* environment needed for the documented chronology	*sǣ*	useful comparative label, but not the Old English-facing input
selected i-stem input	**sáiwiz*	documented regular output: *sǣ*	*sǣ*	exact match between derivational input and Old English target

8.23 sieve — OE *sife*

Derivation: citation reconstruction **síbaz*; form followed here **síbi* > *sife* (early analogy).

Derivation trace

Proto input: **síbi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	PGmc B Allophony	**síβi*
[no change]	OE Med Unstressed I Lowering1	**síβe*
Early Anglo-Frisian		
[no change]		

Old English form: *sife*

Reconstruction and comparative evidence

Kluge-Seebold gives wg. **sibi-* n. ... ae. *sife*, and Campbell groups *sife* 'sieve' with short neuter i-stems such as *spere* (Kluge 2011, 847; Campbell 1959, § 609). The older morphological background is the s-stem **sib-iz* 'sieve', but the derivational input is the normalized i-stem form **síbi* 'sieve'.

Kroonen's nearby **sebjō-* entry belongs to the separate kinship lexeme that yields Old English *sibb* 'sieve', not to the sieve word. Orel's **sibaz* 'sieve' ... OE *sife* 'sieve' preserves a broader handbook notation, but that a-stem shape does not fit the Old English form treated here (Orel 2003, 328).

Old English evidence

Clark Hall gives *sibi* 'sieve' (GL) ... = *sife* 'sieve' and also *sife* 'sieve' n. 'sieve' (Clark Hall 1960, 263). Campbell likewise cites Corpus Glossary *sibi* 'sieve' and treats *sife* 'sieve' as a short neuter i-stem (Campbell 1959, §§ 444, 609). The normalized Old English target is therefore *sife* 'sieve', while *sibi* 'sieve' is an attested earlier spelling rather than a separate lexeme.

Development to Old English

From **síbi* 'sieve', the regular derivation gives **síβi* 'sieve' > **síβe* 'sieve' > *sife* 'sieve'. Medial *b* is realized as a spirant and later written *f*, while the final unstressed *i* lowers to *e*. The older s-stem background **sib-iz* 'sieve' explains the morphology, but the derivational input **síbi* is the immediate pre-Old-English form.

Stem comparison

The comparison below sets the relevant forms side by side. It distinguishes the accepted i-stem line from its rejected competitors.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
ja-stem kinship line	Kroonen **sebjō-* / comparator **sibja*	OE *sibb* 'sieve'	*sibb* 'sieve'	separate lexeme, not the target treated here
a-stem handbook line	**síbaz*	expected *sif*	*sif*	wrong ending for the attested noun
selected i-stem line from older **sib-iz*	**síbi*	documented regular output: *sife* 'sieve'	*sife* 'sieve'; early spelling *sibi* 'sieve'	exact match between derivational input and Old English evidence

8.24 spare — OE *sparian*

Derivation: citation reconstruction **sparēną*; form followed here **spárōjaną* > *sparian* (early analogy).

Derivation trace

Proto input: **spárōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærōjaną*
[no change]	OE A Restoration	**sparōjaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**sparōjan*
[no change]	OE Secondary Nasalization	**sparōjąn*
	OE I Umlaut	**sparējąn*
	OE Unstressed Long Vowel Shortening	**sparejąn*
	OE Weak Tail Reduction	**sparejan*
	OE Intervocalic J Vocalization	**spareian*
	OE Unstressed EI Contraction	**sparian*

Old English form: *sparian*

Reconstruction and comparative evidence

Kroonen keeps the inherited verb under class-III **sparēn-* (Kroonen 2013, 465). Orel similarly preserves **sparēnan* 'spare' (Orel 2003, 362). Ringe and Taylor, however, reconstruct **sparai-* ~ **sparja-* for the English branch and derive the citation verb from a class-II line (Ringe and Taylor 2014, 162, 191). The derivational input **spárōjaną* 'spare' therefore represents the refashioned class-II formation behind Old English *sparian* 'spare', while the citation reconstruction **sparēną* 'spare' remains the inherited comparative headword.

Old English evidence

Campbell says that *sparian* 'spare' does not show the ordinary class-III characteristics, but the Ritual forms, normalized here as *spæria* 'spare', *spær* 'spare', and *spærede* 'spare', together with Vespasian Psalter *spearad* 'spare', point to primitive Old English forms both with and without back vowels (Campbell 1959, § 764). Brunner likewise records Northumbrian *spæria* 'spare', *spærede* 'spare' beside common Old English *sparian* 'spare' and Vespasian Psalter *spearad* 'spare' (Brunner 1965, § 364 Anm. 11). The citation form treated here is *sparian* 'spare'; the Anglian forms are relics of the older formation, not alternative headwords of equal status.

Development to Old English

Once the class-II formation **spárōjaną* 'spare' is chosen, the remaining development is regular. The regular derivation shows brightening, restoration of *a* before the back vocalism of the suffix, later i-mutation within the weak ending, weak-tail reduction, and contraction to *sparian* 'spare'. By contrast, Brunner's rule against further apocope of final *-e* explains why Ritual *spær* 'spare' cannot be the regular continuation of inherited **spárē* 'spare' (Brunner 1965, § 150).

Formation comparison

The comparison below sets the relevant forms side by side. It contrasts the inherited class-III formation with the refashioned class-II one that yields the citation verb.

Formation / comparison	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited class-III infinitive	_*spárēną_	paradigm comparison / probe output: _sparen_	_sparian_ ‘spare’	wrong class and wrong ending for the citation verb
inherited class-III imperative singular	_*spárē_	paradigm comparison / probe output: _spære_	Ritual _spær_ ‘spare’	loss of final _-e_ is not regular, so the relic form cannot control the entry
inherited class-III finite present	_*spárēθi_	paradigm comparison / probe output: _spæreþ_	_spearad_ ‘spare’	attested form is mixed, not a direct continuation of the inherited cell
selected class-II formation	_*spárōjaną_	documented regular output: _sparian_ ‘spare’	_sparian_ ‘spare’	exact match between derivational input and Old English citation form

8.25 staff — OE _stæf_

Derivation: citation reconstruction _*stábiz_; form followed here _*stábaz_ > _stæf_ (early analogy).

Derivation trace

Proto input: _*stábaz_

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		PWGmc Final Bare A Loss	_*stáb_
EAF Final Z Deletion	_*stába_	EAF Brightening	_*stæb_
Early Anglo-Frisian		PGmc B Allophony	_*stæβ_
[no change]			

Old English form: _stæf_

Reconstruction and comparative evidence

The comparative dictionaries do not give one uniform stem class. Kroonen reconstructs an a-stem _*staba-_ (Kroonen 2013, 471). Orel writes _*stábiz_ ‘staff’ ~ _*stábaz_ ‘staff’ (Orel 2003, 368). A direct i-stem input in _*-iz_ ‘staff’ would predict i-mutation in Old English, whereas the attested noun keeps _æ_.

Old English evidence

The Old English noun itself is the ordinary citation form _stæf_ ‘staff’. Luick lists _stæf_ ‘staff’ among closed monosyllables with _æ_ (Luick 1914--1940, 176). Ringe and Taylor pair singular _stæf_ ‘staff’ with plural _stafas_ ‘staffs’ (Ringe and Taylor 2014, 193). The normalized form here is therefore _stæf_; later English _staff_ with _a_ belongs to a later stage of the word’s history.

Development to Old English

With the selected a-stem input, the development is regular. Final **-z* disappears, bare final *-a* is lost, Anglo-Frisian brightening gives *æ* in the closed monosyllable, and medial *b* surfaces as a fricative written *f*. The documented chain is **stábaz* 'staff' > **stába* 'staff' > **stáb* 'staff' > *stæb* 'staff' > *stæf* 'staff'. A direct continuation of **stábiz* 'staff', by contrast, would produce i-mutated *stefe* 'stave, staff' rather than the attested singular.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the rejected i-stem line from the selected a-stem input.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
comparative i-stem line	**stábiz*	expected *stefe* after i-mutation	*stæf*	wrong vowel for the attested singular
mixed comparative notation	Orel **stabiz* ~ **stabaz*; Kluge **stabi-*/a-	source-level stem-class uncertainty	*stæf*	useful comparative background, but not a single OE-facing input
selected a-stem input	**stábaz*	documented regular output: *stæf*	*stæf*	exact match between derivational input and Old English target

8.26 stem — OE *stefn*

Derivation: citation reconstruction **stámnaz*; form followed here **stámniz* > *stefn* (early analogy).

Derivation trace

Proto input: **stámniz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**stæβni*
EAF Final Z Deletion	**stámni*	OE I Umlaut	**steβni*
PNWGmc Mn Dissimilation	**stáβni*	OE High Vowel Apocope	**steβn*
Early Anglo-Frisian			
[no change]			

Old English form: *stefn*

Reconstruction and comparative evidence

Orel reconstructs the Proto-Germanic source as **stámnaz* 'stem, trunk' (also variant **stamniz* 'stem, trunk'), citing Old Norse *stafn* 'stem of a ship', Old Frisian *stevene* 'stem of a ship' (fem.), and Old Saxon *stamn* 'stem' as its main continuants (Orel 2003, 371).

This word is etymologically unrelated to the Old English homonym *stefn* 'voice, sound' / *stemn* 'voice, sound', which descends from a distinct Proto-Germanic **stebnō* 'voice, sound' / **stemnō* 'voice, sound' etymon. The two OE lexemes are distinguished in Clark Hall, Bosworth-Toller, Brunner, and Luick; they collide on the surface forms *stefn* and *stemn* through historically independent developments.

Kroonen relates the 'stem' family to the *stam(m)* 'stem, trunk' group attested in Old High German, Dutch, and German, without reconstructing a separate 'prow' proto-form (Kroonen 2013, 479–80).

Old English evidence

Clark Hall records *stefn* 'stem, trunk, root, prow, foundation', the weak n-stem *stefna* 'prow or stern of a ship', and the related *stofn* 'trunk, stem, branch, shoot' (Clark Hall 1960, 276, 341).

Luick §211 explicitly distinguishes OE *stefn* 'stem, trunk' / *stemn* 'stem, trunk' — noting cognates with Old Saxon *stamn* 'stem' and Middle English *stam* 'stem' — from the separate OE voice/sound word (*stefn* 'voice, sound' / *stemn* 'voice, sound') treated in §75 (Luick 1914--1940, § 211). Brunner §205 lists *stefn, stemn Stamm* among words showing the *fn*/*mn* alternation (Brunner 1965, § 205).

The primary comparison form used here is *stefn* 'stem, trunk, root, prow', the form attested most directly in the semantic range relevant to English *stem*. The n-stem *stefna* 'prow/stern' is a closely related but more narrowly attested nautical specialization.

Development to Old English

The derivation of OE *stefn* 'stem, trunk' from the i-stem input **stámniz* 'stem, trunk' (Orel's attested i-stem variant; the citation reconstruction remains **stámnaz* 'stem, trunk') is now regular and modelled, with multiplicity 1. Live trace under the corrected literal adjacent-*mn* SC022:

step	form
proto input	**stámniz*
EAF Final Z Deletion	**stámni*
PNWGmc Mn Dissimilation (adjacent mn > βn)	**stáβni*
EAF Brightening (á > æ)	**stæβni*
OE i-Umlaut (æ > e, from the i-stem ending)	**steβni*
OE High Vowel Apocope	**steβn*
Old English Orthography (β > f)	*stefn*

The consonantal change mn $\rightarrow$ *βn*/*fn* is attested comparatively: Old Norse *stafn* 'stem of a ship' and Old Saxon *stamn* 'stem' preserve the *fn*/*mn* variants expected from this family. The handbooks treat the change as a pre-Old-English development (Luick §211; Brunner §205), and the FST now encodes it as the historical adjacent *mn* > *βn* (the rule *PNWGmcMnDissimilation* / SC022). The *e* of *stefn* is regular from *á* via brightening (*á* > *æ*) plus i-umlaut triggered by the i-stem ending (*æ* > *e*), so both the vowel and the consonant follow by regular sound change once the i-stem input is selected.

Note the contrast with *heaven*. Here SC022 (*PNWGmcMnDissimilation*) fires directly inside the selected derivation, deriving the labial of *stefn* from the adjacent *-mn-* cluster of **stámniz*. In *heaven*, by contrast, the same historical change *mn* > *βn*/*fn* operated only in the deeper cluster-bearing oblique prehistory; the resulting labial was then generalized into the vowel-bearing **hebun-* stem, so the selected CAPR path **xébun* > *heofon* begins *after* that analogy and does not itself contain SC022 (see `dossier-heaven-paradigm-history-2026.md` §§13–15 and `audits/heaven-sc022-implementation-2026.md`). The earlier cross-syllable *mV...n* proxy — which had formerly fabricated a labial from an intervocalic *m* — has been retired in favour of this literal adjacent *mn* > *βn*. Selecting the i-stem input **stámniz* over the a-stem citation **stámnaz* is a pre-OE input selection; once selected, the Old English development is regular. Hence the classification *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply), not *known_unmodelled*.

Homonym note

Old English *stefn*/*stemn* is a surface homonym for two etymologically unrelated words:

1. *stefn*/*stemn* 'voice, sound': from PGmc **stebnō* (Ringe-Taylor, Orel); via *bn* → *fn* (b-allophony), giving *stefn* 'voice, sound'. This is a different row from 2216.
2. *stefn*/*stefna*/*stofn*/*stemn* 'stem, trunk, prow': from PGmc **stámnaz* 'stem/trunk' (Orel); via the pre-OE cluster change *mn* → *fn*. This is row 2216.

The form followed here, **stébnō*, used in earlier versions of this entry was the wrong homonym's transponent and must not be used here.

Source comparison

Form or label	Status	OE relation	Result
**stámnaz* 'stem, trunk'	Orel's PGmc a-stem citation for the stem/trunk/prow family	lexeme-level citation	retained as PROTO (citation)
**stámniz* 'stem, trunk'	Orel's attested i-stem variant	**selected derivational input**	regular OE output *stefn* (mult 1)
stefn III 'stem, trunk, root, prow'	primary OE target (strong masc.)	stem/trunk sense per Clark Hall	target form
stefna 'prow/stern'	OE n-stem specialization	nautical sense per Clark Hall	comparison form
stofn 'trunk'	OE *o*-grade variant or earlier stage	trunk/stem sense	comparison form
stemn 'trunk'	late West Saxon *fn* → *mn* doublet	secondary form	comparison form
**stébnō* (voice word)	**wrong homonym** — belongs to voice/sound dossier	no relation to stem/trunk sense	**must not be used as row 2216's derivational input**

8.27 swan — OE *swanes*

Derivation: citation reconstruction **swánaz*; form followed here **swánas* > *swanes* (early analogy).

Derivation trace

Proto input: **swánas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**swánæs*
[no change]	OE Unstressed AE Merger	**swánes*
Early Anglo-Frisian		
[no change]		

Old English form: *swanes*

Reconstruction and comparative evidence

The Germanic noun is ordinarily cited as the masculine a-stem **swánaz* 'swan' (Orel 2003, 367). The derivational input **swánas* 'swan' is not a competing lexeme reconstruction. It is the genitive singular **swánas* of the same paradigm.

The question here is therefore one of paradigm cell rather than stem history. The citation form remains **swánaz* 'swan' > *swan* 'swan'; the comparison form is the genitive singular **swánas* 'swan' > *swanes* 'swan'.

Old English evidence

Bright's glossary records the ordinary noun as *swan* 'swan', m. and also gives the exact inflected form *swanes* 'swan', citing the phrase *swanes feðre* (Bright 1971, 441).

The target is therefore an attested Old English genitive singular, not a reconstruction or the ordinary citation lemma.

Development to Old English

From **swánas* 'swan', Anglo-Frisian brightening gives **swánæs* 'swan', and subsequent merger of unstressed *æ* with *e* yields *swanes* 'swan'. The comparison is straightforward once the genitive singular is chosen as the relevant cell.

Paradigm-cell comparison

The comparison below sets the relevant forms side by side. It separates the ordinary citation form from the selected inflected cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**swánaz*	OE headword *swan* 'swan'	*swan* 'swan'	ordinary lexeme line
genitive singular	**swánas*	regular output: *swanes* 'swan'	*swanes* 'swan'	attested cell

8.28 thousand — OE *þūsend*

Derivation: citation reconstruction **θūs-undī*; form followed here **θūs-èndi* > *þūsend* (early analogy).

Derivation trace

Proto input: **θūsèndi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Strip Secondary Stress	**θūsend*
PWGmc Early I Apocope	**θūsènd*		
Early Anglo-Frisian			
[no change]			

Old English form: *þūsend*

Reconstruction and comparative evidence

Kroonen reconstructs the Germanic numeral as **þūsundī-* and cites Old English *þūsend* 'thousand' among its continuations (Kroonen 2013, 554). The derivational input **θūs-èndi* 'thousand' is not the same claim. It is an OE-oriented transponent with the second-member vowel already resolved to *e* and the final high vowel already shortened for apocope.

The chronology must explain why Old English shows *þūsend* ‘thousand’ while related languages such as Old Saxon and Old High German keep *u* in the second syllable? (Kroonen 2013, 554).

Old English evidence

Old English *þūsend* ‘thousand’ is an ordinary citation form, not a selected oblique or paradigm cell. Campbell treats it as a neuter noun with normal case forms (Campbell 1959, § 689). The problem lies in the internal history of the word, not in its lexical status.

Development to Old English

If the old final -*ī* had remained long enough to trigger ordinary double umlaut, Campbell’s rule would point toward a form of the **þȳsend* ‘thousand’ type rather than attested *þūsend* ‘thousand’ (Campbell 1959, § 203). Preserved root *ū* therefore argues that the umlaut-triggering vowel was lost or neutralized before the ordinary OE umlaut could develop.

That early loss, however, does not by itself explain the medial *e*. Luick compares the word with *ærende* ‘errand/message’ and later groups *thousand* with forms reshaped on that pattern (Luick 1914--1940, §§ 198, 492). Viredaz is more cautious, arguing that Old English *e* in this weak position may simply write schwa and so need not prove a unique *ærende* ‘message’-type analogy (Viredaz 2025, § 2.1.4).

The selected transponent **θūs-èndi* ‘thousand’ captures the OE-side state from which the regular derivation reaches *þūsend* ‘thousand’.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the secure chronology from the more interpretive account of the second-syllable vowel.

Stage / interpretation	Candidate form	OE relation	Result
surviving -*ī* with ordinary double umlaut	**þūsundī-* treated as still umlaut-active in OE	would point toward **þȳsend*	excluded by preserved *ū*
early loss of the trigger without further reshaping	**þūsund-* type	explains *ū*, but not why OE alone has medial *e*	incomplete account
selected OE-oriented transponent	**θūs-èndi*	regular output: *þūsend* ‘thousand’	selected modeling input

8.29 timber — OE *timber*

Derivation: citation reconstruction **tímrą*; form followed here **tímbrą* > *timber* (early analogy).

Derivation trace

Proto input: **tímbrą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tímbr*
[no change]	OE Epenthetic Vowel	**tímber*
Early Anglo-Frisian		
[no change]		

Old English form: *timber*

Reconstruction and comparative evidence

Kroonen reconstructs the noun as **timbra-* and cites Old English *timber* 'timber' among its continuations (Kroonen 2013, 517). Ringe and Taylor instead state the history from **timra* 'timber' through West Germanic **timbr* 'timber' to Old English *timber* 'timber' (Ringe and Taylor 2014, 327).

The difference is therefore not over the Old English noun itself. It concerns whether medial *b* belongs in the comparative citation form or appears in an early pre-Old-English stage of the cluster.

Old English evidence

Clark Hall lemmatizes the noun as *timber* 'timber' and also records *timbor* 'timber' as a variant spelling (Clark Hall 1960, 294). The Old English form is thus an ordinary citation noun, not a selected oblique cell or a reconstructed convenience form.

Development to Old English

With the consonantal frame **timbr-* 'timber' in place, the rest of the development is straightforward. Loss of final *-ą* leaves **tímbr* 'timber', and epenthetic *e* in the final cluster yields *timber* 'timber'. Ringe and Taylor's *timra* > *timbr* > OE *timber* and the handbook treatment of this epenthetic vowel point to the same Old English result (Ringe and Taylor 2014, 327; Campbell 1959, §§ 463–464).

Formation comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing consonantal input.

Formation or notation	Candidate form	OE relation	Result
Kroonen's comparative citation	**timbra-*	already matches the consonantal frame of OE *timber* 'timber'	closest comparative support for the derivational input
Ringe-Taylor citation line	**timra* > **timbr* 'timber'	reaches the same OE noun through early cluster expansion	compatible comparative background
modeled input	**tímbrą*	regular output: *timber* 'timber'	Old English-facing input

8.30 wake — OE *wacan*

Derivation: citation reconstruction **wakēną*; form followed here **wákaną* > *wacan* (early analogy).

Derivation trace

Proto input: **wákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wækaną*
[no change]	OE A Restoration	**wakaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**wakan*
[no change]	OE Secondary Nasalization	**wakąn*
	OE Weak Tail Reduction	**wakan*

Old English form: *wacan*

Reconstruction and comparative evidence

Kroonen gives the strong verb as **wakan-* with Old English *wacan* 'wake' (Kroonen 2013, 568). Ringe and Taylor separately derive Old English *wacian* 'wake' from weak **wakai-* ~ **wakja-* (Ringe and Taylor 2014, § 3.3.2).

The difference is therefore lexical and class-based, not graphic. Strong *wacan* 'wake' 'wake up, arise' and weak *wacian* 'wake' 'be awake, watch' belong to related but distinct histories.

Old English evidence

Clark Hall lists *wacan* 'wake' and *wacian* 'wake' as separate headwords (Clark Hall 1960, 338). Bosworth-Toller cautions under *wacan* 'wake': the simplex infinitive itself does not occur, its place seeming to be taken by *wæcnan* 'wake' (Bosworth and Toller 1898, 226).

The target *wacan* 'wake' is therefore best understood as a normalized strong headword for the verb family, not as a directly quoted simplex infinitive. It still remains the correct Old English comparison form for the strong branch.

Development to Old English

With strong **wákaną* 'wake', Anglo-Frisian brightening first gives a form of the **wækaną* 'wake' type. A-restoration then returns *a*, and the ordinary tail reductions yield *wacan* 'wake'. The weak verb *wacian* 'wake' belongs to a different prehistory and is not the expected outcome of this input.

Class comparison

The comparison below sets the relevant forms side by side. It separates the strong and weak verb lines.

Formation / class	Candidate input	OE outcome or comparison	Result
weak class-III / class-II branch	**wakēną*, **wakai-* ~ **wakja-*	OE *wacian* 'wake' and related weak forms	related lexeme, but not the target of this entry
strong class-VI branch	**wákaną*	regular output: *wacan* 'wake'	Old English-facing input
strong normalized headword	*wacan* 'wake'	dictionary comparison form beside attested strong-family forms	correct Old English comparator, though not a directly quoted simplex infinitive

8.31 water — OE *wæter*

Derivation: citation reconstruction **wátną*; form followed here **wátōr* > *wæter* (early analogy).

Derivation trace

Proto input: **wátōr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**wætær*
PWGmc Final Or Lowering	**wátar*	OE Unstressed AE Merger	**wæter*
Early Anglo-Frisian			
[no change]			

Old English form: *wæter*

Reconstruction and comparative evidence

Kroonen reconstructs a heteroclitic noun **watar-* ~ **watan-* and states that the Proto-Germanic material points to **watōr* 'water' and **watenaz* 'water' (Kroonen 2013, 616). Ringe and Taylor likewise start from singular **wator* 'water' before the Old English branch (Ringe and Taylor 2014, § 3.1.4).

The generalized comparative label is therefore broader than the singular form that actually corresponds to Old English *wæter* 'water'. The relevant comparator is the inherited nominative-accusative singular **wátōr* 'water'.

Old English evidence

Bright gives the noun as *wæter* 'water' with the regular paradigm *wæteres* 'water (gen.)', *wætere* 'water (dat.)', *wæter(u)*, *wætera* 'water (gen.pl.)', *wæterum* 'water (dat.pl.)' (Bright 1971, 29). Ringe and Taylor add the dialectal contrast between West Saxon *weeter* 'water' and Mercian *weter* 'water' (Ringe and Taylor 2014, § 6.5.2).

The target is therefore an attested Old English citation form within a normal paradigm. The complication lies on the comparative side of the lexeme, not in Old English attestation.

Development to Old English

From **wátōr* 'water', pre-final *ō* becomes *a*, yielding **watar* 'water' (Ringe and Taylor 2014, § 3.1.4). Anglo-Frisian brightening then gives **wætær* 'water', and merger of unstressed *æ*/*e* yields *wæter* 'water'.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the generalized lexeme label from the singular input that matches the Old English citation form.

Stage or notation	Candidate form	OE relation	Result
generalized comparative label	**wátną*	broader lexeme shorthand, not the singular that corresponds directly to *wæter* 'water'	useful background only
heteroclitic stem notation	**watar-* ~ **watan-*	source-faithful comparative reconstruction	explains why a singular comparator is needed
inherited singular input	**wátōr*	regular output: *wæter* 'water'	Old English-facing input

8.32 whale — OE *hwæl*

Derivation: citation reconstruction **wálaz*; form followed here **xwálaz* > *hwæl* (early analogy).

Derivation trace

Proto input: **xwálaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xwál*
EAF Final Z Deletion	**xwála*	EAF Brightening	**xwæl*
Early Anglo-Frisian			
[no change]			

Old English form: *hwæl*

Reconstruction and comparative evidence

The comparative sources are not uniform. Orel gives **xwalaz* 'whale' and notes some mixed **xwaliz* 'whale' evidence (Orel 2003, 197). Kroonen instead cites **hwali-* (Kroonen 2013, 262).

Both notations agree on inherited initial *hw-/xw-*, but they differ in stem label. The a-stem-like input followed here is closer to Orel's notation than to Kroonen's citation form.

Old English evidence

Clark Hall lemmatizes the noun as *hwal* 'whale', and Bosworth-Toller preserves plural *hwalas* 'whales' (Clark Hall 1960, 170; Bosworth and Toller 1898, 326). The comparison form is normalized here as *hwæl* 'whale' for the singular citation form with Anglo-Frisian fronting.

The plural *hwalas* 'whales' supplies control evidence. It shows the same lexeme with *a* in an open syllable beside singular *hwæl* 'whale' in a closed monosyllable.

Development to Old English

From **xwálaz* 'whale', final *-z* disappears and bare final *-a* is lost. Anglo-Frisian fronting then yields *æ* in the closed monosyllable, and Old English orthography writes *hwæl* 'whale'.

Formation comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative notations from the normalized Old English singular.

Comparative line	Candidate form	OE relation	Result
Orel's citation	**xwalaz*	same stem notation as the modeled singular line	closest comparative support for the derivational input
Kroonen's citation	**hwali-*	same initial cluster, different stem label	important comparative rival, but not the notation followed here
modeled input	**xwálaz*	regular output: *hwæl* 'whale'	Old English-facing input
plural control	*hwalas* 'whales'	attested open-syllable plural beside singular *hwæl* 'whale'	confirms that the lexeme also preserves an *a*-vocalism branch

8.33 whine — OE *hwīnan*

Derivation: citation reconstruction **wainōjaną*; form followed here **xwī́naną* > *hwīnan* (early analogy).

Derivation trace

Proto input: **xwī́naną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xwī́nan*
[no change]	OE Secondary Nasalization	**xwī́nąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**xwī́nan*
[no change]		

Old English form: *hwīnan*

Reconstruction and comparative evidence

The citation reconstruction preserved in the header belongs to the lament-family verb seen in German *weinen* 'whine' and Old English *wānian* 'wane'. Kroonen instead separates Old English *hwīnan* 'whine' under **hwinan-* (Kroonen 2013, 267). Orel likewise distinguishes strong **xwinanan* 'whine' from weak **wainōjanan* 'whine' (Orel 2003, 201). Ringe and Taylor make the same split at the Northwest Germanic level, linking Old Norse *hvina* 'whine' and Old English *hwinan* 'whine' to the same strong verb (Ringe and Taylor 2014, 130).

The two families also differ phonologically and morphologically. The lament family has initial *w*-, diphthongal *ai*, and weak-II morphology, whereas the verb behind Old English *hwīnan* 'whine' has initial *hw-/xw-*, long *ī*, and strong-verb inflection. The derivational input **xwī́naną* 'whine' therefore represents a competing comparative identification rather than a hidden cell of **wainōjaną* 'whine'.

Old English evidence

Clark Hall records *hwinan* 'whine' with the gloss 'to hiss, whizz, whistle' (Clark Hall 1960, 171). Seebold keeps the verb among the strong verbs and notes that only a present-tense attestation is directly preserved (Seebold 1970, 280).

The Old English form is normalized here as *hwīnan* 'whine'. That normalization adds the usual vowel length marking to the dictionary spelling *hwinan* 'whine'; it does not turn an unattested verb into a reconstructed one.

Development to Old English

Once the strong-verb input **xwī́naną* 'whine' is selected, the path to Old English is straightforward. The compact trace shows heavy-syllable nasal apocope, secondary nasalization, and weak-tail reduction, after which the form surfaces as *hwīnan* 'whine'.

No special paradigm maneuver is needed for this verb. The comparison is between two different Germanic verb families: the Old English form belongs with the strong verb **hwīnan-*, not with the weak lament verb.

Verb-family comparison

The comparison below sets the relevant forms side by side. It separates the competing comparative labels that stand behind the inherited Old English forms.

Verb family / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lament-family weak verb	**wainōjaną*	comparative continuation in OE *wānian* ‘wane’	*wānian* ‘wane’	competing citation reconstruction, but not the source of *hwīnan*
selected strong verb	**xwī́naną*	regular output: *hwīnan* ‘whine’	*hwīnan* ‘whine’	exact match between derivational input and OE verb
comparative North Germanic cognate	Northwest Germanic strong verb behind ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	*hwīnan* ‘whine’	supports the strong-verb identification

8.34 withy — OE *wīþiġ*

Derivation: citation reconstruction **wáiθiz*; form followed here **wī́θagą* > *wīþiġ* (early analogy).

Derivation trace

Proto input: **wī́θagą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wī́θægą*
[no change]	OE Heavy Syllable Nasal Apocope	**wī́θæg*
Early Anglo-Frisian	OE Velar Palatalization	**wī́θæʤ*
[no change]	OE Unstressed AE Merger	**wī́θeʤ*
	OE Late Unstressed Ag Suffix	**wī́θiʤ*

Old English form: *wīþiġ*

Reconstruction and comparative evidence

The comparative evidence groups the word with Germanic forms of the **wīþja* ‘withy’ (or **wīþjō*-variant) or **wiþ-* type (Orel 2003, 503). These forms establish the cognate set but do not by themselves explain the Old English suffix of *wīþiġ* ‘withy’.

For Old English, the relevant point is the suffix history. Campbell’s account of OE -*ig*, including forms such as *hunig*, supports an analysis in which the -*iġ* of *wīþiġ* ‘withy’ continues a derivational **-ag-* sequence rather than a heavy ja-stem **-ij-* (Campbell 1959, §§ 275, 376). The derivational input **wī́θagą* ‘withy’ is thus a formation choice rather than a mere respelling of the comparative headword.

Old English evidence

Clark Hall records the noun as *wiðig* ‘withy’ (Clark Hall 1960, 358). The form used here, *wīþiġ* ‘withy’, is a normalized Old English spelling with macrons and palatal ġ marked explicitly.

The relevant comparison form is therefore not a reconstructed dictionary convenience but an established Old English noun. What requires explanation is why the selected Proto-Germanic input is **wī́θagą* 'withy' rather than a comparative headword of the **wīþja* 'withy' type.

Development to Old English

From **wī́θagą* 'withy', Anglo-Frisian brightening gives a fronted vowel in the suffixal syllable, and, on the Campbell analysis adopted here, the later Old English development of **-ag-* yields *-iġ* (Campbell 1959, §§ 275, 376). Palatalization supplies the final *ġ*, and the full development reaches *wīþiġ* 'withy'.

This derivation is regular for the form compared here. The central claim of the entry is therefore morphological: Old English *wīþiġ* 'withy' belongs with an **-ag-* derivative, whereas the comparative **wīþja* 'withy' label belongs to a different way of presenting the cognate family.

Formation comparison

The comparison below sets the relevant formations side by side. It distinguishes the comparative headword from the Old English-facing formation that actually yields the attested noun.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative family label	**wáiθiz*	broader cognate-set headword	OE family context	useful lexeme label, but not the direct source of *wīþiġ*
heavy ja-stem analysis	**wīþja* 'withy' type	Campbell/Adamczyk-style heavy ja-stem *-e* / zero outcome	*wīþiġ* 'withy'	does not account cleanly for the OE suffix
**-ag-* derivative followed here	**wī́θagą*	regular output: *wīþiġ* 'withy'	*wīþiġ* 'withy'	exact match between formation and target

8.35 world — OE *weorold*

Derivation: citation reconstruction **wíra-àldiz*; form followed here **wír-àldu* > *weorold* (early analogy).

Derivation trace

Proto input: **wíràldu*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Inter Stress Raising	**wéruldu*
PNWGmc I Lowering	**wéràldu*	OE Med Unstressed U Lowering	**wéroldu*
Early Anglo-Frisian		OE Back Mutation	**wéoroldu*
[no change]		OE High Vowel Apocope	**wéorold*

Old English form: *weorold*

Reconstruction and comparative evidence

The word is the old compound 'age of men'. Orel and the **wira-* tradition reconstruct the older *i*-vocalism, while Ringe and Taylor discuss the lowered form **weraldiz* 'world' and its pre-Old-English chain **weraldu* 'world' > **weruld* 'world' (Orel 2003, 501; Ringe and Taylor 2014, 341). Kluge-Seebold likewise gives the compound **wira-aldō* 'world' and explicitly includes Old English *weorold* 'world' (Kluge 2011, 981).

The derivational input **wír-àldu* 'world' therefore combines the older **wir-* vowel of the comparative headword with the early shift of the compound into the ō-stems that Ringe and Taylor note for this lexeme (Ringe and Taylor 2014, 341). The early analogical step lies in that stem-class reassignment; the later phonological developments can then run regularly.

Old English evidence

Old English does not preserve a single isolated form. Ringe and Taylor give West Saxon *weorold* 'world' ~ *worold* 'world', Mercian *weoruld* 'world', Northumbrian *woruld* 'world', and Kentish *wiarald* 'world' (Ringe and Taylor 2014, 341). Sievers-Brunner and Bright present the same wider set, including the syncopated *world* 'world' and later rounded *wurold* 'world' (Brunner 1965, § 113; Bright 1971, 465).

The Old English form used here is the West Saxon form *weorold* 'world'. It is an attested Old English form within that broader variant cluster, not the only form the lexeme ever shows.

Development to Old English

From the derivational input **wír-àldu* 'world', Northwest Germanic *i*-lowering gives **wér-àldu* 'world'. Inter-stress raising then changes the medial *a* to *u*, producing **wér-uldu* 'world'. In the Old English branch that unstressed *u* lowers to *o*, and back mutation yields **wéor-oldu* 'world'; final high-vowel apocope then gives *weorold* 'world'.

This sequence matches the comparative background in Ringe and Taylor's literature-stage chain for the word while preserving the **wir-* notation of the comparative label (Ringe and Taylor 2014, 341). The modeled Old English form therefore stands at the meeting point of an early stem-class reshaping and later regular sound change.

Stage comparison

The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing stage chosen for the derivation.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
comparative compound with older first-element vowel	**wíra-àldiz*	citation reconstruction / lexeme label	preserves the older **wir-* tradition of the compound
literature-stage lowered compound after early stem-class shift	**weraldiz* > **weraldu* > **weruld*	Ringe-Taylor background chain to OE *weorold* 'world' ~ *worold* 'world'	explains the older comparative literature cited for the word
Old English-facing input	**wír-àldu*	regular output: *weorold* 'world'	exact match for the West Saxon form used here
broader OE variant cluster	—	*worold* 'world', *weoruld* 'world', *woruld* 'world', *wiarald* 'world', *world* 'world'	real attested comparanda that remain outside that West Saxon line

8.36 youth — OE _ġeoguþ_

Derivation: citation reconstruction _*júgunθiz_; form followed here _*júgunθ_ > _ġeoguþ_ (early analogy).

Derivation trace

Proto input: _*júgunθ_

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	_*júguθ_
EAF Nasal Spirant Lengthening	_*júgūnθ_		
EAF Nasal Spirant Loss	_*júgūθ_		
Early Anglo-Frisian			
[no change]			

Old English form: _ġeoguþ_

Reconstruction and comparative evidence

The wider etymological tradition reconstructs an earlier form of the word as _*ju(w)unþi-_. The comparative label _*júgunθiz_ 'youth' already stands at a later Germanic stage with _g_, and the derivational input _*júgunθ_ 'youth' is later again: it represents the form after final _-i_ has been lost.

Ringe and Taylor explicitly give the sequence _*jugunþi_ 'youth' > _*juguþ_ 'youth' > OE geoguþ ~ iuguþ (Ringe and Taylor 2014, 141). The derivational input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final _-i_.

Old English evidence

The Old English noun is attested with varying spellings. Ringe and Taylor cite _geoguþ_ 'youth' ~ _iuguþ_ 'youth' (Ringe and Taylor 2014, 141). The form is normalized here as _ġeoguþ_ 'youth': the initial palatal is written with _ġ_, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun _ġeoguþ_ 'youth' itself.

Development to Old English

The decisive early step is the loss of final _-i_ before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted _y_-type vowel instead of the attested form (Ringe and Taylor 2014, 141).

From the derivational input _*júgunθ_ 'youth', the later development is regular: palatal fronting yields _*jéugunθ_ 'youth'; nasal-spirant lengthening and loss give _*jéogūθ_ 'youth' (Fulk 2018, 109); unstressed long-vowel shortening then produces _*jéoguθ_ 'youth', which surfaces as _ġeoguþ_ 'youth'. Campbell preserves _u_ after accented _u_ in forms such as _duguþ_ 'troop' and _munuc_ 'monk' (Campbell 1959, § 374). Brunner likewise cites _iuzuð_ 'youth' and _munuc_ 'monk' under the same development (Brunner 1965, § 150.3).

Stage comparison

The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the later stages relevant to the Old English noun.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
earlier etymological headword	**ju(w)unþi-*	comparative family background	older comparative reconstruction of the lexeme
later g-bearing comparative label	**júgunθiz*	citation reconstruction / lexeme label	preserves the later Germanic stage behind the selected entry
Old English-facing input	**júgunθ*	regular output: *ġeoguþ*	exact match for the Old English form used here
full -*i* stage retained too long	**jugunþi*	expected over-umlauted *y*-type result	negative control showing why early -*i* loss must precede the OE umlaut stage

Chapter 9

Late analogy and paradigm-cell selection

These Old English forms continue a particular paradigm cell or a later analogical remodeling, not the unaltered citation form. The phonology may be regular once the proper morphological history has been identified.

9.1 ban — OE *bannes*

Derivation: citation reconstruction **bánną*; form followed here **bánnas* > *bannes* (late analogy).

Derivation trace

Proto input: **bánnas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**bánnæs*
[no change]	OE Unstressed AE Merger	**bánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *bannes*

Reconstruction and comparative evidence

Orel cites a bann-noun under **bannan* ‘ban’, while Seebold distinguishes bann-stems of both masculine and neuter type and gives Old English *gebann* ‘ban’ as the noun reflex (Orel 2003, 35; Seebold 1970, 89). The citation reconstruction **bánną* ‘ban’ names the lexeme, but the comparison here turns on the genitive singular **bánnas* ‘ban’.

The analysis therefore depends on medial, not final, gemination.

Old English evidence

Old English lexicographic evidence securely supports the noun itself. Bosworth-Toller records the noun under nominative *ge-bann* ‘ban’, with oblique usage such as *gebanne* ‘ban’ (Bosworth and Toller 1898, 303). The exact unprefixed genitive *bannes* ‘ban’ is less directly cited in the dictionaries, so it is best treated here as the regular genitive form used for comparison rather than as a dictionary headword.

Development to Old English

From **bánnas* ‘ban’, the geminate remains medial before the case ending and the unstressed vowel develops regularly to give *bannes* ‘ban’. The paradigm comparison therefore sets the genitive against nominative *ban* ‘ban’, the ordinary nominative form of the same noun, rather than against a directly cited genitive headword.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows why the genitive singular is the conservative cell used for the entry.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bánną*	regular output: *ban*	ban	regular nominative outcome, but not the Old English form here
genitive singular	**bánnas*	regular output: *bannes*	*bannes*	direct match for the conservative genitive

9.2 berry — OE *berġes*

Derivation: citation reconstruction **bázją*; form followed here **bázjas* > *berġes* (late analogy).

Derivation trace

Proto input: **bázjas*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		EAF Brightening	**bærjæs*
Rhotacism	**bárjas*	OE I Umlaut	**berjæs*
Early Anglo-Frisian		OE Unstressed AE Merger	**berjes*
[no change]			

Old English form: *berġes*

Reconstruction and comparative evidence

Kroonen reconstructs the berry noun as **basja-* ~ **bazja-* (Kroonen 2013, 54). The derivational input **bázjas* ‘berry’ is therefore not a rival lexeme headword, but a specific genitive singular cell drawn from that paradigm.

The relevant point is that **rj* did not geminate in Proto-West Germanic. Ringe and Taylor’s *here* ‘army’, *herges* ‘army’s’ comparison shows the same *rj* environment in an Old English paradigm without any hidden gemination repair (Ringe and Taylor 2014, 181).

Old English evidence

Campbell cites feminine *berige* ‘berry’ and notes that *-j-* is retained after *r* in this type (Campbell 1959, 250). The reviewed evidence therefore supports the citation form more directly than the exact genitive *berġes* ‘berry’, which is best read here as the regular genitive comparison form rather than as a dictionary headword.

Development to Old English

Citation **bázją* 'berry' gives *bere* 'berry (variant)', not the Old English form here. The genitive singular **bázjas* 'berry', however, gives *berġes* 'berry', with medial *-rġ-* preserved in the same way that Ringe and Taylor cite *herges* 'army's' beside *here* 'army' (Ringe and Taylor 2014, 181). This points to paradigm choice rather than to an extra phonological rule.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bázją*	regular output: *bere*	*berige* / *berġe*	useful citation-form background, but not the Old English form here
genitive singular	**bázjas*	regular output: *berġes*	*berġes*	exact match for the conservative cell

9.3 bow — OE *bēag*

Derivation: citation reconstruction **béuganą*; form followed here **báug* > *bēag* (late analogy).

Derivation trace

Proto input: **báug*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Fronting	**báeug*
[no change]	OE Diphthong Leveling	**bēag*
Early Anglo-Frisian		
[no change]		

Old English form: *bēag*

Reconstruction and comparative evidence

The inherited verb belongs to the class-II strong-verb family **béuganą* 'bow' (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The derivational input **báug* 'bow' is the singular preterite cell, whereas the citation form **béuganą* is the infinitive.

Campbell's account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag* 'bowed', whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* 'to bow' (Ringe and Taylor 2014, 55).

Old English evidence

Bosworth-Toller and Clark Hall both record *bēag* 'bowed' as a preterite form of *būgan* 'to bow' (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan* 'to bow', but the relevant comparison form for this entry is the singular preterite *bēag* 'bow, ring' 'bowed'. That is the paradigm cell in which the inherited **au* grade is preserved most directly.

Development to Old English

From **báug* 'bow', Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* 'bow, ring' (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan* 'bow, bend'. The derivational input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**béuganą*	inherited present-stem history behind *būgan*	*būgan*	establishes the lexeme, but not the Old English form here
singular preterite	**báug*	regular output: *bēag*	*bēag*	exact match between input, output, and attested cell
past participial branch	participial **bugan-* type	later participial outcomes	bogen-type evidence	relevant to the paradigm, but not the clearest match for this entry

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag* 'bow, ring', while the citation form *būgan* 'bow, bend' belongs to a paradigm whose present stem has already undergone later leveling.

9.4 cow — OE *cȳ*

Derivation: citation reconstruction **kōz*; form followed here **kūi* > *cȳ* (late analogy).

Derivation trace

Proto input: **kūi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**kȳi*
[no change]	OE High Vowel Apocope	**kȳ*
Early Anglo-Frisian		
[no change]		

Old English form: *cȳ*

Reconstruction and comparative evidence

Kroonen reconstructs a root noun with the inherited alternation **kō- ~ *ku-*, explicitly nom. *kōz 'cow', obl. **kū-* (Kroonen 2013, 299). The citation form therefore belongs to the nominative singular, whereas the derivational input **kūi* 'cow' belongs to the oblique stem.

Ringe and Taylor also posit a later PNWGmc nominative **kūaz* 'cow' > **kūz* 'cow' behind Old English *cū* 'cow' (Ringe and Taylor 2014, § 3.1.3). That nominative history accounts for the headword, whereas the form cȳ 'cow' depends on the oblique **kū-* stem and a following **i*.

Old English evidence

Clark Hall lemmatizes the noun under *cū* 'cow' and records gen.sg. *cū* 'cow' (or *cū(e)*), *cȳ* 'cow', *cūs* 'cow (gen.)', dat.sg. *cȳ* 'cow', nom.-acc.pl. *cȳ* 'cow', and dat.pl. *cūm* 'cow' (Clark Hall 1960, 67). Ringe and Taylor likewise state that the root-noun *cū* 'cow' exhibits dat.sg., nom.-acc.pl. *cȳ* 'cow' < **cūi* 'cow', dat.pl. *cūm* < **cūm*, and apparently gen.sg. *cā* 'cow (gen.)' < **cūiz* 'cow (gen.)' (Ringe and Taylor 2014, § 6.6.1).

The dictionary headword is therefore *cū* 'cow', but *cȳ* 'cow' is an established Old English paradigm form rather than a convenient reconstruction. It serves as dative singular and also as nominative-accusative plural. The genitive singular is less stable, with *cā* 'cow (gen.sg.)', *cȳ* 'cow', *cū(e)*, and *cūs* 'cow (gen.sg.)' all appearing in the local source record.

Development to Old English

**kūi* 'cow' is the dative singular of the oblique **kū-* stem. The following **i* triggers i-umlaut, so *ū* becomes *ȳ*, and loss of the final high vowel leaves *cȳ* 'cow'. The development is **kūi* 'cow' > **kȳi* 'cow' > **kȳ* 'cow' > cȳ.

This is the regular oblique-cell path recognized by Ringe and Taylor's paradigm statement (Ringe and Taylor 2014, § 6.6.1). It also explains why *cȳ* 'cow' is the cleanest comparison form for the present entry, even though the ordinary headword is *cū* 'cow'.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*kōz	OE headword *cū* 'cow' belongs to the nominative history of the lexeme	*cū*	useful background, but not the chosen comparison for *cȳ* 'cow'

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
later generalized nominative	PNWGmc *kūaz* > kūz	inferred nominative *cū* 'cow'	*cū*	explains the leveled headword, not the oblique target
dative singular oblique	*kūi	regular output: *cȳ* 'cow'	*cȳ*	exact match between input, output, and paradigm cell
genitive singular oblique	*kūiz	Ringe-Taylor: apparently *cā* 'cow (gen.sg.)'; Hall: *cū(e)*, *cȳ*, *cūs*	gen.sg. variable	too unstable to control the entry

The dative singular is the relevant comparison form. It gives a regular path to attested *cȳ* 'cow', while the broader Old English paradigm shows how far the oblique **kū-* grade spread beyond that one cell.

9.5 find — OE *fundene*

Derivation: citation reconstruction **fínθanq̨*; form followed here **fúnðanǭ* > *fundene* (late analogy).

Derivation trace

Proto input: **fúnðanǭ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Fronting Early	**fúndœnǭ*
PWGmc Dental Hardening	**fúndanǭ*	OE Unstressed Long Vowel Shortening	**fúndœnœ*
Early Anglo-Frisian		OE Unstressed AE Merger	**fúndene*
[no change]			

Old English form: *fundene*

Reconstruction and comparative evidence

The inherited verb is the strong verb **fínθanq̨* 'find', continued by Old English *findan* 'find' (Ringe and Taylor 2014, 344). The form followed here, **fúnðanǭ* 'find', belongs to the past-participial paradigm rather than to the infinitive. It represents an oblique singular form of the participle.

The familiar dictionary form *funden* 'found' is not the form compared here. The derivational input instead models an attested participial form directly, rather than treating the infinitive or the ordinary dictionary headword as primary. It therefore reaches *fundene* 'found (p.p.)' in the form where the trace and the attested evidence match directly.

Old English evidence

Bosworth-Toller records *fundene* 'found (p.p.)' 'found' under *findan* 'find', citing the form in *Beón þá herigeata swa fundene* (Bosworth and Toller 1898, 219). Clark Hall likewise preserves the participial stem in forms such as *funden* 'found' and *tō-fundennes* 'finding' (Clark Hall 1960, 124).

The ordinary dictionary headword for the participle is *funden* ‘found’, but the relevant comparison form for this entry is the attested oblique *fundene* ‘found (p.p.)’ ‘found’. It is an Old English form in its own right, not a merely convenient probe.

Development to Old English

From **fúnðanǭ* ‘find’, the participial oblique develops through regular loss and weakening of the final ending, yielding *fundene* ‘found (p.p.)’. In that cell both the consonantism and the medial vowel history remain regular.

The broader participial paradigm fixes the interpretation. The more familiar nominative *funden* ‘found (p.p.)’ is the ordinary dictionary form, whereas the oblique form *fundene* ‘found (p.p.)’ is the attested form compared here.

Paradigm comparison

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**fínθaną*	inherited verb *findan*	*findan*	establishes the lexeme, but not the form compared here
nominative participial line	**fúnðanaz*	ordinary dictionary *funden* type	*funden*	important paradigm background, but not the form compared here
oblique participle compared here	**fúnðanǭ*	regular output: *fundene*	*fundene*	exact match between input, output, and attested cell

The oblique participle is the relevant comparison form. It matches the derivational input and Old English form directly, while the nominative participial headword remains a different presentation cell within the same paradigm.

9.6 fright — OE *fyrhte*

Derivation: citation reconstruction **furxtīn*; form followed here **fúrxtīnaz* > *fyrhte* (late analogy).

Derivation trace

Proto input: **fúrxtīnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fúrxtīn*
EAF Final Z Deletion	**fúrxtīna*	OE I Umlaut	**fyrxtīn*
Early Anglo-Frisian		NWGmc In Stem N Loss	**fyrxtī*
[no change]		OE Unstressed Long Vowel Shortening	**fyrxti*
		OE Med Unstressed I Lowering1	**fyrxte*

Old English form: *fyrhte*

Reconstruction and comparative evidence

The noun belongs to the inherited in-stem abstract **furxtīn* 'fright', the same family as Gothic *faurhtei* 'fear, fright' (Orel 2003, 120). The derivational input **fúrxtīnaz* 'fright' is not a different lexeme but an oblique singular cell within that in-stem paradigm.

Ringe and Taylor treat the later nominative forms with *-u* or *-o* as analogically remodeled (Ringe and Taylor 2014, 395–96). I therefore use the oblique in-stem form, which continues the older formation rather than the remodeled nominative.

Old English evidence

Bosworth-Toller records *fyrhte* 'fright' with textual attestation, and it also records nominative forms such as *fyrhtu* 'fright' and *fyrhto* 'fright' (Bosworth and Toller 1898, 160). Clark Hall lists adjective and verb material separately under *fyrht* / *fyrhtan* (Clark Hall 1960, 141).

The relevant comparison form is therefore the attested oblique *fyrhte* 'fright'. The nominative lemma forms remain part of the Old English evidence, but the Old English form here of this entry is the oblique cell.

Development to Old English

From **fúrxtīnaz* 'fright', the oblique in-stem develops through the loss and weakening of the final ending, yielding *fyrhte* 'fright'. The form compared here therefore follows the ordinary Old English reduction of the abstract ending in this paradigm.

The later nominative forms with *-u* or *-o* belong to a subsequent analogical reshaping of the paradigm. The Old English form here is earlier in that sense: it is the attested OE cell in which the inherited in-stem development remains most transparent.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested oblique form from the later remodeled nominative line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation in-stem headword	**furxtīn*	broader noun-class label	wider family context	useful lexeme label, but not the cell compared here
remodeled nominative line	nominative in-stem forms	*fyrhtu* 'fright' / *fyrhto* 'fright' type lemma forms	*fyrhtu* 'fright' / *fyrhto* 'fright'	genuine OE evidence, but later remodeled
selected oblique singular	**fúrxtīnaz*	regular output: *fyrhte* 'fright'	*fyrhte* 'fright'	exact match between input, output, and attested cell

The oblique in-stem form is the relevant comparison form. It yields attested *fyrhte* 'fright' directly, while the more familiar nominative forms belong to a later analogical layer.

9.7 hammer — OE *hameres*

Derivation: citation reconstruction **xámaraz*; form followed here **xámaras* > *hameres* (late analogy).

Derivation trace

Proto input: **xámaras*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xámæræs*
[no change]	OE Unstressed AE Merger	**xámeres*
Early Anglo-Frisian		
[no change]		

Old English form: *hameres*

Reconstruction and comparative evidence

The inherited noun is the masculine a-stem **xámaraz* 'hammer', reflected in Old English citation forms such as *hamor* 'hammer' and *hamer* 'hammer' (Kroonen 2013, 206; Orel 2003, 197; Clark Hall 1960, 160). The derivational input **xámaras* 'hammer' is the genitive singular of that same noun rather than a different lexeme.

The citation tradition is already mixed in its unstressed vowel, while the genitive singular gives a closer comparison form. This is a cell choice within one paradigm, not a change of stem class.

Old English evidence

Bosworth-Toller directly records *hameres* 'hammer' in an Old English genitival phrase (Bosworth and Toller 1898, 78). Clark Hall preserves the simplex headword as *hamer* 'hammer' / *hamor* 'hammer' (Clark Hall 1960, 160).

Sievers-Brunner gives a paradigm line *hamor* 'hammer' — *hamores* 'hammer', which shows that the oblique tradition itself was not entirely uniform (Brunner 1965, § 245). The relevant comparison form here is the attested genitive singular *hameres* 'hammer'.

Development to Old English

From **xámaras* 'hammer', Anglo-Frisian brightening and the subsequent merger of unstressed *æ* with *e* yield *hameres* 'hammer'. The derivation of that oblique form is straightforward once the genitive singular cell is selected.

The noun as a whole retains a mixed citation tradition in *hamor* 'hammer' and *hamer* 'hammer', and the selected oblique cell avoids making that variation carry the argument of the entry.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the attested genitive singular from the less stable citation tradition.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	_*xámaraz_	regular citation form _hamer_ / _hamor_	_hamor_ / _hamer_	good lexical background, but not the Old English form here
genitive singular	_*xámaras_	regular output: _hameres_	_hameres_	exact match between input, output, and attested cell
later oblique tradition	oblique a-stem forms	_hamores_ type evidence	_hamores_	attested background variant, but not the chosen comparison form

9.8 have — OE _hæfeþ_

Derivation: citation reconstruction _*xabēną_; form followed here _*xábēθi_ > _hæfeþ_ (late analogy).

Derivation trace

Proto input: _*xábēθi_

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	_*xæbǣθ_
PWGmc Early I Apocope	_*xábēθ_	OE Velar Fricative Palatalization	_*çæbǣθ_
PNWGmc Long E Lowering	_*xábǣθ_	PGmc B Allophony	_*çæβǣθ_
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	_*çæβæθ_
[no change]		OE Unstressed AE Merger	_*çæβeθ_

Old English form: _hæfeþ_

Reconstruction and comparative evidence

The verb belongs to the inherited class-III weak paradigm usually cited under _*xabēną_ ‘have’ and Old English _habban_ ‘have’ (Kroonen 2013, 237; Ringe and Taylor 2014, 93). Within that paradigm, however, the infinitive and the singular present indicative do not continue the same stem. Ringe and Taylor distinguish a _-ja-_ stem in the infinitive from a non-geminating -ai- / _-ē-_ stem in the 2sg and 3sg present forms (Ringe and Taylor 2014, 93).

The derivational input _*xábēθi_ ‘have’ is therefore the 3sg present cell rather than a rephrasing of the infinitive. For the present analysis, that finite cell is the cleaner comparator for the inherited non-geminating stem.

Old English evidence

The ordinary Old English headword is _habban_ ‘have’ (Clark Hall 1960, 157). Campbell’s Anglian paradigm includes unsyncopated 3sg forms of the _hæfed_ ‘has’ type, and the present paradigm therefore shows forms of the _hæf-_ type that support the normalized target _hæfeþ_ ‘has’ (Campbell 1959, § 762).

The target form is therefore a normalized finite cell rather than the ordinary dictionary lemma. It represents the inherited non-geminating present stem more directly than *habban* 'have' does.

Development to Old English

From **xábēθi* 'have', the finite form yields *hæfeþ* 'has' regularly. Ringe and Taylor discuss this non-geminating present stem under *habban* 'have' (Ringe and Taylor 2014, 364). Campbell's Anglian paradigms include unsyncopated 3sg forms of the *hæfeþ* 'has' / *hæfed* 'has' type (Campbell 1959, § 762).

The wider lexeme is less straightforward only because the infinitive *habban* 'have' shows later leveling. The 3sg present cell is therefore the closer comparison form.

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the analogically leveled citation form from the regular 3sg present line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*-ja-* stem of **xabēną*	citation form *habban*	*habban*	important headword, but shaped by later leveling
3sg present	**xábēθi*	regular output: *hæfeþ*	*hæfeþ*	exact match between input, output, and finite form compared here
syncopated finite tradition	same present stem	*hæfþ* type evidence	*hæfþ*	genuine later OE finite form, but not the normalized target used here

9.9 live — OE *lifeþ*

Derivation: citation reconstruction **libēną*; form followed here **líbēθi* > *lifeþ* (late analogy).

Derivation trace

Proto input: **líbēθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PGmc B Allophony	**líβǣθ*
PWGmc Early I Apocope	**líbēθ*	OE Unstressed Long Vowel Shortening	**líβæθ*
PNWGmc Long E Lowering	**líbǣθ*	OE Unstressed AE Merger	**líβeθ*
Early Anglo-Frisian			
[no change]			

Old English form: *lifeþ*

Reconstruction and comparative evidence

The inherited verb belongs to the class-III weak family cited by Kroonen under **libēn-*, reflected in Old English *libban* 'live' (Kroonen 2013, 336). Ringe and Taylor show that the paradigm also contained a separate 3sg present stem, continued in late Northumbrian *lifed* 'lives', which they treat as an archaism (Ringe and Taylor 2014, 364).

The derivational input **líbēθi* 'live' therefore represents a finite present cell rather than the citation infinitive. The later lemma tradition also includes remodeled forms such as *lifian* 'live'.

Old English evidence

The ordinary lemma tradition centers on *libban* 'live' and, in later remodeling, *lifian* 'live'. For this entry, however, the relevant comparison form is the archaic 3sg present attested as *lifed* 'lives', here normalized as *lifeþ* 'lives' (Ringe and Taylor 2014, 364; Campbell 1959, § 762).

The target is thus a normalized finite form, not the ordinary dictionary lemma. It preserves the older present-stem history more clearly than the remodeled lemma tradition does.

Development to Old English

From **líbēθi* 'live', regular reduction of the final syllable and later weakening of the unstressed vowel yield *lifeþ* 'lives'. The attested spelling *lifed* 'lives' belongs to the same finite form in late Northumbrian orthography (Campbell 1959, § 762; Ringe and Taylor 2014, 364).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the archaic finite cell from the ordinary infinitival and later remodeled lemma lines.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive line	**libēną*	OE *libban* headword tradition	*libban*	establishes the lexeme, but not the Old English form here
3sg present	**líbēθi*	regular output: *lifeþ*; attested *lifed*	*lifed*, normalized here as *lifeþ*	selected archaic finite cell
later remodeled present tradition	later class-II-type forms	*lifian* and related finite remodeling	*lifian*	genuine OE development, but secondary to the cell compared here

9.10 man — OE *mannes*

Derivation: citation reconstruction **mánnaz*; form followed here **mánnas* > *mannes* (late analogy).

Derivation trace

Proto input: **mánnas*

Earlier Germanic changes	**Old English changes**	
Northwest and West Germanic	EAF Brightening	**mánnæs*
[no change]	OE Unstressed AE Merger	**mánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *mannes*

Reconstruction and comparative evidence

The lexeme-level reconstruction is not uniform. Kroonen cites **mannan-*, and Orel has **mannz* ‘man’ (Kroonen 2013, 354; Orel 2003, 299). The derivational input **mánnas* ‘man’ belongs to a different level: it is the genitive-singular cell chosen for the Old English comparison.

The target is not the ordinary citation form. Its medial geminate precedes the genitive ending.

Old English evidence

Campbell gives the paradigm *mann* ‘man’, *man* ‘man’ / *mannes* ‘man’s / *menn* ’men’ (Campbell 1959, § 621). Sievers-Brunner likewise cites *man* ‘man’ *mannes* ‘man’s (Brunner 1965, § 226). He also explains that word-final simplification underlies forms such as *man* ’man’ beside inflected *monnes* ’man’s (Brunner 1965, § 231). Clark Hall keeps the dictionary headword under *mann* (Clark Hall 1960, 197).

The relevant comparison form is therefore the attested genitive singular *mannes* ‘man’s, not the citation lemma *mann* ’man’.

Development to Old English

Campbell’s paradigm *mann*, *man* / *mannes* / *menn* ‘men’ confirms the selected genitive singular *mannes* (Campbell 1959, § 621). In the present analysis, **mánnas* ‘man’ develops through Anglo-Frisian brightening and later unstressed merger to *mannes*. In this cell the geminate remains medial before *-es*. The citation form behaves differently because word-final gemination was simplified in Old English (Brunner 1965, § 231).

Paradigm comparison

The comparison below sets the relevant forms side by side. It separates the citation-form line from the genitive singular.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mannăz*	expected citation-form outcome *man*	*mann* / *monn*	establishes the lexeme, but not the Old English form here
accusative singular	**mannǫ*	expected *man*	*man*	same word-final simplification as the nominative
dative singular	**mannăi*	expected *manne*	*manne*	preserves medial *nn*, but not the chosen cell

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
genitive singular	**mánnas*	regular output: *mannes*	*mannes*	exact match between input, output, and attested comparator

9.11 meed — OE *meorde*

Derivation: citation reconstruction **mizdō*; form followed here **mízdai* > *meorde* (late analogy).

Derivation trace

Proto input: **mízdai*

Earlier Germanic changes	
Northwest and West Germanic	
PNWGmc Unstressed Ai Monophthongization	**mízdē*
PNWGmc I Lowering	**mézdē*
Rhotacism	**mérdē*
Early Anglo-Frisian	
[no change]	

Old English changes	
OE Breaking	**méordē*
OE Unstressed Long Vowel Shortening	**méorde*

Old English form: *meorde*

Reconstruction and comparative evidence

The lexeme-level reconstruction is **mizdō* ‘meed’, but the derivational input **mízdai* ‘meed’ is a dative-singular cell rather than the citation form. The Old English evidence for the *meord* ‘meed, reward’ side is oblique.

The wider history of competing *mēd* ‘meed, reward’ remains disputed. Kroonen and Fulk explain it through some form of *z*-loss and compensatory lengthening (Kroonen 2013, 410; Fulk 2018, 69), while Orel keeps a doublet analysis (Orel 2003, 311). The comparison here concerns the attested oblique line *meorde*.

Old English evidence

The directly attested forms are obliques: *meorde* ‘meed’ as a dative singular and *meorda* ‘meed’ as a genitive plural (Bright 1971, 328; Bosworth and Toller 1898, 647). Lexicographers reconstruct a bare nominative *meord* ‘meed’ from those obliques, while West Saxon prose more commonly shows the competing doublet *mēd* ‘meed’ (Clark Hall 1960, 214; Bosworth and Toller 1898, 647).

The target of this entry is therefore the attested oblique *meorde* ‘meed’, not the reconstructed lemma *meord* ‘meed’ and not the better-known West Saxon citation form *mēd* ‘meed’.

Development to Old English

Ringe and Taylor give the broader noun history as **mizdo* ‘reward’ > **mizdu* ‘reward’ > OE meord ~ méd (Ringe and Taylor 2014, 99). The fuller oblique-cell path modeled here spells out the intermediate rhotacism, monophthongization, lowering, breaking, and unstressed-shortening steps needed for the selected dative-singular comparison.

This entry therefore follows the attested oblique line within the broader development. It does not depend on a full decision about the history of the competing *mēd* ‘meed, reward’ tradition.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the attested oblique target from the broader lemma history.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mizdō*	inferred lemma outcome *meord* ‘meed’	*meord* ‘meed’	useful background, but the bare lemma is reconstructed rather than directly attested
selected dative singular	**mízdai*	regular output: *meorde* ‘meed’	*meorde* ‘meed’	exact match between derivational input and attested target
genitive singular	**mizdōz*	regular output: *meorde* ‘meed’	*meorde* ‘meed’	converges on the same attested string, but the dat.sg. has the clearest direct support
genitive plural control	plural oblique line	attested *meorda* ‘meed’	*meorda* ‘meed’	confirms the broader oblique tradition, but not the chosen singular target

9.12 night — OE *niht*

Derivation: citation reconstruction **náxtz*; form followed here **náxti* > *niht* (late analogy).

Derivation trace

Proto input: **náxti*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**næxti*
[no change]	OE Breaking	**neaxti*
Early Anglo-Frisian	OE I Umlaut	**niexti*
[no change]	OE Ws Palatal Umlaut	**nixti*
	OE High Vowel Apocope	**nixt*

Old English form: *niht*

Reconstruction and comparative evidence

Ringe and Taylor cite gen.sg. **nahtiz* ‘night’, dat.sg. **nahti* ‘night’, and nom.pl. **nahtiz* ‘night’ for the high-vowel side of the paradigm, and derive West Saxon *niht* ‘night’ from that side (Ringe and

Taylor 2014, 240). The citation reconstruction **náxtz* 'night' therefore belongs to the nominative-like headword, while the derivational input **náxti* 'night' represents the dative-singular cell.

The word later became the model for endingless datives. Ringe and Taylor explicitly explain forms such as *dæg* 'day' by analogy with dat.sg. niht < **nahti* 'night' (Ringe and Taylor 2014, 380).

Old English evidence

Clark Hall lemmatizes *niht* 'night' and cross-references forms such as *neaht* 'night', *neht* 'night', and *nieht* 'night' (Clark Hall 1960, 215). Campbell likewise preserves the fluctuation between *neaht* 'night' and *niht* 'night', giving genitive *nihte, nihtes*, dative *niht, nihte*, nominative plural *niht* 'night', and the contrasting plural-side forms represented by *neahtas* 'night' (Campbell 1959, § 628.3).

The comparison form used here is therefore an attested Old English *niht* 'night', not a reconstructed substitute. The broader lexical record still preserves the non-umlauted side of the paradigm in *neaht* 'night'-type forms.

Development to Old English

Ringe and Taylor derive West Saxon *niht* 'night' from **nahti* 'night' via **nehti* 'night' and **neahti* 'night' (Ringe and Taylor 2014, 240). Campbell and Brunner preserve the contrasting non-umlauted *neaht* 'night'-type forms elsewhere in the paradigm (Campbell 1959, § 628.3; Brunner 1965, § 284).

The modeled path is therefore **náxti* 'night' > **næxti* 'night' > **neaxti* 'night' > **niexti* 'night' > **nixti* 'night' > niht.

Paradigm comparison

The comparison below sets the relevant forms side by side. It identifies the inherited cell that matches the attested Old English form.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**náxtz*	expected non-umlauted outcome *neaht*	*neaht*	useful background, but not the comparison used for *niht*
selected dative singular	**náxti*	regular output: *niht* 'night'	*niht* 'night'	exact match between input, output, and paradigm cell

9.13 rest — OE *ræste*

Derivation: citation reconstruction **rastō*; form followed here **rástōz* > *ræste* (late analogy).

Derivation trace

Proto input: **rástōz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Surviving Bimoric O Unrounding	**rástā*
EAF Final Z Deletion	**rástō*	EAF Brightening	**ræstǣ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**ræstæ*
[no change]		OE Unstressed AE Merger	**ræste*

Old English form: *ræste*

Reconstruction and comparative evidence

Kroonen treats the noun as a feminine ō-stem **rastō-*, continued by Old English *ræst* 'rest' (Kroonen 2013, 445). The derivational input **rástōz* 'rest' therefore does not replace the lexeme-level headword. It identifies one oblique singular cell on the side of the paradigm that yields *ræste* 'rest'.

The source tradition used here labels that cell specifically as genitive singular, but the broader local synthesis of the ō-stem paradigm shows that the oblique singulars converge on the same front-vocalic *ræste* 'rest' side, in contrast to a nominative singular that would remain *rast* 'rest'.

Old English evidence

The ordinary Old English citation form is *ræst* 'rest' (Kroonen 2013, 445). Clark Hall likewise gives *ræst* 'rest' (Clark Hall 1960, 239). Bosworth-Toller also preserves oblique uses of *ræste* 'rest', including prepositional examples such as *on ræste* 'at rest' and *tó ræste* 'to rest' (Bosworth and Toller 1898, 121).

The comparison form used here is therefore an attested oblique *ræste* 'rest', not a reconstructed surrogate. The dictionary headword *ræst* 'rest' remains an equally real part of the Old English record.

Development to Old English

Once final **z* is lost, the derivational input moves through the front-vocalic oblique side of the paradigm rather than the back-vocalic nominative side. In the modeled derivation, the surviving final long vowel is first exposed, unrounded, fronted, shortened, and then reduced to the final *-e* of *ræste* 'rest'.

The key point is the paradigm split. Nominative **rastō* 'rest' yields a regular *rast* 'rest', whereas the selected oblique input yields *ræste* 'rest'. The later citation form *ræst* 'rest' is best explained as leveling from that oblique *ræst-* stem.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the nominative citation form from the oblique singular chosen here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**rastō*	expected regular outcome *rast*	*ræst* 'rest'	useful background, but not the cell that matches attested oblique *ræste*

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
selected oblique singular	**rástōz*	regular output: *ræste* ‘rest’	*ræste* ‘rest’	exact match between derivational input and attested OE oblique form

9.14 shoulder — OE *ṡculdrum*

Derivation: citation reconstruction **skuldrō*; form followed here **skúldramiz* > *ṡculdrum* (late analogy).

Derivation trace

Proto input: **skúldramiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Sk Palatalization	**ʃúldrum*
PNWGmc A To U Before M	**skúldrumiz*		
PWGmc Early I Apocope	**skúldrumz*		
EAF Final Z Deletion	**skúldrum*		
Early Anglo-Frisian			
[no change]			

Old English form: *ṡculdrum*

Reconstruction and comparative evidence

The handbooks do not agree on the reconstruction of the Germanic word. Orel gives **skuldr(j)ō* as a feminine ō-/jō-stem and explicitly notes that Old English *sculdor* ‘shoulder’ is masculine beside OFrisian *skulder* ‘shoulder’, Middle Low German schulder ‘shoulder’, and Old High German *scultra* ‘shoulder’, *scultirra* ‘shoulder’ (Orel 2003, 345). Kroonen reconstructs **skuldra-*, a masculine a-stem, and derives the Old High German feminine forms from **skuldrjōn-* (Kroonen 2013, 478). Ringe and Taylor cite PWGmc **skuldru* ‘shoulder’ for the Old English branch (Ringe and Taylor 2014, 142).

These forms imply different stem classes and different expectations for the Old English inflection. The question is which inflectional cell best aligns with the Old English evidence.

A dative/instrumental plural form **skúldramiz* ‘shoulder’ aligns with the inherited plural ending that later yields Old English *-um*, and it corresponds directly to the attested dative plural discussed below.

Old English evidence

The ordinary Old English headword is *sculdor* ‘shoulder’. Clark Hall lemmatizes *sculdor* ‘shoulder’ as the normal dictionary form (Clark Hall 1960, 257). Bosworth-Toller also preserves the dative plural *sculdrum* ‘shoulder’ (Bosworth and Toller 1898, 85).

Bosworth-Toller’s Supplement records a weak-feminine *sculdra* ‘shoulder’ (Bosworth and Toller 1898, 699), so *sculdra* ‘shoulder’ belongs to the Old English record beside the stronger masculine paradigm headed by *sculdor* ‘shoulder’. Brunner and Luick also record later spellings such as *sceoldor* ‘shoulder’ and the i-mutated dative plural *scyldrum* ‘shoulder’, which reflect secondary phonological and analogical reshaping within Old English (Brunner 1965, § 92.2.a; Luick 1914--1940, 230).

The singular and plural evidence point to different parts of the paradigm. The relevant comparison form here is the attested dative plural *sċuldrum* ‘shoulder (dat.pl.)’. The spelling with *sċ-* is a normalized representation of the same Old English initial cluster.

Development to Old English

Proto-Germanic **skúldramiz* ‘shoulder’ can be interpreted as a dative/instrumental plural form. In this environment the post-tonic *a* before *m* is raised to *u*, giving a form of the **skúldrumiz* ‘shoulder’ type. Unstressed *u* is regularly preserved before *m*, especially in the dative plural ending *-um*: Campbell states this explicitly, and Hogg formulates the same condition for the dative plural inflexion (Campbell 1959, § 373; Hogg 1992, § 3.3.1.3). Brunner points in the same direction by excluding *m* from the environments in which medial *o* became general in West Saxon (Brunner 1965, § 44 Anm. 7).

Subsequent reduction of the ending removes the final **i* and **z*, so that the inflectional ending appears in Old English as *-um*. The initial cluster is written here as *sċ-*, and the development is **skúldramiz* ‘shoulder’ > **skúldrumiz* ‘shoulder’ > **skúldrum* ‘shoulder’ > *sċuldrum* ‘shoulder’.

Paradigm comparison

A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
singular-oriented citation input	**skúldrō*	probe output: *sċoldor*	*sculdor* ‘shoulder’	fails: the singular output has root *o*, not the attested *u*
serious plural-based singular alternative	**skúldru*	probe output: *sċuldor*	*sculdor* ‘shoulder’	close formally, but it compares a plural-stage input with a singular form
dat./inst.pl. input	**skúldramiz* ‘shoulder’	regular output: *sċuldrum* ‘shoulder (dat.pl.)’	*sculdrum* ‘shoulder’	matches both the output and the dative plural comparison form
later weak-feminine singular	—	OE *sculdra* ‘shoulder’	*sculdra* ‘shoulder’	secondary doublet, useful as a control rather than the inherited target

The dative plural line is decisive because it matches both the output and the paradigm cell of Old English *sculdrum* ‘shoulder’. Singular-oriented candidates either lower the root vowel or compare unlike cells.

9.15 shove — OE *sċēaf*

Derivation: citation reconstruction **skéubaną*; form followed here **skáub* > *sċēaf* (late analogy).

Derivation trace

Proto input: **skáub*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Fronting	**skáeub*
[no change]	OE Diphthong Leveling	**skēab*
Early Anglo-Frisian	PGmc B Allophony	**skēaβ*
[no change]	OE Sk Palatalization	**ʃēaβ*

Old English form: *sċēaf*

Reconstruction and comparative evidence

Kroonen reconstructs the strong verb as **skeuban-* ~ **skūban-* and cites Old English present forms *scēofan* 'shove', *scūfan* 'shove' (Kroonen 2013, 444). Those present-system forms belong to the same verb family, but the comparison here uses the singular preterite **skáub* 'shove', not the infinitive.

Old English evidence

The ordinary dictionary verb is *scūfan* 'shove'/*scēofan* 'shove', but the preterite itself is well attested. Bright gives the principal parts *scufan* 'shove', *sceaf* 'shove', *scufon* 'shove', *scofen* 'shove' (Bright 1971, 347). Sweet gives the same paradigm (Sweet 1953, 29). The normalized form here is *sċēaf* 'sheaf', regularizing the attested spellings *sceaf* 'shove' and prefixed *āsceaf* 'shove'.

Development to Old English

From **skáub* 'shove', the development is straightforward. **au* fronts and levels to *ēa*, final **b* becomes a fricative and is written *f*, and initial **sk-* undergoes the usual Old English palatalized spelling in this environment. The derivation therefore gives **skáub* 'shove' > **skáeub* 'shove' > **skēab* 'shove' > **skēaβ* 'shove' > *sċēaf* 'sheaf'.

Paradigm comparison

A paradigm comparison is needed here because the ordinary citation verb and *sċēaf* 'sheaf' belong to different cells of the same strong paradigm. The comparison below is manual.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation infinitive	**skéubaną*	inherited infinitive line *sċēofan*; present system also leveled *scūfan* 'shove'	*scēofan* 'shove' / *scūfan* 'shove'	necessary background, but not the comparison used for *sċēaf* 'sheaf'
1/3 sg. preterite	**skáub*	documented regular output: *sċēaf* 'sheaf'	*sċēaf* 'sheaf'	direct match for the singular preterite
preterite plural	**skúbun*	later leveled plural *scufon* 'shove' beside expected *sċufun* under the corrected cascade	*scufon* 'shove'	poorer comparison for the singular-preterite target

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
past participle	**skúbanaz*	attested participial line *scofen* ‘shove’	*scofen* ‘shove’	valid alternative cell, but not the form compared here

9.16 span — OE *spanne*

Derivation: citation reconstruction **spannō*; form followed here **spánnai* > *spanne* (late analogy).

Derivation trace

Proto input: **spánnai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**spánne*
PNWGmc Unstressed Ai Monophthongization	**spánnē*		
Early Anglo-Frisian			
[no change]			

Old English form: *spanne*

Reconstruction and comparative evidence

Seebold gives Old English *spann* ‘span’ under this noun family (Seebold 1970, 450). The form followed here, **spánnai* ‘span’, is therefore not a rival headword, but the specific dative singular form compared on the model of the feminine ō-stem paradigm (Brunner 1965, § 252; 1965, § 255.2).

Old English evidence

The reviewed lexicographic evidence more directly supports the citation noun *spann* ‘span’ than the exact form *spanne* ‘span’. Clark Hall gives *spann* ‘span’ (Clark Hall 1960, 286), and *spanne* ‘span’ is accordingly treated as the regular dative singular form compared here rather than as a dictionary headword.

Development to Old English

Citation **spannō* ‘span’ yields *span* ‘span’. The oblique cell **spánnai* ‘span’ therefore supplies the conservative comparison form: it preserves the medial geminate and yields *spanne* ‘span’, while citation **spannō* gives the nominative background form.

Paradigm comparison

The comparison below sets the citation form beside the dative singular form compared here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**spannō*	regular output: *span* ‘span’	*spann* ‘span’	useful citation-form background, but not the form compared here
dative singular compared here	**spánnai*	regular output: *spanne* ‘span’	*spanne* ‘span’	exact match for that conservative form

9.17 thistle — OE *þistles*

Derivation: citation reconstruction **θéstilaz*; form followed here **θístilas* > *þistles* (late analogy).

Derivation trace

Proto input: **θístilas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**θístilæs*
[no change]	OE L Adjacent Syncope	**θístlæs*
Early Anglo-Frisian	OE Unstressed AE Merger	**θístles*
[no change]		

Old English form: *þistles*

Reconstruction and comparative evidence

Orel prints **þe(x)stilaz* ‘thistle’ for the lexeme (Orel 2003, 458). The comparative label **θéstilaz* ‘thistle’ therefore remains in view as the lexeme-level headword, while the derivational input **θístilas* ‘thistle’ is a specific genitive singular cell.

Old English evidence

The ordinary simplex headword tradition is broken *þistel* ‘thistle’ / *ðistel* ‘thistle’. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The Old English form here here is the genitive singular *þistles* ‘thistle’, which preserves the same stem in an oblique form where the cluster is medial.

Development to Old English

Campbell’s discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn* ‘token, sign’, *wépn* ‘weapon’, and *botm* ‘bottom’ beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The genitive singular **θístilas* ‘thistle’ therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *þistles* ‘thistle’, while the simplex nominative belongs to the broken headword tradition *þistel* ‘thistle’.

Paradigm comparison

The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**θéstilaz*	computed output: *þistl*	*þistel* ‘thistle’	useful citation-form background, but not the Old English form here
genitive singular	**θístilas*	computed output: *þistles* ‘thistle’	*þistles* ‘thistle’	exact match for the conservative cell

9.18 make (iptv.2sg) — OE *maca*

Derivation: citation reconstruction **makōną*; form followed here **mákô* > *maca* (late analogy).

Derivation trace

Proto input: **mákô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**mækô*
[no change]	OE A Restoration	**makô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**maka*
[no change]		

Old English form: *maca*

Reconstruction and comparative evidence

The make-family belongs to the Old English class-II weak verbs. Campbell cites *lapian, macian* ‘make’ among verbs with restored *a* (Campbell 1959, § 159). Ringe and Taylor place the Germanic verb in the same class, comparing West Germanic continuants such as Old Frisian *makia* ‘make’, Old Saxon *makon* ‘make’, and Old High German *mahhon* ‘make’ (Ringe and Taylor 2014, 191).

The derivational input **mákô* ‘make (iptv.2sg)’ is not the citation form of the lexeme but a finite paradigm cell. Ringe and Taylor give the class-II weak imperative singular as -a < **-ō*, which makes this cell the relevant comparison point for the Old English form treated here (Ringe and Taylor 2014, 314).

Old English evidence

The dictionary headword is *macian* ‘make’ (Clark Hall 1960, 193). The form compared here in this entry is therefore not the lemma but the imperative singular *maca* ‘make’, chosen as a paradigm form beside the headword *macian* ‘make’ and the related finite form *macaþ* ‘makes’.

The lexical history is still that of *macian* ‘make’, but the finite cell isolates the regular outcome of trimoric **ō* more cleanly than the citation form does.

Development to Old English

From **mákô* 'make (iptv.2sg)', Anglo-Frisian brightening first gives **mækô* 'make (iptv.2sg)'. Campbell cites *macian* 'make' among the class-II verbs with restored *a* (Campbell 1959, § 159). Ringe and Taylor's class-II weak imperative singular -a < **-ō* then supports the later finite ending (Ringe and Taylor 2014, 314).

The same development explains why earlier fronted forms of the *mæċa* 'make' type do not control the entry. Once trimoric **ô* is treated as a back-vocalic trigger for restoration, the imperative singular falls into line with the broader *macian* 'make' family.

Paradigm comparison

A paradigm comparison identifies which finite cell of the make-family matches the Old English form chosen here. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**mákōjanąą*	comparative continuation *macian* 'make'	*macian* 'make'	ordinary headword of the verb, but not the finite form compared here
imperative singular	**mákô*	regular output: *maca* 'make'	*maca* 'make'	exact match between input, output, and selected paradigm form
present third singular companion	**mákōθi*	comparative companion *macaþ*	*macaþ*	useful family control, but not the target of this entry

9.19 make (3sg) — OE *macaþ*

Derivation: citation reconstruction **makōną*; form followed here **mákōθi* > *macaþ* (late analogy).

Derivation trace

Proto input: **mákōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mækōθ*
PWGmc Early I Apocope	**mákōθ*	OE A Restoration	**makōθ*
Early Anglo-Frisian		OE Late O Shortening	**makaθ*
[no change]			

Old English form: *macaþ*

Reconstruction and comparative evidence

Kroonen derives the Old English verb from **makōjan-* on the make-family base **maka-* (Kroonen 2013, 350). Ringe and Taylor likewise derive Old English *macian* 'make' from PWGmc **makon* 'make' through **mekojan* 'make' (Ringe and Taylor 2014, 191).

The derivational input **mákōθi* 'makes' is therefore a finite 3sg cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *macian* 'make' (Clark Hall 1960, 193). The relevant comparison form here is normalized present-third-singular *macaþ* 'makes', set beside the dictionary headword and the related imperative singular *maca* 'make'.

Campbell's class-II paradigm gives ordinary 3sg ending *-aþ*, while his dialect survey allows secondary *-e-* spellings in some traditions (Campbell 1959, § 356.4; 1959, § 757). *Macaþ* 'makes' is thus the regular comparison form for the non-*j* 3sg cell.

Development to Old English

After early loss of final *-i*, **mákōθi* 'makes' → **mákōθ* 'makes' after loss of final -i. Anglo-Frisian brightening gives **mækōθ* 'makes', but Campbell lists *macian* 'make' among class-II verbs with restored *a*, so the stem returns to *mak-* before the ending is reduced (Campbell 1959, § 159).

The ending then follows ordinary class-II 3sg development: **makōθ* 'makes' > **makaθ* 'makes' > macaþ_. Campbell's_lufas, -aþ (< -ōsi, -ōþi)_and Ringe and Taylor's discussion of stable class-II_a_in finite non-_j' cells support this sequence (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the make-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**makōjaną*	dictionary headword *macian* 'make'	*macian* 'make'	family background, but not the cell compared here
3sg present	**mákōθi*	regular output *macaþ* 'makes'	*macaþ* 'makes'	exact match
imperative singular companion	**mákô*	related finite form *maca* 'make'	*maca* 'make'	useful control, but not the target

9.20 bore (iptv.2sg) — OE *bora*

Derivation: citation reconstruction **burōną*; form followed here **búrô* > *bora* (late analogy).

Derivation trace

Proto input: **búrô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**bóra*
PNWGmc U Lowering	**bórô*		
Early Anglo-Frisian			
[no change]			

Old English form: *bora*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* 'bore' among its continuants (Kroonen 2013, 85). Ringe and Taylor give the class-II weak imperative singular ending as -a < **-ō* (Ringe and Taylor 2014, 314).

The derivational input **búrô* 'bore (iptv.2sg)' is therefore an imperative cell of the same family, not the citation form of the verb.

Old English evidence

Clark Hall lemmatizes the verb as *borian* 'bore' (Clark Hall 1960, 48). The comparison form here is the normalized imperative singular *bora* 'bore (iptv.2sg)', used beside the headword and the related 3sg form *boraþ* 'bores'.

The imperative is thus a paradigm form rather than a replacement for the dictionary lemma. It is the most direct Old English comparator for the non-*j* finite cell represented by **búrô* 'bore (iptv.2sg)'.

Development to Old English

Northwest Germanic lowering first gives **bórô* 'bore (iptv.2sg)' from **búrô* 'bore (iptv.2sg)', and late shortening of the unstressed long vowel then yields **bóra* 'bore (iptv.2sg)', whence *bora* 'bore'.

Ringe and Taylor's class-II imperative singular -a < **-ō* points to exactly this type of outcome (Ringe and Taylor 2014, 314). The form compared here therefore isolates the regular finite-cell development more cleanly than the remodelled infinitive does.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the bore-family lemma and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
imperative singular	**búrô*	regular output *bora* 'bore'	*bora* 'bore'	exact match
3sg present companion	**búrōθi*	related finite form *boraþ* 'bores'	*boraþ* 'bores'	useful control, but not the target

9.21 bore (3sg) — OE *boraþ*

Derivation: citation reconstruction **burōną*; form followed here **búrōθi* > *boraþ* (late analogy).

Derivation trace

Proto input: **búrōθi*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Late O Shortening	**bóraθ*
PWGmc Early I Apocope	**búrōθ*		
PNWGmc U Lowering	**bórōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *boraþ*

Reconstruction and comparative evidence

Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* 'bore' among its reflexes (Kroonen 2013, 85). The form compared here isolates the finite 3sg cell **búrōθi* 'bores' rather than the infinitive.

Campbell's class-II pattern lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor's account of stable class-II *a* in finite forms make this 3sg cell the relevant comparison form for the ending (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Old English evidence

Clark Hall lemmatizes the verb as *borian* 'bore'; the imperative singular *bora* and present-third-singular *boraþ* 'bores' are the relevant comparison forms (Clark Hall 1960, 48).

Campbell's dialect survey allows secondary *-e-* and *-o-* spellings in 2sg and 3sg class-II forms, but the basic ending remains *-aþ* (Campbell 1959, § 757). The regular comparison form for this non-*j* 3sg cell is *boraþ* 'bores'.

Development to Old English

From **búrōθi* 'bores', loss of final *-i* first gives **búrōθ* 'bores'. Northwest Germanic lowering then produces **bórōθ* 'bores', and late shortening of unstressed *ō* yields **bóraθ* 'bores'. *Boraþ* 'bores' is therefore the regular comparison form for this non-*j* 3sg cell.

Campbell's class-II ending evidence and Ringe and Taylor's discussion of stable *a* in finite non-*j* cells support this sequence (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the bore-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
3sg present	**búrōθi*	regular output *boraþ* 'bores'	*boraþ* 'bores'	exact match
imperative singular companion	**búrô*	related finite form *bora* 'bore'	*bora* 'bore'	useful control, but not the target

9.22 learn (iptv.2sg) — OE *liorna*

Derivation: citation reconstruction **liznōjaną*; form followed here **líznô* > *liorna* (late analogy).

Derivation trace

Proto input: **líznô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**líornô*
Rhotacism	**lírnô*	OE Unstressed Long Vowel Shortening	**líorna*
Early Anglo-Frisian			
[no change]			

Old English form: *liorna*

Reconstruction and comparative evidence

Ringe and Taylor give Old English *liornian* 'learn' and *leornian* 'learn' from a learn-family base of the **lizn-* type (Ringe and Taylor 2014, 38), and Kroonen likewise keeps the weak verb as **liznōn-* (Kroonen 2013, 380). Fulk cites the same Old English family from **liznō-* (Fulk 2018, 127).

The derivational input **líznô* 'learn (iptv.2sg)' is a finite imperative cell of that family, not the citation form of the verb.

Old English evidence

Clark Hall gives the ordinary headword as *leornian* (Clark Hall 1960, 186). Brunner explicitly records *leornian, nordh. auch liorna*, and Campbell notes Northumbrian forms with *io* beside *leornian* (Brunner 1965, § 417 Anm. 10; Campbell 1959, § 123 n. 2).

Liorna can therefore be treated as an attested Northumbrian finite form, while *leornian* 'learn' remains the better-known dictionary headword.

Development to Old English

The form compared here develops regularly as **líznô* 'learn' > **lírnô* 'learn' by rhotacism, then **líornô* 'learn (iptv.2sg)' by breaking before *rn*, and finally **líorna* 'learn' > *liorna* by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's explicit statement that no form of *liornian* stood in an i-umlauting environment support this stem shape (Campbell 1959, § 123 n. 2; Ringe and Taylor 2014, 247). The West-Saxon-looking *eo* forms belong to a different dialectal presentation of the same family.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the learn-family infinitive and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjaną*	Northumbrian *liornian* ‘learn’; dictionary headword often *leornian* ‘learn’	*liornian* ‘learn’ / *leornian* ‘learn’	family background, but not the cell compared here
imperative singular	**líznô*	regular output and Brunner’s Northumbrian *liorna*	*liorna*	exact match
3sg present companion	**líznōθi*	related finite form *liornaþ*	*liornaþ*	useful control, but not the target

9.23 learn (3sg) — OE *liornaþ*

Derivation: citation reconstruction **liznōjaną*; form followed here **líznōθi* > *liornaþ* (late analogy).

Derivation trace

Proto input: **líznōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**líornōθ*
PWGmc Early I Apocope	**líznōθ*	OE Late O Shortening	**líornaθ*
Rhotacism	**lírnōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *liornaþ*

Reconstruction and comparative evidence

Ringe and Taylor give Old English *liornian* ‘learn’ and *leornian* ‘learn’ from a learn-family base of the **lizn-* type (Ringe and Taylor 2014, 38), and Kroonen likewise keeps the weak verb as **liznōn-* (Kroonen 2013, 380). The derivational input **líznōθi* ‘learns’ is the finite 3sg cell of that family, not the citation form of the verb.

For the ending, Campbell’s class-II *-aþ* (< *-ōþi)* and Ringe and Taylor’s discussion of stable class-II *a* in finite forms make the non-*j* 3sg cell the relevant comparison point (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80).

Old English evidence

Clark Hall gives the ordinary headword as *leornian* ‘learn’ (Clark Hall 1960, 186). Brunner records Northumbrian finite forms in *liorn-*, including *liorna* ‘learn’ and *liornes* ‘learning’, beside the West-Saxon-oriented *leornian* tradition (Brunner 1965, § 417 Anm. 10). Campbell likewise notes Northumbrian forms with *io* beside *leornian* (Campbell 1959, § 123 n. 2).

The relevant comparison form here is the normalized 3sg *liornaþ* ‘learns’. The directly cited Old English evidence supports the finite stem *liorn-*; the *-aþ* ending follows the regular class-II 3sg pattern.

Development to Old English

The form compared here develops as **líznōθi* 'learns' > **lírnōθi* 'learns' by rhotacism, then **lírnōθ* 'learns' after early apocope of final *-i*, then **líornōθ* 'learns' by breaking before *rn*, and finally **líornaθ* 'learns' > *liornaþ* 'learns' by late shortening of the unstressed long vowel.

Campbell's Northumbrian *io* evidence and Ringe and Taylor's statement that no form of *liornian* stood in an i-umlauting environment support the stem, while Campbell's class-II ending evidence supports the final *-aþ* (Campbell 1959, § 123 n. 2; Campbell 1959, § 356.4; Ringe and Taylor 2014, 247).

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the learn-family infinitive and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**liznōjanǫ*	Northumbrian *liornian* 'learn'; dictionary headword often *leornian* 'learn'	*liornian* 'learn' / *leornian* 'learn'	family background, but not the cell compared here
3sg present	**líznōθi*	regular output *liornaþ* 'learns'	*liornaþ* 'learns'	exact match
imperative singular companion	**líznô*	regular output and Brunner's Northumbrian *liorna*	*liorna*	useful control, but not the target

9.24 lick (iptv.2sg) — OE *licca*

Derivation: citation reconstruction **likkōną*; form followed here **líkkô* > *licca* (late analogy).

Derivation trace

Proto input: **líkkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**líkka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *licca*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **ekkōn* 'lick (3sg.)' continuing as Old English *liccian* 'lick', Old Saxon *likkon* 'lick', and Old High German *lecchon* 'lick' (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* 'lick' with the same Old English continuation (Orel 2003, 285).

Campbell's weak class-II discussion gives present forms such as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, § 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The form treated here is therefore not that remodeled infinitive but a finite cell in bare trimoric **-ō*.

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* 'lick' (Bosworth and Toller 1898, 614). Campbell cites *liccian* 'lick' among Old English forms with preserved geminate *cc* (Campbell 1959, § 398.1). Brunner likewise cites *liccian* 'lick' (Brunner 1965, § 45 Anm. 3). The Old English evidence therefore establishes the verbal headword and its consonantal frame securely.

The Old English form here in this entry is the imperative singular *licca* 'lick'. It is a paradigm form chosen beside the headword *liccian* 'lick' and the related present *liccaþ* 'licks', not a separately lemmatized citation word.

Development to Old English

With the stem *licc-* established, the remaining development is brief. Campbell's class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support late *-a* in this finite cell (Campbell 1959, § 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). The same stem consonantism that appears in *liccian* 'lick' is preserved here, giving *cc* throughout the finite form.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* 'lick'	*liccian* 'lick'	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**líkkô*	regular output *licca* 'lick'	*licca* 'lick'	exact match between the derivational input and the Old English form here
present third singular companion	**líkkōθi*	regular output *liccaþ*	*liccaþ*	useful family control, but not the target of this entry

9.25 lick (3sg) — OE *liccaþ*

Derivation: citation reconstruction **likkōną*; form followed here **líkkōθi* > *liccaþ* (late analogy).

Derivation trace

Proto input: **líkkōθi*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Late O Shortening	**líkkaθ*
PWGmc Early I Apocope	**líkkōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *liccaþ*

Reconstruction and comparative evidence

Ringe and Taylor give PWGmc **ekkōn* 'lick (3sg.)' continuing as Old English *liccian* 'lick', Old Saxon *likkon* 'lick', and Old High German *lecchon* 'lick' (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* 'lick' with the same Old English continuation (Orel 2003, 285).

The form compared here in this entry is the non-*j* present third singular **líkkōθi* 'licks', not the remodeled infinitive. Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi, -ōþi)* (Campbell 1959, § 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80).

Old English evidence

Bosworth-Toller lemmatizes the verb as *liccian* 'lick' (Bosworth and Toller 1898, 614). The same consonantal frame appears in Campbell's and Brunner's grammatical citations of *liccian* 'lick' (Campbell 1959, § 398.1; Brunner 1965, § 45 Anm. 3). The Old English headword is therefore clear even though the entry here is not about the citation form.

The form treated here is the present third singular *liccaþ* 'licks'. It is a selected paradigm form beside the lemma *liccian* 'lick' and the related imperative *licca* 'lick', not a separately lemmatized headword.

Development to Old English

**líkkōθi* 'licks' first loses final *-i*, giving **líkkōθ* 'licks'. Campbell's class-II present endings lufas, *-aþ* (< *-ōsi, -ōþi)* support the regular 3sg outcome *-aþ* (Campbell 1959, § 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). Because this ending never contains *-j-*, the form does not pass through an i-umlauted *-eþ* stage.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* 'lick'	*liccian* 'lick'	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**líkkô*	regular output *licca* 'lick'	*licca* 'lick'	useful family control, but not the target of this entry

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
present third singular	**líkkōθi*	regular output *liccaþ* ‘licks’	*liccaþ* ‘licks’	exact match between the derivational input and the Old English form here

9.26 show (iptv.2sg) — OE *sċēawa*

Derivation: citation reconstruction **skawōną*; form followed here **skáwô* > *sċēawa* (late analogy).

Derivation trace

Proto input: **skáwô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skǽawô*
[no change]	OE Sk Palatalization	**ʃǽawô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**ʃǽawa*
[no change]		

Old English form: *sċēawa*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* ‘show (iptv.2sg)’ and cites Old English *sceáwian* ‘show’ beside Old Frisian *skawia* ‘show’, Old Saxon *skawōn* ‘show’, and Old High German *scouwōn* ‘show’ (Orel 2003, 337). The derivational input in this entry is not that infinitive but the imperative singular **skáwô* ‘show (iptv.2sg)’, a finite class-II cell with imperative -a < **-ō* (Ringe and Taylor 2014, 314).

The imperative singular provides the direct comparison with the Old English form. The lexical history still belongs to the *sceáwian* ‘show’ verb, but the cell compared here isolates the finite *-a* outcome more clearly than the citation form does.

Old English evidence

Bright lists *scēawian* ‘show’ and explicitly gives the imperative singular *scēawa* ‘show’ under that headword (Bright 1971, 346). The form treated here is therefore an attested finite paradigm form, not a reconstructed convenience form.

The spelling used in this entry is normalized *sċēawa* ‘show’, while Bright’s glossary gives source spelling *scēawa* ‘show’. The ordinary Old English headword remains *scēawian* ‘show’; *sċēawa* ‘show’ is the imperative singular chosen beside it.

Development to Old English

Campbell lists *scéawian* ‘show’ under the West Germanic **auw* ‘show (iptv.2sg)’ developments (Campbell 1959, § 120). Ringe and Taylor’s class-II weak imperative singular -a < **-ō* supports the late finite ending that yields *sċēawa* 'show' (Ringe and Taylor 2014, 314). The result is therefore the expected finite singular form of the *scēawian* 'show' family rather than an analogical replacement of the headword.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	regular output *sċēawian*	*scēawian* 'show'	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**skáwô*	regular output *sċēawa* 'show'	*scēawa* 'show' / normalized *sċēawa* 'show'	exact match between the derivational input and the Old English form here
present third singular companion	**skáwōθi*	regular output *sċēawaþ* 'show'	*sċēawaþ* 'show'	useful family control, but not the target of this entry

9.27 show (3sg) — OE *sċēawaþ*

Derivation: citation reconstruction **skawōną*; form followed here **skáwōθi* > *sċēawaþ* (late analogy).

Derivation trace

Proto input: **skáwōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Aw Long Diphthong	**skḗawōθ*
PWGmc Early I Apocope	**skáwōθ*	OE Sk Palatalization	**ʃḗawōθ*
Early Anglo-Frisian		OE Late O Shortening	**ʃḗawaθ*
[no change]			

Old English form: *sċēawaþ*

Reconstruction and comparative evidence

Orel reconstructs the verb as **skawōjanan* 'show (3sg)' and cites Old English *sceáwian* 'show' beside Old Frisian *skawia* 'show', Old Saxon *skawōn* 'show', and Old High German *scouwōn* 'show' (Orel 2003, 337). The derivational input in this entry is the present third singular **skáwōθi* 'show (3sg)', a finite class-II cell with stable *a* in the 3sg ending (Ringe and Taylor 2014, 80).

Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, § 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The relevant comparison is therefore the 3sg cell itself, not an i-umlauted alternative.

Old English evidence

Bright lists the simplex headword *scēawian* 'show' and the imperative *scēawa* 'show', and under *geond-scēawian* 'survey' also records a third singular *sceawað* 'show' (Bright 1971, 383). The evidence thus establishes the *scēaw-* / *sceawað* finite-cell pattern directly.

The form written here as *sċēawaþ* 'shows' is the normalized simplex comparison form for that weak class-II pattern. It is therefore not a dictionary headword but a finite comparison form aligned with the attested *scēaw-* evidence and the directly cited *sceawað* 'show' ending pattern.

Development to Old English

Campbell lists *scéawian* 'show' under the same West Germanic **auw* 'show (3sg)' development (Campbell 1959, § 120). **skáwōθi* 'show (3sg)' therefore belongs to the *scēaw-* family before the class-II 3sg ending is applied. Campbell's chronology and Ringe and Taylor's stable-*a* discussion show that the class-II 3sg ending gives *-aþ*, not *-eþ* (Campbell 1959, § 356.4; Ringe and Taylor 2014, 80). Because the ending never contains *-j-*, no i-umlaut applies.

Paradigm comparison

The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjaną*	regular output *sċēawian*	*scēawian* 'show'	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**skáwô*	regular output *sċēawa* 'show'	*scēawa* 'show'	useful family control, but not the target of this entry
present third singular	**skáwōθi*	regular output *sċēawaþ* 'shows'	normalized *sċēawaþ* 'shows'; source-side pattern *sceawað*	exact match for the finite form compared here

Chapter 10

Reconstructed Old English comparators

Direct attestation does not supply the required comparator in these entries. The target is an explicitly reconstructed Old English form and carries the corresponding evidential burden.

10.1 knob — OE **cnobba*

Derivation: citation reconstruction **knúppaz*; form followed here **knúbbô* > **cnobba* (reconstructed Old English comparator).

Derivation trace

Proto input: **knúbbô*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**knóbba*
PNWGmc U Lowering	**knóbbô*		
Early Anglo-Frisian			
[no change]			

Old English form: **cnobba*

Reconstruction and comparative evidence

The wider knob-family is not uniform. Kroonen's discussion of the Germanic n-stems points to related voiced and voiceless branches within this group (Kroonen 2011, 297). The citation reconstruction **knúppaz* 'knob' represents the broader cognate-set headword, while the form followed here, **knúbbô* 'knob', represents the voiced weak-noun branch treated here.

The Old English record is uneven. The better attested OE material belongs to the voiceless branch, but the present entry represents the reconstructed OE form that would continue the voiced branch behind later English knob.

Old English evidence

Clark Hall preserves Old English evidence of the *cnoppa* 'knob' type (Clark Hall 1960, 79). Those forms are genuine Old English evidence, but they belong to the voiceless branch of the family.

The target **cnobba* ‘knob’ is a reconstructed Old English form, not a directly attested one. I use it for the voiced branch because attested *cnoppa* ‘knob’ continues the voiceless branch and therefore represents a different prehistory.

Development to Old English

From the weak-noun form followed here, **knúbbô* ‘knob’, the regular Old English outcome is **cnobba* ‘knob’, with Proto-Germanic *kn-* represented in Old English as *cn-* and with the expected weak-noun ending.

The entry therefore does not claim that **cnobba* ‘knob’ is attested. Its claim is different: if the voiced weak-noun branch is the one to be represented, then **cnobba* is the regular Old English form corresponding to that branch.

Reconstruction status

Form	Status	Relevance to this entry
*knúbbô > *cnobba	reconstructed OE form; regular derivation	reconstructed Old English form compared here
cnopp / *cnoppa*	attested OE branch	important control form, but belongs to the voiceless branch
cnæp	attested OE form from another family	not part of the present lexeme line

This remains the most review-sensitive item here, because the choice between reconstructed **cnobba* ‘knob’ and attested *cnoppa* ‘knob’ is still a comparator-policy question rather than a settled point of OE attestation.

10.2 reek — OE **rēac*

Derivation: citation reconstruction **ráukiz*; form followed here **ráukaz* > **rēac* (reconstructed Old English comparator).

Derivation trace

Proto input: **ráukaz*

Earlier Germanic changes		**Old English changes**	
Northwest and West Germanic		OE Au Fronting	**ráeuka*
EAF Final Z Deletion	**ráuka*	OE Diphthong Leveling	**rēaka*
Early Anglo-Frisian		PWGmc Final Bare A Loss	**rēak*
[no change]			

Old English form: **rēac*

Reconstruction and comparative evidence

The wider noun family is represented by **ráukiz* ‘reek’ / **rauki-*, with Old English *rēc* ‘reek’ as the attested noun reflex in the comparative dictionaries (Kroonen 2013, 446; Orel 2003, 338). The derivational input **ráukaz* ‘reek’ is therefore not the lexeme-level headword, but the form used here for the Old English derivation.

Old English evidence

The attested noun is *rēc* ‘reek’, not **rēac* ‘reek’. Clark Hall records *rēc* ‘reek’ as the noun and also preserves related forms such as *rēcels* ‘incense’; Kroonen likewise gives OE *rēc* ‘reek’ under the noun family (Clark Hall 1960, 255; Kroonen 2013, 446). Clark Hall and Seebold also record verbal *rēac* ‘reek’ as the preterite of *rēocan* ‘reek’, but that verbal form is separate from the noun treated here (Clark Hall 1960, 254; Seebold 1970, 380).

The Old English form here **rēac* ‘reek’ is therefore a reconstructed West Saxon noun form, not a directly attested manuscript headword.

Development to Old English

From **ráukaz* ‘reek’, the regular West Saxon development gives **rēac* ‘reek’. The attested noun *rēc* ‘reek’ belongs to the same lexical family, but reflects a later smoothed surface form rather than the regular noun target represented here.

Form note

The distinction here is between an attested noun headword *rēc* ‘reek’ and a reconstructed regular West Saxon target **rēac* ‘reek’. The latter is treated as the modelling target, while the former remains philological background.

10.3 strew — OE **strīeġan*

Derivation: **stráwjaną* > **strīeġan* (reconstructed Old English comparator).

Derivation trace

Proto input: **stráwjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Awj Glide Formation	**stráujaną*
[no change]	OE Au Fronting	**stráeujaną*
Early Anglo-Frisian	OE Diphthong Leveling	**strēajaną*
[no change]	OE Heavy Syllable Nasal Apocope	**strēajan*
	OE Secondary Nasalization	**strēająn*
	OE I Umlaut	**strīejąn*
	OE Weak Tail Reduction	**strīejan*
	OE J Strengthening After Front Diphthong	**strīeʒan*

Old English form: **strīeġan*

Reconstruction and comparative evidence

Kroonen cites the inherited weak verb as **straujan-* and gives Old English *streowian* ‘strew’ as its dictionary continuation (Kroonen 2013, 483). Ringe and Taylor separate the inherited class-I branch from the remodelled class-II branch: Anglian *strēgan* ‘strew’ continues the inherited verb, while West Saxon *streowian* ‘strew’ is remodelled (Ringe and Taylor 2014, § 6.1 n. 27).

Luick groups **strauwjan* ‘strew’ with the same set as **hauwja-* and **kauwjan* ‘chew’, yielding Anglian *strēzan* ‘strew’ beside West Saxon forms of the *hīez* ‘hew’, *ciezan* ‘choose’ type (Luick 1914--1940, § 98). Fulk likewise allows an early West Saxon **striegan* ‘strew’ directly from Proto-Germanic **straujana* ‘strew’ (Fulk 2018, § 4.10 n. 1).

Old English evidence

The attested inherited Old English form is *strēgan* ‘strew’ in Anglian. The attested West Saxon citation forms are *strewian* ‘strew’, *streowian* ‘strew’, and *strēawian* ‘strew’, which belong to the remodelled class-II branch (Ringe and Taylor 2014, § 6.1 n. 27; Campbell 1959, § 753.7).

The target **strīeġan* ‘strew’ is therefore a reconstructed Old English form, not an attested manuscript lemma. It represents the inferred West Saxon reflex of the inherited class-I branch.

Development to Old English

From **stráwjaną* ‘strew’, the inherited West Saxon line passes through **straujaną* ‘strew’, then a fronted **strēajan-* stage, then **strīejan* ‘strew’ by i-umlaut, with retained/strengthened glide after the front diphthong to yield reconstructed **strīeġan* ‘strew’.

This differs from Anglian *strēgan* ‘strew’, where smoothing removes the diphthongal sequence, and from West Saxon *strewian* ‘strew’ / *streowian* ‘strew’ / *strēawian* ‘strew’, where the verb has already been remodelled into class II (Fulk 2018, § 4.10 n. 1; Campbell 1959, § 753.7).

Reconstruction status

Form or branch	Status	Relevance to this entry
strēgan ‘strew’	attested Anglian inherited class-I form	proves that the inherited verb survived into Old English
**strīeġan* ‘strew’	reconstructed West Saxon inherited class-I form; trace-supported	Old English form here
strewian / *streowian* ‘strew’ / *strēawian*	attested remodelled West Saxon class-II forms	genuine OE evidence, but not the inherited branch modeled here

Chapter 11

Known but unmodelled developments

The historical development is sufficiently supported by comparative and philological evidence, but the current transducer cascade does not yet implement the phonological or morphological process required to derive the target. Naming the gap explains the status without pretending that no historical account exists.

11.1 fire — OE *fȳre*

Derivation: **fūri* yields regular *fȳr*; the Old English form here is *fȳre* (known but unmodelled development).

Derivation trace

Proto input: **fūri*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**fȳri*
[no change]	OE High Vowel Apocope	**fȳr*
Early Anglo-Frisian		
[no change]		

Regular outcome: *fȳr*

Old English form: *fȳre*

Reconstruction and comparative evidence

Kroonen places the lexeme in a heteroclitic family **fōr* ‘fire’ ~ **fun-* and explains the front-mutated West Germanic forms from an oblique form of the **fu(w)eri* type (Kroonen 2013, 151). The derivational input **fūri* ‘fire’ therefore does not function as an arbitrary substitute for the headword: it represents the specific inherited cell that supplies the *i* needed for i-umlaut.

The Old English target combines a regular inherited form *fȳr* ‘fire’ with an attested analogical surface form *fȳre* ‘fire’.

Old English evidence

Bosworth-Toller records *fyr* ‘fire’ as the noun ‘fire’ and also preserves oblique *fyre* ‘fire’ in the Old English record (Bosworth and Toller 1898, 288). The first is the regular inherited outcome of the phonological development from the selected input; the second shows the later restoration of a final *-e* within the paradigm.

The entry therefore concerns the relation between a regular inherited oblique input and an attested Old English surface form that has undergone later morphological remodeling.

Development to Old English

From **fūri* 'fire', i-umlaut changes *ū* to *ȳ* (Hogg 1992, § 3.3.3.1). Subsequent loss of the final high vowel after a heavy syllable yields *fȳr* 'fire' (Campbell 1959, § 345). The inherited phonology is complete at that point.

fȳre 'fire' is later than that inherited output. Its final *-e* belongs to analogical restoration in the Old English paradigm rather than to the original Proto-Germanic ending. The form therefore remains a known but unmodelled remodelling: the deterministic phonology is regular, but the attested surface form includes later morphological rebuilding.

Paradigm comparison

The comparison below sets the relevant forms side by side. It distinguishes the lexeme-level etymological background from the inherited cell that actually produces the front-mutated form and from the later analogical surface result.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
lexeme-level heteroclitic headword	**fōr* 'fire' ~ **fun-*	comparative background only	fire family	explains the wider lexeme, but not the selected oblique input
inherited oblique cell	**fūri*	regular output: *fȳr* 'fire'	*fȳr* 'fire'	regular inherited output from the derivational input
later analogical surface form	—	attested *fȳre* 'fire' with restored *-e*	*fȳre* 'fire'	genuine OE target, but not the direct phonological output

11.2 tap — OE *tæppa*

Derivation: **táppô* yields regular *tappa*; the Old English form here is *tæppa* (known but unmodelled development).

Derivation trace

Proto input: **táppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**tæppô*
[no change]	OE A Restoration	**tappô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**tappa*
[no change]		

Regular outcome: *tappa*

Old English form: *tæppa*

Reconstruction and comparative evidence

Orel gives the noun under **tappòn* ‘tap’ and already connects it with Old English *tæppa* ‘tap’ (Orel 2003, 402). The derivational input **táppô* ‘tap’ is therefore the inherited noun itself; the entry does not depend on a different lexeme-level proto or a different inherited noun cell.

Old English evidence

The Old English noun family is well attested. Orel gives *tæppa* ‘tap’, and Clark Hall records *tæppa* ‘tap’ together with derivatives *tæppere* ‘tapster’ and *tæppestre* ‘tapster’ (Orel 2003, 402; Clark Hall 1960, 305). The target is therefore a real Old English noun form, not a reconstructed convenience spelling.

Development to Old English

From **táppô* ‘tap’, the regular inherited noun path gives *tappa* ‘tap’. The attested target *tæppa* ‘tap’ therefore stands outside that regular phonological development.

The mismatch is historically intelligible, but it is not solved here by a new inherited input. A related j-verb pathway would give *teppan* ‘tap’, not the noun target *tæppa* ‘tap’. The entry accordingly remains a known but unmodelled case.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**táppô*	regular output: *tappa* ‘tap’	regular output, but not the target
attested OE target	—	*tæppa* ‘tap’	genuine target form, but analogically remodelled in the present classification
related j-verb background	**táppjaną*	*teppan* ‘tap’	related formation, but not the noun target

Chapter 12

Unexplained or deliberately unmodelled exceptions

No sufficiently supported account yet reconciles the regular output with the Old English form. An ad hoc sound law would conceal rather than explain the mismatch.

12.1 buck — OE *bucc*

Derivation: _*búkkaz_ yields regular *bocc*; the Old English form here is *bucc* (unexplained exception).

Derivation trace

Proto input: _*búkkaz_

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	_*bókk_
PNWGmc U Lowering	_*bókkaz_		
EAF Final Z Deletion	_*bókka_		
Early Anglo-Frisian			
[no change]			

Regular outcome: *bocc*

Old English form: *bucc*

Reconstruction and comparative evidence

Kroonen and Orel both reconstruct the word with a geminate stop, _*bukkaz_ ‘buck’ (Kroonen 2013, 121; Orel 2003, 61). Orel also preserves parallel n-stem material behind Old English *bucca* ‘buck’ (Orel 2003, 62). The derivational input therefore remains identical with the lexeme label: no alternative inherited cell accounts for the form.

Old English evidence

Old English preserves a mixed lexical picture. Campbell cites *bucca* ‘buck’ in the exception set for this phonological environment (Campbell 1959, § 115). Clark Hall and Bosworth-Toller show that Old English has both *bucca* ‘buck’ and *bucc* ‘buck’ (Clark Hall 1960, 53; Bosworth and Toller 1898, 122). The a-stem citation form *bucc* ‘buck’ is the target treated here; *bucca* ‘buck’ supplies genuine philological background from the same lexical family.

Development to Old English

From **búkkaz* 'buck', the regular inherited path gives *bocc* 'buck'. That is the form expected under the ordinary lowering pattern in this environment. *bucc* 'buck' therefore remains outside the deterministic phonology.

No accepted inherited cell repairs the mismatch. A high-vowel alternative would introduce i-umlaut and produce a *byċċ* 'buck'-type form rather than the target. *bucc* 'buck' is therefore best treated as a documented exception, not as a regular paradigm-cell survival.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**búkkaz*	regular output: *bocc* 'buck'	regular output, but not the target
attested OE target	—	*bucc* 'buck'	genuine target form, but unexplained in the present classification
parallel OE lexical background	—	*bucca* 'buck'	related n-stem form, not the present target

12.2 fowl — OE *fugol*

Derivation: **fúglaz* yields regular *fogol*; the Old English form here is *fugol* (unexplained exception).

Derivation trace

Proto input: **fúglaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fógl*
PNWGmc U Lowering	**fóglaz*	OE Epenthetic Vowel	**fógol*
EAF Final Z Deletion	**fógla*		
Early Anglo-Frisian			
[no change]			

Regular outcome: *fogol*

Old English form: *fugol*

Reconstruction and comparative evidence

The noun is the ordinary Germanic a-stem **fúglaz* 'fowl', continued by forms such as Old Norse *fugl* and Old High German *fogal* 'bird, fowl' (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the derivational input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

Old English evidence

Old English dictionaries record the noun as *fugol* 'fowl', with variant spelling *fugel* 'fowl' (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fu-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

Development to Old English

From **fúglaz* 'fowl', the regular cascade yields *fogol* 'fowl': the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol* 'fowl', not *fogol* 'fowl'. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914--1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

Expected and attested forms

The comparison below sets the relevant forms side by side. It distinguishes the regular prediction from the attested Old English noun.

Form	Status	Relevance to this entry
fogol	computed regular output from **fúglaz*	establishes the expected inherited outcome
fugol	attested Old English form	Old English form here; preserves unexplained root *u*
fugel	attested variant spelling	secondary spelling variant of the attested noun

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from **fúglaz* 'fowl' to attested *fugol* 'fowl'.

12.3 rust — OE *rust*

Derivation: **rústō* yields regular *rost*; the Old English form here is *rust* (unexplained exception).

Derivation trace

Proto input: **rústō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**róst*
PNWGmc U Lowering	**róstō*		
PNWGmc Final Long O Raising	**róstu*		
Early Anglo-Frisian			
[no change]			

Regular outcome: *rost*

Old English form: *rust*

Reconstruction and comparative evidence

The comparative dictionaries do not support a single citation reconstruction uniformly. Orel cites **rustaz* 'rust' (sb.m./f.) with Old English *rust* 'rust' and Old Saxon and Old High German *rost* 'rust' (Orel 2003, 308). The form **rústō* 'rust' therefore stands here as a competing citation reconstruction rather than as the best-supported inherited headword.

That disagreement does not remove the central problem. Whether one starts from **rústō* 'rust' or from source-supported **rustaz* 'rust', the regular citation-form history points toward *rost* 'rust', not toward the attested Old English noun.

Old English evidence

The Old English noun is attested, not reconstructed. Clark Hall gives *rūst* 'rust' (m.) (Clark Hall 1960, 245), and Bosworth-Toller records *rúst* 'rust' (also *rust*) (Bosworth and Toller 1898, 677). The form is normalized here as *rust* 'rust' from that attested record.

Those dictionary entries identify a masculine noun, which aligns better with Orel's **rustaz* 'rust' than with the competing **rústō* 'rust' preserved in the header.

Development to Old English

Under Campbell's regular lowering of stressed *u* before a following mid or low vowel, the citation-form input gives *rost* 'rust', not *rust* 'rust' (Campbell 1959, § 115). The same lowering would also affect comparative citation-form reconstructions such as **rustaz* 'rust'.

A high-vowel comparator such as instrumental-type **rústu* 'rust' would yield *rust* 'rust' regularly, but that does not explain the attested citation form of the noun. No accepted regular pathway from the citation form to attested *rust* 'rust' has been established.

Expected and attested forms

The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**rústō* > *rost*	computed regular output from one competing citation reconstruction	shows that this citation reconstruction does not reach attested *rust*
**rustaz* > *rost*	expected regular output from the source-supported citation reconstruction	shows that correcting the stem class does not solve the vowel problem
**rústu* > *rust*	regular high-vowel comparator	useful negative control, but not a defensible citation-form solution
rust	attested Old English noun, normalized from *rūst* / *rúst* / *rust*	attested Old English form; the citation-form development remains unexplained

12.4 wolf — OE *wulf*

Derivation: **wúlfaz* yields regular *wolf*; the Old English form here is *wulf* (unexplained exception).

Derivation trace

Proto input: **wúlfaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wólf*
PNWGmc U Lowering	**wólfaz*		
EAF Final Z Deletion	**wólfa*		
Early Anglo-Frisian			
[no change]			

Regular outcome: *wolf*

Old English form: *wulf*

Reconstruction and comparative evidence

The inherited noun is an a-stem: Kroonen gives **wulfa-*, and Ringe and Taylor list PGmc **wulfaz* 'wolf' among the inherited words that preserve *u* in Old English beside Old High German *wolf* 'wolf' (Kroonen 2013, 638; Ringe and Taylor 2014, 47). Campbell accordingly names Old English *wulf* as an exception to the regular lowering of stressed *u* before a following non-high vowel (Campbell 1959, § 115).

The older literature often notices that the exceptional words cluster near labials. Bülbring lists *full*, *wulle*, and *wulf* together, but he also says that the ordinary rule still gives *o* in comparable forms such as *folc* and *bolt* (Bülbring 1902, § 116). Luick rejects a categorical labial blocker on the same grounds and prefers a lexical or analogical account instead (Luick 1914--1940, 148).

Old English evidence

Campbell treats *wulf* 'wolf' as part of the exceptional *u* set (Campbell 1959, § 115). Sievers-Brunner notes that oblique *wulfe* 'wolf (dat.)' continues *wulfi* 'wolf (dat.)' or older *wulfai* 'wolf (dat.)' (Brunner 1965, § 160).

The surviving oblique forms do not supply a regular route back to bare *wulf* 'wolf'. They belong to the same lexeme, but they do not remove the explanatory problem presented by the citation form.

Development to Old English

Under the ordinary Northwest Germanic lowering of stressed *u* before a following non-high vowel, the citation-form input would point toward *o*-vocalism (Ringe and Taylor 2014, 42–44). The compact trace shows exactly that path: **wúlfaz* 'wolf' > **wólfaz* 'wolf' > **wólfa* 'wolf' > wolf.

A high-vowel oblique input would behave differently. There the following high vowel would block the lowering of *u*, but the same environment would also trigger i-umlaut, so the regular control result would be *wylf* 'wolf' or *wylfe*, not bare *wulf*. The attested noun therefore remains unexplained at the citation-form level.

Expected and attested forms

The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**wúlfaz > wolf*	computed regular output from the citation form	shows the regular development expected from the inherited a-stem
OHG wolf	comparative regular cognate	confirms that the *o*-vocalism is the ordinary outcome
**wúlfi / *wúlfis > wylf / wylfe*	expected high-vowel control forms	shows why oblique high-vowel cells do not solve the noun's vowel history

Form / interpretation	Status	Relevance to this entry
wulf	attested Old English noun	attested Old English form; the preservation of *u* remains unexplained

12.5 wool — OE *wull*

Derivation: **wúllō* yields regular *woll*; the Old English form here is *wull* (unexplained exception).

Derivation trace

Proto input: **wúllō*

Earlier Germanic changes

Northwest and West Germanic

PNWGmc U Lowering	**wóllō*
PNWGmc Final Long O Raising	**wóllu*

Early Anglo-Frisian

[no change]

Old English changes

OE High Vowel Apocope	**wóll*

Regular outcome: *woll*

Old English form: *wull*

Reconstruction and comparative evidence

The inherited form is a feminine ō-stem **wúllō* 'wool'. In the ordinary phonological history of West Germanic, stressed *u* lowers before a following non-high vowel, so the regular Old English outcome is an *o*-form. Campbell's discussion of the parallel adjective *full*, with OHG foll 'full' as the regular comparator, shows that the handbooks treat this as a genuine exception cluster rather than as a place where the rule itself is doubtful (Campbell 1959, § 115).

Bülbring likewise lists *wulle* 'wool' among the traditional *u*-preserving exceptions (Bülbring 1902, § 116). The comparative evidence therefore establishes two things at once: the regular result should be *woll*, and Old English still has a lexical exception of the *wull* 'wool' / *wulle* 'wool' type.

Old English evidence

The Old English target is given here as *wull* 'wool', a normalized lexeme form. Handbook discussion often cites *wulle* 'wool', the feminine weak form of the noun (Bülbring 1902, § 116). Both point to the same lexical item and to the same exceptional preservation of root *u*.

The OE evidence therefore does not remove the problem. It confirms that the language has a *u*-form where the regular phonology would have produced *o*.

Development to Old English

From **wúllō* 'wool', the regular sequence is lowering of stressed *u* before a non-high vowel, followed by the ordinary later reductions of the ending. The regular outcome is therefore *woll*.

That regular derivation is not the attested Old English form. Luick rejects a simple phonological rule that would protect this word alone, and Ringe and Taylor state the larger problem plainly: we do not really know why **u* failed to lower in forms of this sort (Luick 1914--1940, 148; Ringe and Taylor 2014, § 2.3.1).

What remains unexplained

The comparison below sets the regular result beside the attested lexical exception.

Form	Status	Relevance to this entry
**wúllō* > *woll*	regular output	shows what the deterministic sound laws produce
wull / *wulle*	attested OE exception	attested Old English form to be recorded
high-vowel escape from another paradigm cell	unsupported for this noun	rejected because the feminine ō-stem paradigm supplies no suitable escape cell

References

Adamczyk, Elżbieta. 2001. 'Old English Reflexes of Sievers' Law'. *Studia Anglica Posnaniensia: An International Review of English Studies* 36: 61–72. http://hdl.handle.net/10593/18335.

Bammesberger, Alfred. 1990. *Die Morphologie Des Urgermanischen Nomens*. Vol. 2. Untersuchungen Zur Vergleichenden Grammatik Der Germanischen Sprachen. Carl Winter.

Bammesberger, Alfred. 1997. 'Die Vorform von Altenglisch hærfest'. *Anglia: Zeitschrift für Englische Philologie* 115 (2): 223–30. https://doi.org/10.1515/angl.1997.115.2.223.

Bosworth, Joseph, and T. Northcote Toller. 1898. *An Anglo-Saxon Dictionary*. Clarendon Press.

Bright, James W. 1971. *Bright's Old English Grammar and Reader*. 3rd edn. Edited by Frederic G. Cassidy and Richard N. Ringler. Holt, Rinehart; Winston.

Brunner, Karl. 1965. *Altenglische Grammatik nach der angelsächsischen Grammatik von Eduard Sievers*. 3rd edn. Max Niemeyer.

Bülbring, Karl D. 1902. *Altenglisches Elementarbuch. 1: Lautlehre*. Carl Winter.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Clark Hall, J. R. 1960. *A Concise Anglo-Saxon Dictionary*. 4th edn. Cambridge University Press.

Crist, Sean J. 2001. 'Conspiracy in Historical Phonology'. Ph.{D}. dissertation, University of Pennsylvania.

Crist, Sean J. 2002. *An Analysis of *z Loss in West Germanic*. Paper presented at the Linguistic Society of America Annual Meeting.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Hogg, Richard M. 1992. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Hulden, Mans. 2009. 'Foma: A Finite-State Compiler and Library'. *Proceedings of the Demonstrations Session at EACL 2009* (Athens), 29–32.

Kaplan, Ronald M., and Martin Kay. 1994. 'Regular Models of Phonological Rule Systems'. *Computational Linguistics* 20 (3): 331–78.

Kilday, Douglas G. 2024. 'Crist's Law, Smith's Law, and English *Wizen*'. Unpublished manuscript.

Kluge, Friedrich. 2011. *Etymologisches Wörterbuch der deutschen Sprache*. 25th edn. Edited by Elmar Seebold. De Gruyter.

Kroonen, Guus. 2011. *The Proto-Germanic n-Stems: A Study in Diachronic Morphophonology*. Vol.

18. Leiden Studies in Indo-European. Rodopi.

Kroonen, Guus. 2013. *Etymological Dictionary of Proto-Germanic*. Vol. 11. Leiden Indo-European Etymological Dictionary Series. Brill.

Lloyd, Albert L. 1966. 'Is There an a-Umlaut of i in Germanic?' *Language* 42 (4): 738–45. https://doi.org/10.2307/411829.

Luick, Karl. 1914--1940. *Historische Grammatik der englischen Sprache*. Tauchnitz.

Orel, Vladimir. 2003. *A Handbook of Germanic Etymology*. Brill.

Osthoff, Hermann, and Karl Brugmann. 1878--1910. *Morphologische Untersuchungen auf dem Gebiete der indogermanischen Sprachen*. 6 vols. S. Hirzel.

Ringe, Don. 2017. *From Proto-Indo-European to Proto-Germanic*. 2nd edn. Vol. 1. A Linguistic History of English. Oxford University Press.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.

Seebold, Elmar. 1970. *Vergleichendes und etymologisches Wörterbuch der germanischen starken Verben*. Mouton.

Sims-Williams, Patrick. 2018. 'Mechanising Historical Phonology'. *Transactions of the Philological Society* 116 (3): 555–73. https://doi.org/10.1111/1467-968X.12138.

Stausland Johnsen, Sverre. 2015. 'Vowel Reduction in Verbs in King Alfred's *Pastoral Care*'. In *Historical Linguistics 2013: Selected Papers from the 21st International Conference on Historical Linguistics, Oslo, 5–9 August 2013*, edited by Dag T. T. Haug. Current Issues in Linguistic Theory 334. John Benjamins. https://doi.org/10.1075/cilt.334.02jon.

Stiles, Patrick V. 1985. 'The Fate of the Numeral "4" in Germanic (1)'. *NOWELE: North-Western European Language Evolution* 6: 81–104. https://doi.org/10.1075/nowele.6.06sti.

Sweet, Henry. 1953. *An Anglo-Saxon Primer*. 9th edn. Clarendon Press.

Versloot, Arjen P. 2017. 'Proto-Germanic *Ai in North and West Germanic'. *Folia Linguistica Historica* 38: 281–324. https://doi.org/10.1515/flih-2017-0010.

Viredaz, Rémy. 2025. *Germanic, Slavic and Baltic 'Thousand' Once More*. Unpublished manuscript (unfinished), Geneva, 20 October 2025. https://www.academia.edu/144462167/2025_Germanic_Slavic_and_Baltic_thousand_once_more_full_text_unfinished_.

Index verborum

Proto-Germanic